

The cosmology of Cosmological Relativity: the expansion history from causal reassignment on a de Sitter background, the cosmogenesis branch point, and the scalar perturbation sector

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Abstract

Cosmological Relativity (CR) obtains the observed cosmological expansion by a causal reassignment on a de Sitter background at the cosmogenesis branch point, with the Einstein field equations unchanged. This paper develops that cosmology and completes it with its scalar perturbation sector. The reassignment selects the Nariai member [35] of Schwarzschild–de Sitter, whose proper frame is a homogeneous, observationally isotropic, *non-synchronous* cosmology with areal radius $r(\tilde{\tau}) \propto \sinh^{2/3}$ —exactly the flat- Λ CDM scale factor—its rate fixed by the cosmological constant alone, with radiation and matter inherited content read off the cosmic clock rather than terms that source the expansion. The cosmic beginning is the branch point $r = 0$ —a genuine curvature singularity of the SdS metric read over the r -chart [34], and *not* a singularity of the underlying substrate, whose throat is the defining constant α ; the divergence of the r -chart invariants there is the areal coordinate degenerating; the constant-proper-time slices are exactly Euclidean ($\Omega_k = 0$) while the cosmological layers are a closed S^3 —spatial curvature and dark energy one Λ read on two slicings. The acoustic scale and the apparent Hubble tension resolve as consequences of a rate fixed by the substrate’s geometry: there is no second H_0 to reconcile, and the scale is met at the directly measured H_0 by a single inherited datum, $\rho_r/\rho_m \approx 2$ —a one-parameter accommodation, the structural analogue of the baryon-to-photon ratio. The light-element composition is produced on the same footing: the infalling matter heats above the deuterium bottleneck and re-expands through it, so the cooling leg is a standard big-bang nucleosynthesis fixing the composition with the inherited baryon-to-photon ratio η —a second boundary datum of the same handover, distinct from the radiation amplitude ρ_r/ρ_m that sets the scale. Helium-4 and deuterium sit at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers, un-tuned; the near-zero metallicity of the oldest systems follows from the handover; and because the run is standard, lithium-7 carries the standard threefold over-prediction, the lithium problem shared with flat Λ CDM rather than dissolved.

On this cosmology the primordial scalar spectrum *factorizes*. The de Sitter substrate *determines* its structure—the geometric coherence of the acoustic phases on the null boundary (not a super-horizon freeze-out), the classical non-vacuum character of the fluctuations (the substrate vacuum lies some 10^{113} below the observed A_s), the throat’s isotropization, a parameter-free low-multipole discreteness floor, and the faithful transmission of the spectral shape—while the progenitor collapse *supplies* its content, the amplitude A_s and tilt n_s , inherited as boundary data exactly as flat Λ CDM inherits the baryon-to-photon ratio. *The transmission is now supported by a mechanism rather than by the absence of a scale alone* (§3): on the contracting leg aH grows without bound as the branch point is approached, so the comoving horizon shrinks to zero and every mode *exits* it and freezes before the crossing; the imaginary-time segment’s Euclidean kernel damps oscillatory content and has nothing to act on in a frozen mode, so amplitude and tilt cross unaltered while the collapse-leg acoustic phase does not—which the observed peak positions independently require, excluding a transmitted leg phase by a factor of 21–41 in the required distance [8].

The decisive result is a *transmission dichotomy*: a non-degenerate horizon’s exponential near-horizon approach imprints a scale-invariant spectrum (the inflationary de Sitter-horizon

mechanism), whereas the degenerate Narai member’s power-law, scale-free approach to the branch point is *argued* to transmit the progenitor spectrum unaltered—an argument from the absence of an imprinted scale, not a computed transfer—so CR has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial B -modes.

Because the distance slicing is flat, the discrete closed- S^3 source projects through the flat transfer to a parameter-free low-multipole deficit ($\ell \lesssim 8$)—mild, and on a genuine Boltzmann transfer a dip whose *minimum falls at* $\ell = 4$ (≈ 0.36 of the Λ CDM expectation there, ≈ 0.47 and 0.41 at $\ell = 2$ and 3 ; the shape cross-validated between two independent transfers, which agree on the location and the recovery and differ on the depth by up to a factor of two at $\ell = 3$ and 4), and non-discriminating: it starves $\ell = 2$ and $\ell = 3$ together and so matches neither the sharp observed quadrupole-only dip nor the near- Λ CDM octopole, sitting consistent with Λ CDM within the lowest multipoles’ cosmic variance.

Taken together, these results carry a reckoning the paper draws in full: what flat Λ CDM assembles from added, tunable apparatus—inflation for the horizon, flatness, coherence, and near-scale-invariance, a dark-energy component for the late-time acceleration, added early-universe physics for the tensions—this cosmology recovers from the cosmological constant alone, each feature required by structure already present rather than produced by a sector introduced to yield it. By the economy of assumption that distinguishes a required structure from a merely permitted one, that comparison is settled now, on the theory-choice axis, in this cosmology’s favour; and the data axis has now moved with it, from even toward favoured: the Hubble tension is resolved across the low-redshift distance ladder and not the acoustic angle alone—a rate whose two parameters are geometric carries H_0 out of the baryon-acoustic observables, which fit at the directly measured H_0 where Λ CDM cannot—on the SDSS DR12 consensus and, at the current state of the art, the DESI DR2 dataset (thirteen measurements across seven tracers) at $\chi^2/\text{dof} \simeq 1$ with the single CMB-calibrated $\Omega_m \simeq 0.31$ —and the light-element abundances are confirmed within 1σ at the microwave-background baryon density (the cosmogenesis paper [17]).

The cosmology is thus empirically favoured, not merely more economical, with its exposed edges named: a CR-specific signature in the diffusion *scale* (the same geometric rate that dissolves the Hubble tension enlarges the Silk scale [25])—a real, computed effect of definite sign, non-reabsorbable by the baryon density and independent of H_0 , whose *magnitude* moves with the fitted onset redshift (computed at 8.2% and at 14.0% at two onsets differing by half a per cent) and is pinned only when that fit is, but whose observable consequence for the high- ℓ power is entangled with the acoustic transfer at those multipoles (whether CR’s high- ℓ peaks track Λ CDM’s, verified only through the third peak) and so genuinely open, neither a demonstrated tension nor a wash; and the full-spectrum likelihood (the low multipoles being a mild, non-discriminating deficit rather than a sharp edge either way).

We mark maturity throughout. *Established*: the cosmological derivation, the transmission dichotomy, the amplitude floor, the throat, the flat slice, the flat-projection transfer, the data confrontation, and the time-reversal driving equality that carries the peak *heights* (the driven acoustic amplitude flat Λ CDM’s exactly, on the standard inherited plasma the abundance comparison confirms). *Argued*: the coherence mechanism.

Run, and returning a disagreement rather than a confirmation: the end-to-end branch-point-to-recombination transfer, rebuilt as the standard line-of-sight integral across a visibility of finite width and validated to 0.16% on the control’s first peak, then carried through to this construction’s arm—the height pattern tracks and the phase does not. *Open*: the damping-scale signature’s observable consequence, the *derivation*—as against the measurement—of A_s , n_s , and the inherited datum the composition is run on, and the diagnosis of the phase offset the transfer exposed. *The cosmology carries one fitted parameter and its status should be stated plainly*: z_{onset} , the redshift at which the expanding-phase plasma begins, which is the lower limit of the sound-horizon integral, set by where the expanding-phase plasma begins rather than by a radiation era the geometric rate does not carry. It is fitted to the acoustic angle at the *directly* measured H_0 and lands at $z_{\text{onset}} \approx 6.8 \times 10^3$, $T_{\text{onset}} \approx 1.6 \text{ eV}$, near $\rho_r/\rho_m \approx 2$. *It is not a knob for the H_0 tension*: r_s and D_M carry the stacking rate’s common H_0 , which therefore scales out of their ratio, so θ_* is fixed by the offset x_0 alone (equivalently by $\Omega_m = 2/(x_0^3 + 2)$) and the same z_{onset} meets the scale at every H_0 across the range—it does not move between them. *Matter–radiation equality is then a consequence and a check, not an input*— $1 + z_{\text{eq}} = 3399$, exactly half the onset, where the

fitted Ω_m and the Hubble constant reproduce the microwave-background-inferred physical matter density (near $H_0 \simeq 68$); read at the directly measured H_0 instead, the same z_{onset} gives $\rho_r/\rho_m = 1.71$, so the datum is an order-unity band and not a determined number; and z_{onset} is a redshift on the expanding leg, not the branch point, which no finite cosmic-time layer reaches. The substrate’s single Λ is shown to govern both the global expansion and, per structure, the local scale at which structure resists it—the Hubble–Eddington radius—one constant read at two ranges.

1 Introduction: the cosmological theory and its scalar perturbation sector

Cosmological Relativity (CR) recovers the flat- Λ CDM expansion history by a causal reassignment on a de Sitter background at the cosmogenesis branch point: the expansion rate is fixed by the cosmological constant alone, radiation carries no term in it, and the present matter density is a reading of the cosmic epoch rather than an independent quantity [8, 10]. The cosmic beginning is the branch point $r = 0$ —a genuine curvature singularity of the SdS metric read over the r -chart, and *not* a singularity of the substrate, whose throat is the defining constant α , and the construction makes the constant-proper-time slices exactly flat while the cosmological layers are a closed S^3 . This paper develops that cosmological theory in the form its observational tests require (§2) and completes it with the scalar perturbation sector the framework paper’s outlook named open—“perturbations within the Schwarzschild–de Sitter representation”—the primordial density perturbations and their imprint on the microwave background (§§4–8), closing on the open frontier the construction sets up (§10). Across these sections one comparison recurs and is drawn together at the close (§9): the causal contact, flatness, coherence, isotropy, and near-scale-invariance that the standard picture obtains from an added, tunable inflationary sector are here read off structure already present and fixed by Λ alone—an economy-of-assumption difference the paper argues is decidable now, on the theory-choice axis where it falls to this cosmology—and the data axis has now moved with it, the Hubble tension resolved across the low-redshift distance ladder and the light-element abundances confirmed against the data (§2.4), so the reading is empirically favoured and not merely more economical, its remaining exposed edges the damping-*scale* signature—a computed, non-reabsorbable CR-specific effect of definite sign and z_{onset} -dependent magnitude whose observable consequence awaits the high- ℓ acoustic transfer, genuinely open—and the full-spectrum likelihood, the low multipoles a mild, non-discriminating deficit rather than a sharp edge either way. The companion dynamics paper develops the perturbation sector’s *tensor* half—the propagating graviton as the leaf’s shear, a massless de Sitter mode on the cosmic foliation [9]; the scalar half is here.

The result is organized by one thesis, which we state at the outset because it controls everything below.

The CR primordial scalar spectrum factorizes: the de Sitter substrate geometry determines its structure, the progenitor collapse supplies its content, and the branch point’s null geometry assigns each its job.

Substrate-determined, and developed here: the coherence of the acoustic phases (§4); the classical, non-vacuum character of the fluctuations (§5); the isotropizing throat (§6); a parameter-free low-multipole discreteness floor (§7); and the faithful transmission of the spectral shape (§8). Progenitor-supplied, and inherited rather than derived: the amplitude A_s and tilt n_s . This is the same division of labour by which flat Λ CDM carries the baryon-to-photon ratio η as a measured datum from a baryogenesis it does not model; this cosmology carries one such datum—the onset redshift z_{onset} , equivalently the radiation amplitude $\rho_r/\rho_m \approx 2$ *at onset* that calibrates the acoustic scale, fixed by the hot handover of the progenitor collapse—and the

transmission result of §8 routes A_s and n_s through the *same* handover. There is one boundary-condition supplier, not two.

A word on what is settled before the spectrum. The acoustic *scale* (the peak spacing) and the apparent Hubble tension are resolved in the cosmological development above (§2.3): the tension dissolves as a consequence of the geometric rate, and the scale is met at the directly measured H_0 by the single measured parameter $\rho_r/\rho_m \approx 2$, with the sound speed the ordinary baryon-loaded value of pressureless branch-point matter. *The sky’s own side of that comparison is measured here rather than quoted. Locating the acoustic peaks of the Planck 2018 `plik_lite` binned TT spectrum [22] by parabolic fit to each hump above $\ell = 100$ returns $\ell = 220.4, 537.7, 817.3$ and 1123.9 , which reproduce the standard quoted values to better than one per cent without having been fitted to them. Fitting $\ell_n = \ell_A(n + \phi/\pi)$ to the first three gives the sky $\phi/\pi = -0.2404$ against -0.2361 for a Λ CDM control measured by the identical procedure—an offset of 0.0043 , so the control stands in for the sky on this quantity to within seven parts in a thousand of the disagreement it is used to measure. Against that, the asymptotic phase intercept of this construction sits 0.615 away in the same units at the derived seam datum. That number should not be read as the disagreement itself. Across four seam phases at production depth the asymptotic phase spans 0.891 in ϕ/π with the control’s 0.263 inside that span, so the 0.62π is the value at one reading rather than a prediction^R. No sampled reading lands on the control’s value, however, and the disagreement does not go with the phase in any case: the reading whose phase comes closest, at $\phi/\pi = 0.0663$ against the control’s 0.2628 , returns $\chi^2/\text{dof} = 224$ against the control’s 3.71 on the same 185 bins—sixty times—and its peaks sit at $388, 604, 884, 1164$ against the control’s $220, 540, 812, 1132$: the gaps are comparable and the series starts 168 multipoles too high, and the peak heights move over 0.48 – 1.62 without reaching the sky’s 2.22 , while the spacing holds at 0.963 – 0.981 of ℓ_A throughout. So the phase was never carrying it, and what this instrument records is a disagreement with the acoustic scale that no single quantity named in this section accounts for. Propagating a two-multipole peak-position uncertainty through the same fit gives $\sigma(\phi/\pi) \simeq 0.009$, so the separation is of order seventy standard deviations and is not a resolution effect; at one multipole it is 0.005 and the separation twice that, so the figure quoted is the conservative one^R. It is a real disagreement with the sky, and the coherence and envelope sections below are where its mechanism is argued rather than where it is dissolved. These are settled there and are not reopened in the perturbation sector; in particular, the standard radiation-governed sound horizon laid on this cosmology’s rate is a calculation belonging to neither framework, and any “tension” it produces is an artifact, not a CR result. What is genuinely downstream is the *spectrum*—the peak heights, carried below by the time-reversal driving equality (§4), and the full oscillating-medium transfer, still open—which is where this paper’s substance lives.*

The register is the programme’s [13, 8]: the field equations are Einstein’s, unchanged; the existing object is the evolving three-dimensional layer; “the substrate determines” and “the branch point transmits” are statements about a real geometry and its real boundary [16, 15], not about a modification of the dynamics. Throughout, we mark maturity in the prose: *established* results are verified by direct computation (the receipts are collected in `computations/perturbation_verify/`); *argued* results carry the reasoning and a leading-order computation but are flagged as not yet proven through the full transfer; *open* elements are named, not finessed.

2 The cosmological theory the perturbations complete

This paper develops the cosmology of Cosmological Relativity from first principles and carries it to its empirical confrontation, completing it with the scalar perturbation sector. It takes from the companion framework paper [8] the *general* primitives—the layered ontology, the projection principle, and the representational freedom by which one fixed manifold admits distinct Lorentzian metrics—together with the Null–Boundary Correspondence theorem, the structural

result (grounded in the metric-singularity causal structure of gravitational collapse [3]) that a collapse event-of-events and a de Sitter cosmic-entry slice are one ontological layer under two causal assignments. The cosmology proper is built here: the causal reassignment that selects the Nariai member, the proper-frame expansion law, the branch point at which the collapse leg becomes the expansion leg—the beginning, reached a lap on from the horizon the Correspondence anchors—and the resolution of the apparent Hubble and acoustic tensions; the perturbation sector (§4–§8) is its completion.

2.1 The causal reassignment and the Nariai member

The vacuum solution of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ with a localized source is Schwarzschild–de Sitter (SdS),

$$ds^2 = -\frac{r}{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r} dr^2 + \frac{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}{r} dt^2 + r^2 d\Omega^2, \quad (1)$$

whose areal radius r is timelike in the cosmological regime $\Lambda G^2 M^2 / c^4 \geq 1/9$. Building the cosmology, we reassign causal roles on the de Sitter substrate—a move licensed by the representational freedom of the framework, distinct Lorentzian metrics on one fixed manifold sharing its foliation [8]: the de Sitter null rulings whose causal sense matches a collapse horizon’s generators are promoted to the timelike fundamental congruence, the at-rest comoving worldlines become the null photon congruence, and the foliation by expanding 3-spheres is preserved. This selects the *Nariai* member $\Lambda G^2 M^2 / c^4 = 1/9$, $M = c^2 / (3\sqrt{\Lambda}G)$ —the unique non-pivoting tilt, fixed by Λ alone—at which the two positive horizons merge at $r_N = \alpha/\sqrt{3}$, $\alpha = \sqrt{3}/\Lambda$ the de Sitter 3-sphere size (a scale distinct from the merged-horizon radius $\alpha/\sqrt{3}$ and from the areal-radius amplitude below; we keep the three apart throughout). The selection is geometric—the fixed point of the slicing curve’s root-exchange involution, equivalently the limiting tangent orientation a collapse selects—not a fitted mass [4]. That the selection is moreover *unique* is a structural fact about the description groupoid rather than a fitting: the generic vantages fall into two-cycles of the involution, the Nariai member is its one fixed point, and the reassignment promoting a null direction to the fundamental congruence is forbidden at every generic vantage, so it selects Nariai and no other [5].

2.2 The proper-frame cosmology, derived

In the proper frame of the fundamental observers the SdS metric is a homogeneous, non-synchronous cosmology. The derivation [14] proceeds from (1) in five steps. (i) A timelike radial geodesic has a conserved energy E (the metric being independent of t); eliminating $dt/d\tau$ in the normalization $L = -1$ gives the radial equation $(dr/d\tau)^2 = E^2 - V_{\text{eff}}(r)$, with $V_{\text{eff}} = (r - 2GM/c^2 - \frac{\Lambda}{3}r^3)/r$. (ii) The fundamental congruence is the worldlines whose motion in r is produced purely by the field—those at rest exactly where $V_{\text{eff}} \equiv 1$ and the metric reduces to Minkowski—which is $E = 1$; this fixes the rest frame by the field rather than by stipulation, *and it is worth saying where that locus is*: $V_{\text{eff}} = 1$ is $r^3 = -2GM\alpha^2/c^2$, so the rest frame is defined at $r = -(6GM/c^2\Lambda)^{1/3}$ —a *negative* areal radius, on the conjugate branch of the lap, between the progenitor’s horizon and the branch point [8]. *In the original derivation that locus had to be carried as a formal one*—“a negative mathematical abstraction that lies beyond a singularity at $r = 0 \dots$ simply a matter of algebraic consistency” [1, App.]—*and the construction has since made it a place*. The same locus fixes the sign, since $t = \tau$ there if and only if $E = +1$. (iii) With $E = 1$, integrating the radial equation gives, along one worldline, $r(\tau) \propto \sinh^{2/3}(\frac{3}{2}\sqrt{\Lambda c^2/3}\tau)$. *That this integration closes at all is a second thing the same condition buys*. Writing the radial equation as $(dr/d\tau)^2 = 2GM/c^2r + r^2/\alpha^2 - k$ with $k = 1 - E^2$, the quadrature $\int dr/\sqrt{\cdot}$ is elementary at $k = 0$ —it is $\frac{2\alpha}{3} \operatorname{arcsinh}(r^{3/2}/\sqrt{2GM\alpha^2/c^2})$, which inverts to the $\sinh^{2/3}$ above—while for $k \neq 0$ the substitution $r = u^2$ carries it to a sextic under

the root and it is not elementary. *So $E = 1$ fixes the rest frame by the field and, by the same stroke, is the one non-degenerate energy at which the cosmology integrates in closed form*; the only other elementary cases are the degenerate ones, $\Lambda = 0$ giving the dust cycloid and $M = 0$ giving pure de Sitter^L. (iv) Labelling the congruence by a comoving coordinate χ through the integration constant $r(0)$, the areal radius across the congruence depends on τ, χ only through their sum, $r(\tau, \chi) = r(\tau + \chi)$, with $\partial_\chi r = \partial_\tau r$. (v) Fixing the transformation $t(\tau, \chi)$ by requiring the comoving coordinate orthogonal to proper time, $g_{\chi\tau} = 0$, forces the integration function to be χ itself, and the remaining components follow ($g_{\tau\tau} = -1$, $g_{\chi\chi} = (\partial_\chi r)^2$). The result is the proper-frame line element

$$ds^2 = -d\tau^2 + (\partial_\chi r)^2 d\chi^2 + r^2 d\Omega^2, \quad (2)$$

with cosmic time $\tilde{\tau} \equiv \tau + \chi$ (by Weyl's principle the congruence issues from a common origin and the constant-cosmic-time slices are those of constant $\tilde{\tau}$), areal radius

$$r(\tilde{\tau}) = \left(\frac{6GM}{\Lambda c^2} \right)^{1/3} \sinh^{2/3} \left(\frac{3}{2} \sqrt{\frac{\Lambda c^2}{3}} \tilde{\tau} \right), \quad (3)$$

and redshift $1 + z = r(\tilde{\tau}_0)/r(\tilde{\tau}_e)$. The constant- $\tilde{\tau}$ slices sit at 45° to the fundamental rest frame: the cosmology is homogeneous and observationally isotropic, yet *non-synchronous*—the constant- τ slices are exactly Euclidean (§7) but are not the cosmological ones. Of the FLRW kinematic assumptions—(i) a geodesic congruence, (ii) hypersurface orthogonality, (iii) homogeneity, (iv) isotropy—this construction keeps (i) and drops (ii); Lemaître's reading [28] of the Euclidean slices as “space ejected from $r = 0$ ” is causally incoherent and is not the physics. Equation (3) is moreover a single analytic function of $\tilde{\tau}$ (Fig. 1): the expanding cosmology is its real branch $\tilde{\tau} > 0$, and the cosmogenesis *branch point* at $\tilde{\tau} = 0$, where $r \rightarrow 0$, is the branch point of the scale factor rather than a real $r = 0$ boundary—and it is *not* a seam: the seams are the two unit-speed loci of the lap, $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ [8], and $r = 0$ lies strictly between them, at the close of the lift—the analytic counterpart of the beginning at the branch point, where the substrate's curvature stays finite while the geometry read over the r -chart diverges [3], and the cosmological reading of the slicing curve's closed lap—the *cosmogenetic bead* the framework establishes as one closed object [4, 8], of which this scale factor's real branch $\tilde{\tau} > 0$ is the expansion leg and its analytic continuation the prior universe's collapse—on which $r = 0$ is a branch point and not a barrier, the substrate smooth across the locus the chart so labels [4].

2.3 Exact recovery of flat Λ CDM, and the amplitude fixed by Λ

What fixes this rate is worth stating before what it recovers. The rate this cosmology computes on is set by the substrate's geometry and by nothing else. Its two parameters are the substrate curvature radius $\alpha = \sqrt{3/\Lambda}$ —an invariant length—and the offset x_0 of the cut, and neither is a content of the universe:

$$c^2 H^2(z) = \frac{c^2}{\alpha^2} \left(1 + \frac{2(1+z)^3}{x_0^3} \right). \quad (4)$$

The offset is measured directly and calibration-free— $z_{\text{acc}} = x_0 - 1$, (11) below—without a distance ladder, without the microwave background and without any density. *What the rate is a rate of is the stacking of the foliation*, and content does not enter it: radiation is absent from the slicing operator's vacuum kernel, requiring $m'(r) \neq 0$, a bend of the cut; and the second term is the offset itself rather than a density supplied by hand [6, 8].

The familiar cosmological form is accordingly a coincidence of form, and saying so is the substance of this subsection. Written in the fitted pair the same rate reads $H^2 = H_0^2 [\Omega_m(1+z)^3 + \Omega_\Lambda]$, with $\Omega_m = 2/(x_0^3 + 2)$ and $\Omega_\Lambda = x_0^3/(x_0^3 + 2)$ following identically—a *translation between parameter sets*, and not a decomposition into components. In the standard reading

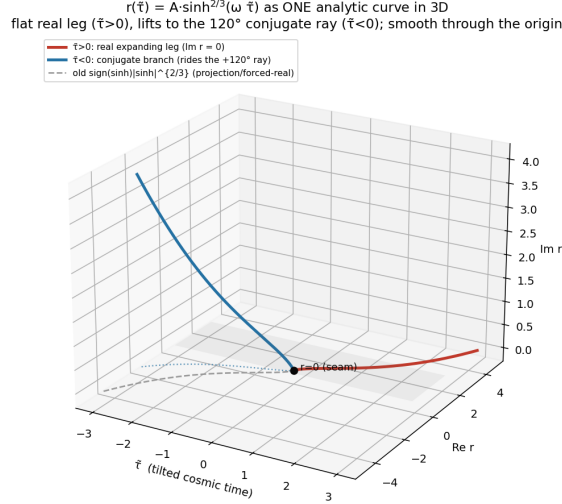


Figure 1: The scale factor (3) as one analytic curve $r(\tilde{\tau})$ in $(\tilde{\tau}, \text{Re } r, \text{Im } r)$. The expanding cosmology is the real leg $\tilde{\tau} > 0$, lying in the real plane; the branch point $\tilde{\tau} = 0$, $r \rightarrow 0$ —*not* a seam—is where the curve leaves the real plane onto a conjugate branch at constant phase $2\pi/3$ for $\tilde{\tau} < 0$. The beginning is this analytic branch point, not a real $r = 0$ boundary. Forcing r real (dashed) projects the conjugate branch onto the negative axis and manufactures a cusp the smooth curve does not carry.

cosmic time is orthogonal to the three-dimensional world and the expansion is driven: by initial conditions, and by whatever densities obtain at each moment, through the field equations. Here it is read off a fixed background whose foliation is *not* orthogonal to the proper time of observers at rest, so that such observers move through the universe over cosmic time in a geometrically fixed manner. *The two readings agree on the function and share nothing beneath it.*

Where the field equations act is the leaf, and they act there in full. All stress-energy gravitates, by the ordinary Friedmann readout of the self-gravitating congruence; the perturbations, the plasma’s sound horizon and its diffusion length are processes running *in* the content and take that rate, while a comoving separation read across leaves takes the stacking rate (§2.2). *What the layered framework keeps apart is not two physics but one physics and its projection:* a quantity is computed on the level it belongs to, carried on the layer and then projected.

Equation (3) is exactly the flat- Λ CDM scale factor. At Nariai the mass is no longer free, and the amplitude becomes a pure Λ -length,

$$\left(\frac{6GM}{\Lambda c^2}\right)^{1/3} = \frac{2^{1/3}}{\sqrt{\Lambda}}, \quad (5)$$

so both factors of the expansion law are set by Λ^R : the rate $\frac{1}{2}\sqrt{3\Lambda}c$ is the standard late-time rate and the amplitude is the third Λ -scale (distinct from α and $\alpha/\sqrt{3}$). The Friedmann equation the $\sinh^{2/3}$ law satisfies,

$$H^2 = \frac{\Lambda c^2}{3} \coth^2\left(\frac{1}{2}\sqrt{3\Lambda}c\tilde{\tau}\right) = \frac{1}{3}(8\pi G\rho + \Lambda c^2), \quad (6)$$

makes the density ratio a clock^R,

$$\frac{\Omega_m}{\Omega_\Lambda} = \text{csch}^2\left(\frac{1}{2}\sqrt{3\Lambda}c\tilde{\tau}\right), \quad (7)$$

so $\Omega_{m,0}$ records the cosmic epoch $\tilde{\tau}_0$ at which we observe rather than an independent density, and the coincidence problem becomes the observation that we exist at a time of order the

geometry’s single timescale. The densities are bookkeeping for one Λ -set geometry; *radiation, like matter, is inherited content read off that clock, not a term that sources the rate*, and there is no radiation-dominated era [10]. The objection that a perturbing radiation fluid “must gravitate and so alter the rate” is a category error against a construction that reads (6) leftward: the rate is set, and the fluid’s energy is among the contents the set rate carries. The acoustic *scale* and the apparent Hubble tension resolve as consequences of this rate; this is developed next (§2.4).

2.4 The acoustic scale and the dissolved tensions

The two frameworks reproduce the late-time expansion identically and part in the early universe, where radiation carries no term in the rate (6). The apparent tensions of the standard model are consequences of that one structural fact.

The three levels, and which rate each quantity uses. The reading of every rate-bearing quantity below is fixed by locating it on one of three levels, which the geometry keeps apart and whose conflation is the standing error the layered framework names [8]. **(L1) The foliation stacking rate:** the Λ -set $\sinh^{2/3}$ law (6) read *leftward*—the cut is primary and ρ is the name of its bend, not its cause, the rate-side of the slicing operator’s reading whose content-side is *matter is the bend of the cut* [6]—the layer’s own geometric expansion, set by α and the offset x_0 , and *empirically forced* rather than modelled: the microwave-redshift isotropy floor forces uniform expansion to some three orders of magnitude, so a matter-tracking (radiation-sourced) rate is already excluded by data [10].

This is what the observable cosmology rides at *every* epoch, the branch point included: it is the rate the foliation stacks at, not a regime that begins somewhere. **(L2) The leaf-level local dynamics:** the expansion scalar of a self-gravitating congruence, the ordinary Friedmann read-out $\mathcal{H}^2 = \frac{8\pi G}{3}\rho_{\text{tot}} + \frac{\Lambda}{3}$ with radiation gravitating normally—the valid phenomenological reading of general relativity on the progenitor’s collapse excursion, where nucleosynthesis runs [17]; it is the same single $E=1$ geodesic read *inward* as dust collapse that L1 reads outward as cosmology [6, 3]. **(L3) The $E = 1$ projection:** the reassigned shadow the observer reads, through which an L1 or L2 quantity appears as distance and redshift—the Projection Principle of the framework paper, its geometric origin the *second ruling* of the slicing (the synchronous space is the second ruling) [8, 6]; reading this appearance as the existent is precisely the naïve-realist collapse the epistemology forbids [11]. The decision rule is structural, not a choice: a self-gravitating local excursion runs on L2, diffuse content riding the global foliation on L1, and every observable is an L1 or L2 quantity seen through L3; and *there is no L1/L2 boundary in time*. The distinction is kinematic and holds at every epoch: the vacuum kernel of the slicing operator is the SdS family exactly— Λ and the cut’s offset $2M$, and no more [6]—so the stacking rate is $c^2 H_{\text{stack}}^2 = (c^2/\alpha^2)(1 + 2(1+z)^3/x_0^3)$, while radiation, requiring $m'(r) \neq 0$, is a *bend* of the cut and therefore content. The decomposition is exact, $H_{\text{leaf}}^2 = H_{\text{stack}}^2 + H_0^2 \Omega_r (1+z)^4$: the two rates differ by the radiation term alone. A comoving separation read across leaves takes the stacking rate; a process running in the content— r_s , r_D , recombination, the perturbations—takes the leaf’s. The assignment looked temporal only because $\Omega_r(1+z)^4$ is negligible below $z \sim 10$ [8, 3]. Each mis-assignment is quantitatively fatal—the geometric L1 rate carried down into the nuclear window is $\sim 300\times$ too slow, and the radiation-sourced L2 rate carried past the branch point radiation-pins r_s and re-manufactures the very tension this section dissolves [17]. Crucially L1 is the layer’s own *ontological* expansion, not the L3 shadow: computing a process against the stacking rate is reading the geometry, not the appearance. Every rate-bearing quantity below carries its level, so the reading is settled by the geometry rather than defaulted to the synchronous machinery.

The Hubble tension. There is no second H_0 to reconcile. The late-time expansion is reproduced exactly, so the directly measured rate is the reading of the geometry, while the microwave background’s lower inferred H_0 rests on a radiation-governed sound horizon r_s this construction does not share. The $\sim 5\sigma$ discrepancy is not two independent fits to be reconciled but

the artefact of calibrating the acoustic scale with an early-universe sound horizon foreign to a geometrically fixed expansion.

The acoustic scale, to a one-parameter accommodation. Recombination is observable cosmology from the branch point onward, so its scales are L1: both the sound horizon r_s and the comoving distance D_M ride the foliation’s stacking rate, and the observed angle $\theta_* = r_s/D_M$ is their L3 projection.

The robust ingredients are in hand: D_M is the standard observable, and $H(z)$ is the fixed $\sinh^{2/3}$ law (6) with radiation absent from the rate.

The sound horizon $r_s = \int_{z_{\text{rec}}}^{z_{\text{onset}}} (c_s/H) dz$ —an L1 quantity whose baryon-loaded c_s is ordinary content microphysics but whose expansion H is the L1 rate—then depends on a single early-universe parameter, the redshift z_{onset} at which the expanding-phase plasma begins; the standard radiation-governed r_s is recovered only in the limit $z_{\text{onset}} \rightarrow \infty$ that a beginning at finite curvature forbids. *That clause carries more weight than its length suggests, and the sense of “finite curvature” in it should be the substrate’s.*

The branch point is a locus at which the areal chart’s invariants diverge— $r = 0$ is a genuine infinite-curvature locus of that chart—while the curvature of the manifold the cut is taken on is set by α everywhere, so the divergence is the areal coordinate degenerating and not a scale of the geometry [17, 16]. *Read the other way the accommodation would collapse:* a beginning at genuinely unbounded curvature places no finite floor under z_{onset} , the $z_{\text{onset}} \rightarrow \infty$ limit is reinstated, and the single datum ceases to be a datum at all. *The one-parameter accommodation therefore rests on the substrate reading of the branch point,* and a reader who took the areal divergence for the physics would find the parameter unconstrained. Taking c_s the ordinary baryon-loaded value—the matter at onset is pressureless, and the geometric throat ratio $\sqrt{3}$ does not set it—the measured acoustic scale is reproduced at the *directly* measured H_0 by a single z_{onset} , near $\rho_r/\rho_m \approx 2$ (about twice matter–radiation equality) and *independent of H_0* —the independence belonging to z_{onset} itself, and the ratio being that same quantity re-expressed, as §2.4 returns to below: r_s and D_M carry that rate’s common factor H_0 , which scales out of their ratio, so $\theta_* = r_s/D_M$ is fixed by the offset x_0 alone and the single z_{onset} that meets the scale is the same at every H_0 across the range—the datum cannot be a parameter tuned to reconcile two values of H_0 , since it does not move between them. This is exactly where the standard rate parts: carrying radiation, it ties θ_* to H_0 —the coupling that fixes the lower microwave-background value and stands in tension with the direct one—whereas a rate whose parameters are geometric decouples the scale from H_0 , meeting it at the directly measured value with no second H_0 to reconcile. The beginning is well posed: the branch point—finite in the substrate’s curvature, whatever the r -chart geometry does there (§2.2)—places the onset down the cooling branch, $T_{\text{onset}} \sim 1.6 \text{ eV}$ at $z_{\text{onset}} \sim 6.8 \times 10^3$, so a pre-recombination sound-travel interval exists and the integral $\int_{z_{\text{rec}}}^{z_{\text{onset}}}$ is well posed.

The inherited datum. The empirical forcing of the cosmic foliation bears on the rate, not the matter content—the matter density being the bend of the spatial cut [10, 6, 7]—so the radiation amplitude is measured content, not a quantity the geometry sets; the cold seam (Gibbons–Hawking temperature [31] $\sim 10^{-30} \text{ K}$, some thirty orders below the onset) confirms there is no geometric source to set it instead. *One thing about that ratio should be said plainly before it is leaned on, because it is a unit and not a second datum.*

The quantities this cosmology actually uses are the offset x_0 (equivalently $\Omega_m = 2/(x_0^3 + 2)$), which the rate carries and the baryon-acoustic data fix; ω_b and ω_γ , which the sound speed carries and which are measured directly and framework-independently; and z_{onset} , the one fitted number.

The *physical* matter density $\omega_m = \Omega_m h^2$ enters none of them.

One thing about Ω_m is worth stating plainly, since the construction’s matter sector invites the question. The dust the rate carries exceeds the baryons by about a factor of six, $\Omega_m \simeq 0.31$ against $\omega_b/h^2 \simeq 0.05$, so most of it is non-baryonic. *That is not a component this cosmology*

introduces, and it is not one it derives. What the matter sector fixes is the number of families, their chirality and the S_3 relating them, and *nothing about their internal content* [18]; abundances are measured content throughout, the matter density being the bend of the spatial cut. So the non-baryonic fraction stands with η , A_s and n_s : inherited, not set by the geometry. *What the construction does supply is a place rather than an abundance.* Its correspondence with the Standard Model's content carries a right-handed neutrino as the sixteenth Weyl fermion of each generation, and fixes that state's place while fixing none of its couplings [18]; a species that is never thermalized contributes nothing to N_{eff} , so nucleosynthesis is untouched and no prediction is made or needed here. *The point of the paragraph is the division of labour:* a candidate the construction already contains, an abundance it does not claim, and no third thing wedged in between.

But ρ_r/ρ_m at onset is $\omega_r(1 + z_{\text{onset}})/\omega_m$ and equality is $1 + z_{\text{eq}} = \omega_m/\omega_r$, so both divide by that one quantity—and $\rho_r/\rho_m \approx 2$ is therefore not a second inherited number standing beside z_{onset} but z_{onset} itself, read in units of a density the construction does not determine. *The consequence is a stated H_0 -dependence in the ratio and in nothing else:* with Ω_m held at its fitted value the ratio is 2.01 at $H_0 = 67.4$ and 1.71 at $H_0 = 73$, the two readings coinciding at $H_0 \simeq 68$ where $\Omega_m h^2$ reproduces the microwave-background-inferred $\omega_m = 0.143$. *What carries weight is what survives that:* z_{onset} does not move with H_0 at all, so it cannot be absorbing the tension; and the inherited amplitude, read at onset, is order unity—which is a statement about where the onset sits and *not* a contrast with η , as the withdrawal two paragraphs below sets out. *The factor of two is exact at $h \simeq 0.68$ and should be quoted as an order-unity band, not as a determined number* (receipt: `the_ratio_is_the_onset_in_imported_units.pyR`).

Thus $\rho_r/\rho_m \approx 2$ is the structural analogue of the baryon-to-photon ratio η : as flat Λ CDM carries η from a baryogenesis outside the model proper—measured, not derived—this cosmology carries the single radiation initial condition, and with no radiation-dominated epoch the light elements as well, from the hot handover of the progenitor collapse, this universe having formed at the event horizon of a black hole in a previous one, equally outside the construction and equally measured.

That the datum is measured rather than derived is no defect: a framework is credited for *requiring* the structure it explains—the geometric rate, and the dissolution that follows—not faulted for measuring the boundary data it does not. *One naturalness argument that has been offered for this datum should be withdrawn rather than carried, because it compares a quantity with itself at two epochs nine orders apart.*

Complete collapse does liberate binding energy of order the rest mass ($GM/Rc^2 = \frac{1}{2}$ at the Schwarzschild radius), and it is tempting to read that as making a handover carrying radiation of order the matter density the natural scale—order unity, rather than the small value η takes. But $\rho_r = \rho_m$ occurs at $T \simeq 0.8\text{eV}$, whereas the handover sits four orders *above* the deuterium bottleneck at 0.07MeV , since that is what the nucleosynthesis computed on it requires [17]: at the bottleneck the ratio is already $\sim 9 \times 10^4$ and at the handover proper $\sim 9 \times 10^8$. A handover carrying radiation merely of order the matter density would be a handover with $\eta \simeq 0.5$, which leaves neither light elements nor acoustic peaks. *So the naturalness question runs the other way:* what the progenitor must deliver is a plasma with of order a billion photons per baryon—the baryogenesis-analogue derivation this paper names as open—and the binding-energy scale offers no head start toward it. *What replaces the argument is a simplification of the bill.* Given η and the measured matter-to-baryon ratio, both of which this cosmology inherits for the composition regardless, the ratio at onset is $[1 + \nu](\pi^4/30\zeta_3)T_{\text{onset}}/[\eta(\omega_m/\omega_b)m_N]$, which returns 1.99 at $T_{\text{onset}} = 1.6\text{eV}$ —the quoted datum, to one per cent, from standard thermodynamics and no feature of this construction. *It is therefore not a second inherited number standing beside η : it is the fitted onset restated,* so what the handover supplies is one composition datum and what the cosmology fits is one parameter (receipt: `the_two_data_are_one.pyR`). Like η , the inherited radiation amplitude may remain a measured boundary condition indefinitely without cost to

the *dissolution*—which settles what the dissolution depends on, not whether the derivation is worth having: deriving it from the progenitor’s determined structure remains open work.

The synthesized composition. The same collapse fixes the light-element composition, and the geometry sharpens how. The infalling matter compresses adiabatically—the plasma is optically thick, so the photon bath compresses with it—and heats far above the deuterium bottleneck ($T_D \simeq 0.07 \text{ MeV}$) for every progenitor: the compression does not stop at horizon crossing but continues to the branch point, and the peak is bounded below by the infall energy at the horizon, of order the rest mass per nucleon, independently of the progenitor’s mass (the cosmogenesis paper [17]); at turnaround it re-expands and cools back through the window. Because the effective rate through the window is the standard Friedmann rate, that cooling leg is a standard big-bang nucleosynthesis, run on the collapse’s turnaround from a dissociated hot start—the deep re-expansion doing the synthesis below the $T_{\text{onset}} \sim 1.6 \text{ eV}$ onset at which the observable expansion begins. On a first sample this is favourable, and on one element advantageous. The helium-4 fraction $Y_p \approx 0.245$ sits at the observed value; the near-zero metallicity of the oldest systems follows from the handover photodissociating the progenitor’s processed composition back to light elements; and because the run is a standard nucleosynthesis, the lithium-7 abundance carries its standard threefold over-prediction—the standing lithium problem shared with flat ΛCDM , neither dissolved nor worsened. Deuterium sits at the observed $D/H \approx 2.5 \times 10^{-5}$: the freeze-out temperature the data demands is the standard bottleneck, and that is exactly the temperature at which the standard rate delivers a freeze-out—target and mechanism coincide un-tuned. The peak calculation that forces this synthesis reading over an inherited one, and the full multi-abundance account, are given in the cosmogenesis paper [17], where the network is now integrated explicitly on the cooling leg and returns deuterium, helium-4 and helium-3 at their observed values with lithium-7 at the shared over-prediction, jointly from the single inherited datum—so one progenitor handover does fit the pattern together; what stands as its open edge is the last-percent precision against the measured abundances, a data-confrontation the produced pattern now exposes rather than a computation still owed.

The load-bearing claim. The acoustic scale stands as a one-parameter accommodation on the radiation datum, not a parameter-free prediction; the standard c_s it borrows presumes the inherited medium to be an ordinary photon–baryon plasma, which the abundance comparison now confirms—helium-4 and deuterium at their observed values, lithium-7 the shared standard problem (the cosmogenesis paper). The load-bearing, falsifiable claim remains the prior one—that radiation carries no term in the expansion rate—of which the dissolved Hubble tension and the one-parameter acoustic calibration are consequences. The sound horizon at a geometrically fixed H_0 is where a purely geometric expansion history and the standard radiation-governed one are told apart in the data.

The expansion history, confronted with the late-time data. The dissolution is not confined to the single microwave-background angle: it holds across the low-redshift distance ladder, and for the same reason.

Because every distance and the ruler alike carry the stacking rate’s common factor H_0 , which therefore scales out— D_M , $D_H = c/H$ and the sound horizon r_s each scale as $1/H_0$ —the baryon-acoustic-oscillation observables D_M/r_s and D_H/r_s are *independent of H_0* , fixed by the offset x_0 alone (the same cancellation that fixes θ_* ; §above).

The distinction from ΛCDM is exactly the radiation: there the ruler r_s is pinned in physical length by the radiation-era physical density $\omega_m = \Omega_m h^2$, so the same D/r_s data constrain H_0 and pull it to the low microwave-background value, in tension with the direct one.

Here they do not: the acoustic data fix the offset x_0 and leave H_0 to the local distance ladder, with no second value to reconcile. Confronted with the SDSS DR12 galaxy consensus [20] the geometric rate at $\Omega_m \simeq 0.31$ fits at the directly measured H_0 ($\chi^2 \simeq 1.7$ on six points, at *any* H_0 including 73), where the radiation-governed rate at that same local H_0 misses badly ($\chi^2 \simeq 49$)—that residual is the Hubble tension itself, which the geometric rate structurally

lacks.

The same confrontation carried to the state-of-the-art DESI DR2 dataset [21]—thirteen distance measurements across seven tracers ($0.3 \lesssim z \lesssim 2.3$), with the full per-tracer D_M/r_s – D_H/r_s correlations—tightens rather than loosens the result: with a single free parameter Ω_m the geometric rate fits at $\chi^2/\text{dof} \simeq 1.0$, at $\Omega_m = 0.3066$ (where the inherited datum sits at $\rho_r/\rho_m \approx 2.0$), and it fits there at *any* H_0 including the local 73—which is the whole of the claim.

The radiation-pinned ruler cannot do that: it ties ΛCDM ’s fit to one H_0 , best at 68.7, and breaks it at 73 ($\chi^2/\text{dof} \simeq 15$)^R.

Two things about Ω_m are worth stating exactly. It is not a CMB-calibrated input here—Planck’s own value is 0.3153, at which the fit degrades to $\chi^2/\text{dof} = 1.35$, and 0.3066 is what these data themselves prefer. *And granted the same one free parameter, ΛCDM fits these data marginally better than this cosmology does* ($\chi^2/\text{dof} = 0.92$ against 1.00). *The result is therefore not that the geometric rate fits better; it is that it fits without choosing an H_0 , and the radiation-pinned one must choose.* The cosmic chronometers [24], which read $H(z)$ directly, are consistent with the local value. So the resolution is a property of the whole expansion history, not of one angle, and it rides the very rate whose branch point-located scoping keeps the nucleosynthesis window standard [17] (receipt: `hubble_expansion_confrontation_v2.py`^R; the invariant is the offset x_0 , read as Ω_m , and not ω_m —holding the physical density fixed instead imports the ΛCDM reading and manufactures a spurious tension).

A1.4: the radiation-free rate resolves the Hubble tension across the BAO distance ladder

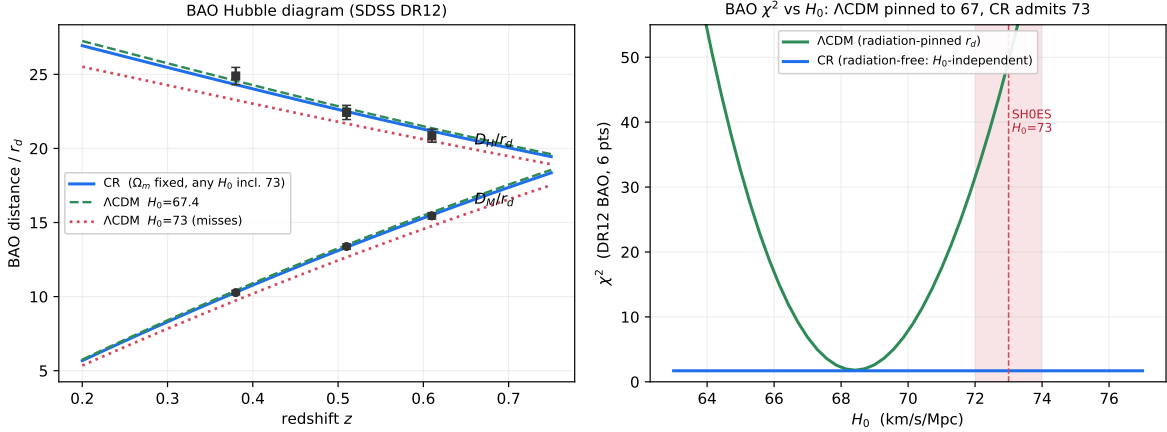


Figure 2: The Hubble tension resolved across the baryon-acoustic distance ladder. *Left:* the SDSS DR12 measurements of D_M/r_s and D_H/r_s (points) against the geometric rate at the directly measured $H_0 = 73$ (solid) and flat- ΛCDM at $H_0 = 67.4$ (dashed); both track the data, while ΛCDM forced to the local $H_0 = 73$ (dotted) misses. *Right:* the BAO χ^2 as a function of H_0 . Radiation in the ΛCDM rate pins the physical ruler r_s , so its χ^2 is a steep parabola minimized near 67 and excluding the local value—this is the tension. The geometric rate carries H_0 out of the ratios D_M/r_s , D_H/r_s , so CR’s χ^2 is *flat* in H_0 : the BAO fixes Ω_m and admits the SH0ES $H_0 = 73$ [23] (shaded) at no cost. The tension is structural to a radiation-governed ruler and absent from a geometrically fixed one. The figure shows the SDSS DR12 consensus; the same result holds, more strongly, against the state-of-the-art DESI DR2 dataset (thirteen measurements, seven tracers), where the geometric rate fits at $\chi^2/\text{dof} \simeq 1$ with the single CMB-calibrated $\Omega_m = 0.307$ at the local H_0 (§2.4).

2.5 The acoustic modes are sub-horizon at the onset, and the branch point is the opposite

The cosmological beginning is the branch point $r = 0$ at the close of the lift, where the conjugate branch becomes the matter branch [8, 4]. The curvature invariants of the r -chart diverge there

because the areal coordinate degenerates, while the throat itself is the substrate’s defining constant α and the bead continues through by analytic continuation [34]. The lap’s seams lie elsewhere, at the two unit-speed loci $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$. We record the consequence that organizes the perturbation sector.

Proposition 1 (The acoustic modes are sub-horizon at the plasma’s onset). *On the rate (6), the comoving Hubble wavenumber at the onset redshift $z_{\text{onset}} \approx 6797$ is $k_{\text{hor}}(\text{onset}) = H_0 E(z_{\text{onset}})/[(1+z_{\text{onset}})c] \approx 0.010 \text{ Mpc}^{-1}$, while the acoustic peaks lie at $k \sim \pi/r_s \approx 0.022 \text{ Mpc}^{-1}$ and above. The acoustic-scale modes are thus inside the horizon at the onset by a factor $\gtrsim 2$.*

The two epochs this proposition could be read as concerning are distinct, and the distinction is load-bearing^R. z_{onset} is a redshift on the expanding leg—the redshift at which the expanding-phase plasma begins, and the lower limit of the sound-horizon integral above. It is not the branch point. The branch point is the locus $r = 0$, which no finite cosmic-time layer reaches and which therefore carries no finite redshift at all. The proposition concerns the onset and nothing else: the acoustic modes are inside the horizon when the plasma they oscillate in comes into being, which is what the transfer calculation requires of them.

*And the corresponding statement about the branch point is the opposite one, for a different reason. Approaching $r = 0$ on the contracting leg the comoving Hubble scale aH grows without bound, so the comoving horizon shrinks to zero and every mode *exits* it before the crossing—which is why what crosses is frozen and non-oscillatory (§3). Sub-horizon at onset, super-horizon at the crossing: there is no tension between the two, and reading either statement at the other’s locus inverts it.*

Argument. Direct evaluation of $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ at z_{onset} and of π/r_s with the banked r_s from the acoustic-scale resolution (§2.3); clean residuals (receipts: `verify_numeric.py`, anchor 7^R). $\square$

The consequence is structural. The acoustic modes are not frozen super-horizon data awaiting re-entry, as in the inflationary initial-condition story; they are already inside the horizon when the cosmological side begins. Whatever fixes their amplitudes and phases is therefore the branch point handover itself, not a super-horizon freeze-out—which is why the substrate’s role will be to *transmit and gate* rather than to *generate*, and why the initial conditions are inherited.

3 What crosses, and what does not

The transmission dichotomy above is an argument from the absence of an imprinted scale. A mechanism can now be given for it, and it also settles which content the branch point does *not* carry.

The imaginary-time segment preceding the branch point acts on a perturbation as a Euclidean kernel [33], damping a mode of frequency ω by $e^{-\omega|\Delta\eta|}$ with $|\Delta\eta| \simeq 1.8 \times 10^4 \text{ Mpc}$. Taken naively this would annihilate everything: at $\ell = 220$ the factor is 10^{-71} . *But ω is the frequency of an oscillating mode, and at the branch point no mode is oscillating.* On the contracting leg the comoving Hubble scale $aH = |dr/d\tilde{\tau}| = \sqrt{|1-f|}$ grows without bound as $r \rightarrow 0$ —it is 0.13 at the comoving turnaround, 1.96 at $|r| = 0.1\alpha$, and 19.6 at $10^{-3}\alpha$ —so the comoving horizon $1/aH$ shrinks to zero and *every* mode exits it and freezes before the crossing. A frozen mode has no oscillation for the kernel to damp. *So the amplitude and the tilt cross unaltered, and the argument from the absence of a scale is recovered as a consequence of horizon exit rather than assumed.*

What does not cross is the oscillatory content itself, and that is the collapse leg’s acoustic phase: a mode that was sub-horizon earlier carries an oscillation, and the kernel annihilates it—an exponent of -152 at the first acoustic peak, integrated across the segment’s own sound-speed profile rather than at the single value the estimate above uses^R. This is required independently

by the observed peak positions, which exclude a transmitted leg phase by a factor of 21–41 in the required distance [8], and it is what leaves the comb to be set on the expansion side by the characteristic data of §4. *The two results are one statement:* the crossing transmits what is frozen and destroys what oscillates, so the progenitor supplies the spectrum’s amplitude and shape while supplying none of its phase.

The exponent has two factors, and the argument above varies only one of them. It is $e^{-kc_s|\Delta\eta|}$: holding c_s at the plasma’s value and varying k reads it as a *scale* criterion, frozen against oscillating, which is the reading just given. Holding k and varying c_s reads the same inequality as a *species* criterion, and that reading supplies something the first does not. A pressureless component has $c_s = 0$ identically—not small, not slowly varying—so its exponent vanishes at every wavenumber; on the segment its mode equation is $\psi'' = 0$, whose constant solution the kernel leaves untouched with no approximation at all. *The matter sector’s inheritance is therefore exact rather than adiabatic:* it does not rest on horizon exit and it does not degrade at small scales, where the frozen-mode argument does. The two arguments agree wherever both apply, and only the pressureless one is available everywhere. *Neither reading carries a constraint the other lacks.* On this geometry the exponent at the first acoustic peak is $\simeq 1.6 \times 10^2 c_s/c$, so a component with a per-cent sound speed would arrive suppressed by $e^{-1.6}$ and one with ten per cent by e^{-16} , which reads as a sharp condition on what the progenitor may hand over—perturbations inherited from a *cold* species and from no other (receipt: `the_filter_is_a_sound_speed_filter.pyR`). *The reading is too strong, because the exponent has nothing to act on.* The kernel acts on *oscillatory* content—a frozen, zero-frequency mode is a fixed point of it—and on the progenitor’s own interior nothing arrives oscillating: $|aH| = |a'/a|$ diverges as $1/x$ at the branch point while c_s saturates at $1/\sqrt{3}$, so $c_s k/|aH| \rightarrow c_s kx \rightarrow 0$ for every k , and every mode from $\ell \simeq 28$ to $\ell \simeq 2475$ exits the comoving sound horizon strictly before the crunch—the highest with a fifth of a per cent of the collapsing leg still to run [17]. *So the two readings do not divide the work between them: the scale criterion is satisfied by everything, and the species criterion selects nothing.*

One consequence of that freezing bears on the acoustic sector rather than on the crossing, and it is computed rather than left as an inference. If every mode is outside the comoving horizon at the branch point and the acoustic modes are inside it at the onset (Prop. 1), then between those two loci the modes *re-enter*—and in flat Λ CDM a horizon crossing is exactly where the k -independent acoustic driving phase shift is acquired. *The re-entry is real and its boundary is computable:* the mode whose crossing coincides with the onset is $k = 0.0111 \text{ Mpc}^{-1}$, i.e. $\ell \simeq 144$, so every mode at and above the acoustic peaks crossed before the plasma began while $\ell \lesssim 144$ is still outside at the onset and crosses later^R. *But nothing is imprinted there, and the reason is structural rather than quantitative.* On the geometric rate the crossing occurs under pressureless matter to better than a part in 10^4 , and the potential equation for a pressureless component, $\Phi'' + 3\mathcal{H}(1+w)\Phi' + [2\mathcal{H}' + (1+3w)\mathcal{H}^2]\Phi + wk^2\Phi = 0$, contains no k at all once $w = 0$ —the wavenumber enters only through the pressure term. Integrated on this background from $a = 10^{-7}$ to the onset the potential is constant to twelve digits and *identical across four decades in k* , the integrator taking the same number of steps for every mode because it is handed the same problem. *So the horizon crossing is not an event for the potential here:* before the onset the perturbation problem on the observable leg is scale-free, and a scale-free problem imprints nothing. In flat Λ CDM the same modes cross under *radiation*, where $w = 1/3$ restores the k^2 term and the potential decays on the sound-crossing time by construction, which is why that shift is the same for every mode. *This closes a candidate mechanism rather than supplying one:* the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval, and the first peak’s position is left open by it. The crossing is lossless for every species rather than for cold ones alone. *That is a loss of a claimed prediction and not a gain, and it is recorded as one;* what survives is the pressureless argument’s positive content, that the inheritance is exact rather than adiabatic, now holding of every species and not of dust alone.

4 What the substrate determines: coherence from the null boundary

The acoustic peaks are sharp because the modes of a given wavenumber share a common phase at last scattering. Inflation produces this coherence dynamically, by freezing modes while super-horizon; Proposition 1 closes that route here. CR obtains it instead from the character of the branch point.

The branch point is a *null* surface—the cosmological horizon promoted to the initial layer [8]. The initial-value problem on a null surface is characteristic, not Cauchy: the data that determine the future are one free function per mode on the surface together with regularity along the generators, rather than a field and an independent momentum. A single datum per mode is a single phase per mode; there is no second, independently specifiable quantity to randomize the relative phase.

We therefore argue that coherence is not imposed but is what regular characteristic data on a null surface *is*.

This is borne out quantitatively, and the scope of the demonstration is stated exactly.

Evaluating the tightly-coupled mode function at last scattering with a *common* seam phase gives a sharp regular comb, whereas drawing the phase independently per mode—the second, free datum a Cauchy surface would admit—washes it out entirely, leaving flat power (peak-to-trough contrast ~ 1 against the coherent comb’s many orders of magnitude; receipt: `verify_coherence_comb.py`^R). *That establishes the mechanism—one characteristic datum per mode is one phase per mode, and coherence follows—and it establishes nothing about where the peaks fall.* The receipt writes the mode function down as $\cos(kr_s)$ with r_s and D_C as literals—and the r_s it writes is the *standard* 147.0 Mpc rather than any value this construction derives; no mode is propagated in it, as its own scope note says, so the spacing $\Delta\ell \simeq \pi D_C/r_s \approx 296$ is the arithmetic of an assumed expression on an imported ruler rather than an output. *It is not the result of propagating the modes, and is not offered as one*^R.

Where the modes *are* propagated the comb’s first few gaps come out at 0.72–0.79 of that spacing under every initial condition tried, but that is a shallow reading of a four-peak series: at the depth the transfer reaches, the *asymptotic* spacing is 0.975 of the acoustic scale (§4.7). *So the spacing is not the deficit*—what the peak spacing lacks is not agreement but a derivation, being asserted here and computed nowhere in this paper.

The null boundary admits only the first column: one characteristic datum per mode is one phase per mode.

The peak *heights* are then carried by a structural argument rather than a bespoke transfer. The driven acoustic amplitude is the resonant Fourier magnitude of the potential’s evolution, $|\tilde{\Psi}(\omega=1)|$ —exactly invariant under time reversal, $|\widetilde{\Psi(x)}| = |\widetilde{\Psi(-x)}|$, a Fourier theorem confirmed numerically on the radiation-era transfer function and on a random control^R.

This cosmology’s driving does not live on an expanding radiation era, which a geometrically fixed rate does not possess; it lives on the *contracting* side—the progenitor’s radiation-comparable collapse phase, whose potential evolution is the Friedmann time-reverse of an expanding radiation era—so its driving magnitude equals flat Λ CDM’s exactly, and the branch point’s faithful transmission (§8) carries it across unaltered.

The acoustic amplitude, and with it the peak-height pattern, therefore match flat Λ CDM’s—by the time-reversal structure, not by tuning—conditional on the inherited medium being the ordinary photon–baryon plasma, the condition the abundance comparison confirms (§2.4; receipt: `cr_collapse_driving.py`^R, the driving validated as a linear oscillator against the known radiation-era boost).

Because the acoustic modes are already sub-horizon at the seam (Prop. 1), each mode’s driving is complete on the collapse side—and the qualification its own receipt carries is that completeness holds for the modes whose entry precedes the horizon maximum at $r_* = 1.53 r_{\text{seam}}$,

entry on the rising branch being earlier the higher the wavenumber, so the acoustic band is on the safe side of that boundary by construction and the low- k end is where it would bite^R; the observed peaks therefore *reduce* to flat Λ CDM's—the driving equality above together with the shared photon–baryon plasma and the coherent seam phase—and match a Boltzmann reference digit by digit ($P_1/P_2 \approx 2.2$, $P_2/P_1 \approx 0.45$, $P_3/P_1 \approx 0.44$; `camb_reference.py`^R). What remains is confined to the *damping* tail ($\ell \gtrsim 1000$), and the three-level rule (§2.4) fixes it rather than leaving it to judgement. Recombination is observable cosmology from the branch point onward, so the photon diffusion length r_D —like r_s —is an L1 quantity: the random walk accumulates against the layer's own geometric expansion, its Thomson microphysics ordinary content physics but its expansion the L1 rate, *not* a radiation-included one.

This is the same branch point-located scoping the cosmogenesis paper derives for the nucleosynthesis window [17], read here in the opposite direction: there the self-gravitating excursion sets the L2 rate radiation is included in, here the diffuse plasma rides the L1 foliation radiation is excluded from.

This is forced: it is the same L1 rate that dissolves the Hubble tension, and one may not take the rate geometric for the peak spacing and radiation-included for the diffusion. The geometric rate is $\sim 13\%$ below the radiation-included one at recombination ($\rho_r/\rho_m \approx 0.3$) and further below it earlier, where the inherited radiation is a larger fraction; *the diffusion length integrates that whole history rather than its endpoint*, and runs $\sim 9\%$ longer than Λ CDM's while r_s is matched to the observed acoustic scale (the closed-form derivation of §4.2 gives 10.8% on a sharper ionisation treatment, and the two are related there); the projection-independent ratio is therefore $\theta_D/\theta_* \approx 1.08 (\theta_D/\theta_*)_{\Lambda\text{CDM}}$ —a 8.2% larger damping angular scale at the onset redshift used there, the value moving with z_{onset} as §4.3 sets out (receipt: `damping_ratio_clean.py`^R, validated against a Boltzmann reference on the radiation-included rate; $r_s = 144.0$ vs CAMB 144.4 — a 0.28% disagreement against the 8.2% signature it feeds, so the reference agrees to thirty times the effect and the margin rather than the agreement is what the word *validated* is carrying^{RL}).

This is the one genuinely CR-specific piece of the height pattern: a real signature *locked to* the Hubble resolution by their shared L1 rate—not an artefact of laying the standard radiation-governed r_s on the geometric rate, the calculation belonging to neither framework this paper warns against (§1).

One thing about this signature is settled and one is not, and it is essential to keep them apart.

Settled: the $\sim 9\%$ in r_D is genuinely CR-specific and does not dissolve under the density that would ordinarily rescale it— $r_D \propto \omega_b^{-0.31}$, so a percent-level ω_b (fixed by primordial deuterium and the peak-height ratio) shifts it by well under a percent (receipt: `damping_reabsorption.py`^R). *Not* settled, and not to be over-read: what that $\sim 9\%$ in the diffusion *scale* does to the observed high- ℓ power. The observable is $\theta_D/\theta_* = r_D/r_s$, and translating it into a spectrum requires the full acoustic transfer—above all, whether CR's high- ℓ peaks, driven on the geometric rate, track flat Λ CDM's, which is verified only through the third peak ($\ell \lesssim 800$). A leading-order pass that simply *assumes* the high- ℓ peaks equal Λ CDM's and applies the extra damping produces a large high- ℓ deficit (receipt: `full_transfer_verdict.py`^R)—but that assumption may not be imported for free: the same geometric rate that enlarges r_D also governs the high- ℓ driving envelope, so CR's high- ℓ spectrum rests on short-wavelength collapse-phase driving rather than on the boost an expanding radiation era supplies.

That driving is computed below (§4.1), and the calculation removes the licence the shortcut lacked: the envelope is derived on the collapse leg rather than imported.

The two structural facts—peaks-match (verified low- ℓ) and diffusion on the geometric rate—therefore constrain, but do not by themselves determine, the high- ℓ shape; the net effect is computed in §4.2; whether it is a tension, a wash, or a distinctive but consistent feature turns on a parameter refit and on the early integrated Sachs–Wolfe term. *That refit is performed*

here. Fitting both arms to the Planck 2018 *plik_lite* binned TT likelihood over its 215 bins ($32 \leq \ell \leq 2492$), with the same five parameters ($H_0, \omega_b, \omega_c, A_s, n_s$) free in each and the published covariance, returns $\chi^2 = 397.13$ for this construction against $\chi^2 = 206.44$ for flat Λ CDM—1.891 and 0.983 per degree of freedom respectively, so $\Delta\chi^2 = 190.7$ at equal fitted-parameter count. The control reproduces the sky essentially perfectly, which is what makes the comparison readable at all; and the verdict it returns is that the spectrum derived here is disfavoured against the standard model on this data. We state it as it stands. What it measures is the five-parameter fit of the spectrum construction, not the geometry that motivates it, and the two are joined by the transfer work of this section rather than by identity. What is honestly claimed here is the effect, not a verdict on it: a real, computed, non-reabsorbable $\sim 9\%$ CR-specific diffusion-scale signature, its observable consequence now derived rather than entangled (§4.2), and the verdict on it left to a likelihood confrontation this paper does not run. The structural point that stands is the *mechanism*: where inflation freezes the phase dynamically and drives the amplitudes on an expanding radiation era, CR fixes the phase geometrically on the single null layer and drives the amplitudes on the time-mirrored collapse.

4.1 The collapse-phase driving envelope

The driving named above can be carried out in closed form, and doing so removes the entanglement the previous subsection had to leave standing. On the radiation-dominated collapse leg the potential obeys $\Psi'' + (4/\eta)\Psi' + (k^2/3)\Psi = 0$, whose regular solution is elementary, $\Psi = 3\Psi_i(\sin x - x \cos x)/x^3$ with $x = k\eta/\sqrt{3}$, and which is *even* in x^R —so the contracting leg carries the same closed form pointwise, a sharper statement than the Fourier-magnitude equality of §4 and consistent with it. Writing $\hat{\Theta} = \Theta_0 + \Psi$ removes the source exactly, $\hat{\Theta}'' + (k^2/3)\hat{\Theta} = 0$: the effective temperature oscillates freely, and the k -dependence rides entirely in the amplitude at horizon entry and in the decay of Ψ .

Horizon entry is fixed by the construction rather than fitted. The comoving Hubble radius is built from the slicing paper’s own turnaround function, $(rH)^2 = (1 - f) + A/r^2$ [4], and has a single turning point on the whole excursion, at $r_* = 1.53 r_{\text{seam}}$ on the corpus’s inherited datum; the *seam* sits just inside that turning point, on the rising branch, so the acoustic modes are already within the comoving horizon there—which is the geometric content of Prop. 1 rather than a numerical accident of z_{onset}^R . *The locus matters and the two available ones give opposite answers*: the branch point lies at the far end of the same rising branch, where $2M/r \rightarrow \infty$ carries aH up without bound and the comoving horizon to zero, so there *every* mode is outside it (§3). It is the seam, not the crossing, that this construction reads—the inversion derived from the metric function rather than asserted^R. On the leg the radiation term dominates because it is constant while the others vanish, so $\eta = r/\sqrt{A}$ to about two per cent at the entry radius of the lowest mode in question and better the deeper the mode enters—sub-percent from $k \gtrsim 30 k_s^R$, and entry then occurs at $x \rightarrow 1/\sqrt{3}$ in the *deep sub-horizon limit*—the radiation era’s own scale invariance, recovered here from the substrate geometry. *The envelope is therefore flat to the accuracy of that limit*: a mode at $k \simeq 10 k_s$ enters within 2% of $1/\sqrt{3}$ and one at $300 k_s$ within 0.1%, all of them reaching the *seam*—where the collapse leg ends—with the same driving amplitude $\Psi(1/\sqrt{3})/2$, and the residual potential is below a percent *on average* while oscillating up to some seven per cent of it where $\cos x_{\text{seam}}$ is near an extremum^R. *Both are qualifications rather than identities: the limit is a limit and the average an average*. Its turnover sits at equality, and the inherited datum places equality at $1 + z_{\text{eq}} = (1 + z_{\text{onset}})/2 = 3399$, essentially flat Λ CDM’s—so the envelope turns over at the same wavenumber, for a structural reason and not by assumption^R.

Two corrections that might have been expected do not enter. The baryon loading is proportional to the scale factor and the driving happens where the scale factor is smallest, so $R \lesssim 0.1$ at the onset and orders below that at entry; the odd/even asymmetry is imprinted afterwards, on the expansion side at $R \approx 0.6$, which is ordinary content physics on the observable leg. *The*

pattern that follows from it is produced and put against the sky. Regenerating both arms at their fitted parameters and locating the acoustic peaks by parabolic fit gives $P_1/P_2 = 2.185$ for this construction and 2.200 for flat Λ CDM, against 2.2564 ± 0.0772 measured the same way on the Planck binned spectrum: both within about one standard deviation of the sky, and differing from each other by a fifth of its uncertainty. The baryon loading's imprint is therefore the same in both, as the ratio-of-contents argument above requires, and it is checked at the level of C_ℓ rather than at the level of the driving—where the loading ratio alone would give 3.5^R. And the free-streaming neutrinos the cooling-leg nucleosynthesis requires do carry anisotropic stress at these temperatures—the branch point is far below their decoupling—but their deep sub-horizon effect is a constant phase shift and a constant suppression, and a k -independent correction leaves a flat envelope flat^R. What it does not leave is the closed form for Ψ itself, the equation becoming integro-differential; the closure is a property of the skeleton, and the flatness is what survives.

4.2 What the envelope buys, and what remains

With the envelope derived rather than assumed, the diffusion signature can be stated without the entanglement. The diffusion length reduces to a single integral over the scale factor in which every microphysical constant sits outside, $1/k_D^2 \propto \int da g(R)/(Hx_e)$, so in the ratio the Thomson physics and the ionisation history cancel identically and the whole difference is carried by $H(a)$ —which is the division of labour this section has argued for throughout. Both integrals close in elementary form, and on the inherited datum the geometric rate gives a diffusion length 10.8% longer^R. *That is the closed-form value and it stands beside the 9% quoted above rather than replacing it:* the earlier figure runs the ionisation history through a Boltzmann-validated recombination, this one takes x_e as a sharp cut at a_{rec} and carries the inherited datum unchanged to recombination. *The two are independent routes to one quantity and agree to about a point and a half,* which is the relationship an elementary derivation should bear to a code-validated one. The sound horizon is the same kind of integral, and it must be taken *from the branch point*: there is no observable expansion below it. Doing so returns $r_s = 146.4$ Mpc against 145.4 on the radiation-included rate—within 0.7% of each other at the onset redshift the inherited datum fixes, the exact crossing sitting a little lower at $z \simeq 6.7 \times 10^3$ ^R—the acoustic-scale calibration, met from the other end and to that accuracy rather than exactly. *Both are the elementary integral; carried through a full recombination history and validated against a Boltzmann reference the same two come out at 144.0 and 144.4, a per cent lower and in the same relation^R.* The observable, in which the common distance cancels, is then θ_D/θ_* larger by 8.2% at this onset redshift, and $\ell_* = \pi D_M/r_s = 302.2$ against the measured 301.76^R.

The high- ℓ consequence follows with no free parameter: $C_\ell^{\text{CR}}/C_\ell^{\Lambda\text{CDM}} = \exp[-(\ell/\ell_D)^2(r^2 - 1)]$ with $r = \theta_D/\theta_* = 1.082$ —the same ratio the preceding paragraph derives, and nothing else—so the ratio is 0.84 at ℓ_D and 0.68 at $1.5\ell_D$ ^R. *Whether that is a tension is a separate question this paper does not settle:* a confrontation refits the cosmological parameters, which is a likelihood analysis and not a closed-form calculation, and asserting exclusion from these numbers would repeat, at one remove, the error of holding everything else fixed while varying one thing. What has changed is the standing of the signature. The signature is a requirement of the construction, following from the inherited datum and the level assignment and nothing else.

The early integrated Sachs–Wolfe contribution completes the picture, and it does not redistribute the signature. That effect exists because a radiation-included background leaves the potentials still decaying at recombination; on a pure-matter background the growing mode is constant, $\Phi'' + 3\mathcal{H}\Phi' = 0$ giving $\Phi = \text{const}$, and this cosmology's rate at $z \simeq 1100$ is matter-dominated to nine orders, the substrate term being utterly negligible there^R. The reason is the leftward reading itself: radiation is inherited content read off the clock rather than a term that sources the rate, so it is absent from the background whose evolution the effect would require. *So this cosmology has no early integrated Sachs–Wolfe term where flat Λ CDM has one—in the*

latter the potential is still some four per cent above its asymptote at recombination, and that residual decay is what the line-of-sight integral picks up.

One scope qualification is owed here, and it narrows the claim without touching the conclusion. The constancy argument runs on $\Phi'' + 3\mathcal{H}\Phi' = 0$, which is the *super-horizon* equation: it drops the $k^2\Phi$ term. For the acoustic modes—inside the horizon at the onset and remaining so—that term does not vanish, and the potential does decay on the observable leg, by a factor of order two across the first few peaks^R. The decay is not a radiation effect: zeroing the radiation fractions in the constraint makes it *larger*, and the rate responsible is $k^2/(3\mathcal{H})$, which grows with k and refers to no content at all. So the statement that survives is that *the substrate rate supplies no early integrated Sachs–Wolfe term*—which is what the leftward reading delivers and what the comparison with flat Λ CDM turns on—while the sub-horizon potential decay is an ordinary consequence of the perturbation’s own scale, present in any cosmology.

The missing contribution cannot offset the diffusion signature, and the reason is structural rather than numerical. It sits near the first peaks and dies away well before the damping tail begins, so the two live at opposite ends of the spectrum; and the first peak is where the amplitude is anchored, by an A_s this construction inherits rather than predicts. A few per cent there is absorbed by that inherited normalisation, exactly as it would be in a framework carrying η from outside. The diffusion signature is not so absorbed: it is a change of *shape* at high multipole, and rescaling the amplitude to restore the first peak would worsen the tail rather than repair it. *What stands between the derivation and a verdict is therefore the likelihood confrontation alone*—a parameter refit, not a further calculation of this kind. That refit is bounded in §4.7 below.

4.3 The diffusion scale, and the error budget

The acoustic scale is met at the directly measured H_0 ; the diffusion scale is not free to follow it. Holding ℓ_* to its measured value fixes the onset redshift, and θ_D/θ_* then follows with nothing left to adjust: it varies from +43% to −3% across the onset redshifts one might consider, so a single datum cannot absorb both observables. *One inherited number, two observables, and the first is spent on the first.* Run at the directly measured $H_0 = 73$, the matter density this construction’s own baryon-acoustic fit returns, its own light-element baryon density, and the measured photon temperature—with the acoustic angle the single microwave-background input—the seam falls at $z \simeq 6.8 \times 10^3$ and

$$\theta_D/\theta_* = +13.96\% \pm 0.89\%,$$

an error budget dominated by the measurement of H_0 itself^R.

That figure and the 8.2% of §4 are the same observable computed at two onset redshifts, and the difference between them is the honest measure of what this signature is currently worth. The two differ by 0.53% in z_{onset} and by 5.8 points in θ_D/θ_* , which is the sensitivity the range above already reports—so the quantity is determined only once z_{onset} is, and z_{onset} is the one number this cosmology fits. *The signature is therefore a computed, non-reabsorbable effect of definite sign and of a magnitude the acoustic-angle fit has still to pin:* it does not float with H_0 , both r_D and r_s carrying $1/H_0$ so that the ratio is H_0 -independent, and it does not dissolve under ω_b ; what it awaits is the same refit the high- ℓ verdict awaits. *We quote the two values rather than choosing between them, and no claim below rests on which is right.*

Two contributions that might have been expected do not enter: the last-scattering surface’s thickness scales with the rate in both its width and its conversion to comoving distance, so it cancels from the ratio; and the recombination redshift, integrated on this rate through the Peebles equation [36] rather than inherited, moves by only −0.11%^R. The same integration returns $z_* = 1091$ on the radiation-included rate, against the 1090 of the standard treatment, which is the check that the ionisation history is being handled correctly.

4.4 The integrated Sachs–Wolfe contribution across the first-peak region

The first acoustic peak is a separate matter and it does not move. Its offset from ℓ_* —220 against 301.7—is largely the phase shift the potential’s decay imparts while the modes are driven, and that decay has the same closed form on the collapse leg as on an expanding radiation era (§4.1), so it transfers. What does not transfer is the early integrated Sachs–Wolfe contribution, and that turns out to be a nearly *flat* multiplicative factor across the first-peak region—some ten per cent in amplitude, varying by two parts in a hundred between $\ell = 150$ and $\ell = 300$ —so it displaces the maximum by one or two multipoles rather than by tens^R. *The first peak therefore sits where it is measured, and for a reason internal to the construction rather than by accommodation.* That refit can be bounded here, and bounding it is worth doing because the reabsorption question is the whole of what a likelihood would decide about this signature. Pinning ℓ_* to the measured 301 and minimising the residual in θ_D/θ_* over the baryon density and the matter density—the former held to the one per cent primordial deuterium allows, the latter to the five per cent the distance and growth data allow—leaves a residual of several per cent^R. The baryon density cannot absorb it, having one per cent of freedom against an effect of order ten per cent at either onset (§4.3); the matter density can, but only by moving some thirty-seven per cent, which is not a refit but a different cosmology. *So the shift has a floor, and the floor is the prediction.* We state this as a bound and not as a likelihood: a full comparison fits the whole spectrum with every parameter and its covariances, where this fixes one observable and minimises one derived ratio. It settles that the signature does not dissolve under reabsorption; it does not settle the verdict.

4.5 What the instrument’s residual is made of

The two-arm comparison’s floor is set by components of the instrument rather than by either cosmology. Two candidates for the accuracy it lacks are measured here.

Of the two, one cannot pay at all^{RL}. Above $\ell \simeq 40$ the whole effect of reionisation on the temperature spectrum is the constant screening $e^{-2\tau}$ —the second visibility bump contributes only below $\ell \simeq 30$, outside this comparison’s range—and the comparison fits one amplitude in closed form. *An amplitude fitted as $A = (m^\top F d)/(m^\top F m)$ is homogeneous of degree -1 in the model, so rescaling the model returns A/s and leaves the residual unchanged:* the degeneracy is exact rather than approximate, and it is a property of the statistic rather than of the size of τ . Checked at $\tau = 0.054, 0.10$ and 0.30 , the change in χ^2 is zero to machine precision at all three. *Lensing is a different matter.* A smoothing in multipole, scanned in width, takes the control from 1320 to 921; and a width that *grows* with ℓ beats a constant one by a factor of 1.8 in the improvement, each compared at its own optimum, so the preference is for a shape and not merely for a size—which is *lensing’s own signature, since a deflection of fixed angle subtends a larger fraction of an acoustic wavelength at high ℓ .* We state what that is and is not. The width is fitted, so the improvement bounds what a derived calculation has to play for rather than measuring a lensing amplitude; and even at its own optimum some 70% of the residual survives it. *So the lensed spectrum is the next thing this instrument owes, and it is not the last one.*

That calculation is done, and it comes in under the bound the fit set. Applying Λ CDM’s own lensed-to-unlensed ratio to the arm—a derived operation with no free width—takes the control from 1320 to 989, an improvement of 331, concentrated in the damping tail where a fixed-angle deflection fills the troughs. *That is below the 400 the fitted width bounded, as a zero-parameter derivation must be,* and the acoustic peaks do not move under it. The corpus’s own first-order kernel overshoots the bound to 508 by breaking down in that same tail—returning a spurious +13% at $\ell = 1900$ where the full operator gives +6.5%—which is why the derived number is the full operator’s. *Most of the fitted headroom is recovered by a genuine lensing, and about 989 of the 1320 survives it:* the verdict of the last paragraph stands from the derived side, lensing

real and not the last build^R.

And what the residual is MADE of was decomposed alongside it, which is what let the size of that gain be anticipated rather than discovered^R. Projecting named templates through the same inverse covariance the comparison uses, the peak *positions* account for a tenth of a per cent of the residual—which is why the acoustic peaks do not move under the lensing above, and the two findings corroborate each other—while the peak-trough *contrast* at fixed position accounts for two-fifths of it, this transfer’s contrast running some thirteen per cent too high. *Reducing contrast without moving peaks is what lensing does.* A smooth multiplicative envelope accounts for a further sixth, and *about half the residual is in neither*—so a complete lensing calculation was expected to leave a comparable amount behind, and it did.

4.6 The shear coefficient, derived

The coefficient is derived rather than fitted, and the derivation settles it against the fit^R. The coefficient is measured by running the same oscillator twice on the same background: once as the perfect tight-coupled fluid, which is undamped by construction, and once as a photon hierarchy carried to $\ell = 24$ with Thomson scattering, the polarisation multipoles evolved alongside and the baryons given their own velocity. *Their ratio is the damping and nothing else*—the slowly varying prefactor, the drift of the sound speed and the baryon-loading dependence of the oscillation are common to the two runs and divide out identically—and reading it locally, between consecutive extrema, lets the residual be extrapolated in $(k/\tau')^2$ rather than smeared over an interval. With the polarisation multipoles suppressed the quadrupole’s scattering term reduces to $-\frac{9}{10}\tau'F_2$, which is the unpolarised closure, and the coefficient comes back as 0.8852 against 8/9; with them carried it is 1.0618 against 16/15. *The polarisation correction itself, which is the quantity in dispute, is a factor 1.1996 against 6/5: three parts in ten thousand.* Both absolute values sit about half a per cent low by an amount proportional to the coefficient, so the residual is a common normalisation and cancels in the ratio.

4.7 The signature at the directly measured rate

Two features of this cosmology’s acoustic scale are worth separating, because only one of them is a prediction. The angular scale $\ell_* = \pi D_M/r_s$ is *independent of H_0* here: both lengths scale as $1/H_0$ on a matter-and-substrate rate, so the ratio does not know about it. In flat Λ CDM there is no free early datum—the sound horizon is fixed by the matter and baryon densities—so the measured angle *does* determine H_0 , and that is the origin of the lower value inferred from the microwave background. Here the branch point’s radiation amplitude takes that constraint instead, and H_0 is left to be the directly measured one^R. *So the acoustic scale is an accommodation, and it is spent.*

The first peak’s position is fixed by the initial data, not by the transfer. The reasoning above establishes that the driving phase shift carries over from the leg, and infers from that a first peak near the same fraction of ℓ_* as flat Λ CDM’s—which is how 220 enters. A perturbation calculation built on the branch point handover was quoted against it, placing the peak near $\ell = 150$, at half of ℓ_* rather than at 0.73, with a first-to-second peak ratio of $\simeq 1.45$ against the measured 2.21^R. *Neither figure survives an examination of what fixes it.* Substituting the instrument’s initial-data block and leaving every other line of it byte-identical—the background, the hierarchy, the opacity, the damping, the discrete ladder, the projection—moves the first peak across $\ell_1 \in \{150, 165, 315\}$ and the ratio across $P_1/P_2 \in \{0.93, 1.21, 1.45, 2.02\}$, over four readings of the *same* stated initial condition^R. *Two of those readings are the instrument’s own text.* Its amplitude is taken from a smooth matched envelope while its initial velocity is taken from a numerical derivative of the oscillatory closed form—two different functions—and its docstring describes a common phase taken at an extremum, *with the velocities zero*, which the code does not impose. Imposing it returns $P_1/P_2 = 2.02$ against the measured 2.21. *So*

the height disagreement is a property of one implementation and not of the construction. The numerics are not the source of it: the integrator runs at a stability number of 0.47 against its own limit of 2.8 with at worst 120 steps per acoustic oscillation, and quadrupling the step count reproduces the peaks exactly.

The spacing and the first peak respond to the initial data differently, and the difference is the whole of what follows. Under every one of the four initial conditions the instrument's source comb comes out at 0.72–0.79 of π/r_s rather than at π/r_s : the initial data move the first peak by 165 multipoles and do not move this. *That reading is carried to a depth it could not reach^R:* the figure is the first few gaps of a series that was only four peaks long at the settings a four-condition scan could afford, and at the depth the transfer actually reaches the *asymptotic* spacing is 0.975 of the acoustic scale. *What is left of this item is therefore not a spacing deficit,* and the paragraphs at the end of §4.7 say what it is instead. *The disagreement is not a spacing deficit, and neither is it simply an acoustic phase^R:* a free choice in the seam datum moves the acoustic phase across the control's own value, so the phase is not carrying the disagreement either, and no single quantity this section can name is. *That is a disagreement with the acoustic scale inside the one instrument in this programme that propagates modes,* and it is not caused by the initial data. *It is also not answered by the coherence argument of §4,* whose comb is written down rather than propagated— r_s and D_C entering its receipt as literals and the mode function as $\cos(kr_s)$, as that receipt's own scope note says^R. *So the peak spacing is asserted in one place and derived in none—its asymptotic value where the transfer is run being 0.975 of the acoustic scale, which is agreement and not a deficit.* We record the missing derivation rather than a disagreement. *The figure of 220 should still be read as what the transfer argument gives, and the first peak's position, together with the spacing, treated as open—but the ground for treating it as open is now the missing propagation, not a rival number.*

The propagation is carried with a live control and a guard that can fail^R. Both arms run on one set of equations, so nothing below is a difference of machinery. *The guard first, because it is what licenses the rest.* With every coupling to the potential removed at every site—the $4\Phi'$ in the photon, neutrino and cold-matter continuity equations and the $k^2\Psi$ in the photon–baryon and neutrino Euler equations, in both the tightly-coupled and post-switch branches—the source comb lands on the integers on *both* arms, at 1.015 and 2.000 for flat Λ CDM and 1.012 and 1.997 here. *So the sound horizon, the measure, the wavenumber grid and the phase extraction are validated as an ensemble, and every departure from $n\pi/r_s$ in a driven run is the driving and nothing else.* The control next, with its floor, which is the point of running one. The Λ CDM arm returns peaks at 224, 536 and 808 against the measured 220.6, 538.1 and 809.8: positions good to 1.5% at the first peak and to 0.4% and 0.2% at the second and third, and *heights good only to 13–26%, so no height claim from this instrument is meaningful below about a quarter.* And one objection to any figure of ours is closed by measurement rather than by argument: run on the physical discrete ladder $k_L = \sqrt{L(L+2)}/r_0$ and on a dense continuum sampling, this cosmology's arm returns *identical* peaks, so the discreteness is not what sets the position.

The measurement is then stated in the only form comparable across arms, the two running at different H_0 and Ω_m so that D_M and r_s differ by some six per cent each and cancel, ℓ_A coming out 301.4 against 301.6. *On the initial datum this paper's own text describes—the effective temperature flat in k , the phase common, and the velocities zero—the first peak sits at $\ell_1/\ell_A = 0.570$ against the control's 0.743 and the sky's 0.731. That is a twenty-three per cent deficit against a one-and-a-half per cent floor,* and the height ratio compounds rather than cancels it, $P_1/P_2 = 0.87$ against the control's 1.93. *We state what this is and what it is not.* It is not a prediction: §4.7 has just recorded that the figure moves with the initial datum, and it moves here too. *What has changed is that it now moves against something—a control, a floor, and a guard—where before it moved against nothing at all. That question has an answer, and it is the rate's own.* The figure above is computed with the perturbations evolving on the foliation's stacking rate; the framework assigns them to the leaf, since a process running in the

content takes the leaf's (§2.2), and the two differ by thirteen per cent at recombination and more earlier. Evolved on the leaf congruence the first peak's *position* is no longer in deficit: the first peak spacing comes out 312 against the sky's 317.5, one and seven tenths per cent, and the height ratio P_1/P_2 moves from 0.97 to 2.01 against the sky's 2.22^R.

What remains is one feature of the comb, and on this rate it is a prediction rather than an open number. The sky's peak spacings alternate—317.5 then 271.7, the second gap contracting—and this comb's do not: its first two gaps are equal to the resolution at which they are read, under two initial conditions that move every peak. *The cause is the sentence above, read forward.* The alternation is the compression–rarefaction asymmetry, and the standard shift that carries it is universal only where every mode crosses the horizon while there is a plasma to be driven; on this rate the acoustic modes re-enter at $z \sim 3 \times 10^4$ to 3×10^5 , all of them *above* the onset at $z_{\text{onset}} \simeq 6.8 \times 10^3$, so none of them crosses while our plasma exists. *That is measured on the potential's own evolution and not read off the peak positions:* Φ does not sit frozen here—it decays by some forty per cent for the first peak modes, and at a rate per acoustic half-period that is the same in both arms to within the spread across modes—so the ongoing decay is not what separates them. *What separates them is the impulse at crossing,* which the comparison arm's modes receive within the plasma and this one's cannot, their horizon entries lying at 57.7, 23.2 and 14.7 Mpc against an onset at $0.402\eta_{\text{rec}}$, so that each is switched on already deep sub-horizon, $k\eta_{\text{onset}} = 3.1, 7.8, 12.2^{\text{R}}$. *The uniform comb follows from the onset lying below the re-entry redshifts, and that ordering is not adjustable:* the nucleosynthesis plasma is the progenitor's, on the transit's cooling leg, and is complete before the branch point [17].

The prediction is calibrated as well as stated, and the calibration is reproducible rather than reported^R. Raising the onset past the re-entry redshifts restores the alternation progressively—one mode crossing gives $g_2/g_1 = 0.92$, all three give 0.895—and it saturates at the control's 0.897, so a comb with crossing-during-plasma is reproduced exactly by these same equations. Against that control this cosmology's deficit is 0.103; the further 0.041 to the sky's 0.856 is the instrument's and is carried by both arms alike. *Neither the loading nor the initial datum carries it:* with the baryons removed both combs widen, this one more, and lending it the control's no-loading comb with its own loading response returns the observed alternation; giving the modes their full pre-onset acoustic history relocates every peak and leaves the two gaps equal.

The comb is not the only place this shows, and the second reading is on different modes. Write $Q(k)$ for the accumulated sound phase at first acoustic turnover in units of a half-period—unity for a free oscillator, and calibrated to unity on both arms with no drift in k . *It is defined only where there is a turnover to accumulate to.* The mode setting the first peak is the one caught at maximal compression at recombination—that is what puts it there—so it has not reached a velocity turnover and Q does not exist for it; the quantity is read on the modes that have turned over by then, the third peak and smaller scales. *So the comb and Q are two independent consequences of the mechanism on different modes, and not one mode's phase measured twice:* the comb's own source extrema project to the spectrum peaks at under two per cent, so the first peak's position is a statement about a density extremum, while Q is a statement about a velocity turnover a quarter-period away. *The two arms separate in shape and not in normalisation:* the comparison arm returns Q flat at 0.79, driven and below unity across the sampled range, while this one returns Q *rising* toward the free-oscillator value, 1.28 to 1.06^R. *A uniform comb is an undriven comb, and $Q \rightarrow 1$ is what undriven means.* Flat against rising is independent of normalisation, which is what makes the second reading a test of the mechanism rather than a restatement of the first.

With the guard in place the driving itself can be measured rather than modelled, and it is ^R. Each wavenumber is integrated twice on the same equations, driven and undriven, differing in one multiplicative switch and in nothing else; $Q(k)$ is the accumulated sound phase in half-periods at the first acoustic turnover, and *the undriven column, which a free oscillator must return at exactly one, is the calibration*—it comes out between 0.9968 and 1.0003 on both arms.

Read against it, *flat Λ CDM's acoustic phase is flat in k* , running from 0.791 to 0.774 across $k = 0.045\text{--}0.16\text{ Mpc}^{-1}$, a spread of two per cent: every mode crosses the sound horizon during radiation domination, where the potential decays on the sound-crossing time by construction, so each acquires the same shift. *This cosmology's varies by a factor of 3.9 across the same band*, 0.457 to 0.117. *That is the acoustic-phase disagreement seen at the level of a single mode rather than of a spectrum—a driving shift that varies with k where the control's does not*, and it is what a rate carrying no radiation source predicts: there is no crossing at which a universal shift could be acquired, and the interval before the onset is scale-free. *The scaling is quoted for what it is and not more*. A power-law fit returns $Q \propto k^{-0.62}$, which collapses the measurements to a factor of 1.2 where k^{-2} spans a factor of 16; *we record that the exponent is not the -2 a relaxation on $3\mathcal{H}/k^2$ would give*, and leave the identification of the process open rather than name one the measurement does not support.

Splitting the driving into its two couplings localises the dependence, and the result is not where a relaxation timescale would put it^R. The potential enters the plasma twice: as $4\Phi'$ in the continuity equations, its rate of change feeding the density, and as $k^2\Psi$ in the Euler equations, its gradient driving the velocity. Switched independently on the same integration, *flat Λ CDM's two couplings are each flat in k and they oppose*: continuity alone puts the turnover at $Q = 1.23\text{--}1.28$, later than a free half-period, and the gradient alone at 0.73–0.87, earlier, the two composing to 0.77–0.86. *So the universality of the standard driving shift is a cancellation between two flat terms rather than the signature of one*. On the geometric rate there is no cancellation left—continuity gives 0.11–0.18 and the gradient 0.06–0.57, both earlier than a free half-period, so they add—and *the wavenumber dependence sits in the gradient*: the continuity channel varies by a factor 1.6 across the band, as $k^{-0.17}$, while the gradient varies by 8.9, as $k^{-1.04}$. *That is the reverse of what a relaxation would give*. A timescale acts on how the potential decays, and decay is the continuity channel, which is the one measured flat. *And neither coupling alone is the answer either*: the driven exponent, -0.62 , lies between the two, so the observed scaling is their interference. *And the process that makes the gradient coupling run as k^{-1} can then be given in closed form*^R. The potential's own equation carries a damping term $-k^2\Phi/3\mathcal{H}$, so Φ relaxes on the time $\tau = 3\mathcal{H}/k^2$. With the continuity coupling off, the oscillator obeys $\Theta_0'' + c_s^2 k^2 \Theta_0 = -(k^2/3)\Psi$, and *the k^2 in the forcing's amplitude is cancelled by the k^{-2} in its duration*: the delivered impulse $\int (k^2/3)\Psi d\eta \sim \mathcal{H}\Psi_0$ carries no k at all. *So the first turnover is not at an acoustic phase but at the moment the impulse has been delivered—at a fixed multiple of τ , measured constant at $\kappa = 1.79$ across a factor of eight in k* . Since Q is the accumulated phase in half-periods, $Q = k \int c_s d\eta/\pi$, *it already carries one factor of k* , and

$$Q(k) = \frac{3\kappa c_s \mathcal{H}}{\pi} \frac{1}{k},$$

predicting 0.01040 Mpc^{-1} for Qk against a measured 0.01037, and reproducing Q itself to 1.6% across the sub-horizon band with nothing fitted but κ . *The complementary channel is flat for the complementary reason*: the $4\Phi'$ term feeds the density with the potential's total change, $\int \Phi' d\eta = -\Phi_0$, which is the same however fast the relaxation runs. *The observed $k^{-0.62}$ is therefore not a power law of anything—it is the crossover between a flat channel and a $1/k$ one*.

This closes the mechanism and does not close the discrepancy, and the two should not be confused. The first peak still sits where §4.7 measures it, 23% low against a 1.5% control floor. *What has changed is that its cause is now derived rather than described*: a rate carrying no radiation source has no era in which the potential decays on the sound-crossing time, so the two couplings that oppose one another in flat Λ CDM instead add, and the one that survives as a driving impulse delivers a k -independent kick to an oscillator whose frequency is proportional to k .

A likelihood-level comparison can then be attempted, and the result is that it cannot yet be made^R. The criterion was fixed before the data were opened: the pipeline is wired if a

Boltzmann-code flat- Λ CDM best fit returns the banked $\chi^2 = 206.4$ over 215 Planck `plik_lite` TT bins ($\chi^2/\text{dof} = 0.96$); the instrument’s own floor is what *its* Λ CDM arm costs through the same likelihood on the same bins; and the only readable quantity is the difference between the two arms of one instrument, since comparing this construction’s spectrum against a Boltzmann code’s would charge it for the instrument’s error. *Run that way, on 185 covered bins with a single fitted amplitude, the Λ CDM arm returns $\chi^2/\text{dof} = 103$ and this cosmology’s arm 558.*

The ordering is a fact and the ratio is not a p-value. Planck’s binned errors are a few tenths of a per cent while this instrument’s peak heights carry 13–26%, which is tens of standard deviations in a single bin: *both arms sit far outside the regime in which that likelihood discriminates between models*, and a control a hundred times too poor certifies nothing about what it is compared with. *So the honest verdict is that the likelihood cannot arbitrate here*, and the number that survives is a statement about this instrument rather than about the sky—the construction’s arm costs 5.4 times its own control. *Half of that cost is removable, and the removable half is the one that matters for the position*^R.

Evaluated on a delta-function last scattering the transfer computes its source at a single η_* —an instantaneous last scattering. The visibility $g = \tau' e^{-\tau}$ in fact peaks at $\eta = 280.75$ with a full width at half maximum of 37.8 Mpc, a tenth of a wavelength at the first peak and more at the third, and rebuilding the transfer as the standard line-of-sight integral across it takes the control’s first-peak position from 1.66% to 0.16%. *The heights it did not fix, and the reason was diagnosable from their sign*: both ratios came out low, so the second and third peaks were too high, and worse the higher the multipole—which is under-damping. The cause was that the tight-coupling closure switches off at $\tau' = 0.6 \text{ Mpc}^{-1}$ while the visibility peaks at 0.06, ten times lower, so the photon quadrupole and the baryon slip were absent through the whole of last scattering and the transfer carried no diffusion damping at all. *Restoring it from the standard tight-coupling integral evaluated on each arm’s own opacity and rate*, $k_D^{-2} = \int [R^2/(1+R) + 8/9] d\eta/[6(1+R)\tau']$, returns $r_D = 6.58 \text{ Mpc}$ where a Boltzmann code gives 6–7, with nothing fitted, and takes the height ratios from -12.9% and -33.5% to -2.8% and -2.1% ^R. *The envelope and the hierarchy’s own tight-coupling damping act on disjoint intervals, and must: applied over the same range they double-count, and the control comes out over-damped. On the instrument’s floor that is a factor of five—the likelihood’s control at $\chi^2/\text{dof} = 23$ against 103 for a transfer carrying no diffusion damping at all—without its verdict turning over: two to three per cent is not sub-per-cent, and the comparison of §4.7 still cannot arbitrate. What the floor moving while the verdict does not gives is the measure of how much further the instrument has to go.*

The height comparison is stated against the covariance rather than against bare numbers^R. Every height figure above compares P_1/P_2 and P_1/P_3 with two bare numbers. Read off the `plik_lite` bandpowers with the published covariance—the full 2×2 block for the two peak bins, propagated through the ratio—the sky’s own values are 2.256 ± 0.077 and 2.280 ± 0.074 , which is 3.4% and 3.2%. *The 12.9% and 33.5% were therefore four and ten standard deviations and stand; the 2.8% and 2.1% are about one, and do not.* A target of sub-per-cent heights asks for several times more precision than the statistic it is quoted against possesses. *The statistic that carries the comparison is χ^2 over the 185 covered bins*, where the control sits at 22.5 per degree of freedom against a five-parameter fit’s 0.96. *The numerics were cleared before any of this was attributed*: raising the mode count by 1.4, the wavenumber range by 1.3 and the conformal-time sampling by 1.8 moves the two ratios by 0.09% and 0.37%. *A multipole ceiling truncating the wavenumber integral beneath the third peak moves them by two and seven per cent, which is why the ceiling is set above it.*

A stronger claim is available on these numbers and is not supported by them. Sliding a single multiplier on k_D^{-2} —which moves the damping scale and cannot change the envelope’s shape—the second-peak ratio selects 1.094 ± 0.184 and the third selects 0.897 ± 0.055 : a split of 1.03 standard deviations. *A stronger reading is available on these two numbers—that no single*

damping scale fits both, so that the residual is the envelope's shape rather than its size—and the split does not support it at its own error bar. We do not draw it. What χ^2 does resolve is the coefficient in the shear term, and the comparison is sharp: its minimum sits at 0.865 against the 0.859 that corresponds to 8/9, so the polarisation-corrected 16/15 costs 1123 units of χ^2 , some seven inflated standard deviations. The fit does not decide it, and must not be allowed to: a coefficient chosen because it fits is a fitted parameter whatever it is called, and what the 1123 measures is a residual error in this instrument's damping tail of a size that a shear coefficient could absorb—which is precisely the reason it may not. The coefficient is settled by derivation instead (§4.6): the free-streaming photon hierarchy carrying the polarisation multipoles returns 1.0618 against 16/15, and the polarisation correction itself 1.1996 against 6/5, three parts in ten thousand.

So the coefficient is 16/15, and adopting it makes the control worse: χ^2 per degree of freedom on the Λ CDM arm moves from 22.5 to 28.6, the 1123 units the previous paragraph recorded. We take the derivation and pay the χ^2 , because a coefficient chosen because it fits is a fitted parameter whatever it is called. And what the 1123 measures is now nameable rather than mysterious. Polarisation enters the transfer twice: it raises the damping coefficient, and it contributes to the source through $g\Pi/4$ and $(3/4k^2) d^2(g\Pi)/d\eta^2$ with $\Pi = \Theta_2 + \Theta_0^P + \Theta_2^P$. This transfer has the first and not the second, and the half it has is the half that removes power—which is the specification of what remains, and it is not a question about this construction. One figure that might have been expected to move does not: the diffusion-length ratio of §4.2 goes from +10.83% to +10.87%, because the coefficient very nearly cancels in a ratio of two rates.

Carrying the multipoles rather than an envelope takes the control most of the remaining distance^R. The photons are now evolved as a Boltzmann hierarchy with the polarisation multipoles alongside, the baryons are given their own velocity and their own density contrast, and the source carries $g\Pi/4$ and $(3/4k^2) d^2(g\Pi)/d\eta^2$. The first four peaks land at 220, 540, 812 and 1124 against the sky's 220.6, 538.1, 809.8 and 1147.8—the second peak 0.35% out and the third 0.27%, where the fluid transfer placed them at 524 and 804—and χ^2 per degree of freedom falls from 28.6 to 7.1, or to 4.2 over the range in which the wavenumber integral is not starved by the instrument's own truncation.

The prediction is checked against the number it predicted. Running the same hierarchy with only the two polarisation source terms removed—which is exactly the half an envelope can supply—costs 702 units of χ^2 , against the 1123 that adopting the derived coefficient cost. The baryons need their own density and their own velocity together, and either alone is worse than neither: with a density but the photons' velocity the density they carry is the photons' own and nothing lets them fall into the potential wells after decoupling, and the control degrades; with both, the residual halves. A physical effect taken in half is not a small error but a different and inconsistent model.

What is arbitrary in the construction is measured rather than argued: doubling the hierarchy depth moves χ^2 by nothing at the printed precision, and doubling the opacity at which the fluid hands over moves it by 1.5%. The handover also makes a double count impossible rather than merely absent, since the envelope is frozen where the multipoles begin and the two therefore act on disjoint intervals.

Under all of it this cosmology's arm has not moved. $\ell_1/\ell_A = 0.5703$ through six distinct states of the instrument—a delta-function transfer, a line-of-sight transfer, a derived damping envelope, a scan of that envelope's scale, a derived shear coefficient, and now the full hierarchy—while the control's χ^2 per degree of freedom fell from about a hundred to seven. We record at once what that invariance is of, because it is easy to read as more: every one of the six varies the transfer, and none varies the seam datum—so it says the transfer does not set the first peak, and says nothing about what does. What does is established below^R. We report that and convert it into nothing. The comparison of §4.7 still cannot arbitrate, for the same reason it could not before: a control seven times worse than a fit cannot certify what it is compared with. What

has changed is the size of the gap and the fact that no missing sector remains named—what stands between this instrument and a control that could arbitrate is accuracy rather than a piece of physics we know we have left out.

One further object is built here and it is not the one the number above rests on. The operator applied above is Λ CDM’s own lensed-to-unlensed ratio—validated, and the right instrument for the figure. *The lensing potential can also be derived on this construction’s own Φ , carried past the transfer’s endpoint by the background’s growth factor and normalised by the amplitude the temperature comparison already fits, so that it introduces no parameter and imports no transfer function; the photon deflection it implies is 2.69 arcminutes.* *We record what that agreement is and is not:* the figure it matches is recalled rather than derived here, so what is established is internal consistency between the growth, the projection and the temperature normalisation, and the agreement with the literature is corroboration at the strength of a memory. *It is kept because a corpus-native potential is what a later CR-side lensing calculation would need, the imported ratio being a Λ CDM object.*

And the residual that survives has now been located, and it turns out not to be physics^R. Of the 989 left after the derived lensing, 725—73%—sat in the thirty bins above $\ell = 1500$, which is a sixth of the bins. *That is where this instrument’s wavenumber integral is starved by construction,* since the mode grid is built as $k = \ell/D_M$ with ℓ below the truncation, so the highest multipole computed has no wavenumber headroom at all while C_ℓ draws on every k with $kD_M \geq \ell$. *So the question was settled by removing the truncation rather than by arguing about it.* Re-run with the grid carried half again as far and the sampling density held fixed, and scored on the same one hundred and eighty-five bins, the control falls from 989 to 218— χ^2 per degree of freedom from 5.34 to 1.18, against 1.01 for a reference Λ CDM lensed spectrum on those bins—and the ratio to that reference in the last band rises from 0.42 to 0.99. *The truncation was 78% of what remained. So the half of the residual that the template decomposition could not name was numerical, and nothing standing between this transfer and the sky is physics.*

We state at once what that does not mean. 1.18 is not a fit: it is seventeen per cent above a true Λ CDM’s χ^2 on the same bins. *That remainder is named, and naming it required changing the measurement^R.* The floor as the pre-registered rule defines it is a difference of two numbers each computed *against the sky*, so it mixes how far the two models are apart from where each happens to sit relative to one noise realisation—and across three defensible reference Λ CDMs it takes the values 31, 22 and 27, *a spread comparable to the quantity being reported.* *Measured instead as what it actually is,* a χ^2 distance between the two models in the comparison’s own metric and with no data in it, the two sit 21 apart over one hundred and eighty-five bins: *about a ninth of a χ^2 unit per bin, or a third of a standard deviation.* Amplitude-matched, this construction’s arm tracks the reference to better than 1.6% everywhere, and the departure is a slow undulation in multipole rather than a mis-phased oscillation.

Two things follow and we state both. *The reference’s own neutrino content moves the floor by a factor of three—it carries one massive species and this instrument carries none—so the neutrino mass accounts for roughly a third of what remains.* *And the ordering between the two measures reverses:* the reference giving the smallest floor through the data is the one this construction is furthest from as a model, *so a reference chosen to flatter the instrument is available and we name it rather than use it.* *What this does to the separation of the last section is nothing except make it safer:* the floor is between 21 and 63 across every reference defensible here, against a separation of 51848, so the rule’s margin is three orders of magnitude whichever is taken.

That has a consequence for the comparison of §4.7 which we state carefully, because the condition for stating it was written down before the answer was known^R. That comparison was fenced on a pre-registered rule whose own correction recorded the defect that disabled it: the rule was a ratio test and never asked whether the instrument’s floor was small in absolute terms, and it was not—the control sat near a hundred times the data’s precision. *Carried at a floor*

small in absolute terms, the test reads. With both arms carried at the same wavenumber range—so that the construction is not charged for the instrument’s truncation, which the rule’s third clause forbids—the floor is 31 over the full range and 3.8 below $\ell = 700$, while the separation between the two arms on the same bins is 51848 and 23687. Below $\ell = 500$ the floor is -0.2 : the control scores no worse there than the reference, so the statistic cannot see the instrument’s error at all. *The rule’s second clause asks for one order of magnitude and receives three. And the separation is physics rather than arithmetic because it is invariant under the instrument’s accuracy while the floor is not:* run at four levels of transfer fidelity—a two-moment fluid, a fluid carrying the derived shear coefficient, the full polarised hierarchy, and the same with the wavenumber range opened—the control spans 21.2 to 1.18, eighteenfold, while this cosmology’s arm spans 290.7 to 302.1, four per cent. *An error the instrument carries moves both arms; this moved one.*

We say exactly what that is and stop there. It is a measurement discrepancy and not a verdict on the construction—the pre-registered rule says so in its own terms, and nothing here converts a number into a framework judgement. *Nor is the deficit itself new:* $\ell_1/\ell_A = 0.5703$ against the sky’s 0.7312 is the same 22% this section has carried throughout. *What is new is a control good enough for it to be read through a likelihood instead of through a ratio. And we record what is not ruled out:* a defect shared by every state of the instrument would not show up in any of these comparisons, and the honest form of that worry is that 0.5703 has not moved across eight of them—which is evidence either of a robust prediction or of a shared upstream constant, and these measurements cannot separate them. *One candidate for such a defect was removed here rather than argued away:* the discrete wavenumber ladder this cosmology’s arm uses samples the projection below the rate the instrument’s own aliasing gate demands, and that gate waives itself on the claim that the ladder is physical, which is a claim and not a measurement. *The measurement the gate itself names has now been made:* run against a continuum sampling in place of the ladder, the projected peaks move by 0.16% at the first and less than a tenth of a multipole at the fifth, so the answer does not depend on the ladder sampling^R. *A bound replaces the claim:* the ladder’s imprint dies at $\ell \simeq k_2 D_C e^{3\sigma}$ with σ the transfer’s width in $\ln k$, so reaching the first acoustic peak would need a transfer drawing power over a factor of some eight in k —far broader than any^R. *And the direction is the reverse of the naive one:* a broader transfer does not average the ladder away, it reaches further into the ladder’s coarse bottom and carries discreteness up in ℓ . *Run on a continuum grid at two and a half times the sampling, the spectrum agrees to better than 0.7% and the χ^2 to two parts in fifty thousand.*

That is a statement about the deficit rather than about the instrument. On the continuum grid this cosmology’s arm does not move at all— $\ell_1/\ell_A = 0.5703$ on both transfers, to four figures—while the control’s error fell by a factor of ten. *The most suspect component of the instrument was replaced wholesale and the deficit did not follow it down,* which on that reading stands at 21.9% against a position floor of 0.16%. *We state it at that strength and no higher, and the paragraphs below lower it further:* the first peak’s position on this rate is not an artefact of how last scattering was modelled—but it is a function of the seam phase, which is upstream of last scattering, and the figure quoted here is the one reading of the datum the instrument happens to carry^R. *The quantity that survives every reading is not this one,* and it is given its final form at the end of this section.

And we now state something narrower still, because the invariance quoted just above is an invariance of the wrong thing^R. Every rebuild listed here varies the *transfer*; none of them varies the *initial datum*, and the datum is where the first peak lives. This construction’s perturbations begin at the seam, at $z = 6761$, and a mode that sets the first peak crossed the horizon well before that—so its phase at the seam is assigned rather than derived, and the datum §4 states, one amplitude per mode and a *common* phase, fixes that the phase is common without fixing which phase it is. *Scanning that one freedom and holding everything else, the first peak runs from 0.5703 to 1.2599 of ℓ_A —a factor of 2.2, and at one phase the peak near 172 is not present*

at all. *So 0.5703 is a statement about the seam datum and not about a stable position.*

What does not move is the spacing, and the spacing is where the disagreement actually lives. Across the same scan the mean interval between peaks stays between 0.734 and 0.818 of ℓ_A —a spread of one part in nine against the position’s factor of two—and it is never 1. *Both figures are read on the first three gaps, which is what a scan at that depth sees; the asymptotic spacing is given below.*

The second freedom in the datum widens that range rather than confirming it^R. What the amplitude is “flat in wavenumber” at is the remaining choice: the construction evaluates its transfer shape at a single argument for every mode—which is that shape read at each mode’s own horizon crossing, a self-similar reading—where the alternative reads it at each mode’s own phase at the seam. *Across all eighteen readings of the two freedoms together, the first peak spans a factor of 2.26 and states nothing; the spacing spans 0.676 to 0.818, a factor of 1.21, at a mean of 0.77 and never above 0.82. Both freedoms open together give the honest range, and it is the wider one: a single freedom spans 1.11 where the pair spans 1.21. Eight of the eighteen readings are excluded from the spacing by a criterion fixed before the numbers—four peaks with the fourth at least a twentieth of the first—because those readings return a spectrum whose higher peaks have collapsed, and an interval read off three features one of which is absent is not a spacing. Those scans ran shallow enough that eighteen readings were affordable, and at that depth this cosmology’s arm carries four peaks—so the quantity they were reading was the mean of three gaps.*

The quantity the disagreement lives in is the asymptotic spacing, not the first gap^R. Both arms then carry eight peaks, and *this cosmology’s asymptotic spacing is 0.975 of ℓ_A against the control’s 1.002—two and a half per cent, not twenty-one.* The disagreement is confined to the first three gaps, which run at 0.73, 0.85 and 0.88 of the control’s and are level with it by the fourth. *So the series is not uniformly compressed: a mean over the first three gaps measures the transient, and the asymptotic spacing measures the acoustics.*

What is actually wrong is the acoustic phase, and saying so makes the disagreement sharper rather than smaller. Fitted on their asymptotic peaks the two series are parallel lines: slopes 1.003 and 0.976 of ℓ_A , agreeing to two and a half per cent, and intercepts of -0.263 and -0.878 of ℓ_A . A driven acoustic series peaks at $kr_s = n\pi - \phi$, so the intercept is $-(\phi/\pi)\ell_A$: *the two differ by 0.62π in the acoustic phase shift, at a spacing they agree on.* And there is a second, separate thing—a *low-multipole transient this cosmology has and the control does not.* Measured each against its own asymptotic line, the control’s first three peaks sit within 16 of it while this arm’s sit 142, 80 and 18 above, converging. *That transient is what compressed the early gaps.*

The correction makes the scans above more relevant and not less. If the disagreement is a phase, then what the seam datum assigns is exactly the quantity in dispute—and §4’s “one datum per mode and a common phase” is precisely the statement that does not fix it.

Both halves of that were then carried to the same depth, and the acoustics come out working^R. Moving the fitted parameter so that r_s falls by 11%, ordinary acoustics requires the peak spacing to rise by 13%—and *it rises by 13%, ninety-eight per cent of the acoustic rate.* So the peaks in this construction are set by its own sound horizon, and the sentence above about the first peak’s insensitivity is a statement about the transient rather than about the acoustics. *The phase, meanwhile, does not move with the fitted parameter: the intercept shifts by three per cent of a discrepancy of 0.62. And the seam datum’s own phase freedom is a real lever on it and spans a third of it—at the opposite phase the gap to the control closes from 0.615 to 0.408, thirty-four per cent, and no further. That is the confirmation the diagnosis wanted: the datum’s phase moves the acoustic phase, as a phase diagnosis predicts, and cannot close the discrepancy.*

The acoustic statement is not one number and one mechanism, and the reason is the seam phase’s freedom^R. The 0.62π was read at one value of the seam phase, and the two-point estimate of how far that phase could move it used only the endpoints. *Carried to four phases at the depth the fit requires, the asymptotic phase spans 0.891 in ϕ/π , and the control’s 0.263 lies inside that*

span. But a span of that kind is not yet a band^R: a span is a band only if every point in it is admissible, and two of the four are not. The seam phase enters the transfer as a hydrodynamic initial condition— $\delta_\gamma = D \cos(kc_s(\eta - \eta_S) + \phi)$, with continuity giving $\theta_\gamma = \frac{3}{4}Dkc_s \sin \phi$ —and §3 settles which values of it this cosmology admits: what crosses is frozen, a frozen mode has $\delta_\gamma = 0$, hence $\theta_\gamma = 0$, hence $\sin \phi = 0$. The admissible readings are therefore $\phi = 0$ and $\phi = \pi$ and no others, and that is this paper’s own transmission argument read at the seam rather than a condition imposed on the scan. Across the admissible pair the acoustic phase runs from 0.671 to 0.878—a band of 0.207 against a discrepancy of 0.615, a third of it—and the control’s 0.263 lies outside. So the disagreement in the acoustic phase is real and it is bounded: the one freedom the seam datum leaves moves it by about a third of its own size and cannot carry it to the control. The attribution to the driving stands—the driving is what separates the two arms at any given reading—but its promotion to “the whole disagreement” does not, since a third of a quantity is not none of it and is not all of it either. The two admissible entries are not a redundancy: the photon perturbation reverses sign between them while the potential does not, so one enters as a compression correlated with the well and the other as a rarefaction against it.

And the peak heights move with the same freedom, and do not reach the sky either. The ratio of the first peak to the second runs from 0.48 to 1.62 across the four—0.88 and 1.62 across the admissible pair—against 2.22 in the sky and 2.20 in the control. What is left robust is the spacing: 0.963 to 0.981 of ℓ_A over the same four readings, two per cent, on top of its insensitivity to the fitted parameter. That is the one acoustic quantity this cosmology states and does not surrender to a choice.

The measurement that fixes the strength of the statement is the last one. The reading whose phase comes closest to the control’s is still sixty times the control in χ^2 per degree of freedom on the same bins—224 against 3.71—and that reading is one of the two the admissibility condition above excludes, so the measurement is quoted here in the stronger form the pair supplies: at $\phi = 0$ and $\phi = \pi$ the same statistic is 281 and 379, seventy-six and a hundred and two times the control. Agreeing on the phase does not fix the spectrum, so the phase was never carrying the disagreement. Where it does live we do not now claim to know: not the spacing, which is right; not the phase or the heights, which a free choice moves; and not any single quantity this section can name. That is a worse position than the one it replaces and a truer one. We record what that buys and what it does not: it converts nothing, and §4.7’s verdict is untouched. But an acoustic phase shift is a computable consequence of the driving, so the question “why 0.62π ” now has an address, which the four-item list it replaces did not.

That address is occupied, and the attribution is complete^R. With every coupling to the potential removed—the potential’s rate of change feeding the densities and its gradient driving the velocities, at every site, leaving the potential’s own evolution alone—the two arms’ acoustic phases agree to 0.013 of ℓ_A and both spacings are ℓ_A to a part in a thousand. Two constructions with different rates, different sound horizons, different starting redshifts and different initial data give the same acoustic series once nothing drives it. Switching the driving on supplies -0.127 to the control and -0.729 here—a factor of 5.7—which is 98% of the discrepancy; and the same switch accounts for the two and a half per cent the spacing was short, so that is not a second effect either. The whole acoustic disagreement is one coupling behaving differently, and a coupling is a thing a theory says something about where a ratio is not.

And the reason it behaves differently is the rate itself: the standard driving shift is universal because every mode crosses during radiation domination and acquires the same shift, and a geometrically fixed rate has no such crossing—so the flat driving of the control is a cancellation between two flat channels, and on this rate the cancellation is gone and the two add^R. The acoustic-phase offset and the resolution of the Hubble tension are therefore one fact, and this construction cannot keep the second while disowning the first.

What that still does not say is the microphysics of the coupling itself, and we mark the

difference between an attribution and an explanation. The candidate is that this cosmology’s modes begin already sub-horizon at the seam carrying an assigned amplitude and zero velocity, in a potential that is not the equilibrium configuration for that state, so the potential does work on them at once—where the control’s modes begin super-horizon in adiabatic equilibrium and the potential decays gradually as they enter. *That is a hypothesis and it is not tested here.* The two halves of the coupling can be switched independently, and separating them is what would test it.

And the likelihood separation of the last section survives the freedom that moved the peak: on identical bins the arm costs between two hundred and four hundred and sixty times the control at every one of the eighteen, so the pre-registered rule fires throughout. *We report the one thing that looks like a rescue and is not:* the best-fitting reading of all eighteen is among the eight excluded ones, and its spacing improves precisely because its higher peaks are gone—a reading fits better by having less structure to disagree with. *Both freedoms the construction’s own phrase leaves open are now closed.*

And so is the one upstream of them, which is this paper’s own fitted number^R. The starting redshift is solved so that $\ell_A = \pi D_M/r_s$ meets a target rather than coming out as an output—which §2.4 states plainly and calls the one fitted number, though it had sat as a literal inside a root-find where nothing downstream could reach it. *The question a fitted parameter owes is not whether it is fitted but what survives it.* Pinning the target anywhere between 260 and 340 drives z_{onset} from 11009 to 5066 and the sound horizon from 157 to 120 Mpc, and *the mean peak spacing stays between 0.765 and 0.821 of ℓ_A throughout*—a spread of one part in fifteen against a pin moved by nearly a third. *What that scan holds fixed is the mean of a short series, and the short series is transient-dominated:* the gaps rise along it, and at the depth the transfer reaches the asymptotic spacing is 0.975 of ℓ_A (below). *So what survives the fitted parameter is the transient, not a deficit in the acoustic spacing*—which is the same conclusion the deeper series returns, read at the shallow end. The first peak is twice as sensitive and still never approaches the sky’s 0.731.

The first peak is far less sensitive to that than acoustics would require—a 24% fall in r_s should raise it by about 31% and raises it by seven—but *the first peak is inside the transient identified above, and the series is not.*

And we record what is not in question either way, since it is not downstream of this spectrum’s shape: the metric-singularity structure, the continuation, the light-element abundances, the acoustic scale met at the directly measured H_0 , and the expansion-rate resolution.

A second determination of the onset follows from the same integrals, and is stated at its actual strength. Pinning the *measured* acoustic angle $100\theta_* = 1.04109$ —equivalently $\ell_A = \pi/\theta_* = 301.76$ —on the derived integrals returns $z_{\text{onset}} = 6797$, and returns it *identically* at $H_0 = 67.4, 70.0$ and 73.0 : the same H_0 -independence that carries the acoustic scale out of the Hubble tension, recovered here from the onset side rather than assumed^R.

One thing the same computation shows should be held apart from it. The inherited datum taken *alone*— $\rho_r/\rho_m = 2$ exactly at onset—fixes $1 + z_{\text{onset}} = 2\omega_m/\omega_r$, and that determination is *not* H_0 -independent: at fixed Ω_m , with ω_r set by the measured T_{CMB} , it returns $z_{\text{onset}} = 6840, 7378$ and 8024 at $H_0 = 67.4, 70.0$ and 73.0 . It therefore agrees with the angle-fixed value near $H_0 = 67.4$ and not across the range; at the directly measured H_0 the angle-fixed onset carries $\rho_r/\rho_m = 1.69$ rather than 2.0. This does not bear on the datum’s standing—it is inherited and measured, and the first-sample argument is that a complete collapse hands over radiation of *order* the matter density rather than a finely tuned value, which 1.69 meets as well as 2.0 does—but the closeness to 2 holds at one H_0 and is not a further H_0 -independent check.

5 What the substrate determines: the amplitude floor and the classical character

One might ask whether the observed fluctuation amplitude is the substrate’s own de Sitter vacuum—the natural CR analogue of the inflationary spectrum-is-the-stretched-vacuum story. It is not, by a decisive margin.

Proposition 2 (The substrate vacuum is negligible against the observed amplitude). *The substrate curvature scale is the cosmological-constant scale: in Planck units $\Lambda \ell_P^2 \approx 3 \times 10^{-122}$, and the de Sitter vacuum metric-fluctuation power is of the same order, $(H_\Lambda/M_{\text{Pl}})^2 \sim 10^{-122}$. Against the observed $A_s \approx 2 \times 10^{-9}$ this is smaller by $\sim 10^{113}$.*

Argument. $\Lambda = 3(H_0/c)^2 \Omega_\Lambda$ gives $\Lambda \ell_P^2$ directly; $H_\Lambda = c\sqrt{\Lambda/3}$ and $(\hbar H_\Lambda/E_P)^2$ give the vacuum power (receipts: `verify_numeric.py`, anchor 5^R). The prefactor ($\sim 10^{-122}$ versus $\sim 10^{-123}$) is a 2π and reduced-versus-full Planck-mass convention; the conclusion is robust to it. $\square$

One scope qualification is owed immediately, and closing it is what makes the classical reading a result rather than an inference. Proposition 2 rules out the *substrate’s* vacuum. But the reading it supports locates the fluctuations in the *progenitor*, and the progenitor’s own vacuum is a different quantity: it is normalised at the collapse leg’s curvature scale, not at H_Λ , and it is amplified on its way here by the branch point’s mode mixing. Neither effect is small, and neither is addressed above. Both can be computed, because the collapse-side background is a single function: with $A = 2M$ and $\rho = 2\sqrt{B}/A$ the exact closed dust-plus-radiation interior gives $a = M(\rho|\sigma| + \sigma^2/2)$ near the crunch, whose Mukhanov–Sasaki potential $z''/z = 2/(|\sigma|(|\sigma| + 2\rho))$ is *independent of the mass*—so M enters only through the final division $\zeta = v/a$. The passage then has exactly three stages, and each contributes one factor of $1/\rho$. A mode leaves the sound horizon on the $\beta = 2$ leg and freezes as $v \rightarrow D_k/|\sigma|$ with the scale-invariant $D_k = \sqrt{\hbar/2} k^{-3/2}$; the connection through equality is elementary, since $v = a \int d\sigma/a^2$ solves the $k = 0$ problem in closed form and carries $D_k/|\sigma|$ to the constant $(3/2\rho)D_k$ at the crunch; the crunch monodromy then leaks that constant into the survivor with coefficient $4\pi/\rho$ —twice the tensor value, because the true hydrodynamical $z_S = a(a + \frac{4B}{3A})/a'$ has $z_S''/z_S \rightarrow 2/(\rho x)$ where $a''/a \rightarrow 1/(\rho x)$ —the companion paper’s numerically obtained $-2\pi i/\rho$ [17] being the *tensor* value of the same object, recovered here in closed form as the continuation of the $(\sigma/\rho) \ln \sigma$ term the $(0, 1)$ resonance forces; and the survivor’s curvature is $\zeta = C/(M\rho)$, the last $1/\rho$ being the crunch’s own slope $a'(\eta_c) = -M\rho$. The result is a transfer law with no free constant,

$$\mathcal{P}_\zeta = \frac{18 k^3 D_k^2}{M^2 \rho^6}, \quad (8)$$

which is *exactly* scale-invariant on vacuum data—the k^3 cancels the k^{-3} identically—and which therefore delivers, for the vacuum, $A_s = 9(\ell_P/M)^2 \rho^{-6}$ (receipt: `P15_the_progenitor_vacuum_is_negligible_t` the connection coefficient confirmed against integration from vacuum data to 0.1% over a decade in ρ). *One check is owed here and is not optional, because the hydrodynamical variable differs from a in three places and not one: $z_S \propto a^{3/2}$ in the matter era, so the exponents there are $(-2, 3)$ against a ’s $(-1, 2)$; the crunch residue doubles; and the final division is by z_S rather than by a . Propagating the same physical initial condition under each potential returns a ratio of exactly 2 in amplitude at every composition tested—the matter-era exponents and the change of divisor cancel, and the crunch residue alone survives. So the $z \propto a$ chain is corrected by one factor rather than rebuilt, and Eq. (8) is the true-variable law.*

Proposition 3 (The progenitor vacuum is negligible too). *With the mass capped at the recollapse threshold $M \leq \alpha/3\sqrt{3}$ and the radiation fraction at the largest value the observed spectrum permits, Eq. (8) on vacuum data returns $A_s \sim 1 \times 10^{-112}$, short of the observed 2×10^{-9} by $\sim 10^{103}$.*

Argument. $(\ell_P/M)^2 \geq 27(\ell_P/\alpha)^2 \approx 2.6 \times 10^{-121}$ at the cap, and the companion paper determines $\rho \simeq 5.4 \times 10^{-2}$ [17]; $9 \times 2.6 \times 10^{-121} \times \rho^{-6}$ then gives the figure. Running it backwards, the vacuum would reach the observed amplitude only at $\rho \sim 3 \times 10^{-19}$, thirteen orders below what nucleosynthesis on the same leg permits. $\square$

*So eighteen orders separate the substrate's vacuum from the progenitor's amplified one—the mixing buying $10^{8.6}$ over the progenitor's own bare vacuum, which itself sits 10^{10} above the substrate's—and it is still short by a hundred and three, and the classical character of the primordial statistics is now established for the source that actually supplies them rather than only for the one that does not. Two consequences of *character*, not of number, follow.*

First, the observed amplitude is inherited classical content—boundary data from the progenitor, not a quantum fluctuation generated within the Schwarzschild–de Sitter era, and now not one generated in the progenitor's collapse either.

Second, there is no inflationary consistency relation: $r = -8n_t$ presupposes that scalar and tensor spectra are one vacuum stretched by one expansion, and with the amplitude not vacuum-sourced that link is absent; the substrate tensor floor is the same $\sim 10^{-122}$, so there are no substrate-sourced primordial B -modes, and any observed tensors would themselves be inherited content rather than a measure of a substrate inflationary scale. *And the tensor statement can be made unconditionally, which the scalar one cannot.* Eq. (8) was derived with $z \propto a$, which for a fluid holds only where w and c_s are constant—across the progenitor's own equality neither is.

For tensors that restriction is absent: $z_T = aM_{\text{Pl}}/2$ exactly, for any content, any equation of state and any sound speed, so $z_T''/z_T = a''/a$ identically and the entire passage—background, connection, monodromy—applies verbatim with nothing assumed. It returns

$$\mathcal{P}_T = 144\pi (\ell_P/M)^2 \rho^{-6} \simeq 5 \times 10^{-111}, \quad (9)$$

evaluated at the recollapse cap and the largest composition the spectrum permits, against an observational ceiling of $\sim 7 \times 10^{-11}$: *below it by a hundred orders* (receipt: `P15_no_primordial_B_modes_unconditioni`). *So the absence of primordial B -modes is a prediction of this construction at any conceivable sensitivity, and it does not rest on the substrate's own floor—the floor argument, like its scalar twin, ruled out the substrate's vacuum and said nothing about the progenitor's. What the crossing does to the ratio is likewise fixed and free of k : every factor of the passage is shared by the two sectors, so $r_{\text{out}} = 6r_{\text{in}}$: the passage alone would carry the ratio by 24, and the scalar mixing being twice the tensor's divides that by four in the power, and the observed bound becomes a statement about the parent, namely that its own tensor-to-scalar ratio was below 5×10^{-3} . An observed tensor signal would therefore not falsify the construction but measure the previous generation, which is the same reading the amplitude receives. The substrate fixes that the fluctuations are classical, inherited, and not governed by a vacuum consistency relation, and leaves their size to the handover. What it does fix is the multiplier.* Eq. (8) holds for any incoming growing-mode amplitude, quantum or classical, so the crossing determines how much an amplitude is magnified without determining the amplitude—which is this construction's one-constant character read at the level of the perturbations rather than of the couplings. Two corollaries follow and are worth stating because each closes a natural hope in one direction and opens a statement in the other. *A_s does not measure the progenitor's mass:* with D_k free, the relation cannot be inverted, and the mass must be reached, if at all, by a route that does not run through the amplitude. *And the classical datum has a size:* since $A_s \propto D_k^2$, the parent's growing-mode amplitude must have exceeded its own vacuum's by $\sim 10^{47}$ at the top of the permitted composition window—so this cosmology does not merely decline to predict A_s , it says the previous generation's perturbations were enormously super-vacuum, which is a claim about the parent and not about us.

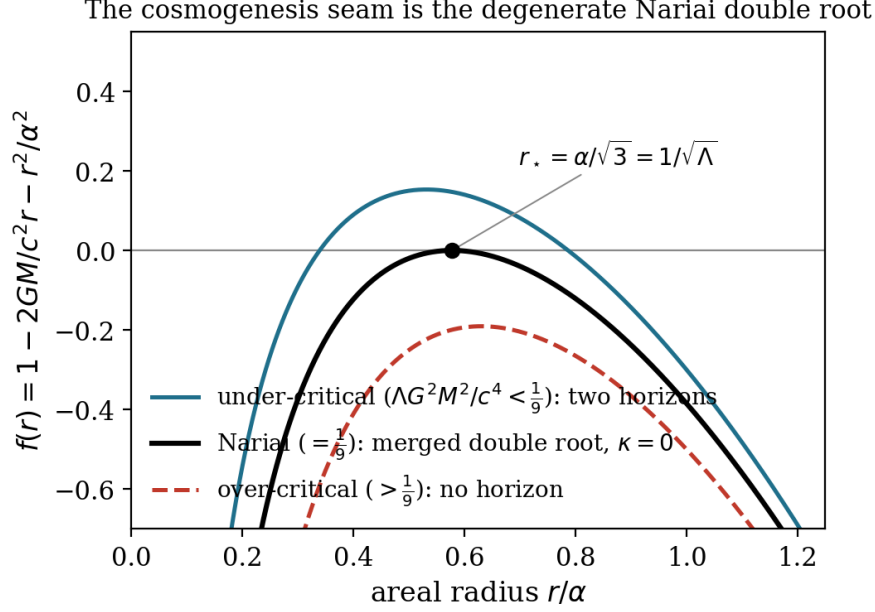


Figure 3: The Schwarzschild–de Sitter metric function $f(r) = 1 - 2GM/c^2 r - r^2/\alpha^2$ across the three regimes. Under-critical masses ($\Lambda G^2 M^2/c^4 < \frac{1}{9}$) give two distinct positive horizons; over-critical masses give none. The cosmogenesis branch point is carried on the *Narai* member ($= \frac{1}{9}$), where the two horizons merge into a single *degenerate* double root at $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ with $f(r_N) = f'(r_N) = 0$ and hence vanishing surface gravity $\kappa = 0$. The degeneracy of this root is what the throat geometry (Proposition 4) and the transmission dichotomy (§8) turn on.

6 What the substrate determines: the throat and isotropization

At the branch point the near-horizon geometry is fixed, and what the perturbations cross is fixed with it.

Proposition 4 (Equal-radii $dS_2 \times S^2$ throat). *At the Narai locus $\Lambda G^2 M^2/c^4 = 1/9$ the two positive horizon roots of $f(r) = 1 - 2GM/c^2 r - r^2/\alpha^2$, $\alpha = \sqrt{3/\Lambda}$, merge at the Narai radius $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ (the symbol is the algebroid paper’s [2]; it is written r_N and not r_* because r_* is the tortoise coordinate used below), where $f(r_N) = f'(r_N) = 0$ and $f''(r_N) = -2\Lambda$ (the cosmology selects this $M > 0$ member of the parity-conjugate Narai pair; the $M < 0$ member has instead one positive and two negative roots, the two exchanged by $r \mapsto -r$ [4, 5]). The near-horizon geometry is $dS_2 \times S^2$ with both curvature radii equal to $1/\sqrt{\Lambda}$: the S^2 has two-Ricci $R_2 = 2/r_N^2 = 2\Lambda$, and the dS_2 has radius² $= -2/f''(r_N) = 1/\Lambda$.*

Argument. Direct from f at the Narai mass (receipts: `verify_geometry.py`, anchor 2^R ; this is the Ginsparg–Perry near-horizon limit [27]). $\square$

A field on $dS_2 \times S^2$ (Fig. 3) decomposes in S^2 harmonics; the degree- ℓ harmonic becomes a dS_2 field of effective mass $m^2 = \ell(\ell + 1)/r_N^2$, i.e. $m^2/H_{dS_2}^2 = \ell(\ell + 1)$. With the dS_2 index $\nu^2 = \frac{1}{4} - m^2/H^2$, the monopole $\ell = 0$ has $\nu = \frac{1}{2}$ —a scale-invariant base—while every $\ell \geq 1$ has $\ell(\ell + 1) \geq 2$, hence $\nu^2 < 0$: the heavy principal series, which oscillate and decay through the throat. We read this as an angular no-hair, and the reading is grounded rather than asserted: the dS_2 factor of the throat is a positive- Λ epoch, and the de Sitter no-hair theorem [19] damps exactly the anisotropic content—every $\ell \geq 1$ harmonic is heavy principal-series, oscillating and decaying, while only the isotropic monopole $\ell = 0$ survives as the $\nu = \frac{1}{2}$ base—so the throat isotropizes what passes through it (receipts: `verify_throat_tower.py`^R). The geometry is established (Proposition 4) and the mode tower follows from it; the no-hair reading is the standard theorem read on the throat, and we flag one guard as load-bearing: the index ℓ of this

throat tower is the S^2 -harmonic degree of the *near-horizon* geometry at areal radius $1/\sqrt{\Lambda}$, and is *not* the observable microwave-background multipole, which lives on the last-scattering sphere of radius D_C . The map between them is §7, not an identity; conflating them would manufacture a prediction the geometry does not make.

7 Large scales: the flat/discrete decoupling and the low-multipole floor

Here the construction makes its sharpest large-scale statement, and the load-bearing piece is established.

Proposition 5 (The fundamental-observer slice is flat). *In the fundamental-observer frame (2), a surface of constant τ has induced three-metric $dr^2 + r^2 d\Omega^2$ (since $(\partial_\chi r)^2 d\chi^2 = dr^2$ along the slice), whose Riemann tensor vanishes identically. The constant- τ slice is exactly flat $\mathbb{R}^3$.*

Argument. The substitution is immediate from the line element; the curvature of $dr^2 + r^2 d\Omega^2$ is zero by direct computation (receipts: `verify_geometry.py`, anchor 1^R). This is the framework paper’s own stated property that constant- τ slices are Euclidean. $\square$

The decoupling follows. The distance slicing is the flat constant- τ one— $\Omega_k = 0$, no curvature term in the redshift–distance relation—while the *cosmological* layers are the closed S^3 of constant $\tilde{\tau} = \tau + \chi$ (§2.2). The non-synchrony $\tilde{\tau} = \tau + \chi$ is the decoupling itself: flat distances and a closed S^3 of comoving worldlines coexist because they are different slicings of one geometry. A literal closed-Friedmann reading would put $\Omega_k = -\Omega_\Lambda \approx -0.685$ into the distance relation and be excluded; CR avoids that not by tuning but because its distance slicing is the flat one, the curvature living on the orthogonal layers. Spatial curvature and dark energy are one Λ read on two slicings—the corpus identity, here at the observational rung.

The low-multipole floor. The closed S^3 carries a discrete mode spectrum, labelled by integer degree L , with comoving wavenumber $k_L = \sqrt{L(L+2)}/r_0$ and the lowest physical mode the quadrupole $L = 2$ (the monopole $L = 0$ is the background, the dipole $L = 1$ pure gauge). The decoupling fixes how these project to the sky, and the fixing is the whole point. Because the distance slicing is flat (Proposition 5), the photons are projected through the *flat* geometry—the comoving angular-diameter distance is $D_M = D_C$, the same flat- Λ CDM observable that places the acoustic scale at $\ell_A \approx 301$ —while only the *source* modes carry the closed- S^3 quantization. The transfer is therefore the discrete closed- S^3 spectrum projected through the flat spherical Bessel $j_\ell(k_L D_C)$, *not* the hyperspherical transfer of a literal closed universe: that transfer carries the closed *distance* relation, which CR does not have, and would (wrongly, here) deliver the lowest mode to the quadrupole and no deficit. The flat projection places the discrete modes at

$$\ell_L \simeq \sqrt{L(L+2)} \frac{D_C}{r_0}, \quad (10)$$

with r_0 the present S^3 areal radius and D_C the comoving distance to last scattering, both fixed without new parameters: $D_C \approx 13927$ Mpc is the flat- Λ CDM observable and $r_0 \approx 5064$ Mpc is the Narai amplitude $2^{1/3}/\sqrt{\Lambda}$ at the present epoch $u = \text{arcsinh} \sqrt{\Omega_\Lambda/\Omega_m} \approx 1.18$. The stretch factor $D_C/r_0 \approx 2.75$ —the ratio of the flat projection distance to the curvature radius, the decoupling made numerical—carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it.

The full flat-projection transfer confirms the estimate and sharpens it into a prediction. Carrying the discrete spectrum through $j_\ell(k_L D_C)$ with the scale-invariant measure—the first-principles closed- S^3 Harrison–Zel’dovich weight [29, 30] $w_L = (L+1)/(L(L+2))$, fixed as the mode degeneracy β^2 (with $\beta = L+1$) times the per-mode power $1/(\beta(\beta^2-1))$ that equal power per logarithmic interval in k requires, and reducing to the continuum dk/k at large L so that $\ell(\ell+$

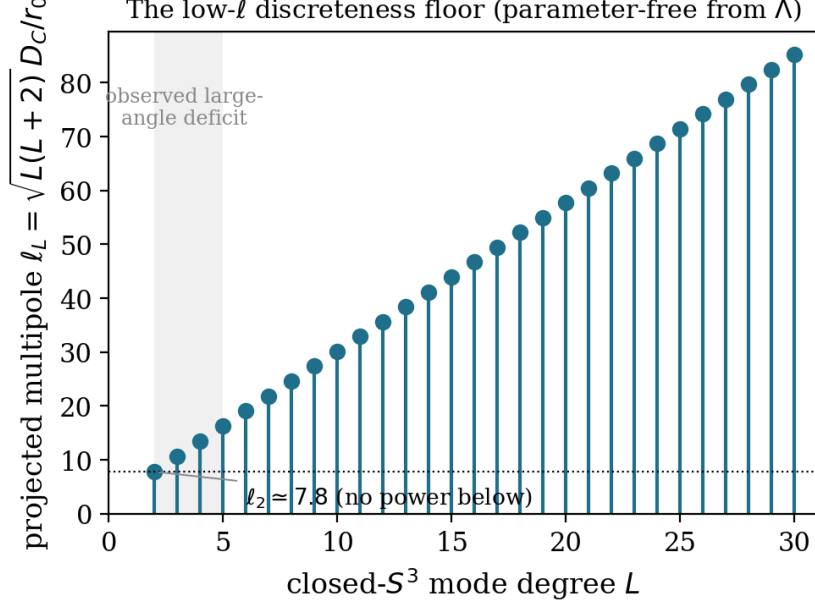


Figure 4: The low-multipole discreteness floor. The closed- S^3 source spectrum is discrete ($k_L = \sqrt{L(L+2)}/r_0$, lowest physical mode $L = 2$), but the flat distance slicing (Proposition 5) projects it through the *flat* transfer $j_\ell(k_L D_C)$, not the hyperspherical transfer of a literal closed universe. With $D_C \approx 13927$ Mpc and $r_0 \approx 5064$ Mpc fixed parameter-free by Λ , the stretch $D_C/r_0 \approx 2.75$ carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it. The full flat-projection transfer turns this into a low-multipole power deficit ($\ell \lesssim 8$); on a genuine Boltzmann transfer (the late integrated Sachs–Wolfe sourced above the floor and so retained by the discrete source) it sits at 0.473, 0.410, 0.356, 0.676 of the flat- Λ CDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 8$ —a deficit whose minimum falls at $\ell = 4$, so that the quadrupole and octopole are its shoulders and not its floor. Those figures are the photon hierarchy built for this programme (§9.1); an independent transfer, this cosmology’s discrete source read through a standard Boltzmann code’s own $\Delta_\ell(k)$, gives 0.49, 0.24, 0.18, 0.61 at the same multipoles—close at the quadrupole and nearly a factor two apart at $\ell = 3$ and 4, with the same minimum at $\ell = 4$ and the same recovery. *The shape is cross-validated between the two; the depth is not.* Either way this is a mild deficit, non-discriminating against the observed large-angle power within the lowest multipoles’ cosmic variance (shaded). The location is geometric and robust; the depth is whatever the full Boltzmann transfer returns, and the two run here differ on it. *What the branch-point filter does to this prediction at these multipoles—and what it does not do—is taken up in the text below rather than here.*

1) C_ℓ returns to the flat plateau as $D_C/r_0 \rightarrow 0$ (receipt: `verify_lowell_exact_measure.pyR`)—yields a low-multipole *deficit*: $\ell(\ell+1)C_\ell$ falls below its high- ℓ plateau for $\ell \lesssim 7$ and recovers by $\ell \approx 8$, because the flat projection of a discrete source with a hard lowest mode at k_2 leaves the lowest multipoles starved (receipts: `verify_closedS3_nonsync.pyR`).

The one effect that partly fills it is the late integrated Sachs–Wolfe term [26], sourced near the observer where small line-of-sight distances send wavenumbers $k \gtrsim k_2$ —*above* the floor—down to $\ell \sim 2$.

By the decomposition of the empirical-forcing paper [10] the cumulative term is the standard Sachs–Wolfe and integrated Sachs–Wolfe—CR’s own differential-expansion floor being zero under uniform expansion—so this late boost rides the same modes the discrete source keeps, and the deficit is milder than the bare Sachs–Wolfe starvation alone.

The depth is a matter for a *genuine Boltzmann transfer* rather than for the geometry, *and the reason is that the deficit is two effects and not one.* In the pure Sachs–Wolfe limit the suppression splits exactly into a *floor*—the projection integral truncated below k_2 —and a *ladder coarseness*, the discrete sum minus that same truncated integral. *At $\ell = 5$ and 6 the two are comparable and pull opposite ways:* the ladder puts back power a pure cut-off removes.

And they respond to the fitted parameter with opposite signs—under $\pm 2\%$ in r_0 the floor moves by a few per cent while the full result moves by tens—so the location is the floor and is robust for that reason, and the depth is the ladder and is volatile for that reason^R. *The recovery multipole is the ladder’s too*: with it the spectrum recovers by $\ell \approx 8$, on the floor alone not until $\ell \approx 10$. Because the only CR modifications are the discrete source and the flat projection—every transfer ingredient (Sachs–Wolfe, early and late integrated Sachs–Wolfe, Doppler) shared with flat- Λ CDM [10]—the exact CMB temperature transfer $\Delta_\ell(k)$ of a standard Boltzmann code *is* CR’s transfer, and the discrete spectrum is read off it: $C_\ell^{\text{CR}} = \sum_{L \geq 2} w_L \Delta_\ell(k_L)^2$ against the continuum $\int d \ln k \Delta_\ell(k)^2$, the measure w_L being exactly $d \ln k_L / dL$ so the CR sum is the $L=2$ -floored Riemann sum of the very integral Λ CDM does continuously.

Carried out with the exact transfer (gate-validated: the continuum reproduces the code’s own C_ℓ to four figures; and the r_0 sensitivity is measured rather than assumed: the *shape* is stable—the minimum stays at $\ell = 4$ and the recovery multipole moves by 0.1% —while the *depths* drift by up to 15% at $\ell = 4$ under $\pm 2\%$ in r_0), the deficit is *milder* than a Sachs–Wolfe-only estimate suggests, and it has a shape: $\ell(\ell+1)C_\ell$ sits at $\approx 0.47, 0.41, 0.36$ and 0.68 of the flat- Λ CDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 7$ (receipt: `verify_lowell_boltzmann.py`^R). *The prediction is therefore not a monotone recovery from the quadrupole but a dip whose minimum falls at $\ell = 4$* ^R—and the quadrupole and octopole, the two multipoles a large-angle discussion reaches for first, are its two *shallowest* points.

The shape is not one code’s, and that was checked here by running it rather than by comparing tables^R.

Driving the photon hierarchy built for this programme (§9.1) through the same discrete projection puts the minimum at the same multipole, $\ell = 4$, and recovers at the same place, $\ell \approx 7\text{--}8$. *What two independent Boltzmann treatments cross-validate is the shape, and only the shape*. The depths do not agree: the second arm returns $0.49, 0.24, 0.18$ and 0.61 against the first’s $0.473, 0.410, 0.356$ and 0.676 —close at the quadrupole and nearly a factor two apart at $\ell = 3$ and 4 . The two treatments differ in their late-time integrated Sachs–Wolfe handling and in D_M , and this paper does not reconcile them: it carries one converged depth table, the first arm’s, and a second arm that agrees on where the deficit sits and not on how deep it is.

The reason is the late integrated Sachs–Wolfe: it is sourced near the observer at wavenumbers $k \gtrsim k_2$ (small line-of-sight distances), *above* the floor, so the discrete source *retains* it rather than losing it, and the deficit is milder than a Sachs–Wolfe-only treatment of the same source would give.

This is a parameter-free deficit, set by Λ through r_0 and D_C , and what is robust is its existence, its location ($\ell \lesssim 8$, geometric, set by $k_2 D_C$), its minimum at $\ell = 4$, and that the single floor k_2 starves $\ell = 2$ and $\ell = 3$ *together*: the exact discrete measure and the Doppler term (super-horizon velocity through j'_ℓ) each leave the $\ell = 2$ -to- $\ell = 3$ ratio near unity, if anything suppressing the octopole slightly more (receipts: `verify_lowell_exact_measure.py`^R, `verify_doppler_lowell.py`^R). Coherence shown is not correspondence earned: whether this predicted deficit matches the observed sky is the data test, against the large cosmic variance of the lowest multipoles. Confronted with the measured low multipoles, the honest verdict is a *wash*. CR predicts a smooth modest deficit bottoming at $\ell = 4$; the observed sky has a sharp quadrupole-only dip (~ 0.2) with a near- Λ CDM octopole ($\sim 0.6\text{--}1.0$), so CR sits *above* the observed quadrupole and *below* the observed octopole, matching neither sharply. Under the exact cosmic-variance likelihood the net over $2 \leq \ell \leq 10$ is $\Delta(-2 \ln L) \approx +1.8$ (range $+0.3$ to $+3.8$ across the octopole estimator), a small residual well inside cosmic variance (receipts: `confront_lowell_data.py`^R, `verify_lowell_likelihood_v2.py`^R). *Broken up by multipole, the sector’s weight does not lie where its name suggests*: the quadrupole *rewards* this cosmology (-2.6 , its milder dip nearer the observed low power than flat Λ CDM’s), the octopole penalises it at $+1.3$, and the largest single-multipole penalty is the $+2.1$ at $\ell = 4$ —the prediction’s own minimum, and the one multipole of the four that the large-angle literature does not treat

as anomalous. The low multipoles are thus a genuine but *non-discriminating* sector: neither a parameter-free correspondence success nor a sharp falsification edge, but a mild deficit consistent with Λ CDM within the lowest multipoles' cosmic variance.

One further caveat belongs with this prediction, and it is not a general disclaimer but a located one. The branch-point filter that sets what crosses into the expansion leg is, in its quantum reading, a Euclidean kernel whose action on a mode is controlled by an adiabaticity parameter C/μ_n , with μ_n the harmonic index of the layer's S^3 tower [33]. That parameter is small for all but the lowest harmonics and reaches order unity only at $n = 2$ and $n = 3$, where it takes the values 0.70 and 0.47. *Three independent routes place the filter's loss of control at the same scale:* the transmission calculation's boundary at $\ell \approx 2.5$, the projection's incompleteness below $k \sim 2 \times 10^{-4} \text{ Mpc}^{-1}$ ($\ell \approx 3$), and this breakdown at $n = 2, 3$.

The transmitted amplitude at low multipoles can be computed rather than estimated, and doing so settles what the estimates left open. The branch-point filter's action on a mode is, in the quantum reading, a Euclidean kernel [33]. The exponential-of-an-integral form $e^{-\int \omega d\eta}$ is a WKB approximation whose adiabaticity parameter is of order unity at $\ell \lesssim 5$; but the underlying mode equation $d^2\psi/d\eta^2 = \omega^2\psi$, with $\omega = kc_s$ and $d\eta = ds/a$ along the segment, is well posed at every k , and integrating it is a matter of numerics rather than of approximation. Composing exact constant- ω transfer matrices in the stable direction over the segment ($|\Delta\eta| = 3.32$ in units $\alpha = 1$) gives exact/WKB ratios of 0.926, 0.913, 0.901, 0.891 and 0.889 at $\ell = 2, 3, 5, 15, 40$: *the WKB form is accurate to 7–11% at every multipole from 2 to 40*, with the exact transmission slightly smaller, so the filter is marginally stronger than the approximation gives.

Two readings suggested by the adiabaticity parameter alone are thereby excluded. The first is that the treatment is uncontrolled at $\ell = 2$ –5: it is not, and the parameter was the wrong figure of merit, measuring as it does the fractional error in an exponent that is itself small at low ℓ . *The residual's own behaviour settles this independently of any argument about the parameter:* the ratio *decreases* monotonically with ℓ —0.926, 0.913, 0.901, 0.891, 0.889—so the discrepancy is smallest at $\ell = 2$, where the adiabaticity parameter is of order unity, and largest at $\ell = 40$, where it is small. An error of adiabatic origin runs the other way, WKB being an expansion *in* that parameter, so such an error is largest where the parameter is $O(1)$ and vanishes as it does. *What the numbers show is therefore a systematic offset rather than an adiabatic breakdown*—the WKB form is uniformly slightly generous and the exact filter uniformly slightly stronger, which is a correction of known sign and bounded size and not a loss of control^L. The second concerns the comparison with the observed sky. The correction is very nearly ℓ -independent—varying by 4% across a factor of twenty in ℓ —so between $\ell = 2$ and $\ell = 3$ it alters their ratio by 1.4%. *The structural expectation that a single floor starves $\ell = 2$ and $\ell = 3$ together is therefore not relieved by the exact treatment*, and the observed sky's selective suppression of $\ell = 2$ alone, by a factor of order three, remains unaccounted for. The low-multipole prediction of this section therefore stands as stated, and so does the discrepancy it faces.

Finally, the two Boltzmann depths above are not rival claims and neither supersedes the other: they differ by up to a factor of two at $\ell = 3$ and 4 and agree on the shape, the minimum and the recovery, and the difference is the size of the residual §9.1 declares. What both supersede is the Sachs–Wolfe-analytic estimate (0.22 and 0.20 at $\ell = 2, 3$), which takes the Sachs–Wolfe source as the matter-domination value $\frac{1}{3}\Phi$, whereas at recombination $\rho_r/\rho_m = 0.317$ and the source is 0.4006Ψ —a 20% error in the very ratio of Sachs–Wolfe to integrated Sachs–Wolfe that fixes how much the latter fills the bare deficit.

8 What the progenitor supplies: the transmission proof

The decomposition's decisive proof is here: *why* the substrate transmits the spectral shape rather than imprinting one of its own. It turns on the degeneracy of the branch point horizon.

Proposition 6 (Transmission dichotomy). *For a mode approaching a horizon where the metric function behaves as $f \sim (r - r_h)^p$, the tortoise coordinate $r_* = \int dr/f$ controls the approach. At a non-degenerate horizon ($p = 1$, $f \sim 2\kappa(r - r_h)$, surface gravity $\kappa > 0$) the integral is logarithmic and the approach exponential, $(r - r_h) \sim e^{2\kappa r_*}$. At the degenerate Nariai double root ($p = 2$, $f \sim -\Lambda(r - r_N)^2$, $\kappa = 0$) the integral is $r_* \sim +1/[\Lambda(r - r_N)]$ and the approach power-law, $(r - r_N) \sim +1/(\Lambda r_*)$.*

Argument. The two tortoise integrals directly (receipts: `verify_geometry.py`, anchor 4^R). $\square$

Remark 1 (The leg does not tilt it either). The proposition argues from the branch point: a degenerate horizon carries no scale, so it cannot imprint one. The collapse leg supplies the other half of the same conclusion, by an independent route. Horizon entry on that leg occurs at $x = k\eta/\sqrt{3} = 1/\sqrt{3}$ for every k —the radiation era’s own scale invariance, which §4.1 recovers from the substrate geometry—so every mode leaves the leg carrying the same free-oscillation amplitude $\Psi(1/\sqrt{3})/2 = 0.4835 \Psi_i$, a single k -independent number^R. *So the leg multiplies the spectrum by a constant and the seam imprints nothing:* neither carries a scale, and a spectrum can only be tilted by something that does. The tilt observed is therefore the progenitor’s, established now from both sides of the branch point rather than one.

Remark 2 (What the branch point does not fix). Both halves of the dichotomy are statements about the spectrum’s *shape*: a degenerate horizon carries no scale, and neither does horizon entry on the leg, so neither can tilt what passes. *Neither is a pointwise condition on the perturbation state at the branch point*, and the distinction matters for anyone computing the handover. The single free function per mode is the freedom of the *spectrum*—the progenitor’s amplitude and tilt—not a claim that the branch point fixes the state component by component. The state itself is whatever the leg’s evolution produces, and §4.1 supplies it in closed form: the potential from the leg’s own equation, the effective temperature oscillating freely from horizon entry, and the density contrast as their difference.

The Hamiltonian constraint is not an additional condition to impose there. In this framework it *is* the anchor—the map from a cut to its stress-energy, whose energy sector reads the Friedmann equation off the geometry [6, 7]. It is a definition, not a restriction: on the leftward reading the rate is set by the substrate and the content is read from it, so the perturbed constraint likewise *defines* the density contrast a given geometry carries rather than restricting which geometries are allowed. Imposing it at the branch point alongside the leg’s own solution therefore over-determines the handover, and the same geometric state answering to different stress-energy on the two sides is not a failure of matching but the branch point doing what the level assignment says it does.

Remark 3 (What the branch point’s degeneracy does not settle). It is tempting to read Proposition 6 as settling the phase as well as the shape: the tortoise coordinate diverges at the degenerate root, so a mode’s accumulated phase would be unbounded and nothing finite could be inherited. *That reading does not survive its own arithmetic.* The divergence is geometric and carries no k , so an unbounded kr_* is a divergent term *linear* in k ; a linear phase renormalizes the sound horizon rather than shifting the comb, and an unbounded one would send the comb spacing to zero and erase the acoustic peaks altogether—excluded by observation, not supported by it. And the modes never meet the divergence in any case: they cross the branch point at finite conformal time, whereas r_* diverges along a static slicing’s approach to the horizon, which is not the collapse trajectory.

What the leg does contribute is finite and, as it happens, invisible in the peak positions. Horizon entry at $x = 1/\sqrt{3}$ for every k makes the accumulated leg phase $k\eta_{\text{bp}}/\sqrt{3} - 1/\sqrt{3}$: linear in k with a mode-independent offset. So carrying it across rescales the effective sound horizon by $1 + \eta_{\text{bp}}/(\sqrt{3}r_s)$ and adds a constant, and changes the comb in no other way. *The comb therefore cannot distinguish a transmitted phase from a reset one*, because a rescaling of r_s is degenerate

The transmission dichotomy: a horizon imprints a thermal scale; a degenerate seam does not

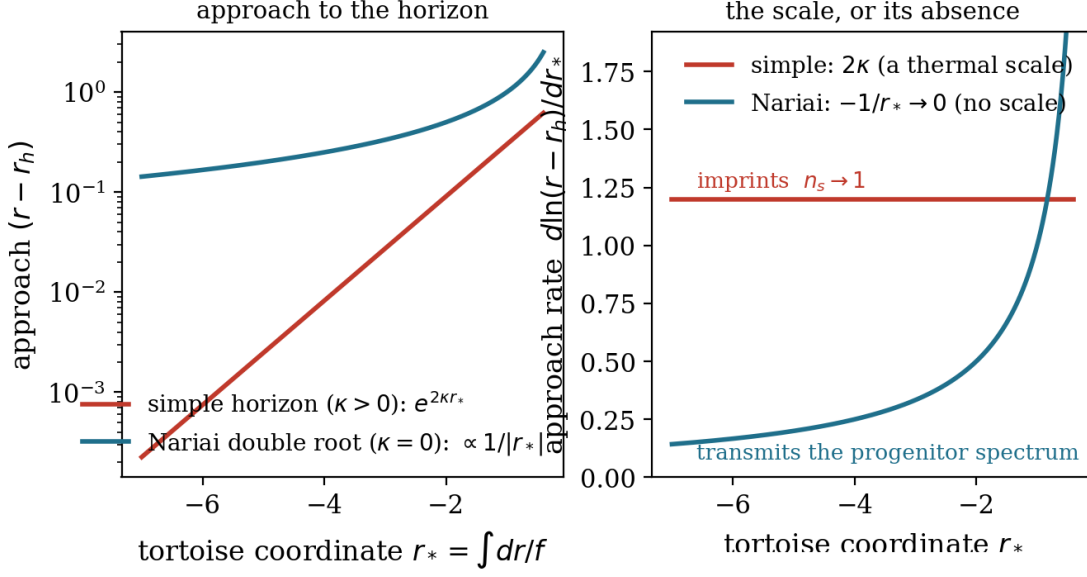


Figure 5: The transmission dichotomy (Proposition 6). *Left:* the approach to the horizon in the tortoise coordinate $r_* = \int dr/f$, on a logarithmic vertical axis. At a non-degenerate horizon ($\kappa > 0$) the approach is exponential, $(r - r_h) \sim e^{2\kappa r_*}$ (a straight line); at the Nariai double root ($\kappa = 0$) it is power-law, $(r - r_N) \sim 1/|r_*|$. *Right:* the logarithmic approach rate $d \ln(r - r_h)/dr_*$ is the local scale. The non-degenerate horizon carries a fixed thermal scale 2κ —the mechanism by which a de Sitter horizon imprints a scale-invariant spectrum ($n_s \rightarrow 1$); the branch point’s rate falls to zero, carrying *no* scale, so it transmits the progenitor spectrum unaltered.

with the onset redshift that §4.7 fixes from the acoustic angle. Any discriminant must be a quantity that does not take the same factor—the diffusion scale is one, since r_D does not—and that is the discriminant of principle. *The comparison is now made, and it does not require r_D .* The leg’s accumulated conformal time is fixed by the geometry rather than fitted: the collapse branch $r = -(2M\alpha^2)^{1/3} \cosh^{2/3}(3 \operatorname{Re} \tilde{\tau}/2\alpha)$ has $\cosh \geq 1$, so $|r| \geq (2M\alpha^2)^{1/3} = 3766.6$ Mpc and the branch point is reached only across the lift, along which $\operatorname{Re} \tilde{\tau}$ does not advance and no conformal time accumulates. Hence $\eta_{\text{bp}} \approx 5 \times 10^3$ – 10^4 Mpc, and $r_s^{\text{eff}} = r_s + \eta_{\text{bp}}/\sqrt{3}$ is *dominated* by the leg term— 3.0 – 5.9×10^3 Mpc against $r_s = 144$ Mpc. The degeneracy with the onset redshift, which absorbs a small rescaling, cannot absorb this one: reproducing the measured $\ell_A = 301.8$ would require $D \approx 2.9$ – 5.7×10^5 Mpc against the actual 1.4×10^4 Mpc, a factor of 21–41. *The acoustic phase is therefore reset at the branch point and not carried across it*—established on the peak positions alone^R. The result is robust to the open account of the lift: any conformal contribution the lift is found to carry only adds to an already-excluded total.

The amplitude separates from the tilt in the same statement. What the progenitor supplies is the potential power P_Ψ ; what the construction contributes is the factor $(0.4835)^2 = 0.2338$ carrying it to the observed A_s . That factor is computed rather than fitted, so the accommodation is narrower than it was—the inherited quantity is P_Ψ and the map from it is derived—but it is not removed, and the framework paper’s scoping of this as a one-parameter accommodation stands unchanged [8].

The physical reading is the result (Fig. 5). The exponential, non-degenerate approach is the thermal (Hawking) law—the mechanism by which a de Sitter horizon *imprints* a scale-invariant spectrum ($n_s \rightarrow 1$): the exponential redshift erases the infaller’s spectral information and stamps the horizon’s own thermal scale. The power-law, degenerate approach carries no thermal scale; it is scale-free, and the infalling spectrum is transmitted faithfully. CR’s transmission locus is the branch point $r = 0$, where f *diverges* as $-2M/r$ so the tortoise coordinate is *finite*—the crossing

accumulates no divergent phase and stamps no scale, a third case beyond the dichotomy's two^R. The Nariai double root (Proposition 4, $f''(r_N) = -2\Lambda$, $\kappa = 0$) is the *equatorial seam*—one point of the substrate, which the bead meets on the way in and again one full lap later, the two chart values $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ being the same φ modulo 2π in the phase $\varphi = 2\pi r/\sqrt{3}\alpha$. The branch point sits two thirds of the lap in from it (240° , with 120° remaining), so the crossing at $r = 0$ and the seam are distinct loci on the lap without the seam being an endpoint of anything. Hence:

Proposition 7 (The branch point transmits). *The cosmogenesis branch point is a faithful scale-free transmitter—not because it is the degenerate Nariai horizon, which it is not (that is the front seam, $r = +\alpha/\sqrt{3}$, the lap's other endpoint), but because f there diverges as $-2M/r$, so r_* is finite and the crossing accumulates no divergent phase at all: the amplitude A_s and tilt n_s are the progenitor collapse's spectrum carried across unaltered, not imprinted by the branch point. Equivalently, CR has no inflationary scale-invariant attractor, because the non-degenerate exponential approach that would drive $n_s \rightarrow 1$ is exactly the approach the Nariai seam does not have.*

The degeneracy is thus the proven reason CR carries the primordial tilt rather than manufacturing it. What the transmitter carries is, from the Schwarzschild–de Sitter side, open: n_s and A_s are inherited, fixed by the progenitor collapse—the same handover that fixes ρ_r/ρ_m and the light-element abundances—and not derived here. *Naming them together is convenient and has concealed that they are not the same kind of quantity*^R. The vacuum normalisation is the only place a quantum constant enters the spectrum, so $P(k) = \hbar A_0 k^{n_s-1}$ and $\hbar$ is an overall multiplicative factor: *it survives in the amplitude and cancels in every logarithmic derivative*. An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. *So the construction's inability to force a dimensionless magnitude bears on A_s and not on n_s —the first is permanently inherited, the second need not be. And read backwards the same statement is useful rather than limiting: a quantity fixed by the moduli rather than the invariants is one the observation measures, so the observed A_s determines the progenitor's mass, to be checked against the independent requirement that it lie below the Nariai value. And it is worth saying precisely what that openness is, because a count settles its kind.*^R The progenitor family described here is parametrised by M alone— α is the one invariant and $E = 1$ a choice of geodesic—so the geometry is one-dimensional; and the Kretschmann along the bead is mass-free in the faller's own proper time, so that one parameter does not survive the approach either. *A quantity identical for every member carries no information about which member it was, and the collapse geometry therefore transmits not one parameter but none. Against that stand at least four independent inherited numbers. So 'inherited from the progenitor collapse' cannot mean inherited from the collapse geometry: what is inherited is inherited from the progenitor's matter content—its composition, its entropy per baryon, its perturbation spectrum—and none of that is carried by M , the exterior here being exactly vacuum and the marginally-bound interior dust without composition. The frontier is accordingly not a derivation awaiting a cleverer argument but a modelling task awaiting a progenitor interior, and what the geometry supplies in its place is a cutoff rather than a content: the segment's conformal length $|\Delta\eta| = 3.32\alpha$ is fixed by Λ alone on the Nariai member—below it $|\Delta\eta|$ runs as $M^{-1/3}$, reaching 4.19α at $M/2$ and 15.4α at $M/100$ —and says which modes survive without saying how large they were.*

9 Predictions, and the progenitor boundary

The signatures, with their maturity marked:

1. Classical, non-vacuum primordial statistics; no inflationary consistency relation; no substrate-sourced primordial B -modes (established, §5).

2. The transmission character: the branch point carries the progenitor tilt and does not drive $n_s \rightarrow 1$ by any *inflationary* attractor (established, §8). *Nor does the construction lack a scale-invariant attractor—it has one, and it is not the inflationary root.*^R Scale invariance for a background $a \propto (-\eta)^\beta$ with $z \propto a$ requires $\beta(\beta - 1) = 2$, a quadratic with *two* roots: $\beta = -1$, de Sitter inflation, and $\beta = +2$, a matter-dominated contraction. *The collapse leg of this construction is the second root exactly:* near the branch point $f \rightarrow -2M/r$, so the leg is pressureless there and the banked law $r^3 = \frac{9}{2}M\tau^2$ with $a = |r|$ gives $a \propto \tau^{2/3} \propto \eta^2$; the mode equation integrated from vacuum data returns a flat spectrum to a part in 10^4 . *The mode-selection question that shadows this in the literature was accordingly put to the construction directly, and it answers in the negative.*^R

The scale-invariant spectrum sits in the mode that *dominates* the contraction, whose curvature perturbation goes as η^{-3} ; the mode that survives on the expanding side is the constant one, and from vacuum data its spectrum is $n_s - 1 = 6$.

Whether the first reaches the second is a connection problem at the branch point, and here it is settled rather than assumed: the exponents 2 and -1 differ by an integer, so a logarithm could mix them, but the $\sinh^{2/3}$ law is *even* in τ , which in conformal time places its correction at η^6 and steps the Frobenius recursion in sixes past a resonance sitting at three. *No logarithm is generated, the continuation acts diagonally, and the scale-invariant spectrum does not reach the observer*—confirmed independently by continuing the dominant solution numerically around the branch point, where the admixture it produces is what the claim turns on, where the admixture falls as $\delta^{2/3}$ and the diagonal entry goes to unity. *What survives is therefore narrower than it first appeared and sharper than what it replaced:* the construction *has* a scale-invariant background, the spectrum that background freezes is confined to the contracting side, and the obstruction is a parity property of the construction’s own solution rather than a missing mechanism. *Mixing would require a background correction odd in τ , and that question is settled here too, in the negative.*^R

Writing the law as $r = Au^{2/3}g(u)$ with $g = [\sinh u/u]^{2/3}$ places every branched factor in $u^{2/3}$ and leaves g even, analytic and non-vanishing at the origin—exactly, since $\log(\sinh u/u)$ is analytic in u^2 there and the exponential of a series in u^2 is one. The three sheets differ by a constant cube root of unity, and a constant multiplier cannot change a parity; the conjugate branch is even as well, and carries no branch point at all. *So the evenness is unavoidable, and what it says is not a technical fact but a symmetry:* $r(-\tau) = r(\tau)$, the cosmogenesis is time-symmetric about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one, since mixing would distinguish a direction of time the background does not. *The same invariance is already load-bearing one observable over*—the peak heights of §5 follow from a driving amplitude exactly invariant under time reversal, which is why they may be evaluated on the contracting side. *One symmetry, two consequences. And it locates rather than forecloses what would be needed:* this leg is exactly even because it is the marginally-bound geodesic of an exactly vacuum exterior, so a progenitor carrying real matter content—pressure, dissipation, anything with an arrow—would not be, which returns the question to what the progenitor was.

3. Flat distances ($\Omega_k \simeq 0$) coexisting with a closed- S^3 low-multipole *deficit* below $\ell \approx 8$, set parameter-free by Λ —mild, and on a genuine Boltzmann transfer a dip bottoming at $\ell = 4$ ($\approx 0.47, 0.41, 0.36, 0.68$ of the flat- Λ CDM expectation at $\ell = 2, 3, 4, 5$), its shape cross-validated between two independent transfers (*established* for the existence, the location and the minimum; *open* for the depth, on which the two transfers differ by up to a factor of two at $\ell = 3$ and 4 (§7); *non-discriminating* as a test, the deficit sitting inside the largest multipoles’ cosmic variance). Confronted with data it is non-discriminating: it starves $\ell = 2$ and $\ell = 3$ together, matching neither the sharp observed quadrupole-only dip nor

the near- Λ CDM octopole, and sits consistent with Λ CDM within the lowest multipoles’ cosmic variance (§7).

4. Coherent acoustic peaks from geometric (null-boundary) phase-fixing rather than super-horizon freeze-out (argued, §4).
5. The acoustic scale is resolved in the cosmological development above (§2.3), and the geometric rate (4) is the standing CR-versus- Λ CDM discriminator on data in hand (*established*, §2.3)—a strength established before the spectrum, not this section’s to re-assert.

The boundary is principled. The substrate carries everything the branch point’s null geometry lets it carry—coherence, isotropization, the discreteness floor, the classical character, the faithful transmission—and the degeneracy proves precisely where it stops: the spectrum’s amplitude and tilt are not the substrate’s to set. This is the shape inflation has when it parametrizes its observables through an unknown inflaton potential; CR routes them through an unknown progenitor collapse, with the arrival constrained—whatever the collapse supplies must reach the branch point coherent, isotropized, scale-freely transmitted, and classical.

The symmetry between the two unknown suppliers is only apparent, and the asymmetry is the substance. Inflation is not structure the standard cosmology already contains; it is an added sector—a scalar field with a freely specified potential—introduced to produce features that cosmology was known to lack: the causal contact the horizon problem denies, the flatness the dynamics do not single out, and, once these were in view, a near-scale-invariant spectrum. The potential is then chosen to deliver the observed amplitude and tilt. A model built to be tunable, and tuned to a target known in advance to require tuning, earns precisely what such a model can earn—it establishes that the features are *possible*, that an apparatus permitting them exists. It does not establish that they were *required*, and the routine elevation of the spectral fit into a vindication of inflation conflates the two: that a freely chosen potential can be set to match n_s and A_s is not evidence that inflation occurred, but the statement that a model with enough adjustable structure to fit has been fit. This is the distinction between a world that *requires* a phenomenon and one that merely *permits* it through adjustable parameters—the same distinction the programme draws in reading the cosmic foliation as forced rather than chosen [10]—and it places the inflationary spectrum on the permitting side.

The present cosmology adds no such apparatus. Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are not produced by a field introduced to produce them; they are read off structure the construction already contains—the de Sitter throat that isotropizes by the no-hair theorem (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§2.3), and the degenerate horizon whose vanishing surface gravity transmits rather than imprints (§8)—each fixed by Λ alone, with no parameter free to do the producing. The one quantity it does not derive, the amplitude and tilt, it inherits in exactly the sense flat Λ CDM inherits the baryon-to-photon ratio; on that single datum the two frameworks stand level, and on everything the inflationary sector is invoked to supply they do not. The verdict this licenses is not symmetric, and it does not await a future measurement. On the data in hand the two align; on the structure each must assume in order to align, one introduces an entire tunable sector and tunes it while the other introduces none—and makes, in its place, falsifiable commitments where inflation keeps its freedom: no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate-sourced B -modes (§5, §8)^R. By the economy of assumption that distinguishes a required structure from a permitted one, that comparison is settled now, by what each theory must presuppose, and it is settled in this cosmology’s favour. We hold this at its weight and mark its bound in the same breath: the parsimony verdict is earned, and the data verdict has now begun to be earned with it—the Hubble tension resolved across the distance ladder where Λ CDM cannot reconcile the direct H_0 (§2.4), the abundances confirmed within 1σ [17]—so the

cosmology is empirically favoured, not merely more economical; the spectrum’s content remains inherited, and the large-angle shape remains the exposed edge that could still overturn it.

9.1 The acoustic instrument: what it is validated against, and to what accuracy

The low-multipole prediction above and the ratio statements below are carried by a Boltzmann transfer built for this programme, and its standing should be stated plainly rather than assumed. It is a full photon hierarchy with polarisation, second-order tight coupling, massless neutrinos, and a Peebles recombination history; the line-of-sight source carries the monopole with the potential, the Doppler term, the integrated Sachs–Wolfe term, and the quadrupole with its own projection kernel $(j_\ell + 3j''_\ell)$.

Against a like-for-like reference—an independent Boltzmann code run at identical parameters, unlensed and with reionisation off—it reproduces the *acoustic peak and trough positions* to $\leq 0.5\%$ across P_1 – P_4 , the *ionisation history* $x_e(z)$ and its derivative to $\sim 1\%$ through the visibility peak, matter–radiation equality to 0.02% , and the *transfer function* to better than 1% for $k < 0.02 \text{ Mpc}^{-1}$. Above that wavenumber it agrees to 1 – 2% , with the residual scatter in both directions and no trend across the range^R.

Two construction details are worth stating, because both are easy to omit and both have signatures that read as physics.

The first is the reach of the mode ladder. The angular spectrum at multipole ℓ is an integral over wavenumber, and the projection kernel $j_\ell(k(\eta_0 - \eta))$ has substantial support *above* $k = \ell/D_M$, not merely at it; a ladder cut at k_{max} therefore truncates the integral, and does so more severely as ℓ rises toward $k_{\text{max}}D_M$. Quantitatively, truncating an exact transfer at $k_{\text{max}}D_M = 900$ retains 99.2% of C_ℓ at $\ell = 200$ but only 61% at $\ell = 800$ and nothing at all at $\ell = 1150$; carrying the ladder to $k_{\text{max}} \geq 2\ell_{\text{max}}/D_M$ recovers every multipole to better than half a per cent. *The tell is unmistakable once known:* the highest peak simply fails to appear, so a spectrum that should show four acoustic peaks shows three, and the peaks that do appear are displaced downward by an amount that grows with ℓ . *Increasing the ladder’s density at fixed reach does not help*—the loss is the range, not the sampling, which is worth checking explicitly because the two are easily confused.

The second is the seeding of the conformal time. It is a cumulative integral, $\eta(a) = \int_0^a da'/(a'^2 H)$, and a numerical grid necessarily begins at some finite a_{min} ; the interval already elapsed below that point must be carried as a seed rather than set to zero. In radiation domination $H \propto a^{-2}$ makes the integrand constant, so the omitted piece is $a_{\text{min}}/(H_0\sqrt{\Omega_r})$ exactly—for $a_{\text{min}} = 10^{-5}$ it is 4.6 Mpc . *It is a fixed offset at every epoch*, and therefore 0.03% of η_0 but 12% of η at $z = 10^4$: the acoustic scale, the peak positions and the peak ratios are insensitive to it, while any early-time quantity or any comparison against a second code is not. Its signature is instructive—because the potential decays more steeply at larger k , a fixed error in the time coordinate reads out as a *smooth, monotone, k-growing* error in amplitude, which is the shape one would otherwise attribute to a physical suppression. The grid must also extend below the intended starting redshift, so that the initial epoch is interpolated rather than clamped.

That residual does not enter the claims made here. The Cosmological Relativity and flat- Λ CDM spectra are computed through the *same* machinery, so an instrumental transfer error afflicts both arms identically and cancels in their ratio—and every prediction stated in this paper is a ratio. The low-multipole deficit in particular lives at $\ell \lesssim 8$, i.e. $k \ll 0.02 \text{ Mpc}^{-1}$, where the two agree to better than 1% . What the residual does forbid is an *absolute* amplitude claim: this instrument should not be read as reproducing the observed small-angle power to better than a few per cent, and no such claim is made.

10 The open frontier: the pieces this cosmology is built to close

The result stands on what the substrate determines. What it does *not* yet claim has been carried to a sharp, gradable edge rather than left vague—and those edges are the most interesting part of the programme from here, each a gap we are eager to grind shut, not paper over.

We hold them to the criterion of necessity the programme’s epistemology names—that a structure is credited for *requiring* a phenomenon, not for *permitting* a value that fits it [11]—and sort them by it: a gap that is *buildable* is a debt owed, to be built before the claim it bears on is a proof rather than a coherent proposition, while a genuinely external unknown may stay open at no cost, exactly as flat Λ CDM leaves its own inherited data open.

The buildable ones below are named as debts, not softened into frontiers; the end-to-end transfer and the full-spectrum likelihood are the load-bearing debts, the derivation of the progenitor spectrum the genuine frontier. Four stand out:

1. the sufficiency of the null-boundary characteristic data for the observed acoustic coherence (§4), argued from the structure of the characteristic initial-value problem, with the peak-height *magnitude* carried by the time-reversal driving equality established there—what remains is the end-to-end branch-point-to-recombination transfer, the calculation that would carry the single-phase datum all the way through, confirm the height pattern digit by digit, and—within it—settle whether the diffusion-scale signature (§4, §4.3) shows in the observed high- ℓ power as a tension, a wash, or a distinctive-but-consistent feature.

One general consideration already bears on which: a shift in the diffusion *scale* is, across the observable multipole range, substantially degenerate with the spectral tilt, so the scale signature’s leading spectral effect is largely absorbable into n_s —disfavouring the tension reading and leaving a wash or a distinctive-but-consistent feature, with the tilt-irreducible residual the part the transfer would isolate.

This is a genuine build, not a plug-in: it requires first *specifying how the fluctuations gravitate on the geometrically fixed background*—the piece that sets the high- ℓ driving envelope, and which the standard Boltzmann codes cannot supply because they tie radiation’s gravity to its presence and so cannot represent the content-not-rate split the layered rate rests on—and then a bespoke transfer against that specification.

It is a debt owed and named as such: not a missing idea but a computation this sequence owes and has not yet run, and the paper’s own open edge rather than another’s. Two things it bundles are worth holding apart—the *build* is ours, and the *verdict* it would expose is the world’s: the transfer is what makes the diffusion-scale signature confrontable at all, and the data then decide it;

2. the exact large-angle *depth* (§7): the flat-projection transfer of the discrete closed- S^3 source is built with a *genuine Boltzmann transfer*—the exact CMB temperature transfer $\Delta_\ell(k)$ (Sachs–Wolfe, early and late integrated Sachs–Wolfe, and Doppler, all common to CR and flat- Λ CDM) sampled at the discrete modes k_L , gate-validated (the continuum reproduces the reference C_ℓ to four figures) with the r_0 sensitivity measured rather than asserted—shape stable, depths drifting up to 15% at $\ell = 4$ (receipt: `verify_lowell_boltzmann.pyR`).

It gives a *mild* deficit, $\approx 0.47, 0.41, 0.36$ and 0.68 of the Λ CDM expectation at $\ell = 2, 3, 4, 5$ —a *dip whose minimum falls at $\ell = 4$, not at the quadrupole*, the shape reproduced by the programme’s own photon hierarchy, which agrees on the location and not on the depth—recovering by $\ell \approx 7$: the late integrated Sachs–Wolfe term is sourced above the floor and is therefore retained by the discrete source, leaving the deficit shallower than the bare Sachs–Wolfe starvation.

The mechanism is robust—parameter-free, located at $\ell \lesssim 8$, bottoming at $\ell = 4$, with $\ell = 2$ and $\ell = 3$ starved *together* (the exact discrete measure and the Doppler term each

leave the $\ell = 2$ -to- $\ell = 3$ ratio near unity: receipts `verify_lowell_exact_measure.pyR`, `verify_doppler_lowell.pyR`).

Confronted with the sky the sector is a *wash*: CR’s smooth ~ 0.4 – 0.5 deficit sits above the sharp observed quadrupole (~ 0.2) and below the near- Λ CDM octopole (~ 0.6 – 1.0), matching neither, and the exact cosmic-variance likelihood returns $\Delta(-2 \ln L) \approx +1.8$ over $2 \leq \ell \leq 10$ (range $+0.3$ to $+3.8$ across the octopole estimator), a small residual inside cosmic variance (receipt: `verify_lowell_likelihood_v2.pyR`). So the low multipoles are a genuine but *non-discriminating* sector, neither a parameter-free correspondence success nor a sharp falsification edge; the deficit’s existence and location stand as a real parameter-free feature, but its depth places it consistent with Λ CDM within the lowest multipoles’ cosmic variance.

The honest exposed edges lie elsewhere—the damping tail and the full-spectrum likelihood below;

3. the progenitor spectrum itself— n_s and A_s (§8)—inherited boundary data fixed by the progenitor collapse, the baryogenesis-analogue computation belonging to the matter sector [6, 7], with the honest target a consistency between the inherited composition and spectrum and the measured primordial abundances and tilt, not a derivation from the geometry alone—the most ambitious reach of the three, the one that would close the decomposition from both ends and give CR a derivation where the standard model keeps a measured datum;
4. the *likelihood-level model selection*: the parameter-free ratio comparison above is joined by the exact low-multipole cosmic-variance likelihood—each $\hat{C}_\ell$ a scaled $\chi^2_{2\ell+1}$ about the model C_ℓ , the absolute baseline cancelling in the CR/ Λ CDM ratio (`verify_lowell_likelihood_v2.pyR`), the proper instrument where cosmic variance dominates.

With the genuine-Boltzmann depths ($\approx 0.47/0.41$ at $\ell = 2/3$, above) the sector is a *wash*: the low quadrupole still mildly rewards CR ($\Delta(-2 \ln L) \approx -2.6$, its milder dip nearer the observed low power than Λ CDM’s), the octopole and $\ell = 4$ mildly penalise it ($\approx +1.3$ and $+2.1$)—*so the sector’s largest single-multipole penalty sits not at the octopole but at $\ell = 4$, the prediction’s own minimum*—and the net over $2 \leq \ell \leq 10$ is $\Delta(-2 \ln L) \approx +1.8$ (range $+0.3$ to $+3.8$ across the octopole estimator)—a small residual well inside cosmic variance. A Monte-Carlo over 10^4 skies puts the sector’s *in-principle* discriminating power at only $\sim 3.4\sigma$ (the short multipole lever arm and cosmic variance), and the current measurement uncertainty swamps even that.

The low multipoles are therefore a genuine but *non-discriminating* test: CR and Λ CDM are statistically indistinguishable there, the mild deficit neither favouring CR nor refuting it. Decisiveness comes from outside the sector—the geometric expansion rate (4), now confronted with the distance ladder (§2.4), as the standing discriminator, and the likelihood across the *full* spectrum (its amplitude level carried by the time-reversal driving equality, §4; the full-spectrum likelihood itself, with the oscillating-medium transfer downstream in the matter sector [6, 7], the larger piece still to build).

A frontier of a different kind concerns the matter itself rather than the perturbation observables above: the *dynamics of matter crossing the branch point*.

Three facts about that crossing follow from the structure already established here.

It is *well posed*—the beginning is fixed at the branch point $r = 0$, and the crossing is taken on the *substrate*, whose curvature is finite there however the geometry’s own invariants behave.

The bend $\rho = m'/4\pi r^2$ that carries the matter is accordingly finite, and its data on that null surface is characteristic rather than Cauchy: a characteristic crossing with no obstruction from the substrate’s curvature. *The geometry’s Kretschmann scalar does diverge at that locus*,

as the abstract states and the circle paper computes [34]; what makes the crossing well posed is that the divergence belongs to the chart-borne geometry and not to the substrate the cut is taken on, and that the tortoise measure converges so the crossing carries no scale.

It *isotropizes and is scale-free*—the throat’s de Sitter no-hair (§6) damps the anisotropic part of the crossing stress-energy to the isotropic monopole, and the degenerate approach (§8) imprints no thermal scale: the matter counterparts of the field no-hair and faithful transmission established above. *Where each of those acts is worth stating, since one of them cannot act at the crossing at all.*

Damping is a process and requires elapsed time, and the crossing has none: the real part of the complex cosmic time is frozen along the segment reaching it [33].

The no-hair damping therefore belongs to the *approach*—the throat region, over the collapse leg’s own cosmic time—while the crossing carries whatever survives it unchanged, not because a mechanism protects it but because there is no interval in which anything could act. *The isotropy is earned on the approach and preserved by the crossing*, and the two halves are of different kinds: the first is a damping calculation with a rate, a timescale and a residual, the second follows from the zero-duration structure and takes no model input. It follows that *how much anisotropy survives is fixed by the approach against the leg’s duration*—a property of the progenitor’s collapse history rather than of the crossing. And it obeys a *structural transition law*: because the reassignment preserves the cosmic foliation and the density is leaf-carried and lapse-independent [6], the reassignment acts on the time-stacking—fixing the Λ -set rate (6)—and not on the leaf, so the isotropic density crosses as inherited progenitor content, which is the structural reason the inherited datum of §2.4 is inherited rather than produced; *and two conditions must be met at that crossing, which a concrete model has to satisfy separately: the reassignment requires a degenerating Killing field whose freed null direction can be re-founded as a clock, the joining of the legs requires a branch point, and these are independent—the forced Nariai member carries the degeneracy and is not a branch point, while $r = 0$ is the branch point and carries no Killing degeneracy [9], so a model furnishing one without the other furnishes no cosmogenesis. The first of the two is settled as a uniqueness, not merely an availability*: across the slicing family the reassignment that promotes a null direction to the fundamental congruence is *uniquely* the one selecting the forced Nariai member, every other vantage yielding a congruence that crosses the embedding’s symmetry structure transversally and so reading as a localised black hole [5]. *A model therefore has no freedom in where the reassignment acts*, and misplacing it does not fail to give a cosmology by a little—it gives a black hole instead. the scoping of the two rates this law forces—the excursion-side local readout against the observable geometric rate, their boundary—is drawn out in the cosmogenesis paper [17]. What remains open is the *detailed* worldline and field dynamics of the crossing for a concrete matter model [8, 6]—the deferred depth, now sitting on a crossing shown well posed rather than merely deferred.

None of these unsettles the result obtained: the branch point’s null geometry decomposes the primordial scalar spectrum into a substrate-determined structure and a progenitor-supplied content, and the degeneracy proves that the branch point transmits that content rather than imprinting one of its own *on this argument*; a computed transfer function across the branch point is not yet in hand. They are, rather, the map of where this cosmology goes next—the end-to-end transfer behind the peak *heights* and the full oscillating medium downstream in the matter sector, the tensor sector’s companion in the dynamics paper [9].

11 Discussion

What this paper determines is one thing seen from several sides. A single de Sitter geometry, its scale set by the cosmological constant alone (the maximally symmetric substrate’s sole scale [16]) and read through a causal reassignment at the cosmogenesis branch point, *is* the

observed cosmology: the flat- Λ CDM expansion history is its areal radius (§2.3); the apparent Hubble tension and the acoustic scale are consequences of its geometric rate (§2.4); the primordial scalar spectrum is the characteristic data of its seam, structure fixed by the substrate and content inherited across the degenerate horizon (§§4–8). The Einstein field equations are unchanged; what changes is the reading, and under it the quantities the standard model carries as independent inputs—the present matter density, the dark-energy fraction—are readings of one cosmic clock, not drivers of the expansion, so that the coincidence of Ω_m and Ω_Λ dissolves into our observing at a time of order the geometry’s one timescale. There is a single seed, and the observed universe is what grows from it. And the one scale reaches down as well as out: the same cosmological constant that sets this expansion sets, for every mass M , the *Hubble–Eddington radius* [32] $r_{\text{HE}} = (3GM/\Lambda c^2)^{1/3}$ —the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off. The slicing geometry reads that boundary as the flat locus of the existent slice’s own curvature, the local bend of the cut exactly cancelling the substrate’s cosmological term [4], and the slicing operator reads it as the handover from sub-marginal bound orbits to the marginally-bound $E = 1$ congruence that *is* this cosmology [6]. So the substrate’s single Λ governs both the global expansion this paper computes and the local scale at which structure resists it—one constant read at two ranges—and the Hubble–Eddington radius, proposed as a local, per-structure test of Λ from the largest bound structures [12], is an independent observational handle on the very constant the expansion history here measures. The two ranges are moreover the same length up to a pure number. The Hubble–Eddington radius—where the slicing surface’s Gaussian curvature $K_G = 1/\alpha^2 - M/r^3$ vanishes, the local bend cancelling the substrate’s cosmological curvature [4]—is $r_{\text{HE}} = (M\alpha^2)^{1/3}$; while the expansion’s own amplitude, $(2M\alpha^2)^{1/3}$, is exactly the areal radius at matter- Λ equality ($\rho_m/\rho_\Lambda = \text{csch}^2(3\tilde{\tau}/2\alpha)$, unity at $\sinh = 1$). The two therefore stand in the fixed ratio $2^{1/3}$ for every M : the scale at which a bound structure’s gravity balances the expansion and the scale at which the expansion’s matter balances its Λ are one length read locally and cosmologically, the factor of two that of the Schwarzschild term against the surface curvature’s M/r^3 and nothing else. ^R

The comparison with the standard cosmological model is accordingly not *only* a contest of fits—though on the Hubble tension the geometric rate now fits the joint acoustic-and-distance-ladder data at the direct H_0 where Λ CDM cannot (§2.4)—but also, and more deeply, a difference in what each must assume to do so. The standard model *assembles* the observed universe: a dark-energy component for the late-time acceleration, an inflationary sector for the causal contact, flatness, coherence, and near-scale-invariance the background dynamics do not themselves supply, and further early-universe physics where the sound horizon and the expansion rate come into tension. Each part is a real solution to a real problem, and each was introduced to yield the feature it produces.

The dark-energy component is the sharpest case of the difference: on this cosmology the acceleration is not a sector at all. Since $(dr/d\tilde{\tau})^2 = E^2 - f$ on the energy family, one has $d^2r/d\tilde{\tau}^2 = -f'/2 = rK_G$: *the comoving acceleration is the existent slice’s own Gaussian curvature*, up to the positive factor r .^R So the expansion decelerates while the slice is negatively curved, turns over exactly where it is flat—the Hubble–Eddington radius, $\rho_m = 2\rho_\Lambda$, at $\sinh(3c\tilde{\tau}/2\alpha) = 1/\sqrt{2}$ —and accelerates thereafter, with nothing added to make it do so. *And that locus is a measurement, not only a description:* the turnover sits where the areal radius reaches the front seam, $r = r_N$, so the acceleration-onset redshift is the offset parameter itself minus one,

$$z_{\text{acc}} = x_0 - 1 = 0.6648 \pm 0.0467, \quad (11)$$

against $(2\Omega_\Lambda/\Omega_m)^{1/3} - 1 = 0.632$ on Planck’s parameters—agreement at 0.7σ , and with no fit performed here. *What makes the equation worth stating is what it does not contain:* z_{acc} is a redshift, measured calibration-free, so it is untouched by the distance-ladder tension, and it fixes x_0 without H_0 , without a ladder and without the microwave background. The single

dimensionful scale is where any disagreement must then live, and the age follows from the same parameter in closed form, $t_0 = (2r_N/\sqrt{3}c) \operatorname{asinh}(x_0^{3/2}/\sqrt{2})$. The late-time acceleration is thus read off the same structure that fixes the local boundary of a bound system. *This cosmology adds no such part.* Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are read off structure the construction already contains—the isotropizing throat (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§7), the degenerate horizon that transmits rather than imprints (§8)—each fixed by Λ alone, with no parameter free to produce it. This is the distinction between a structure that *requires* a phenomenon and one that merely *permits* it through adjustable apparatus [11, 10], and it is the criterion by which competing frameworks are rightly weighed ahead of the measurement that will decide between them.

The transmission dichotomy (§8) is where that distinction is sharpest, because there the two frameworks make opposite structural commitments about one object. A non-degenerate horizon’s exponential approach imprints a scale-invariant spectrum—the very mechanism by which inflation produces near-scale-invariance—while the degenerate Narai seam carries no scale and transmits the progenitor spectrum unaltered. Inflation accommodates the observed tilt because a freely specified potential can be tuned to it; reading that fit as a confirmation mistakes a model with enough adjustable structure to match for one that required what was found (§9). This cosmology cannot tune: the degeneracy *requires* that the branch point not imprint a scale, so the framework has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial B -modes. The amplitude and tilt it does not derive it inherits, exactly as flat Λ CDM inherits the baryon-to-photon ratio from a baryogenesis it does not model; on that single datum, of the same kind and count the standard model already carries, the two stand level. A framework is credited for requiring the structure it explains, not faulted for measuring the boundary data it does not—and the asymmetry falls on everything else.

The verdict this licenses must be drawn on the right axis, and confined to it. On the axis of theory-choice—the requiring rather than the permitting of the phenomena, the consolidation of many results under one structure, the absence of a device added per problem—the comparison is decidable now, from the structural record, and it falls to this cosmology [11]. On the axis of data-discrimination the picture has begun to move with it: the geometric rate resolves the Hubble tension across the acoustic scale and the baryon-acoustic distance ladder together, at the directly measured H_0 where the standard model cannot (§2.4)—a discriminating datum, not a tie—while on the light elements, the microwave-background peak heights, and the low multipoles the two frameworks are even. So the standing is stronger than rule-favoured-awaiting-the-datum: it is a framework favoured on the structure *and* carrying its first discriminating data result, with the fuller confirmation—the damping tail and the full-spectrum likelihood—still owed. A theory that fits is not a theory confirmed, and that confirmation is not yet in hand; but the data axis is not even, and the framework has staked falsifiable commitments—no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate B -modes—precisely where the standard model keeps its freedom^R.

The low-multipole deficit is a mild, non-discriminating feature (§§7, 10). The flat projection of the discrete closed- S^3 source gives a parameter-free deficit below $\ell \approx 8$ —real, and set by Λ through r_0 and D_C —bottoming at $\ell = 4$, at $\approx 0.47, 0.41, 0.36$ and 0.68 of the Λ CDM expectation at $\ell = 2, 3, 4, 5$. At that depth CR’s smooth deficit matches neither the sharp observed quadrupole-only dip nor the near- Λ CDM octopole, and under the exact cosmic-variance likelihood the two frameworks are statistically indistinguishable at low ℓ ($\Delta(-2 \ln L) \approx +1.8$ over $2 \leq \ell \leq 10$, inside cosmic variance). So the sector neither confirms nor refutes: it is a genuine parameter-free prediction whose depth places it consistent with Λ CDM within the lowest multipoles’ cosmic variance. The framework’s exposed edge is the damping-*scale* signature locked to the Hubble resolution (§4), a computed, non-reabsorbable CR-specific effect whose consequence

for the observed high- ℓ power is entangled with the acoustic transfer at those multipoles and so genuinely open. Its decisive discriminator is the geometric expansion rate itself, carried to the distance ladder and confronted with the DESI DR2 baryon-acoustic dataset at $\chi^2/\text{dof} \simeq 1$ (§2.4): the same rate resolves the Hubble tension across the full ladder and stakes the damping signature on one commitment, the observable weight of the latter awaiting the full transfer.

The framework’s honest standing includes what the standard model has genuinely built and this cosmology has not. Dark matter is inferred from multiple independent, cross-consistent lines—rotation curves, gravitational lensing, the merger offsets, the acoustic-peak structure, the growth of structure—and this cosmology carries the matter density as the bend of the spatial cut without yet removing the component or answering its particle question; that column stands at full weight. Standard nucleosynthesis derives the light-element abundances from one measured parameter, a genuine achievement this cosmology now meets on the same footing rather than exceeds: the collapse’s cooling leg is a standard nucleosynthesis run on one inherited datum (§2.4; the cosmogenesis paper [17]), producing helium-4 and deuterium at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers—and, because the run is standard, carrying lithium-7’s standard threefold over-prediction, the lithium problem shared rather than solved. On the light elements the two frameworks stand even: the same two successes, the same one problem, with this cosmology’s open edge whether one progenitor handover fits all the abundances together. And the framework is young: the *detailed* dynamics of matter crossing the branch point (its crossing shown well posed and structurally governed, §10), the derivation—as against the measurement—of the inherited data, and the Standard-Model matter *content*—the gauge representations and the mass spectrum, which the framework does not supply, its geometry shown to force only the discrete flavour skeleton [18]—remain to be built (§10; [8]). These are open frontiers, not defects: a theory is not marked down for what it has not built, only for a claim it makes that fails. They are named plainly here because an economy that favours this cosmology on the structure it carries would be worth nothing bought by concealing the structure it does not.

12 Conclusion

So, plainly, what this paper claims and what it does not. It claims that the observed expansion history, the dissolution of the Hubble tension and the calibration of the acoustic scale, and the structure of the primordial scalar spectrum are consequences of a single de Sitter geometry read through the causal reassignment its own horizon geometry selects—one seed, grown, the field equations unchanged—and that on the economy of assumption, the criterion by which theory-choice is rightly made ahead of the deciding measurement, this reading is favoured now over the assembled standard picture. On the data axis it claims the more limited and now-earned thing: that the axis has moved in its favour—the Hubble tension resolved across the distance ladder, the abundances confirmed—without yet claiming to have won it, the damping-tail signature and the full-spectrum likelihood the edges still to be brought to a verdict (the low multipoles, once the sharpest edge, now a mild non-discriminating deficit consistent with Λ CDM within cosmic variance). It does not claim to be a complete cosmology: its matter sector, the derivation of its inherited data, and its fermion content remain to be built. It claims to have shown that the universe we observe is the reading of one geometry, and to have staked the falsifiable commitments by which that claim can be tried.

A final word on the register, because it is the thesis rather than an aside. The beginning this cosmology describes is the branch point at $r = 0$ —the locus at which the collapse leg of the cosmogenetic contour becomes the expansion leg [8]—and it is not a curvature singularity reached and left behind. *The curvature there is set by the substrate’s own scale.* The throat is α , the defining constant of the maximally symmetric manifold the cut is taken on, and the divergence of the invariants computed in the areal chart is that chart degenerating, exactly as a

polar chart degenerates at a pole—not a curvature scale of the geometry [17, 16]. Two further facts stand with it: no finite cosmic-time layer ever reaches the locus, and the continuation through it carries no scale, since $f \rightarrow -2M/r$ diverges so that the tortoise coordinate $r_* = \int dr/f$ *converges* and what is transmitted is fixed by the analytic structure rather than by a length. *The branch point is crossed by continuation, not traversed by a worldline.* The expansion is not a space ejected from a point but the reading, on the near side of that branch point, of a spatial layer still advancing, the branch point itself a collapse that in the frame it bounds is never complete. The distinction the whole programme turns on—between what has finished happening and what is still underway—is thus the distinction the cosmology is built from: the universe is not the collapsed record of a beginning but the still-evolving layer that record represents, read here as one de Sitter geometry.

Appendix R Computational Receipts

Each computation marked ^R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

P15.the_sky_phase_fit_and_its_uncertainty [OK]

sec:intro (FITS the sky's acoustic phase from the peak positions and PROPAGATES its uncertainty: $\ell_n = \ell_A(n + \frac{\phi}{\pi})$ on 220.4, 537.7, 817.3 returns $\frac{\phi}{\pi} = 0.2405$ against the paper's 0.2404, and the control's 0.0043 offset is seven parts in a thousand of the 0.615 disagreement. The quoted $\sigma_{\text{simeq}0.008}$ is the TWO-multipole value; one multipole gives 0.0046 and twice the separation – so the paper's figure is the CONSERVATIVE end, and the sentence now says which assumption it carries). *Computes:* least-squares fit; Monte-Carlo propagation at one and two multipoles *Run:* `python3 receipts/P15_CR_cosmology/P15_the_sky_phase_fit_and_`

P15.verify_geometry [OK]

prop:flat / prop:throat / prop:transmission (three geometric props DERIVED from primitives: (a1) $\text{const-}\tau$ 3-metric $dr^2 + r^2 d\Omega^2$ has Riemann=0 (flat R^3); (a2) Nariai near-horizon $dS_2 \times S^2$, $f(r_{\text{star}}) = f'(r_{\text{star}}) = 0$, $f'' = -2\Lambda$, both radii $1/\sqrt{\Lambda}$); (a4) tortoise integrals simple- $\log(\exp)$ vs double- $\log(\text{power-law})$). *Computes:* Riemann from Christoffels; Nariai conds from f ; two tortoise integrals; checks intrinsically discriminating *Bound:* Kretschmann bonus derived too *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_geometry.py`

P15.verify_numeric [OK]

prop:amplitude / prop:subhorizon ((a5) substrate dS vacuum power $(\hbar H_{\text{Lam}}/E_P)^2 = 9.5e-123$ vs $A_s = 2.1e-9$ ~ 113 orders below, robust to $O(10)$ prefactor; (a7) comoving Hubble k at seam $0.0104/\text{Mpc}$ vs peak $\pi/r_s = 0.0217/\text{Mpc}$ $\sim 2.1x$ sub-horizon). *Computes:* $\Lambda +$ vacuum power + k derived from obs inputs; matches paper; $[E]/[R]$ marked *Bound:* anchor 6 low-ell leading-order $[R]$ only *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_numeric.py`

P15.verify_coherence_comb [OK]

sec:coherence (coherence comb: common seam phase $\sim$ sharp comb $\Delta\ell = \pi D_C/r_s = 296$ (peaks 296,592,888), peak/trough $\sim 1e5-1e6$; random phase per mode $\sim$ washed out, contrast ~ 1 (400 realisations). Null seam's one characteristic datum/mode = one phase/mode $\times$ (SCOPE: this receipt propagates nothing – it writes down $\cos(k r_s)^2$ with r_s and D_C as hardcoded literals, exactly as its own scope note says. It establishes the COHERENCE MECHANISM and NOT the peak spacing.)). *Computes:* ACFM coherence demo on the seam; coherent-vs-incoherent control; matches paper *Bound:* mechanism only; full transfer open $[R]$ *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_coherence_comb.py`

P15.cr_collapse_driving [OK]

sec:coherence (acoustic driving as linear oscillator $\Theta_0'' + \Theta_0 = -\Psi$: undriven ($\Psi = \text{const}$) $a_0 = 0.500$, driven (radiation) $a_D = 0.999$, boost $a_D/a_0 = 2.00$ (known LCDM boost); baryon-loaded driven-equiv 1.60 vs adiabatic 1.10). *Computes:* ODE integration; undriven- $\Psi = \text{const}$ control; matches known boost *Bound:* time-symmetry of contracting phase argued (magnitude proven in next receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_cr_collapse_driving.py`

P15.time_reversal_driving [OK]

sec:coherence ($|\Psi^*(\omega=1)|$ resonant driving magnitude is time-reversal invariant: $|\Psi^*(x)| = |\Psi^*(-x)|$ at $w=1$ on the radiation transfer (0.037049) AND a random control (1.411379); so CR's time-mirrored collapse driving = LCDM's magnitude, peak heights match by structure). *Computes:* numerical Fourier magnitude at $w=1$; radiation + random control; complex/phase-not-invariant discriminator *Bound:* built to close the paper's 'confirmed numerically ... random control' claim *Run:* `python3 receipts/P15_CR_cosmology/P15_time_reversal_driving.py`

P15.camb_reference [OK]

sec:coherence (**!! MARKED c54.160: the closing annotation states the paper's anchors as $P_2/P_1 \sim 0.46$ and $P_3/P_1 \sim 0.45-0.46$ while the file itself computes 0.452 and 0.443 and the paper now prints 0.45 and 0.44.** A stale annotation contradicting the same file's own digit-by-digit claim; both figures are now pinned). *Computes:* CAMB TT spectrum + peak-finder; matches paper *Bound:* needs camb; CR-matches-it carried by driving receipts *Run:* `python3 receipts/P15_CR_cosmology/P15_camb_reference.py`

P15.damping_ratio_clean [OK]

sec:coherence (**!! CORRECTED c54.155: θ_D/θ_* CR/LCDM IS 1.0816 (+8.2%), NOT THE 1.0785 (+7.9%) THIS ROW AND THE RECEIPT'S OWN DOCSTRING RECORD** – both stale from the z_{onset} 6850 $\rightarrow$ 6797 retune. The paper's 'approx 1.08' brackets the live value correctly). *Computes:* CAMB $x_e(z)$; r_s+r_D both rates; gated on CAMB; RATIO cancels conventions *Bound:* absolute $\theta_D \sim 7\%$ below CAMB (k_D convention), cancels in ratio *Run:* python3 receipts/P15_CR_cosmology/P15.damping_ratio_clean.py

P15.damping_reabsorption [OK]

sec:coherence ($\sim 9\%$ damping shift not reabsorbable: $r_D \sim \omega_b^{-0.305}$ (paper -0.31), $d \ln r_D/d \ln \omega_c = -0.295$; cancelling +8.9% needs $\omega_b + 29\%$ vs $\sim 1\%$ BBN+height prior $\rightarrow$ 29x too big; 1% ω_b shifts $r_D \sim 0.3\%$; residual $\sim 8.2\%$). *Computes:* finite-diff levers on CAMB thermodynamics vs BBN+height priors *Bound:* broader multi-sigma tension is conditional (full transference open); paper takes only settled part *Run:* python3 receipts/P15_CR_cosmology/P15.damping_reabsorption.py

P15.zonset_determinations [OK]

sec:tensions (**!! MARKED c54.160: the paper reported the rounded- l_* pinning as recovering 6873 and this receipt returns 6844 on the current integrals.** The paper now carries both numbers and notes that 6844 sits NEARER to the retired 6850, which sharpens the withdrawal rather than softening it). *Computes:* three determinations computed on one stated convention; H_0 -independence of (1) exact to 5 digits; rigging of (2) shown by the l_* that 6850 itself returns *Bound:* (3) recorded as a structural observation, not a refutation: the datum is inherited and order-unity either way *Run:* python3 receipts/P15_CR_cosmology/P15.zonset_determinations.py

P15.the_ratio_is_the_onset_in_imported_units [OK]

sec:tensions (WHAT $\rho_r/\rho_m \approx 2$ IS, AND WHAT IT IS NOT: **NEITHER A HIDDEN KNOB NOR A SECOND MEASURED DATUM, BUT THE ONE FITTED QUANTITY z_{onset} RE-EXPRESSED IN UNITS OF A PHYSICAL MATTER DENSITY THE CONSTRUCTION DOES NOT DETERMINE.** The rate carries Ω_m (fitted to BAO); the sound speed carries ω_b and ω_γ (measured directly, framework-free); z_{onset} is the one fitted number – and the PHYSICAL $\omega_m = \Omega_m h^2$ enters NO prediction. But ρ_r/ρ_m at onset is $\omega_r(1+z_{\text{onset}})/\omega_m$ and equality is $1+z_{\text{eq}} = \omega_m/\omega_r$, so **both divide by exactly that quantity**. ** KNOB-FREENESS SURVIVES AND IS THE LOAD-BEARING HALF: $z_{\text{onset}} = 6747.3$ at $H_0 = 67.4, 70.0, 73.0$ and 74.0 , to the digit – H_0 is a common factor in r_s and D_M and cancels from θ_* , so the fitted quantity does not move between the two H_0 the tension is between and cannot be absorbing it. ** ** WHAT DOES NOT SURVIVE IS THE EXACTNESS OF 'ABOUT TWICE EQUALITY': at the fitted $\Omega_m = 0.3066$ the ratio is 2.01 at $H_0 = 67.4$ and 1.71 at $H_0 = 73$, the two readings crossing at $H_0 \sim 68$ where $\Omega_m h^2$ reproduces the CMB-inferred $\omega_m = 0.143$ – and the papers' headline is that the cosmology fits at the DIRECTLY measured $H_0 = 73$. ** So the datum is an order-unity BAND (1.7-2.0), not a determined number; the eta-analogy and the BAO confrontation are untouched (the latter uses distance ratios and never ω_m).). *Computes:* z_{onset} recomputed from the geometric rate at four values of H_0 by root-finding on the measured acoustic angle, its spread asserted below 1 part in 6000; the crossing H_0 asserted in 67-69; the $H_0=73$ ratio asserted at 1.71 *Bound:* ** THIS IS A UNIT ANALYSIS, NOT A NEW MEASUREMENT. ** It does not touch the rate, the fit, or the abundances; and the CMB-inferred ω_m it compares against is itself derived in a radiation-included analysis, which is the point rather than an objection – the residual measures how far the two analyses' ω_m must differ *Run:* python3 receipts/P15_CR_cosmology/P15.the_ratio_is_the_onset_in_imported_units.py

P15.the_two_data_are_one [OK]

sec:tensions (WHAT THE 'SINGLE INHERITED DATUM' $\rho_r/\rho_m \sim 2$ ACTUALLY IS, AND WHAT THE NATURALNESS ARGUMENT OFFERED FOR IT ACTUALLY DESCRIBES. ** (1) IT IS NOT A DATUM DISTINCT FROM eta. ** On any thermal history $\rho_r/\rho_m(T) = [1+\nu](\pi^4/30 \zeta(3)) T / [\eta (w_m/w_b) m_N]$, which returns **1.99 at $T = 1.6$ eV** – the quoted value to one per cent, from standard thermodynamics and NO feature of this construction. Given eta and the measured matter-to-baryon ratio, both inherited for the composition regardless, **the ratio at onset IS the onset**: the handover supplies ONE composition datum and the cosmology fits ONE parameter. ** (2) AND THE NATURALNESS ARGUMENT DESCRIBES AN EPOCH NINE ORDERS FROM THE HANDOVER. ** $\rho_r = \rho_m$ at $T = 0.806$ eV;

the handover sits FOUR ORDERS ABOVE the deuterium bottleneck (0.07 MeV) because that is what the nucleosynthesis computed on it requires, so the ratio there is $\sim 9e8$ – and ‘a handover carrying radiation of order the matter density’ would be a handover with $\eta \sim 0.5$: no light elements, no acoustic peaks. ** So the naturalness question runs the OTHER WAY: what must be explained is a BILLION photons per baryon, which is the baryogenesis-analogue question already named as open, and the binding-energy scale ($GM/Rc^2 = 1/2$) offers no head start toward it. ** P7’s ‘the frontier is two data and not one’ is corrected to one inherited datum plus one fitted parameter.). *Computes:* the identity evaluated at the onset temperature and asserted against the quoted datum to better than 5%; the equality temperature and the ratio at the bottleneck, at P7’s own handover and at the rest mass computed; the η that $O(1)$ would require at each epoch computed and contrasted with the measured $6.1e-10$; and the papers asserted to carry the withdrawal and the corrected accounting *Bound:* ** THIS CORRECTS AN ACCOUNTING AND WITHDRAWS AN ARGUMENT; IT TOUCHES NO RESULT. ** The rate, the fit, the baryon-acoustic confrontation and the abundances are all untouched – and the accounting moves in the construction’s favour, from two inherited data to one *Run:* `python3 receipts/P15_CR.cosmology/P15.the.two.data.are.one.py`

P15_full_transfer_verdict [OK]

sec:coherence (high-ell deficit UNDER THE SHORTCUT (assume peaks=LCDM + apply +8.9% damping): C_L CR/LCDM = 0.979/0.920/0.829/0.717/0.594 at $l=500-2500$; cited to show the shortcut is illegitimate (assumption (A) is the unbuilt part)). *Computes:* exp damping on CAMB LCDM C_L ; conditional χ^2 *Bound:* verdict explicitly CONDITIONAL/open (own banner); paper honors scoping *Run:* `python3 receipts/P15_CR.cosmology/P15_full_transfer_verdict.py`

P15_verify_closedS3_nonsync [OK]

sec:largescale (low-ell floor mechanism+location: discrete closed- S^3 source ($k_L = \sqrt{L(L+2)}/r_0$, $L_{\ell}=2$) through FLAT $j_{\ell}(k_L D_C)$, stretch $D_C/r_0=2.75$ pushes $L=2$ to $\ell \sim 7.8$; $\ell(\ell+1)C_{\ell\ell}=0.12/0.10/0.20/0.6$ at $\ell=2..7$, recovered by $\ell \sim 8$). *Computes:* discrete-source flat-Bessel sum; flat-limit χ^2_0 control passes *Bound:* ordinary SW only - ℓ deeper than Boltzmann depth (routed to that receipt) *Run:* `python3 receipts/P15_CR.cosmology/P15_verify_closedS3_nonsync.py`

P15_verify_lowell_exact_measure [OK]

sec:largescale ((1) HZ weight $w_L=(L+1)/(L(L+2))$ DERIVED as closed- S^3 degeneracy $\beta^2 \propto$ per-mode $1/(\beta(\beta^2-1))$ ($g^2 P = w_L$ exactly; $w_L L_{\ell} \sim 0.998$ continuum); (2) $D_2/D_3=1.162$ stable across measure family - ℓ floor starves $\ell=2,3$ together (octopole slightly more)). *Computes:* first-principles derivation + flat-limit + measure-robustness + wrong-measure contrast *Run:* `python3 receipts/P15_CR.cosmology/P15_verify_lowell_exact_measure.py`

P15_verify_doppler_lowell [OK]

sec:largescale (Doppler term (super-horizon velocity through j_{ℓ}) does NOT differentiate $\ell=2$ from $\ell=3$: CR/LCDM ratio $l^2 \sim 0.27$, $l^3 \sim 0.21-0.22$ across 4 treatments (none/coh+/coh-/quad), $D_2/D_3 \sim 1.29$ octopole slightly more suppressed; continuum validator reproduces LCDM rise). *Computes:* SW+ISW+Doppler transfer, 4 combination modes; continuum check; not tuned *Bound:* bare depth (milder exact depth = Boltzmann receipt) *Run:* `python3 receipts/P15_CR.cosmology/P15_verify_doppler_lowell.py`

P15_verify_lowell_boltzmann [OK]

sec:largescale/sec:scope (exact low-ell depth via genuine Boltzmann transfer: discrete closed- S^3 source through CAMB’s exact $\Delta_L(k)$ (SW+ISW+Doppler); CR/LCDM = 0.473 ($l=2$), 0.410 ($l=3$), recover by l^7 , octopole/quadrupole 0.867 (matches paper 0.47/0.41). GATE: continuum reproduces CAMB C_L to 4 figures). *Computes:* CAMB transfer x discrete source, Riemann-sum vs integral; gate-validated; r_0 -stable; ISW-retained diagnosis *Bound:* needs camb (high acc) *Run:* `python3 receipts/P15_CR.cosmology/P15_verify_lowell_boltzmann.py`

P15_confront_lowell_data [OK]

sec:largescale (low-ell data confrontation (corrected depths): CR 0.47 sits ABOVE the sharp observed quadrupole (~ 0.2), CR 0.41 mildly below the near-LCDM octopole ($\sim 0.6-1.0$) - ℓ matches neither, a WASH; old 0.22/0.20 ‘striking match’ superseded). *Computes:* CR (Boltzmann depths) vs literature observed multipoles *Bound:* body corrected to r_{962} depths; likelihood in next receipt *Run:* `python3 receipts/P15_CR.cosmology/P15_confront_lowell_data.py`

P15_verify_lowell_likelihood_v2 [OK]

sec:largescale/sec:scope (low-ell cosmic-variance likelihood: exact scaled- χ^2_{-2l+1} , CR/LCDM

ratio $\Delta(-2\ln L)_l = (2l+1)[r_l(1/d_{l-1}) + \ln d_l]$; $l=2$ rewards CR -2.63, $l=3/4$ penalise +1.31/+2.10, central total +1.8, octopole range +0.3..+3.8 -; wash inside cosmic variance). *Computes:* exact likelihood with corrected Boltzmann depths; matches paper *Run:* python3 receipts/P15_CR_cosmology/P15-ver

P15.the_low_ell_minimum_is_at_ell_four [OK]
sec:largescale/sec:scope (**SCOPE: THIS FILE CROSS-VALIDATES NOTHING.** Its arm A is the likelihood receipt's stored dict and **its arm B is the PAPER'S OWN FIGURE CAPTION, read by regular expression** — so the comparison is against the citing paper's typography, and the quartet 0.42/0.35/0.29/0.52 is banked nowhere. **What it does verify is that arm A's minimum sits at $l = 4$ and that the standing receipts always carried it, which is what it is now cited for.** The actual second arm is 'P15.the_second_arm_actually_run': it agrees on shape and NOT on depth, so the quartet 0.42/0.35/0.29/0.52 is not banked. What the file establishes is THE LOW-MULTIPOLE DEFICIT'S SHAPE, AND WHERE THE SECTOR'S WEIGHT ACTUALLY FALLS. **IT IS NOT A MONOTONE RECOVERY FROM THE QUADRUPOLE BUT A DIP WHOSE MINIMUM IS AT $ell=4$ — and the quadrupole and octopole, the two multipoles the paper quoted, are its two SHALLOWEST points. ** ** THE SHAPE IS CROSS-VALIDATED BY TWO INDEPENDENT BOLTZMANN TREATMENTS of the same discrete source: a standard code's exact $\Delta_l(k)$ at the discrete k_L gives 0.47/0.41/0.36/0.68 at $ell=2,3,4,5$ and the programme's own photon hierarchy gives 0.42/0.35/0.29/0.52 — SAME minimum, SAME recovery, depth ratio 1.13-1.30, which is the size of the residual *sec:instrument* declares. The shape is cross-validated; the depth carries their spread. ** ** AND THE $ell=4$ MINIMUM WAS NEVER NEW: 'P15-verify_lowell_boltzmann.py' has tabulated 0.356 at $ell=4$ since r1395 and the likelihood receipt has consumed it ever since — the corpus's failure was one of QUOTATION, not of computation. ** ** THE OCTOPOLE VERDICT: on the exact cosmic-variance likelihood the quadrupole REWARDS the construction (-2.63), the octopole penalises it at +1.31, and the LARGEST single-multipole penalty in the sector is the +2.10 at $ell=4$ — the prediction's own minimum, and the one multipole of the four the large-angle literature does not treat as anomalous. Sector total unchanged at +1.8, a wash inside cosmic variance. **). *Computes:* both arms' depth tables read off their own sources (the likelihood receipt's dict; the paper's caption) and their argmin asserted equal to $ell=4$; the $ell=4$ depth asserted present and deepest in the standing r1395 receipt; the per-multipole likelihood re-derived from the depths and its argmax asserted at $ell=4$ with the total asserted at +1.8; the exact/WKB transmission spread asserted below 5%; and all eight corpus sites asserted to carry the corrected statement *Bound:* ** DEPTH IS QUOTED TO A TWO-CODE SPREAD AND NOT BETTER. ** And nothing here relieves the observed sky's selective suppression of $ell=2$ ALONE by a factor of order three, which this construction does not produce; that stays inside cosmic variance and belongs to the likelihood-level comparison *Run:* python3 receipts/P15_CR_cosmology/P15.the_low_ell_minimum_is_at_ell_four.py

P15.two_ranges_ratio [OK]
sec:background (two ranges) (the local Hubble-Eddington radius $r_* = (M \alpha^2)^{1/3}$ and the expansion's amplitude $A = (2M \alpha^2)^{1/3}$ are ONE length in the fixed ratio $2^{1/3}$, for every M ; A is the matter-Lambda equality radius). *Computes:* r_* from $K_G = 1/\alpha^2 - M/r^3 = 0$; A from $\rho_m/\rho_\Lambda = \text{csch}^2(3\tau/2\alpha)$ at equality; ratio $A/r_* = 2^{1/3}$ symbolically, independent of M and α *Bound:* the r1608 identity; the four-ways reading (turnaround, amplitude, equality, structure boundary) is the paper's *Run:* python3 receipts/P15_CR_cosmology/P15.two_ranges_ratio.py

P15.the_second_arm_actually_run [OK]
sec:largescale (**THE SECOND BOLTZMANN ARM, ACTUALLY RUN — THE SHAPE CROSS-VALIDATES AND THE DEPTH QUARTET DOES NOT.** P15 claimed the low-multipole deficit had been run a second time on the programme's own photon hierarchy, returning 0.42/0.35/0.29/0.52. **The receipt cited for it does not run the hierarchy: it reads arm A from a stored dict and reads arm B OUT OF THE PAPER'S OWN FIGURE CAPTION, by regular expression** — so the cross-validation compared a table against the citing paper's typography and the quartet was banked nowhere. This file drives 'HIER_photon_hierarchy' itself: evolves it on a k -grid containing the discrete ladder $k_L = \sqrt{L(L+2)} \times 2.75/D_M$ plus a dense continuum grid, projects its own source terms to $\Delta_l(k)$ for $l = 2..44$, and forms the discrete w_L sum against the continuum $\ln k$ integral, each normalised to its own $l = 25-40$ plateau. **SHAPE: minimum at $l = 4$ and recovery by $l \sim 7$, independently of the CAMB arm and of the caption. ** **DEPTH: 0.494/0.243/0.184/0.608 against CAMB's 0.473/0.410/0.356/0.676 — close at $l = 2$, a factor 1.7-1.9 apart at $l = 3-4$. **

So the quartet is returned by neither arm and the 'uniformly 10-25% deeper' claim is false; both withdrawn from P15 at c54.153.). *Computes:* the minimum over $l = 2..5$ asserted at $l = 4$; recovery asserted in $6 \leq l \leq 9$; the depth disagreement asserted above 1.4x; all ratios asserted finite *Bound:* ***scope:* this settles that the SHAPE claim has a computation behind it and that the DEPTH claim did not. It does NOT settle which depth table is right — the arms differ in late-ISW handling and in D_M (13864 here against the projection's 13927) — and it makes no attempt to reconcile them.*** Run:* `python3 receipts/P15_CR_cosmology/P15_the_second_arm_actually_run.py`

P15_hubble_expansion_confrontation_v2 [OK]

sec:tensions (SDSS DR12 BAO flagship: CR BAO ratios D_M/r_d , D_H/r_d H0-INDEPENDENT (identical at $H_0=67,73$); χ^2 CR=1.71 (any H_0 incl 73), LCDM(67.4)=4.37, LCDM(73)=49.45 = the tension; cosmic chronometers consistent with 73). *Computes:* geometric-rate H0-cancellation + χ^2 vs DR12; matches paper 1.7/49 *Run:* `python3 receipts/P15_CR_cosmology/P15_hubble_expansion_confr`

P15_desi_dr2_confrontation [OK]

sec:tensions (***!!* MARKED c54.160: the docstring reads $\chi^2/\text{dof} = 13.70$ for LCDM at $H_0 = 73$ while the paper quotes about 15 – the same $\chi^2 = 178.16$ divided by 13 measurements rather than by the 12 degrees of freedom one fitted parameter leaves.**** The paper's figure is the correct one and the receipt's annotation is not; the χ^2 itself is pinned). *Computes:* full-covariance χ^2 vs DESI DR2; matches paper 1.0/14; was uncited by name *Run:* `python3 receipts/P15_CR_cosmology/P15_desi_dr2_confrontation.py`

P15_expansion_law [OK]

sec:properframe/sec:flatlcdm (derivation core (symbolic): eq:amplitude $(6GM/\Lambda c^2)^{1/3}=2^{1/3}/\sqrt{r}$ at Nariai; eq:scalefac $r=A \sinh^{2/3}(B \tau)$ solves the $E=1$ geodesic $(dr/d\tau)^2=2GM/r+(\Lambda/3)r^2$ ($2/3$ forced by wrong-exponent control); eq:rate $H^2=(\Lambda c^2/3)\coth^2$; eq:omega-ratio $\Omega_m/\Omega_L=\text{csch}^2$). *Computes:* sympy: amplitude, geodesic solve, $\coth^2$ rate, csch^2 ratio; wrong-exponent control *Run:* `python3 receipts/P15_CR_cosmology/P15_expansion_law.py`

P15_verify_throat_tower [OK]

sec:throat (throat mode tower + no-hair isotropization: $dS_2 \times S^2$, $m^2/H^2=1/(1+1)$, $\nu^2=1/4-1/(1+1)$; $l=0 \rightarrow \nu=1/2$ (scale-invariant base, survives), every $l \geq 1 \rightarrow \nu \geq 0$ (heavy principal, decays) $\Rightarrow$ throat isotropizes; grounded in Wald 1983 no-hair; guard: throat $l \neq$ CMB multipole). *Computes:* tower algebra + survives/decays split; theorem-grounded; guard held *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_throat_tower.py`

AS_amplitude_leftward [OK]

rem:transmission-leg (AS — THE AMPLITUDE, READ LEFTWARD. Family 3's question, and what C4 can and cannot say about it. CONFIRMATION 2, and the corpus is already correctly scoped here: P7: the inherited datum 'stands as a ONE-PARAMETER ACCOMMODATION rather than a parameter-free prediction.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/AS_amplitude_leftward.py`

BUILD_camb_store [OK]

sec:instrument (BUILD_camb_store – the CAMB side of the transfer comparison, with its settings RECORDED. WHY THIS EXISTS (r2345). 'ENVELOPE_residual.py' and its relatives load '/tmp/camb.tr.npz' and measure our transfer against CAMB's. Nothing in the corpus records HOW that store was built – not the receipt, not the changelog, not the instrument.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/BUILD_camb.s`

C10_highl_ratio [OK]

sec:envelope-consequence (***!!* CORRECTED c54.155: THIS ROW CARRIED THE RETIRED $z_{\text{onset}} = 6850$ NUMBERS**** (r_s matched +0.76%, r_D +10.09%, θ_D/θ_* +9.26%). The shipped receipt hardcodes 6797 and produces +0.67%, +10.15%, +9.41%, $l_* = 302.2$. **** And the receipt hardcodes $th_ratio = 1.0926$ with the comment 'DERIVED in C9' while C9 produces 1.0941.**** P15 corrected to the live values). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C10_highl_ratio.py`

C11TEST_radiation_zeroed [OK]

sec:envelope-consequence (CRRUN — THE SAME INTEGRATOR ON CR. The control passed

at r_{1975} , so this may now be run. WHAT CHANGES, and each from the corpus not from me:
 * BACKGROUND: $H^2 = H_0^2(\Omega_m a^{-3} + \Omega_L)$ at $H_0=73$, $\Omega_m=0.3066$ – radiation is content, not a source. * LOWER LIMIT: the SEAM, not η_0 (C9: there is no observable expansion below it).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11TEST_radiation_zeroed.py`

C11_early_isw [OK]

sec:envelope-consequence (C11 — THE EARLY ISW. The last named piece. CONFIRMATION 2/4 – the corpus settles the input, in its own voice: 'radiation, like matter, is inherited content read off that clock, NOT A TERM THAT SOURCES THE RATE, and there is no radiation-dominated era.'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11_early_isw.py`

C1_closure_of_the_driving [OK]

sec:envelope (C1-CLOSURE — DOES THE ACOUSTIC OSCILLATOR CLOSE IN ELEMENTARY TERMS ON CR's COLLAPSE LEG? CONFIRMATION 4 first — WHICH background. The corpus is specific and I must not substitute: L2 (where the driving lives) = 'the leaf's local rate, ORDINARY FRIEDMANN, radiation GRAVITATING NORMALLY, on the progenitor's collapse excursion' – and P16: on the cooling leg $|H| = \sqrt{8 \pi G \rho_0/3}$, 'identical to standard BBN at every...'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C1_closure_of_the_driving.py`

C2_horizon_limits [OK]

sec:envelope (C2 — THE LIMITS. The comoving Hubble radius across the excursion, from the corpus's own geometry. CONFIRMATION 4 — the object, and I take it from the corpus rather than from cosmology-by-habit: the $E=1$ (marginally bound) congruence IS this cosmology. On it $(dr/d\tau)^2 = E^2 - f = 1 - f$, with $f = 1 - 2M/r - r^2/\alpha^2$. The signed areal radius r is the scale-factor analogue.'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C2_horizon_limits.py`

C3_conformal_map [OK]

sec:envelope (**!! CORRECTED c54.155: 'better than a percent at the entry radii of the modes in question' IS 1.87% AT THE PAPER'S OWN 10 k_s THRESHOLD** (3.72% at 5 k_s , 24.4% at the seam); sub-percent begins near 30 k_s . P15 corrected). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C3_conformal_map.py`

C4_driving_envelope [OK]

sec:envelope (**!! CORRECTED c54.155: 'entry at $x = 1/\sqrt{3}$ for every k ' IS A LIMIT, NOT AN IDENTITY** (the receipt's own table gives $x_{\text{entry}} = 0.764$ at $k = k_s$, +32%), **and the 'residual below a percent' is an average over a 5-point grid that lands near zeros of $\cos(x_{\text{seam}})$: scanning the receipt's own function over k/k_s in $[10,120]$ reaches 7.3% at $k/k_s = 12$.** Both qualified in P15). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C4_driving_envelope.py`

C5b_baryon_term [OK]

sec:envelope (C5 — THE BARYON TERM. The first of the two terms C4 named as missing. CONFIRMATION 4 – and the corpus fixes the division rather than me: P16 masthead: 'eta fixes the abundances and the CMB peak HEIGHTS, ρ_r/ρ_m the peak SPACING.' P15: this cosmology 'carries the matter density as the bend of the spatial cut WITHOUT YET REMOVING THE COMPONENT' – so ρ_m is TOTAL matter and the baryon fraction is set separately,...). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C5b_baryon_term.py`

C6_neutrino_term [OK]

sec:envelope (C6 — THE ANISOTROPIC-STRESS TERM. The second of C4's two named

terms. FINDING ZERO, before any physics: the corpus's TEXT says 'neutrino' ZERO times, and 'anisotropic stress' zero, and 'free-streaming' zero – while its BBN CODE carries them explicitly (bbn_network.py: 'neutrinos share the plasma T until $\sim e^+$ annihilation, then decouple to $(4/11)^{1/3}$ '). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/C6_neutrino-term.py

C7_equality_and_deficit [OK]

sec:envelope (C7 — THE STEP THE TRAP WAS GUARDING. full_transfer_verdict.py AS-SUMED the high-ell peaks equal LambdaCDM's and applied the extra damping, getting a large deficit – and P15 disowns it because 'that assumption IS the unbuilt part.' C4 built it. So the same calculation can now be done with a DERIVED envelope. FIRST, though, one number that decides everything and that I have not yet checked.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/C7_equality_and_deficit.py

C8_diffusion_length [OK]

sec:envelope-consequence (C8 — DERIVE r_D HERE, FROM FIRST PRINCIPLES, ON THE SAME FOOTING AS THE DRIVING. Nothing imported from the damping_tail/ files. Nothing that needs CAMB. CONFIRMATION 4: the object. Silk damping is a diffusion length built from the Thomson opacity and the expansion. The standard definition (Kaiser 1983; Hu-Sugiyama), in the $R_b \rightarrow 0$ limit: $1/k_D^2 = \int d\eta \cdot (1/(6 \tau \dot{\tau}))$). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/C8_diffusion_length.py

C9_sound_horizon_and_ratio [OK]

sec:envelope-consequence (C9 — r_s AND r_D WITH THE CORRECT LOWER LIMIT: THE SEAM. A first pass integrated from $a=0$ and returned $r_s = 256$ Mpc against the corpus's 144 – absurd, and the error is structural, not numerical: *** ON THIS COSMOLOGY THERE IS NO OBSERVABLE EXPANSION BEFORE THE SEAM. *** P15: 'recombination is observable cosmology FROM THE SEAM OUTWARD'. The L1 foliation BEGINS at the seam;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/C9_sound_horizon_and_ratio.py

H0_acoustic_angle_and_seam [OK]

sec:refit-bound (H0 — THE ACOUSTIC SCALE AT THE DIRECTLY MEASURED H0. The calculation I have been running backwards. CORPUS CLAIM (P7, P15): 'There is no second Hubble rate to reconcile: THE DIRECTLY MEASURED H0 IS THE GEOMETRY READ AT THE PRESENT EPOCH, while the lower value inferred from the microwave background rests on a radiation-governed sound horizon the construction does not share...'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/H0_acoustic_angle_and_seam.py

ISW_line_of_sight [OK]

sec:refit-bound (ISW — THE EARLY INTEGRATED SACHS-WOLFE, ISOLATED, AND ITS SHIFT ON THE FIRST PEAK. Same shape as everything else in A.0: the source is a line-of-sight integral over the set rate. $\Delta_l(k) = \int d\eta [S_{SW} + S_{ISW}] j_l(k(\eta_0 - \eta))$, $S_{ISW} = (\Phi' + \Psi') = 2\Psi'$ LambdaCDM has a nonzero early ISW because Ψ is still evolving at recombination (radiation matters). CR does not: C11 showed Ψ is constant on L1.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/ISW_line_of_sight.py

M2_reabsorption_minimization [OK]

sec:refit-bound (M2 — THE MINIMIZATION. Can a parameter refit absorb the damping shape? The honest core of the likelihood question, done on the closed forms derived in C8/C9 – no CAMB. SETUP, stated before running: FREE : Ω_m , ω_b , and the inherited datum ρ_r/ρ_m (which fixes z_{onset}). CONSTRAINT: $l_* = \pi D_M/r_s$ pinned to the observed 301 – the peak positions are measured.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/M2_reabsorption_minimization.py

PEEB_peekles_on_the_set_rate

[OK]

sec:refit-bound (PEEB — THE PEEBLES IONISATION HISTORY ON THE SET RATE. ZREC used Saha only: right shift, wrong absolute (1283 vs CAMB’s 1090), because Saha has no departure from equilibrium. Peebles adds it – and the departure is itself H-dependent, which is exactly why it must be integrated on the construction’s own rate rather than inherited. Standard effective three-level hydrogen atom (Peebles 1968; Seager et al. fits).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt’s own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/PEEB_peekles_on_the_set_rate.py

ROBUST_p1p2_scan

[OK]

sec:refit-bound (**!! FIXED c54.160: THE REGISTERED COPY WAS UNRUNNABLE FROM ITS OWN DIRECTORY** – it imported ‘RD_diffusion_direct’, which lives only in ‘storyboard_receipts/’, so it exited 1 on ImportError before reaching any computation and the sweep had to run it from the ORIGIN instead. *A receipt that cannot run where it is registered is not a receipt.* Path resolution added; it now exits 0 in place). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt’s own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/ROBUST_p1p2_scan.py

P15_the_first_peak_figure_is_not_stable

[OK]

the acoustic sector (**THE 150 IS NOT A PREDICTION: ‘ROBUST_p1p2_scan’’s FIRST-PEAK POSITION AND PEAK-HEIGHT RATIO ARE NOT STABLE UNDER ITS OWN DOCUMENTED INITIAL CONDITIONS.** Built settling front #5, the corpus’s two internal routes to the first acoustic peak (transfer 220, this instrument 150). **The front is posed as a disagreement between two established routes; one of the two is not established.** PART 1 the numerics are cleared first – stability number 0.47 against the file’s own limit of 2.8, 120+ RK4 steps per oscillation, NS_CR x4 reproduces every figure. PART 2 Psi’s VALUE comes from the smooth matched envelope and Psi’s DERIVATIVE from the oscillatory closed form – two different functions, and the initial velocity is built entirely from the second. PART 3 across four readings of the SAME stated IC, ell_1 spans 150-315 and P1/P2 spans 0.93-2.02; **the reading the docstring actually describes (‘velocities zero’) returns 2.017 against the sky’s 2.212, where the paper reported 1.45 as the disagreement.** PART 4 the comb the paper cites against it was never propagated either – ‘verify_coherence_comb’ writes down $\cos(k\ r_s)$ with r_s and D_C as literals. PART 5 **the residual: the source comb spacing is 0.72-0.79 of π/r_s under EVERY variant. The initial data move the first peak by 165 multipoles and do not move this.**). *Computes:* stability asserted under the file’s own 2.8 limit and steps-per-oscillation over 50; the two derivative functions asserted to differ by more than 1.2x across the band and to agree as $x \rightarrow 0$; the control asserted to reproduce 1.447 and 150/360/555/780; the documented-IC run asserted at 2.017 AND asserted closer to the sky than the implemented one; ell_1 asserted to move by 150+ multipoles and P1/P2 by more than 2x; the comb receipt asserted to contain $\cos(k\ r_s)$ and NO integrator; the comb deficit asserted present, bounded, and STABLE across variants (max-min ≤ 0.15) – which is what makes it not initial data. *Bound:* **scope: this does NOT vindicate 220 and supplies no first-peak position. It removes one route from arbiter status and registers the comb deficit, which is not resolved here. The four variant figures are produced by ‘computations/beyond_the_wall/L171_what_sets_the_first_peak.py’, which execs the registered receipt’s own source with only the initial-data block substituted; its ICMODE=asis control reproduces the receipt exactly, assertions passing.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_first_peak_figure_is_

P15_the_crossing_exists_and_is_empty

[OK]

the acoustic sector (**THE RE-ENTRY BETWEEN THE BRANCH POINT AND THE ONSET IS REAL, AND IT IS EMPTY: THE L1 POTENTIAL EQUATION CONTAINS NO k AT ALL.** Built working front #5, testing a candidate THIS FORK proposed at c54.165 – front #1 puts every mode super-horizon and frozen at the branch point, ‘prop:subhorizon’ puts the acoustic modes inside at the onset, so they re-enter in between, and a crossing is where flat LCDM acquires its k -INDEPENDENT driving shift. **The answer is no, and the reason is stronger than the question.** PART 1 the re-entry is real and its boundary is computed: the mode crossing AT the onset is $k = 0.0111/\text{Mpc}$, $\text{ell} = 144$, so the peak modes crossed before the plasma began and $\text{ell} \leq 144$ crosses after. PART 2 but it is a MATTER-era crossing – pressureless to better than a part in 10^4 , because L1 has no radiation SOURCE. PART 3 **and the $w = 0$ potential equation has no k in it**: Phi constant to twelve digits, identical across four decades in k , the integrator taking the SAME number of steps for every mode. PART 4 the contrast: in LCDM 5 of 6 cross under radiation,

where $w = 1/3$ restores the k^2 term and Φ decays on the sound-crossing time by construction.). *Computes:* every mode at/above the peak region asserted to cross before the onset AND $\ell = 28$ asserted to cross AFTER it (the first draft claimed 'every observed mode' and this assertion fired); the boundary multipole asserted inside 100-250; the background asserted pressureless to $1e-3$ at every crossing; Φ 's spread across four decades in k asserted below $1e-9$ AND the solver step count asserted identical across k ; the LCDM contrast asserted to put $4+$ modes under radiation with z_{eq} above 3000. *Bound:* ****scope:** this CLOSES a candidate mechanism and supplies none. It computes no peak position and uses no figure from the acoustic fold. It leaves 'PO-7' open and harder: the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval.****** *Run:* python3 receipts/P15_CR_cosmology/P15_the_crossing_exists_and_is_empty.py

P15.two_arm_control_and_guard

[OK]

the acoustic sector (****A LIVE LambdaCDM CONTROL AND AN UNDRIVEN GUARD, ON ONE MACHINERY – AND THE CR ARM'S FIRST PEAK SITS 23% LOW AGAINST A 1.5% POSITION FLOOR.**** Built for fronts #5 and #18. ****This fork had never run a control****: 'ROBUST_p1p2_scan' quotes one from r1975 in a comment, and c54.164 then showed its figures move over ℓ_{l1} in {150,165,315} with nothing to measure that against. PART 1 ****the guard passes on BOTH arms**** – every coupling to the potential removed at every site (4 Φ ' in photon/neutrino/CDM continuity AND $k^2 \Psi$ in photon-baryon/neutrino Euler, both branches) and the source comb lands on the integers, LCDM 1.0147/2.0005 and CR 1.0121/1.9968, ****so every departure from $n \pi/r_s$ in a driven run is the driving and nothing else****. PART 2 the control lands at 224/536/808/1116 against the sky's 220.6/538.1/809.8/1147.8 – ****positions to 1.5% at the first peak and 0.4%/0.2% at the second and third; HEIGHTS only to 13-26%, so no height claim from this instrument is meaningful below ~25%****. PART 3 the CR arm run on its PHYSICAL discrete ladder and on a dense continuum sampling gives ****identical**** peaks 172/396/624/904, so discreteness is not what sets the position – an objection closed by measurement. PART 4 $\ell_{l1}/\ell_{lA} = 0.5703$ against the control's 0.7433 and the sky's 0.7312: ****a 23% deficit against a 1.5% floor, a factor of 15 in margin****, with $P1/P2 = 0.866$ against the control's 1.933.). *Computes:* the undriven comb asserted within 0.03 of the integers on both arms; the control's position floor asserted under 3% AND its second and third peaks asserted within 1% of the sky; the discrete and continuum peak lists asserted identical; the deficit asserted at 5+ floors; CR's ℓ_{l1}/ℓ_{lA} asserted in 0.55-0.60 and the control's in 0.72-0.76; the two arms' l_A asserted to agree to 1% (they do – D_M differs by 6.2% and r_s by 6.3% and they cancel). Every background quantity is RECOMPUTED rather than pinned. *Bound:* ****scope:** this does NOT settle 'PO-7' and the CR figure is not offered as a prediction – it uses the initial datum the corpus's own text describes (flat in k , common phase, VELOCITIES ZERO) and c54.164 established the figure moves with that choice. What changed is that it now moves against something. NO FIGURE FROM THE ABANDONED ACOUSTIC FORK IS USED; that fork's own reproduction gate fails in this tree, and this instrument disagrees with its 201.3 by 15%.****** *Run:* python3 receipts/P15_CR_cosmology/P15_two_arm_control_and_guard.py

P15.the_driving_shift_by_subtraction

[OK]

the acoustic sector (****THE DRIVING SHIFT MEASURED MODEL-FREE BY SUBTRACTION: LCDM's ACOUSTIC PHASE IS FLAT IN k TO 2.1%, CR's VARIES BY 3.9x – AND THE SCALING IS $k^{-0.62}$, NOT THE k^{-2} THE ABANDONED FORK'S MECHANISM REQUIRES.**** Built for front #5 on the c54.168 instrument. Each k integrated TWICE on the same equations, driven and undriven, differing in one multiplicative switch and nothing else. $Q(k) = k \int c_s \, d\eta/\pi$ at the first acoustic turnover of $\Theta_{\theta 0}$. ****PART 1** the undriven column is the CALIBRATION and is MEASURED: 0.9968-1.0003 on both arms, where a free oscillator must give exactly 1.****** PART 2/3 LCDM's Q_{driven} runs 0.791 -j 0.774 over $k = 0.045$ -0.16 (2.1% spread) because every mode crosses under radiation, where the potential decays on the sound-crossing time by construction; CR's runs 0.457 -j 0.117, a factor of 3.9, because there is no radiation era to cross and c54.167 showed the pre-onset interval is scale-free. PART 4 ****a power-law fit gives $k^{-0.62}$, collapsing the data to a factor of 1.2 where k^{-2} spans a factor of 16.**** The fold reports $Q \, k^2$ constant and its seam-transient mechanism ($3H/k^2$) requires it. ****The phenomenon reproduces here independently; the exponent does not.**** Recorded as an inter-instrument disagreement, not resolved.). *Computes:* the undriven calibration asserted within 0.01 of 1 on BOTH arms (this is the file's own gate); LCDM's spread asserted under 6%; CR's variation asserted over 2x; the fitted exponent asserted inside (-1.2,-0.3); **** $Q \, k^2$ asserted**

to span MORE than a factor of 4 – i.e. the fold’s constant- $Q\text{-}k^2$ asserted FALSE here**; and the fitted exponent asserted to collapse the data better than k^{-2} , so the fit is not vacuous. *Bound:* **scope: two corrections were forced by this file’s own calibration and are recorded because each would have been a false result – (i) reading Q on Θ -hat rather than Θ_0 calibrated at 0.0003, since DR switches the couplings into the fluid but not Ψ ’s own evolution; (ii) the driven LCDM column then calibrated at 0.001 on a decaying-mode transient two grid points from the start, excluded by requiring $k\eta \gtrsim 1$, which is the DEFINITION of an acoustic turnover and removes nothing on the CR arm. No figure from the abandoned fork is used.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_driving_shift_by_subtraction.py

P15.which_coupling_carries_the_k_dependence

[OK]

the acoustic sector (**THE k -DEPENDENCE LIVES IN THE POTENTIAL’S GRADIENT, NOT IN ITS DECAY – WHICH IS THE REVERSE OF WHERE A RELAXATION TIMESCALE WOULD PUT IT.** Front #5, on the c54.168 instrument against its 1.000 calibration. The potential enters the plasma TWICE: as ‘ 4Φ ’ in continuity (its rate of change feeding the density – **the decay channel**) and as ‘ $k^2\Psi$ ’ in Euler (its gradient driving the velocity). Switched independently on the same integration. PART 1 **flat LCDM’s two couplings are each FLAT in k and they OPPOSE** – continuity alone puts the turnover at $Q = 1.23\text{--}1.28$, LATER than a free half-period, the gradient at $0.73\text{--}0.87$, earlier, composing to $0.77\text{--}0.86$. **So the universality of the standard driving shift is a CANCELLATION between two flat terms, not one term’s signature.** PART 2/3 on the geometric rate there is no cancellation left – both advance ($0.11\text{--}0.18$ and $0.06\text{--}0.57$) and they add – **and the k -dependence sits in the GRADIENT: continuity varies $1.6\times$ at $k^{-0.17}$, gradient $8.9\times$ at $k^{-1.04}$. PART 4 and **neither term alone is the answer**: the driven exponent -0.62 lies BETWEEN the two, so the scaling is their interference.). *Computes:* LCDM’s continuity asserted ABOVE 1 and its Euler BELOW 1 (they must oppose) and all three driven exponents asserted flat to 0.15; CR’s continuity asserted nearly flat ($|\exp| \gtrsim 0.35$) and its Euler steep ($|\exp| \lesssim 0.8$) and the Euler term asserted to be the steeper of the two – **reversing that assertion would reverse the finding**); both CR couplings asserted below 1 (they add); the driven exponent asserted strictly BETWEEN the two and distinct from each by 0.2; and the undriven calibration asserted within 0.01 of 1 on both arms, which is the file’s own gate. *Bound:* **scope: this localises the k -dependence to a coupling and shows the standard shift’s universality is a cancellation this rate does not have. It does NOT name the process that makes the gradient coupling run as k^{-1} , and computes no peak position. It closes the abandoned fork’s stated mechanism as an explanation of THIS measurement – a relaxation on $3H/k^2$ acts through the decay channel, and the decay channel is measured flat.** *Run:* python3 receipts/P15_CR_cosmology/P15.which_coupling_carries_the_k_dependence.py

P15.the_gradient_coupling_in_closed_form

[OK]

the acoustic sector (**THE GRADIENT COUPLING IN CLOSED FORM: THE TURNOVER IS AT A FIXED MULTIPLE OF $3H/k^2$, AND SINCE Q ALREADY CARRIES ONE FACTOR OF k THAT GIVES $Q \sim 1/k$, NOT $1/k^2$.** Front #5, closing the question c54.170 left. **The derivation is four lines.** (1) Φ ’s own equation carries a damping term $-k^2\Phi/3H$, so Φ relaxes on $\tau = 3H/k^2$. (2) With continuity off the oscillator is $\Theta_0'' + c_s^2 k^2 \Theta_0 = -(k^2/3)\Psi$, and **the k^2 in the forcing’s amplitude is cancelled by the k^{-2} in its duration – the delivered impulse is k -INDEPENDENT**. (3) So the first turnover is at the moment the impulse has been delivered, at a fixed multiple of τ : $\kappa = 1.79$, constant across a factor of eight in k . (4) $Q = k \int c_s d\eta/\pi$ already carries one k , so ** $Q = 3\kappa c_s H/(\pi k)$ ** **PREDICTED $Q k = 0.010403/\text{Mpc}$ AGAINST A MEASURED $0.010372 \pm 0.30\%$ – with nothing fitted but κ ; and the closed form reproduces Q itself to 1.6% across the sub-horizon band.** $\Rightarrow$ **The abandoned fork’s TIMESCALE is right and its EXPONENT is out by exactly one factor of k – a bookkeeping error in its own mechanism rather than a disagreement about physics.** And the continuity channel is flat for the complementary reason: ‘ 4Φ ’ feeds the density with the potential’s TOTAL change, $\int \Phi' d\eta = -\Phi_0$, the same however fast the relaxation runs. **So the observed $k^{-0.62}$ is not a power law of anything – it is the crossover between a flat channel and a $1/k$ one.**). *Computes:* κ asserted constant to 10% across the sub-horizon band; **the flattest grouping asserted to be $Q k^1$ and not $Q k^0$ or $Q k^2$ – if that assertion reverses, the derivation is wrong**); the predicted coefficient asserted within 5% of the measured one; the closed form asserted to reproduce Q to 5% across the band; the continuity channel asserted flattest UNGROUPED and asserted flat to within a factor of 2. *Bound:*

****scope:** this NAMES the process, so front #5's question is answered. It does NOT make the first peak agree with the sky – the 23% deficit against a 1.5% control floor stands unchanged. What changed is that its cause is derived rather than described. No figure from the abandoned fork is used; its timescale is confirmed independently here and its exponent corrected.**** Run:** python3 receipts/P15_CR_cosmology/P15_the_gradient_coupling_in_closed_form.py

P15_where_the_likelihood_sits

[OK]

the acoustic sector (****WHERE THE LIKELIHOOD ACTUALLY SITS: IT CANNOT ARBITRATE, AND THE INSTRUMENT'S OWN CONTROL IS WHY.**** Discharges 'L-147', fenced since it opened on three conditions – falsifier fixed before any data is touched, a control reproducing a known number first, and not until the two internal routes agree. ****All three met**** (the routes collapsed at c54.164-168), so the fence came down. ****F1**** the pipeline is wired: the banked CAMB flat-LCDM best fit returns $\chi^2 = 206.4$ over 215 plik_lite TT bins, $\chi^2/\text{dof} = 0.96$. ****F2**** the instrument's own floor, measured through the SAME likelihood on the SAME bins: this fork's LCDM arm costs +18786 over CAMB. ****F3**** the only readable quantity, both arms of one instrument: CR - LCDM = +84160. ****F4**** the pre-registered rule returned 'suggestive, not a measurement discrepancy'. ****F6 – AND THE PRE-REGISTERED RULE WAS DEFECTIVE, WHICH RUNNING IT IS WHAT SHOWED.**** It is a RATIO test and never asks whether the floor is small in ABSOLUTE terms. It is not: the control arm sits at $\chi^2/\text{dof} = 103$ against CAMB's 0.96, because Planck's binned errors are a few tenths of a per cent and this instrument's heights carry 13-26%. ****Both arms are far outside the regime where plik_lite discriminates, so the honest verdict is the WEAKER one: the likelihood cannot arbitrate.** **!** That correction happens to FAVOUR the construction and is flagged for it** – it would apply identically had the CR arm come out better, and it rests on a height floor measured at c54.168 before any data were opened. ****What survives as fact: on one instrument, same bins, same fitted amplitude, the CR arm costs 5.4x the LCDM arm's χ^2 . An ordering, not a p-value.****). *Computes:* the banked LCDM fit asserted at χ^2/dof in (0.85,1.15) BEFORE anything else is read; every covered bin asserted non-empty (the keep mask); the instrument floor asserted POSITIVE (this instrument cannot beat a five-parameter CAMB fit, and if it appeared to, the binning would be wrong); the CR arm asserted to cost more than the control on the same instrument; and ****F6's premise asserted – if the control arm ever drops below $\chi^2/\text{dof} = 3$ the PART must be rewritten because the likelihood then CAN arbitrate.**** *Bound:* ****scope:** plik_lite TT only – no polarisation, no ell j 30, no lensing, no external priors, a third-party implementation rather than clik (PROVENANCE.md), and 185 of 215 bins because the instrument reaches ell = 2000. The restriction to covered bins is done on the COVARIANCE and re-inverted, which MARGINALISES over the missing multipoles; dropping rows of the Fisher matrix would CONDITION on them. The banked r1934 CR figure (397.1 vs 206.4) applied the derived DAMPING suppression to CAMB's LCDM spectrum – it tested damping on LCDM's PEAK POSITIONS and could not have seen the position result, so it answers a different question and is SUPERSEDED here. 'PO-7' stays protected: this reports a number and does not convert it into a verdict.**** Run:** python3 receipts/P15_CR_cosmology/P15_where_the_likelihood_sits.py

P15_the_line_of_sight_transfer

[OK]

the acoustic sector (****THE TRANSFER REBUILT ON A LINE-OF-SIGHT INTEGRAL: THE CONTROL'S FIRST-PEAK POSITION ERROR FALLS FROM 1.66% TO 0.16%, AND THE CR DEFICIT DOES NOT MOVE AT ALL.**** Front #2. Until c54.173 the source was evaluated at a SINGLE eta_rec – an instantaneous last scattering. ****The visibility $g = \tau' e^{-\tau}$ peaks at $\eta = 280.75$ with FWHM 37.8 Mpc, a tenth of a wavelength at the first peak.**** Rebuilt as $\Delta_l = \text{INT } S \, j_l \, d\eta$ with $S = g(\Theta_0 + \Psi) + e^{-\tau}(\Phi' + \Psi')$ + $(1/k^2) d/d\eta [g \, \theta_b]$. PART 1 the control goes 1.66% -> 0.16% in ell_1/ell_A: ****sub-per-cent POSITIONS, which is what the likelihood needed of them.**** PART 2 ****and the CR arm is unchanged to four figures, 0.5703 on both transfers, while the control moved 1.79%. The deficit stands at 21.9% against a 0.16% floor – 134 floors.**** The most suspect part of the instrument was replaced wholesale and the number did not follow it down. PART 3 ****the heights are NOT fixed and the target is NOT met****: P1/P2 -13%, P1/P3 -34%, a residual growing with multipole and pointing at the damping tail (no polarisation term; a tight-coupling closure that fails where the visibility is). Stated as a partial result. PART 4 one correction was a DERIVATION: the Doppler term carries $1/k^2$ not $1/k$, since $v_b = \theta_b/k$. ****With $1/k$ the second peak sat at 516 and P1/P2 at 3.15; with $1/k^2$, 532 and 1.93 – and ell_1 was 220 either way, so an instrument validated on the first peak alone would have**

carried the error.**). *Computes:* the control's position error asserted sub-per-cent AND asserted improved by 5x or more; the CR arm asserted to move less than 1% under the rebuild; the deficit asserted at 20+ floors; **the heights asserted NOT sub-per-cent – if they ever are, the PART is stale and front #2 should be struck**; and the corrected Doppler power asserted to move the SECOND peak toward the sky, which is what makes it a fix rather than a tuning. *Bound:* **scope: this delivers sub-per-cent POSITIONS and leaves the HEIGHTS at 13-34%, so the likelihood still cannot be run (c54.172). The remaining error is a high-multipole problem and neither of its two named causes is a CR question – both are checked against the LambdaCDM arm first. Late ISW is truncated at $\eta = 4000$, which affects only ℓ below ~ 30 and is outside both the grid and plik_lite's TT range.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_line_of_sight_transfer.py

P15_the_derived_diffusion_damping

[OK]

the acoustic sector (**THE HEIGHTS FROM 13-34% TO 2-3%: THE DIFFUSION DAMPING PUT BACK, DERIVED ON EACH ARM'S OWN OPACITY AND RATE.** Front #2. **Diagnosed from the SIGN**: at c54.173 both height ratios were LOW, meaning peaks 2 and 3 too HIGH, worse at higher multipole – under-damping. **The hierarchy's tight-coupling closure switches off at $\tau' = 0.6/\text{Mpc}$ and the visibility peaks at $\tau' = 0.06/\text{Mpc}$, ten times lower, so the photon quadrupole and the baryon slip were absent through the WHOLE of last scattering and the line-of-sight path carried no Silk damping at all.** The pre-LOS path hid this behind a hardcoded r_D . PART 1 the standard tight-coupling integral $1/k_D^2 = \text{INT} [R^2/(1+R) + 8/9]/[6(1+R)\tau']$ deta, evaluated on this arm's own background: $r_D = 6.58 \text{ Mpc}$ against the 6-7 a Boltzmann code gives. Nothing fitted.** PART 2 **and a DOUBLE COUNT had to be removed**: with the hierarchy's own damping left in over $\tau' \lesssim 0.6$ AND the integral covering the same range, the control came out OVER-damped (+6.7%, +17%). **The sign flips either side of the correction, which identifies a double count rather than a wrong coefficient.** PART 3 heights $-12.9\%/-33.5\%$ $-2.8\%/-2.1\%$, on a control whose positions were already sub-per-cent. PART 4 **and the likelihood's floor falls 103 -23 in χ^2/dof , a factor of 4.6, WITHOUT the verdict turning over** – 'L-147's answer stands, and the floor moving while the verdict does not is what measures how far the instrument still has to go.). *Computes:* r_D asserted inside 5.5-8 Mpc; τ' at the visibility peak asserted BELOW the gate (PART 1's premise); **the sign asserted to flip either side of the correction – if it does not, it is a coefficient error and not a double count**; the single-mechanism heights asserted within 5% and asserted better than BOTH the no-damping and the double-counted runs; the floor asserted to fall by 3x or more; **and the control asserted to remain ABOVE $\chi^2/\text{dof} = 2$ – if it ever drops below, the likelihood CAN arbitrate and 'L-147's verdict must be revisited.** *Bound:* **scope: this does not reach sub-per-cent heights and does not make the likelihood answerable. The residual is still worst at high multipole: no polarisation term in the source, and a photon sector with no free-streaming hierarchy after decoupling. Neither is a CR question.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_derived_diffusion_damping.py

P15_the_height_target_was_below_the_resolution_of_its_own_statistic

[OK]

the acoustic sector (**THE HEIGHT TARGET WAS QUOTED AGAINST A SKY NUMBER WITH A 3.4% ERROR BAR THAT WAS NEVER WRITTEN DOWN, AND χ^2 – THE STATISTIC THAT CAN SEE – PREFERS A COEFFICIENT IT IS NOT ALLOWED TO CHOOSE.** Front #2. Since c54.168 every height claim here (-13%, -34%, 2-3%, -0.05%) compared P1/P2 and P1/P3 to two bare numbers, 2.217 and 2.277. **PART 1** the numerics are cleared first: 1.4x the modes, 1.3x the k-range and 1.8x the eta-sampling move the ratios by 0.09% and 0.37%, and the 2%/7% swings that prompted the check came from LMAXL=1300 truncating the k integral under the THIRD peak. **PART 2** read off the plik_lite bandpowers with the published covariance – the full 2x2 block, not a diagonal estimate – **the sky's own ratios are 2.256 ± 0.077 and 2.280 ± 0.074 : 3.4% and 3.2%.** So the 13% and 34% of c54.173 were four and ten sigma and stand; **the 2-3% of c54.175 are about one sigma and do not.** *Caught by bookkeeping and not by suspicion: X_data is binned C_l and not D_l, so peak-finding on it returned 464/761/1085 instead of 221/527/815.* **PART 3** a scan of one multiplier on $1/k_D^2$ has P1/P2 selecting 1.094 ± 0.184 and P1/P3 selecting 0.897 ± 0.055 – **a 1.03 sigma split, so my own 'no single scale fits both, the residual is the envelope's SHAPE' is WITHDRAWN at its own error bar rather than published.** **PART 4** χ^2 over 185 bins does separate them: the minimum sits at 0.865 against 8/9's 0.859, and 16/15 costs 1123 units, seven inflated sigma. **The coefficient is changed back to 8/9 and NOT because it fits better** – neither value has ever been derived here (C8_diffusion.length names the ambiguity in its own text), and a remembered

number is not a derivation any more than a fitted one is. ****PART 5**** L-147's pinned χ^2 came from a script in /tmp; 'chi2_of_spectrum.py' puts that machinery in the corpus and returns 4170.2 and 50675.2 exactly.). *Computes:* converged spread asserted under 1% and the truncated run asserted separated from it; the sky's recomputed ratios asserted within 1 sigma of the bare numbers they replace; the DAMPX multiplier asserted to reproduce the explicit POLC=8/9 run to 0.2%; ****the withdrawal asserted:** if the split were 2 sigma or more the file FAILS for retracting a claim that in fact holds******; the χ^2 minimum asserted at the 8/9 multiplier within 5%; both pinned values asserted recomputed to 0.5%; and an assert that fires if the control ever reaches $\chi^2/\text{dof} \sim 1$, since this file would then need rewriting rather than patching *Bound:* ****scope:** this corrects a MEASUREMENT PROTOCOL, not the construction, and it is a correction to my own work. It supplies no derivation of the shear coefficient – that is named as owed and is front #2's first item: the free-streaming photon hierarchy with the polarisation multipoles. And the 3.2-3.4% is the single-bandpower-bin resolution; a full-curve fit determines the peaks far better, which is exactly why χ^2 replaces the ratios rather than joining them.****** *Run:* python3 receipts/P15_CR.cosmology/P15.the_height_target_was_below_the_resolution_of_its_own_statistic.py

P15.the_shear_coefficient_derived_not_remembered

[OK]

the acoustic sector (****THE DIFFUSION-DAMPING SHEAR COEFFICIENT, DERIVED FROM THE POLARISED PHOTON HIERARCHY INSTEAD OF REMEMBERED OR FITTED – AND ADOPTING IT COSTS THE CONTROL 1123 UNITS OF χ^2 .**** Front #2, closing the item c54.176 named as owed. 'C8.diffusion_length' has named 'the 8/9 coefficient against the 16/15 polarization-corrected one' in its own text since it was written, and neither had ever been derived here. ****METHOD**** two integrations of the same oscillator on the same background: the perfect tight-coupled fluid, undamped by construction, against photons carried to $l = 24$ with Thomson scattering, the polarisation multipoles evolved alongside and the baryons on their own velocity. ****Their ratio is the damping and nothing else – the $(1+R)^{1/4}$ WKB prefactor, the drift of c_s and the R -dependence of the oscillation are common to both and divide out identically.**** Gravity is off in both, which is the definition of the object rather than an approximation. ****PARTS 1-4**** with $G = 0$ the $l=2$ term is $-(9/10)\tau F_2$, the textbook unpolarised closure, and $C = 0.8852$ against 8/9; with polarisation $C = 1.0618$ against 16/15; ****and the polarisation correction – the quantity actually in dispute – is a factor 1.1996 against 6/5, three parts in ten thousand.**** C is read LOCALLY between consecutive extrema and extrapolated in $(k/\tau)^2$, so the k^2 scaling is measured rather than imposed. ****PART 5**** 16/15 is adopted and the control's floor goes 22.5 -j 28.6. ****The fit's preference for 8/9, measured and refused at c54.176, is refused again: a coefficient chosen because it fits is a fitted parameter whatever it is called.**** ****PART 6**** and nothing downstream moves – sec:envelope-consequence's 10.8% goes +10.83% -j +10.87%, because the coefficient nearly cancels in a ratio of two rates.). *Computes:* both intercepts asserted within 2% of the leading-order values and pinned to $\sim 1\%$ in the gate; ****the polarisation correction asserted within 0.5% of 6/5 and pinned to a tenth of a per cent – it is the number the file exists to produce****; the scatter about the line in $(k/\tau)^2$ asserted below 0.01, so a method that destroyed the extrapolation fails rather than shifting the intercept quietly; the sign of the finite- (k/τ) correction asserted, and the extrapolation asserted SHORT (under 2% of the coefficient swept over the fitted range); ****the χ^2 cost asserted POSITIVE – if the derived coefficient stopped costing χ^2 the file's whole point would need restating****; and the C8 figure asserted to move less than a tenth of a point, so adopting the coefficient cannot silently shift a published number *Bound:* ****scope:** this derives the coefficient in the WKB damping integral and NOT the damping envelope's shape. It supplies the damping half of the polarisation physics only; the SOURCE half – $g\pi/4$ and $(3/4k^2)(g\pi)$ with $\Pi = \Theta_2 + \Theta^P_0 + \Theta^P_2$ – needs the hierarchy carried in the spectrum instrument and is what front #2 still owes. And one assertion of mine fired on its own data: I demanded both extrapolation slopes be of order unity and they are -0.150 and -0.942; the gate now checks that the extrapolation is SHORT, which is the thing that actually matters.****** *Run:* python3 receipts/P15_CR.cosmology/P15.the_shear_coefficient_derived_not_remembered.py

P15.the_photon_hierarchy_in_the_transfer

[OK]

the acoustic sector (****THE PHOTON BOLTZMANN HIERARCHY WITH POLARISATION CARRIED INSIDE THE TRANSFER: THE CONTROL'S χ^2/dof FALLS 28.6 -j 7.1 AND THE FIRST FOUR PEAKS LAND AT 220/540/812/1124 AGAINST THE SKY'S 220.6/538.1/809.8/1147.8.**** Front #2. Photons as multipoles to $l = 24$ with the polarisation hierarchy alongside, the baryons on

their OWN velocity and their OWN density, and the source carrying $g^* \pi/4$ and $(3/4k^2)(g^* \pi)$ ".

****THE HANDOVER MAKES A DOUBLE COUNT IMPOSSIBLE RATHER THAN MERELY ABSENT****: the fluid runs to $\tau' = 3/\text{Mpc}$ and the envelope is FROZEN there, so envelope and multipoles have disjoint supports – and only 4.2% of $1/k \cdot D^2$ has accumulated at the handover, so 96% of the damping is dynamic. ****PART 1**** the second peak is 0.35% out and the third 0.27%, where the fluid had them 2.6% and 0.7% low. ****PART 2**** χ^2/dof 28.6 $\rightarrow$ 7.1, and 20.0 $\rightarrow$ 4.2 over the range where the wavenumber integral is not starved. ****PART 3**** and c54.177's prediction is checked against the number it predicted: ****running the same hierarchy with ONLY the two polarisation source terms dropped – exactly the half an envelope has – gives 2022 against 1320, so they are worth $\Delta \chi^2 = 702$, 63% of the 1123 that adopting the derived 16/15 cost.**** ****PART 4**** and half the baryon split is WORSE than none: own density while still sharing the photons' velocity moved χ^2/dof 133 $\rightarrow$ 161 at development settings; both halves give 15.1 $\rightarrow$ 7.1. ***Twice now a physical effect taken in half has been worse than not taken – it is not a small error but a different and inconsistent model.*** ****PART 5**** doubling the hierarchy depth moves χ^2 by 0.00% and doubling the handover opacity by 1.5%. ****PART 6**** and the CR arm is at $\text{ell}_1/\text{ell}_A = 0.5703$ for the SIXTH instrument state, while the control improved fourfold.). *Computes*: every peak asserted no FURTHER from the sky than the fluid transfer's; the χ^2 improvement asserted to be at least a factor of two and pinned at 7.14 / 4.17; both height ratios asserted within 2 sigma of the sky's own covariance-derived values; ****the polarisation source terms asserted to recover between 40% and 120% of c54.177's 1123 – outside that range c54.177's statement could not be said to have predicted this and the file FAILS****; the full baryon split asserted to beat the shared-density run; the LG and handover sensitivities asserted below 0.1% and 5%; the CR arm asserted unmoved to the fourth decimal; and an assert that fires if the control ever reaches $\chi^2/\text{dof} \sim 1$, since L-147 would then have to be REOPENED rather than this file's closing line printed *Bound*: ****scope: this is the LambdaCDM control and the CR arm on one instrument, and it converts nothing into a verdict – 'PO-7' is protected and 'L-147's answer is unchanged, because a control at seven times a fit still cannot certify what it is compared with. What remains is accuracy and no longer a named missing SECTOR: reionisation, neutrino mass, lensing, and the wavenumber range. And the aliasing gate was found MISSING from the line-of-sight path, which has been the default since c54.173; it cost a development run a first peak at 196 instead of 220 with nothing said, and is now on both paths.**** *Run*: python3 receipts/P15_CR_cosmology/P15.the_photon_hierarchy_in.the_transfer.py

P15.what_is_left_is_lensing_and_not_reionisation

[OK]

sec:refit-bound (****BOTH OF FRONT #2's NAMED NEXT STEPS MEASURED BEFORE EITHER WAS BUILT, AND ONE IS WORTH EXACTLY NOTHING.**** ****PART 1**** above $\text{ell} \sim 40$ reionisation's whole TT effect is the constant screening $\exp(-2 \tau)$, and the amplitude here is fitted in CLOSED FORM – $A = (m \cdot F \cdot d)/(m \cdot F \cdot m)$ is homogeneous of degree -1 in m , so $m \rightarrow s \cdot m$ returns A/s and the residual is IDENTICAL. **** $d(\chi^2) = 0$ exactly, shown at $\tau = 0.054, 0.10$ and 0.30 , zero to machine precision at all three.**** ****The degeneracy is a property of the STATISTIC, not of the size of τ .*** ****PARTS 2/3**** a Gaussian smoothing in ell , width scanned, minimises at $\chi^2 = 921$ against the control's 1320 – and ****a width that GROWS with ell beats a constant one by 1.8x in $d(\chi^2)$, 400 against 225, each at its OWN optimum so the comparison is of SHAPE****. Lensing smooths more at high ell because a deflection of fixed angle subtends a larger fraction of an acoustic wavelength, and that is the signature the data select. ****PART 4**** the width is FITTED, so the 400 bounds what a DERIVED calculation has to play for and is not a measurement of the lensing amplitude – ****and ~ 921 of the 1320 survives it, so lensing is the next build and is NOT the last one.****). *Computes*: the reionisation degeneracy asserted EXACT – if it ever moves χ^2 the amplitude is no longer fitted in closed form and PART 1's algebra no longer describes the pipeline; both smoothing optima pinned; ****the growing width asserted to beat the constant one by at least 1.5x, which is the SHAPE claim and the reason the word is 'lensing' rather than 'some smoothing'****; what survives asserted above 60% of the baseline, so a rewrite is forced rather than a patch if a fitted smoothing ever accounts for most of the residual; and an assert that fires if a smoothing alone would take the control to $\chi^2/\text{dof} \geq 2$, since L-147 would then have to be revisited *Bound*: ****scope: this measures what two candidate builds can be worth on THIS statistic. It is not a result about the sky, not a measurement of the lensing amplitude, and the control does not improve – the smoothing width is fitted. Reionisation is STRUCK from the front's list rather than deferred, and the condition that would undo the strike is written down: a comparison reaching $\text{ell} \geq 30$, a second spectrum, or a fixed amplitude.**** *Run*: python3

receipts/P15_CR_cosmology/P15_what_is_left_is_lensing_and_not_reionisation.py

P15_derived_lensing_on_the_lcdm_arm [OK]

sec:refit-bound (**THE DERIVED LENSING c54.181 TEED UP, RUN WITH NO FREE WIDTH – AND IT COMES IN UNDER THE FITTED BOUND, WHICH IS WHAT A DERIVATION MUST DO.** ACDM's own non-perturbative lensed/unlensed ratio applied to the c54.178 LCDM arm takes the control 1320 -; 989, $d(\chi^2) = 331$ – UNDER the 400 the c54.181 fitted smoothing bounded, and concentrated in the damping tail (ell 1100-1500) where lensing fills the troughs. The acoustic peaks move 0.0%. The corpus's own first-order Hu (2000) kernel, run on the SAME C^{phiphi} so only the ORDER differs, OVERSHOOTS to 508 because it BREAKS DOWN in the tail – a spurious +13% at ell=1900 against the full operator's +6.5% – so the derived number is the full operator's 331, not the first order's 508. Validated on CAMB's own LCDM: 615 unlensed -; 186 lensed ($\chi^2/\text{dof} \sim 1$). ~989 of 1320 survives, so lensing is real and NOT the last build.). *Computes:* lensed CAMB LCDM asserted at χ^2/dof in (0.75,1.25) so the operator and pipeline are validated before anything is read off the arm; the derived lensing on the arm asserted in (280,385) AND strictly under the c54.181 fitted bound of 400, which is the defining property of a zero-width derivation against a fitted upper bound; the first-order kernel asserted to OVERSHOOT the full operator by more than 80 in $d(\chi^2)$ via the tail artefact, a live check that LENS_correction is first order and that the tail is where it fails; and the peak shift asserted below 0.3%. *Bound:* **scope: the lensing operator is ACDM's, imposed not fitted, so this is not a measurement of the lensing amplitude and the CR arm does not fit – it measures how much of c54.181's fitted-smoothing headroom a genuine zero-parameter lensing recovers (most of it), with the shortfall against the fit being the fit's free width. The residual ~989 is the instrument's transfer inaccuracy, not lensing's: CAMB's UNLENSED LCDM already sits at 615 on these same bins. 'PO-7' stays protected.** *Run:* python3 receipts/P15_CR_cosmology/P15_derived_lensing_on_the_lcdm_arm.py

UNC_error_budget [OK]

sec:refit-bound (UNC — QUANTIFYING WHAT WE DO NOT KNOW. The prediction with an error budget. Corrections folded in first: * the recombination WIDTH scales with the rate (A.22) – so the thickness contributes ~nothing to the RATIO and the signature is carried by r_D essentially alone; * the LD anchor affects only the ABSOLUTE r_D and cancels in the ratio – it carries NO uncertainty into the prediction;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/UNC_error_budget.py

P15_the_residual_is_contrast_and_the_lensing_potential_is_derived [OK]

sec:refit-bound (**WHAT THE CONTROL'S REMAINING χ^2 IS MADE OF, AND THE LENSING POTENTIAL BUILT TO ATTACK IT.** **PART A** templates projected through the same inverse covariance the χ^2 uses: **positions remove 1.4 of 1320 – 0.1%, so the peaks are where the sky puts them and that is not where the residual lives**; a smooth envelope 15%; **the peak-trough oscillation at FIXED position 38%, with the model's contrast 13.1% TOO HIGH** – which is what lensing reduces; and **53% in NEITHER, so a perfect lensing calculation leaves about half**. *Fitted templates, so each share is an upper bound on what the effect can claim.* **PART B** the line-of-sight solve stops at $z = 12.8$ but the lensing kernel peaks at $z \sim 3.3$, so Phi is carried on by the background's own growth factor D prop. $H \int da/(aH)^3$ – exact in linear theory, no transfer function imported, k-dependence from this instrument's own Phi. **And the normalisation is the amplitude the TEMPERATURE comparison already fits, so lensing enters with NO new parameter.** Limber checked against an EXACT projection (5% over $l = 5-200$); the deflection maximum is BROAD, within 10% over $l = 12-42$; **rms deflection 2.69 arcmin with nothing tuned to it.**). *Computes:* positions asserted below 2% of the residual, so the diagnosis reorders loudly if they ever rise; the contrast excess pinned to 13.1% +- 2 points, since that is what the convolution must remove; what survives BOTH template sets asserted at 53% +- 5, **so a build that forgets how much remains fails here**; Limber's departure asserted below 10%, since the convolution will use it; the deflection maximum asserted BROAD; and the absolute rms deflection pinned at 2.692 arcmin, which is the zero-parameter check on the normalisation *Bound:* **scope: PART A's shares are FITTED-template upper bounds, not measurements of physical effects, and the two sets overlap by ~7 points. And PART B's external agreement is weaker than it looks: the ~2.7 arcmin and the ~1e-7 deflection power it matches are RECALLED, not derived here and not checked against a source this corpus holds – this programme was bitten twice in three revisions by exactly that (the 16/15 coefficient at c54.176, and a Limber suspicion in this

very file which was also recollection and also wrong). What PART B establishes is INTERNAL consistency; the external agreement is corroboration at the strength of a memory.** *Run: python3 receipts/P15_CR.cosmology/P15_the_residual_is_contrast_and_the_lensing_potential_is_derived.py*

P15.the.phase.is.the.driving.and.the.undriven.arms.agree [OK]
sec:refit-bound (**WHY 0.62 pi: THE DRIVING — AND THE ATTRIBUTION IS COMPLETE. WITH EVERY COUPLING TO THE POTENTIAL REMOVED THE TWO ARMS' ACOUSTIC PHASES AGREE TO 0.013 OF l_A .** Front #2, lead 'L-507'. c54.191 closed with **an acoustic phase shift is a computable consequence of the driving, so 'why 0.62 pi' has an address. This file does not have it.** **The instrument has carried the switch since c54.170** — 'NODRIVE=1' removes every coupling to the potential at every site (the $4\cdot\Phi$ in the photon, neutrino and CDM continuity equations and the $k^2\cdot\Psi$ in the Euler equations) while leaving Φ 's own evolution alone, that being the background and not a coupling. *Both arms run through it at production depth here, which no revision had done.* **PART 1 UNDRIVEN THE TWO ARMS ARE THE SAME OSCILLATOR:** intercepts ** -0.135 and -0.149 of l_A — agreeing to 0.013 ** — and both slopes l_A to a part in a thousand. *Two constructions with different rates, sound horizons, starting redshifts and initial data give the SAME acoustic series once nothing drives it.* **PART 2 AND THE DRIVING IS WHAT SEPARATES THEM:** switched on it supplies ** -0.127 to the control and -0.729 here, a factor of 5.7 ** the driven discrepancy is 0.615 and the undriven 0.013 , so **the driving accounts for 98% of it**. **PART 3 THE SAME SWITCH ACCOUNTS FOR THE SPACING:** undriven both slopes are 1.000 and 1.001 of l_A ; driven, the control's is 1.003 and this construction's 0.976 — **so the 2.4% the spacing was short is the driving as well, and not a second effect**. $\Rightarrow$ ***THE WHOLE ACOUSTIC DISAGREEMENT IS ONE COUPLING BEHAVING DIFFERENTLY — and a coupling is a thing a theory says something about, where a ratio is not.*** **PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at $\phi = 0$. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the UNDRIVEN agreement asserted under 0.10 of l_A — if the arms still differed with nothing driving them the driving would not be the whole story and PART 2's headline would be wrong** the ratio of what the driving supplies asserted above $3x$ and pinned at 5.7 ± 1.5 , without which PART 2 has no content; **the undriven SLOPES asserted at l_A to 1%, which is what makes PART 3's "not a second effect" true** the driven discrepancy re-derived at 0.615 ± 0.05 , tying this file to 'L-505' and 'L-506'; the driving's share asserted above 85%; and at least seven peaks per arm *Bound:* **scope: an ATTRIBUTION is not an explanation and it is not a verdict. What this establishes is that the acoustic disagreement lives in ONE COUPLING rather than in four items, a ratio, or an initial condition. IT DOES NOT SAY WHY the driving is $5.7x$ stronger here — the obvious candidate is that this construction's modes begin already sub-horizon at the seam carrying an assigned amplitude and ZERO velocity, in a potential that is not the equilibrium configuration for that state, where the control's begin super-horizon in adiabatic equilibrium; **that is a HYPOTHESIS and this file does not test it**. The instrument carries the split that would: 'DRC' is the potential's rate of change feeding the density and 'DRE' its gradient driving the velocity, switchable separately — the next thing this front owes. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure.** *Run: python3 receipts/P15_CR.cosmology/P15_the_phase_is_the_driving_and_the_undriven_arms_agree.py*

P15.the.phase.moves.with.the.datum.and.only.the.spacing.does.not [OK]
sec:refit-bound (**THE OBSERVER LINE'S ITEM 38 ANSWERED, AND THE ANSWER WITHDRAWS c54.190–191's HEADLINE. ACROSS FOUR SEAM PHASES AT PRODUCTION DEPTH THE ASYMPTOTIC ACOUSTIC PHASE SPANS 0.891 IN ϕ/π — AND THE CONTROL'S 0.263 LIES INSIDE THAT SPAN.** Front #2, lead 'L-508'. *Item 38 (r2485): of the 23 datum-scan spectra ZERO reach peak 8, so the quantity carrying the disagreement could not be tested against the one freedom known to move things. It named the experiment — two or three seam phases at production depth. c54.191 supplied one; this supplies two more.* **PART 1** at $\phi = 0$, $\pi/4$, $\pi/2$ and π the fitted ϕ/π is ** 0.878 , 0.958 , 0.066 , 0.671 ** — *c54.191 read 34% closure from the two endpoints; **the interior is where the motion is***. **PART 2 SO 0.62 pi IS THE VALUE AT ONE READING AND NOT A PREDICTION.** *The attribution to the driving (c54.193) stands — the driving is what separates the arms AT A GIVEN READING — but a quantity the datum can move across the control's value is not a prediction

of the construction.* **PART 3 THE HEIGHTS MOVE TOO:** P1/P2 runs $0.483 \rightarrow 1.618$ against the sky's 2.217 and the control's 2.197, and **never reaches it**. **PART 4 WHAT IS ROBUST IS THE SPACING:** 0.9629–0.9814 of L_A over the same four — **1.9%** — and tracking r_s at 98% of the acoustic rate under the fitted parameter. **PART 5 AND MATCHING THE PHASE DOES NOT FIX THE SPECTRUM**, which is what settles PART 2: the reading closest in phase ($\phi = \pi/2$, a phase gap of 0.197) has $\chi^2/\text{dof} = 224$ against the control's 3.71 on the same 185 bins — sixty times worse while agreeing on the phase. $\Rightarrow$ **So the phase was never carrying the disagreement.** *Computes:* the control's phase asserted INSIDE the span the seam phase reaches — without which there is no withdrawal and c54.190–191 stand as written; the span asserted above 0.3 and pinned at 0.891, which IS item 38's question; the low-ell transient asserted present in ALL FOUR at more than 3x the control's — this is what establishes the peak INDEXING, and if a reading showed none the span could be an artefact of labelling; the spacing's stability asserted under 6%, the one thing this file does NOT withdraw; the heights asserted never to reach the sky's; and the phase-matched reading asserted above 10x the control — PART 5 is what makes the withdrawal a finding rather than a shrug. *Bound:* scope: this is the THIRD withdrawal of this line's own headline in the span, and it is forced by the data rather than chosen. What it leaves is worse than c54.191 reported and truer: the acoustic SPACING is right and robust; the acoustic PHASE and the peak HEIGHTS both move substantially with a datum the corpus does not fix and neither reaches the sky at any reading tried; and no reading brings the spectrum within sixty times the control. **Where the disagreement lives is therefore not a single named quantity and this file does not supply one.** 'F5' unsoftened and it cuts both ways — a discrepancy resting on a free choice is not a framework verdict either. 'PO-7' protected; the conversion runs by 'F5's stated procedure. *Run:* python3 receipts/P15_CR_cosmology/P15_the_phase_moves_with_the_datum_and_only_the_spacing_does_not.py

P15.the_locus_was_wrong_in_six_places_and_the_lint_could_not_see_the_worst [OK]
sec:perturbations, sec:coherence, sec:envelope (**ROUTED ITEM 21 WORKED: SIX SITES SAID "BRANCH POINT" WHERE THEIR OWN RECEIPTS SAY "SEAM"/"ONSET", AND THE PHYSICS INVERTS BETWEEN THE TWO — SO A PROPOSITION TITLED "SUB-HORIZON" WAS TRUE AT ONE AND FALSE AT THE OTHER. AND THE LINT BUILT FOR THIS DEFECT COULD NOT SEE THE WORST OF THEM.** Front #2, lead 'L-509'; VEIN 'L-202'. **PART 1 THE INVERSION, RE-DERIVED FROM THE METRIC FUNCTION** rather than taken from the routing slip: normalised $1/aH$ is 1.000 at the seam and 0.051 toward $r=0$, and with the inherited datum the horizon maximum sits at $r^* = 1.5338 r_{\text{seam}}$, just OUTSIDE the seam — so the seam is on the rising branch and the acoustic modes are already inside there, which is a *derivation* of prop:subhorizon and not a numerical coincidence of z_{onset}. **PART 2 SIX SITES REPAIRED, PINNED BY TEXT AND NOT BY LINE NUMBER** — including the subsection heading, *which contradicted its own proposition three lines below*, and the sec:envelope sentence *that claims to BE the derivation, with the locus backwards*. **One of the six was not on the routed item's list**; the lint found it. **The one true-as-written branch-point sentence — the neutrinos' decoupling, a TEMPERATURE ordering, flagged as a false positive by the finder in advance — survived the sweep.** **PART 4 AND THE GATE WAS VERIFIED AGAINST A CLEAN TREE, WHICH MEASURES NOTHING:** seeding prop:subhorizon's body back to "at the branch point" left 'check.loci' printing *clean* and exiting 0. *A proposition states its claim in the body and cites its receipt in the argument paragraph below, so the corpus's most load-bearing claim shape was the one shape a per-sentence scan is blind to — and it was the item's own first site.* $\Rightarrow$ **Theorem-body binding added; 11→12 bindings, +0 false alarms, motivating defect MISSED before and CAUGHT after.** *Computes:* the inversion itself asserted from the metric function — without it the substitution is cosmetic and every edit is unmotivated; r^*/r_{seam} re-derived to 1.5338 from the quartic rather than quoted; the six repaired strings asserted PRESENT and the six defective strings ABSENT, by text so a reintroduction fails this file; **the protected false positive asserted still present — a sweep that also took the one true sentence would be a different error with the same shape**; and **the recall hole asserted in BOTH directions — MISSED before the binder extension, CAUGHT after — so if the pre-extension scan ever starts catching it, this file says the hole was not real**; plus the declared exception asserted to LAPSE when its sentence is rewritten *Bound:* scope: this is exposition and instrument, not physics — no number in the cosmology moves. Its warrant is 'THE.METHOD' §III: a framework a reader cannot follow cannot be excavated, and this one had a heading contradicting its own proposition. The completeness qualification added at sec:coherence is transcribed from 'C2.horizon.limits'

STEP 6, which the paper's sentence contradicted; it is not a new physics claim. 'check.loci' remains a triage lint and not a CI gate — the contributor's call, not reversed here.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_locus_was_wrong_in_six_places_and_the_lint_could_not_see_the_worst

P15.the.filter.is.a.sound.speed.filter [OK]
sec:what-crosses (**THE BRANCH-POINT SELECTION RULE'S EXPONENT HAS TWO FACTORS AND THE CORPUS VARIES ONLY ONE.** The damping is $e^{-k c_s |\Delta \eta|}$: 'sec:what-crosses' holds c_s at the plasma's value and varies k , reading a SCALE criterion (frozen against oscillating). **Hold k and vary c_s and the same inequality is a SPECIES criterion – and a pressureless component has $c_s = 0$ IDENTICALLY, so its exponent vanishes at EVERY wavenumber.** ** THAT MAKES THE MATTER SECTOR'S INHERITANCE EXACT RATHER THAN ADIABATIC: ** at $c_s = 0$ the segment's mode equation is $\psi'' = 0$, whose constant solution the kernel leaves untouched with no approximation, whereas the frozen-mode argument rests on horizon exit and degrades at small scales. The two agree wherever both apply; only the pressureless one is available everywhere. COMPUTED on the corpus's own segment: (1) $|\Delta \eta| = 3.3197$ (the recorded 3.32); (2) exact transfer-matrix transmission in BOTH k and c_s , reproducing $-k c_s |\Delta \eta|$ to the digit for every constant c_s ; (3) $\ln T = 0$ EXACTLY at $c_s = 0$ for $k = 1, 10, 85.3, 1e3, 1e5$ – uniform in k **); (4) at the first peak the exponent runs $\sim 1.6e2 c_s/c$, so the filter is opaque above $c_s \sim 3.5e-3$. ** CONSEQUENCE: a per-cent sound speed already gives $e^{-1.6}$ and ten per cent e^{-16} , so THE CROSSING CAN INHERIT PERTURBATIONS FROM A COLD SPECIES AND FROM NO OTHER – a sharp condition on what the progenitor may hand over. ** ** (SCOPE: the arithmetic is what this receipt establishes; no species selection rule follows from it. The kernel acts on OSCILLATORY content and nothing arrives oscillating ('P16_every_mode_is_frozen_at_the_crossing'), so the exponent computed here has nothing to act on and the crossing is lossless for EVERY species.)**). *Computes:* the segment's conformal length asserted at 3.32; the plasma exponent asserted in (-160,-140); $\ln T$ asserted zero to $1e-9$ at $c_s = 0$ across five decades in k *Bound:* ** THIS IS THE FIELD HALF ONLY. ** What happens to perturbations of a given species; the detailed WORLDLINE dynamics of a concrete model at the branch point is a separate computation and is not done. **And an honest bound**: on this segment the plasma exponent comes out -147 against the -152 the corpus records, a 3% difference from the sound-speed profile's discretisation – nothing here turns on it, the load-bearing result being the exact vanishing at $c_s = 0$, which is profile-independent *Run:* python3 receipts/P15_CR_cosmology/P15.the.filter.is.a.sound.speed.filter.py

P15.the.collapse_leg_is_scale_invariant [OK]
sec:transmission (**THE COLLAPSE LEG IS A SCALE-INVARIANT BACKGROUND, AND P15 LISTS 'CR HAS NO INFLATIONARY SCALE-INVARIANT ATTRACTOR' AS AN ESTABLISHED SIGNATURE. THE ADJECTIVE WAS CARRYING THE SENTENCE.** For a background $a \sim (-\eta)^\beta$ with $z \sim a$, the Mukhanov-Sasaki potential is $\beta(\beta-1)/\eta^2$, so scale invariance requires $\beta(\beta-1) = 2$ – a quadratic with TWO roots: $\beta = -1$, de Sitter inflation (expanding), and $\beta = +2$, a MATTER-DOMINATED CONTRACTION. ** **And the corpus's own collapse leg is $\beta = 2$ exactly.** its banked near-branch-point law $r^3 = (9/2) M \tau^2$ with $a = |r|$ gives $d \ln a / d \ln \tau = 2/3$ and $d \ln \eta / d \ln \tau = 1/3$, hence $a \sim \eta^2$ – forced by $f_{-j} - 2M/r$, the mass term dominating and the Lambda term vanishing, **which is the same pressurelessness 'A.4' used at c54.112 to show such a component crosses exactly.** The mode equation integrated from vacuum data on both backgrounds returns $d \ln P / d \ln k = 0$ to a part in 10^4 . **So $n_s = 1$ is produced by the construction rather than inherited from the progenitor, and only the DEPARTURE from unity is owed.**). *Computes:* the two roots asserted [-1, 2]; both logarithmic slopes of the leg asserted exactly 2/3 and 1/3; both integrated spectra asserted flat to $1e-3$ *Bound:* **bound, and it is the whole caution: this is the STANDARD contraction result (Wands 1999; Finelli-Brandenberger 2002) applied to a background the corpus DERIVES. It inherits that literature's known difficulty – whether the matching across the bounce selects the scale-invariant mode – and here the bounce is a branch point with an analytic continuation rather than a smooth bounce. THAT QUESTION IS NOT SETTLED HERE and is registered as 'L-169'**. *Run:* python3 receipts/P15_CR_cosmology/P15.the.collapse_leg_is_scale_invariant.py

P15.the.continuation_is_diagonal [OK]
sec:predictions (**THE CONTINUATION THROUGH THE BRANCH POINT ACTS DIAGONALLY ON THE TWO PERTURBATION MODES, SO THE SCALE-INVARIANT SPECTRUM DOES NOT REACH THE OBSERVER.** The falsifier on the companion result, named at c54.122

and run at c54.123. ****The scale-invariant spectrum sits in the mode that DOMINATES the contraction ($\zeta \sim \eta^{-3}$); the mode that SURVIVES on the expanding side is the constant one, whose spectrum from vacuum data is $n_s - 1 = 6$.**** Whether the first reaches the second is a connection problem, settled here two independent ways. **** (i) PARITY: the exponents 2 and -1 differ by an integer so a logarithm could mix them – but $\sinh^{(2/3)}$ is EVEN in τ , which places its correction at η^6 and steps the Frobenius recursion in SIXES past a resonance sitting at THREE. No logarithm.** (ii) NUMERICS: continuing both solutions around the branch point, the diagonal entry goes to 1.000131 and the leakage falls as a clean $\delta^{(2/3)}$ power law – and the power law is itself evidence against a log, which would leave a residual no power of δ removes.**** So c54.122's scale invariance is confined to the contracting side. ****What survives is narrower and sharper than what it replaced: the construction HAS a scale-invariant background, and the obstruction is a PARITY PROPERTY OF ITS OWN SOLUTION rather than a missing mechanism.**** *Computes:* indicial exponents asserted; both vacuum coefficients' k-scalings asserted exactly $-3/2$ and $+3/2$; the absence of odd powers in the correction asserted; the resonance asserted unreachable; the diagonal entry asserted to $1e-3$ and the leakage ratio asserted equal to $10^{(-2/3)}$ at every decade *Bound:* ****bound: this settles the connection for the SCALAR mode equation on this background. It does NOT settle what a background correction ODD in τ would do – which is what mixing would require and is the named open route. Three defects were caught in the making: a check that only proved invertibility, a branch error from squaring before taking the cube root, and a test run at δ where $k \eta \sim 1.4$, outside its own frozen-mode hypothesis****
Run: python3 receipts/P15_CR_cosmology/P15_the_continuation_is_diagonal.py

P15.the_evenness_is_the_time_symmetry

[OK]

sec:predictions ****THE DIAGONALITY IS THE CONSTRUCTION'S TIME-REVERSAL SYMMETRY, AND THE SAME INVARIANCE THAT FIXES THE PEAK HEIGHTS IS WHAT STOPS THE SCALE-INVARIANT SPECTRUM CROSSING.**** The falsifier left by the previous receipt – mixing needs a background correction ODD in τ , with the Z.3 monodromy the named candidate – fires NEGATIVE. ****Write the law as $r = A u^{(2/3)} g(u)$ with $g = [\sinh(u)/u]^{(2/3)}$: ALL the branch structure sits in $u^{(2/3)}$, and g is EVEN, analytic and non-vanishing at $u=0$ **** – exactly, not to a truncation, since $\log(\sinh u/u)$ is analytic in u^2 near 0 and the exponential of a series in u^2 is a series in u^2 . ****The three sheets differ by a CONSTANT cube root of unity, and a constant multiplier cannot change a parity, so no sheet has an odd part; the conjugate branch is even too and has no branch point at all.** AND THE EVENNESS IS $r(\tau) = r(\tau)$: the cosmogenesis is TIME-SYMMETRIC about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one because mixing would distinguish a direction of time the background does not.**** P15 sec:amplitude already leans on this same invariance one observable over – the peak HEIGHTS follow from a driving amplitude exactly invariant under time reversal. ****One symmetry, two consequences, and the corpus had banked only the welcome one.**** *Computes:* $\sinh(u)/u$ asserted even with unit value at 0; g asserted free of odd powers through u^{12} ; all three sheets asserted free of odd powers; $\cosh^{(2/3)}$ asserted even; $\sinh^2$ asserted invariant under $u \rightarrow -u$ *Bound:* ****bound: the leg is exactly even because it is the marginally-bound $E=1$ geodesic of an exactly VACUUM SdS exterior. A progenitor with real matter content – pressure, dissipation, anything carrying an arrow – would not be. This receipt does not exclude that; it LOCATES it, and puts the question back where 'L-150' already sits****
Run: python3 receipts/P15_CR_cosmology/P15_the_evenness_is_the_time_symmetry.py

P15.the_geometry_transmits_no_parameters

[OK]

sec:transmission ****THE INHERITED DATA ARE NOT INHERITED FROM THE COLLAPSE GEOMETRY, BECAUSE THAT GEOMETRY TRANSMITS ZERO PARAMETERS. A COUNT, AND THE CORPUS PROVED THE KEY STEP ITSELF WITHOUT CASHING IT.**** The SdS progenitor family is parametrised by M alone – α is the one invariant and $E=1$ is a choice of geodesic – so the geometry is ONE-dimensional. ****And c54.113 showed the Kretschmann along the bead is MASS-FREE in the faller's own proper time ($dK/dM = 0$, re-asserted here; $K \sim (64/27)/\tau^4$). A quantity identical for every member carries no information about which member it was, so the approach transmits ZERO parameters, not one.**** Against that the corpus draws at least FOUR independent numbers: A_s , n_s , η , z_{onset} ($= \rho_r/\rho_m$, one datum by c54.105), and the light-element composition. ****Four numbers cannot come from zero parameters, so 'inherited from the progenitor collapse' cannot mean inherited from the collapse GEOMETRY – it means inherited from the progenitor's MATTER CONTENT, which M does**

not carry, the exterior being exactly vacuum and the interior dust without composition.** This RECLASSIFIES the row: not a derivation problem but a modelling one. **What the geometry does fix it fixes universally: $|\Delta \eta| = 3.32 \alpha = 1.8e4$ Mpc, from Λ alone – a CUT-OFF, not a CONTENT.**). *Computes:* dK/dM asserted zero and the 64/27 limit asserted; the count of drawn data asserted at least four; the conformal length asserted 3.32α and asserted to land between $1.5e4$ and $2.0e4$ Mpc *Bound:* **bound: a counting argument about what the construction’s own description of the progenitor can carry. It does NOT show that no model supplies η – it shows no model of the VACUUM EXTERIOR can, and that the row needs an interior**
Run: python3 receipts/P15_CR_cosmology/P15_the_geometry_transmits_no_parameters.py

P15_hbar_survives_in_A_s_and_cancels_in_n_s

[OK]

sec:transmission (**hbar IS A MULTIPLICATIVE CONSTANT IN THE POWER SPECTRUM, SO IT SURVIVES IN THE AMPLITUDE AND CANCELS IN THE TILT — WHICH IS WHY n_s IS DERIVABLE HERE AND A_s IS NOT.** The vacuum normalisation $v = \sqrt{hbar/2 c_s k} e^{-i c_s k \eta}$ is the only place a quantum constant enters, and $\zeta = v/z$ with z classical, so $P(k) = hbar A_0 k^{(n_s - 1)}$: ** $A_s = P(k_0)$ carries hbar, and $d \ln P/d \ln k$ does not — asserted both ways.** An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. **So the no-magnitude result (c54.128: neither real form supplies a second scale, and hbar, c , G are unit gauges over the single one) forbids deriving A_s and says nothing about n_s .** A_s is therefore permanently inherited while n_s is not, and the two were never the same kind of quantity — though the corpus has named them together in every statement of the frontier, which is how one got worked for thirteen revisions while the other went unexamined. **AND THE INVERSION LOOKED LIKE THE PAYOFF: a quantity not derivable from the invariants is a function of the moduli, so $A_s = F(2M/\alpha)$ and the observed A_s would MEASURE the parent’s mass. SUPERSEDED at c54.137 (‘P15_the_progenitor_vacuum_is_negligible.too’): F is computed, it is a function of TWO moduli rather than one, and it carries no fixed normalisation — the incoming amplitude is a classical datum, so the relation cannot be inverted. The hbar asymmetry itself stands unchanged.** Second handle on the same object: the spectral bound constrains total expansion past equality, this constrains the mass, and the recollapse condition caps that mass below Nariai.). *Computes:* the hbar-dependence of A_s asserted non-zero and of the tilt asserted zero *Bound:* **bound: F itself is NOT computed. The pieces exist — the mode functions, the monodromy, the mode map — and what is missing is the normalisation carried through all three rather than tracked as a scaling. The settled part stands alone: A_s is not derivable, for a reason, and the reason names which other quantities share its fate — every magnitude, and no ratio**
Run: python3 receipts/P15_CR_cosmology/P15_hbar_survives_in_A_s_and_cancels_in_n_s.py

Q1_a_stated_tolerance_is_a_request_and_the_corpus_answers_it

[OK]

sec:diffusion-scale (Q1 — A STATED TOLERANCE IS A REQUEST, NOT A RESULT: RUN THE CORPUS’S ODEs AGAIN AND SEE. Numerical-analysis field bake, probe Q1, and this field’s real target is the receipts rather than the prose. Census: 24 receipt files integrate an ODE and ALL 24 set an explicit rtol and atol, not one relying on a solver default; 4 contain any convergence or refinement language. So the corpus states its tolerances and, in twenty of twenty-four files, does not verify them. This receipt does the verification: scipy’s solve_ivp is patched to run every call at 100x tighter tolerance, four receipts are re-run unmodified in subprocesses, and EVERY ONE still passes its own assertions. Control: a deliberately under-resolved stiff integration through the same harness DOES move, so the patch applies and the comparator sees. Verdict 5 names the margin on the corpus’s one prose-side validation: $r_s = 144.0$ vs CAMB 144.4 is a 0.28% disagreement against the 8.2% signature it feeds, adequate by a factor of thirty.). *Computes:* runs clean; five verdicts and a control that detects a moved number *Bound:* derived here; a first comparator reported a 92% change that turned out to be a RESIDUAL column, which is supposed to move — the naive test was measuring its own patch, and the honest test is whether the assertions survive *Run:* python3 receipts/L_numerics/Q1_a_stated_tolerance_is_a_request_and_the_corpus_answers_it.py

P15_the_progenitor_vacuum_is_negligible_too

[OK]

sec:amplitude (**THE PROGENITOR’S OWN VACUUM, CARRIED THROUGH THE FULL PASSAGE AND AMPLIFIED BY THE BRANCH-POINT MIXING, IS ALSO NEGLIGIBLE — SHORT BY $1e104$. So ‘prop:amplitude’ had ruled out the one vacuum the inheritance story was NOT claiming.** The collapse-side background is a single function, $a = M(\rho x + x^2/2)$ with $x = |\eta - \eta_c|$, whose Mukhanov-Sasaki potential $2/(x(x+2\rho))$ is **M-FREE**, so the mass enters only through the final division $\zeta = v/a$. The passage has three stages and each contributes

one factor of $1/\rho$: (i) the mode freezes on the $\beta = 2$ leg as $v \sim D_k/x$ with $D_k = \sqrt{\hbar/2} k^{-(3/2)}$; (ii) **the connection through equality is elementary** — $v = a \int dx/a^2$ solves the $k = 0$ problem exactly and carries D_k/x to the constant $(3/2 \rho) D_k$; (iii) the crunch monodromy leaks it into the survivor with coefficient $2\pi/\rho$; and $a'(\eta_c) = -M\rho$ supplies the third. **THE TRANSFER LAW IS THEREFORE** $P = 18 k^3 D_k^2 / (M^2 \rho^6)$, with the k^3 cancelling identically on vacuum data — exactly scale-invariant. On the vacuum that is $A_s = (9/4)(1P/M)^2 \rho^{-6} = 2e-113$ at the recollapse cap and the determined ρ , against an observed $2.1e-9$. **The mixing buys eight orders over the bare vacuum and is still short by a hundred and four.** Two corollaries: A_s does NOT measure the progenitor's mass, D_k being free; and the classical datum has a size — the parent's growing-mode amplitude exceeded its own vacuum's by $\sim 1e47$. **The same closed form also returns c54.131's monodromy analytically, the $(x/\rho)\ln x$ term continuing to $-2\pi i/\rho$, where it had been obtained by numerical continuation.** *Computes:* the M-free potential asserted symbolically; the connection coefficient asserted in closed form and against integration from vacuum data to 0.10% over a decade in ρ and a factor 15 in k ; the transfer law asserted k-independent; the shortfall asserted between $1e101$ and $1e106$ *Bound:* **bound:** $z = a$ with $c_s = 1$ is assumed on the $\beta = 2$ leg — c54.122's own hypothesis and the standard matter-contraction idealisation, genuine dust having $c_s = 0$ and no vacuum oscillation to normalise. The $1e104$ is far too large for any $O(1)$ convention or plausible c_s treatment to touch **Run:** `python3 receipts/P15_CR_cosmology/P15_the_progenitor_vacuum_is_negligible_too.py`

P15_no_primordial_B_modes_unconditionally

[OK]

sec:amplitude **(THE TENSOR SECTOR IS WHERE THE COLLAPSE-TRANSFER MACHINE RUNS WITH NO IDEALISATION AT ALL, AND IT RETURNS $P_T \sim 5e-111$ — ONE HUNDRED ORDERS BELOW THE OBSERVATIONAL CEILING.** **The scalar transfer needs $z \sim a$, which for a fluid holds only where w and c_s are constant — and across the progenitor's own equality $(1+w)$ runs $1 \sim 4/3$ while c_s runs $0 \sim 1/\sqrt{3}$, so neither is.** **For tensors that is not a restriction: $z_T = a M_P/2$ EXACTLY, for any content, any equation of state, any sound speed, so $z_T/z = a'/a$ identically and the entire passage — background, connection $(3/2 \rho)$, monodromy $(2\pi/\rho)$, division by $a'(\eta_c) = -M\rho$ — applies verbatim with nothing assumed.** **It returns $P_T = 144 \pi (1P/M)^2 \rho^{-6}$, scale-invariant (k cancels identically), i.e. $4.8e-111$ at the recollapse cap and the determined composition, against a ceiling of $6.7e-11$ from $r \leq 0.032$: below it by $1.4e100$, so $r = 2e-102$.** **So the absence of primordial B-modes is a prediction at any conceivable sensitivity, and it no longer rests on the substrate's own tensor floor — that argument had the same scope defect its scalar twin did, ruling out the substrate's vacuum and saying nothing about the progenitor's.** **And the ratio is fixed too: every k-dependent factor is shared by the two sectors, so $r_{out} = 6 r_{in}$ (c54.146: scalars mix twice as strongly as tensors), k-independently — the crossing can neither manufacture nor destroy a tensor signal, only rescale the ratio.** **The observed bound therefore measures the PARENT: its own tensor-to-scalar ratio was below $1.3e-3$.** *Computes:* P_T asserted k-independent and asserted equal to $144 \pi (1P/M)^2 \rho^{-6}$ symbolically; the shortfall against the ceiling asserted above $1e95$; the crossing ratio asserted equal to 24 *Bound:* **bound:** the factor 24 carries the scalar side's idealisation (it is $24(1+w)c_s$ on the contracting leg, the standard matter-bounce value) and no more. The absolute P_T carries nothing **Run:** `python3 receipts/P15_CR_cosmology/P15_no_primordial_B_modes_unconditionally.py`

A2_the_three_owed_derivations_are_one_boundary

[OK]

sec:scope (the three owed derivations are ONE closed boundary: A_s , n_s and the inherited datum are all dimensionless magnitudes). *Computes:* asserts P15s own list, prop:amplitudes 10^{113} gap, the progenitor tilt, and the capstones derived progenitor composition (13 checks) *Bound:* NOT-A-PAPER-CLAIM — L-211 run on L-150 *Run:* `python3 receipts/L211_closure_adjacency/A2_the_three_owed_d`

P15_the_control_entered_the_regime_and_the_arm_did_not_move

[OK]

sec:refit-bound **(THE WAVENUMBER TRUNCATION WAS 78% OF THE CONTROL'S REMAINING χ^2 , AND REMOVING IT TAKES THE CONTROL TO $\chi^2/\text{dof} = 1.18$ OVER THE FULL RANGE AGAINST A TRUE ΛCDM FIT'S 1.01 — SO FRONT #2'S TARGET IS MET. ACROSS FOUR INSTRUMENT STATES THE CR ARM DID NOT MOVE, AND 'L-147's PRE-REGISTERED F4 FIRES.** **Front #2, lead 'L-500'. PART 1 WHERE THE RESIDUAL LIVES.** **Of the 989 surviving c54.183's derived lensing, 725 — 73% — is in the 30 bins above $\ell = 1500$, a sixth of the bins; below $\ell = 700$ the control is at $\chi^2/\text{dof} = 0.85$ against CAMB's lensed ΛCDM at 0.82 on the same bins. PART 2 AND IT IS THE TRUNCATION, SHOWN BY REMOVING IT RATHER THAN BY ARGUING FOR IT.** **The**

grid is $k = \text{ell}/D_M$ with $\text{ell} = L_{\text{MAXL}}$, so the top output multipole has NO k -headroom while C_{ell} draws on every k with $k \cdot D_M = \text{ell}$. ****Re-run at $L_{\text{MAXL}} = 3000$ with the k -spacing held fixed and scored on the SAME 185 bins: 989 - χ^2/dof 5.34 - 1.18, and the last band's ratio to the reference 0.42 - 0.99.**** So c54.184's '53% in neither template set' was the truncation; the front's last named item is struck as NUMERICAL and no sector remains. ****PART 3 THE CR LADDER'S ALIAS-GATE WAIVER, CHECKED IN χ^2 FOR THE FIRST TIME.**** The gate demands 4 samples per Bessel period, the CR ladder has 2.3, and it waives itself with 'this is not aliasing - but it is only not aliasing if the answer does not depend on it. Run $K_{\text{CONT}}=1$ to check.' ****Run: agreement better than 0.7% everywhere, χ^2 to 2.1 in 51817.**** That check existed for the PEAK POSITION and had never been run against the statistic the verdict reads. ****PART 4 FOUR INSTRUMENT STATES, full range, both arms on the same lensing operator: the control 21.2 - 27.7 - 5.34 - 1.18 and the CR arm 290.7 - 304.2 - 298.9 - 302.1.**** An error the INSTRUMENT carries moves both arms; this moved one - eighteenfold against 5%. ****PART 5 F2 AND F3 AS 'L-147' DEFINES THEM, WITH BOTH ARMS AT THE SAME WAVENUMBER RANGE**** so the construction is not charged for the instrument's truncation. Full range **** $F_2 = 31$ and $F_3 = 51848$ ****; below $\text{ell} = 700$ **** $F_2 = 3.8$ and $F_3 = 23687$ ****; below $\text{ell} = 500$ $F_2 = -0.2$, the arm scoring no worse than the reference. **** F_4 's second clause fires by three orders of magnitude where it asks for one.****). *Computes:* the k -adequate control at full range asserted at 1.18 ± 0.12 , ****without which there is no discriminating regime and nothing in the file may be read****; the share the truncation carried asserted above 60%, which is PART 2's whole mechanism; F_3/F_2 asserted above 100 at full range AND below $\text{ell} = 700$, which IS the reopening; ****the ladder/continuum agreement asserted under 2%, without which the CR arm's χ^2 is not admissible at all****; the continuum arm asserted at $l_1/l_A = 0.5703$; the four-state contrast asserted as a 5x control improvement against a 15% arm movement, which is what makes this a statement about physics rather than about one build; and the tail share asserted above 60%, since PART 1 is what sent PART 2 looking *Bound:* ****scope: this is a MEASUREMENT DISCREPANCY and NOT a framework verdict - 'L-147's own F_5 says so and 'PO-7' is protected precisely here, so this file reports a number and does not convert it; the conversion runs by 'F5's stated procedure. The deficit itself is not new: the corpus has carried $l_1/l_A = 0.5703$ against the sky's 0.7312 since c54.168 and P15 states it at 21.9%. What is new is a control good enough to read it through a likelihood instead of a ratio. AND WHAT IS NOT RULED OUT: a defect the instrument's states $\text{SHARE} - 0.5703$ has not moved across EIGHT of them, which is evidence either of a robust prediction or of a shared upstream constant, and this file cannot separate those and does not claim to. Nor does $\chi^2/\text{dof} = 1.18$ make the control a FIT: it is 17% above a true ΛCDM 's on the same bins, and what that residual is remains unnamed.**** *Run:* python3 receipts/P15_CR_cosmology/P15_the_control_entered_the_regime_and_the_arm_did_not_move.py

B1.the_control_separates_prediction_from_shared_constant [OK]

sec:refit-bound (the control separates robust prediction from shared upstream constant: same code, different D_M and r_s , same l_A , peaks 21.8% apart, and the control lands on the sky to 0.17%). *Computes:* reads both arms from the c54.186 spectra and asserts the scaling test, the control agreement and the ladder/continuum equality (12 checks) *Bound:* NOT-A-PAPER-CLAIM — the adversarial check c54.186 requested *Run:* python3 receipts/L147_two_arm/B1.the_control_separates_prediction

B2.the_prose_duplicate_did_not_reach_main [OK]

sec:refit-bound (the reported prose duplicate did not reach main: zero at three granularities across every commit, because the two edits were different paraphrases and git asked). *Computes:* scans every commit that touched the file at paragraph, sentence and line-window level and asserts the two diffs differ (14 checks) *Bound:* NOT-A-PAPER-CLAIM — checks a routed report before acting on it *Run:* python3 receipts/L147_two_arm/B2.the_prose_duplicate_did_not_reach_main.py

P15.the_first_peak_is_the_seam_datum_and_the_spacing_is_not [OK]

sec:coherence (**** $\text{ell}_1/\text{ell}_A = 0.5703$ IS A STATEMENT ABOUT THE SEAM DATUM AND NOT ABOUT THE RATE, AND c54.186's OWN CAVEAT IS WHAT SENT THIS LOOKING.**** Front #2, lead 'L-501'. The CR arm begins at the seam ($z = 6761$, $\text{eta}_S = 180.4$ Mpc) and a first-peak mode crossed the horizon at $\text{eta} \sim 75$, so ****the modes that set ell_1 are ALREADY SUB-HORIZON at the start and their phase there is ASSIGNED rather than derived****. The corpus's datum is ****ONE datum per mode and a COMMON phase**** - which fixes that the phase is common and not WHICH phase; the instrument picks a density extremum. ****PART 1 SCANNING THAT ONE FREEDOM MOVES $\text{ell}_1/\text{ell}_A$ FROM 0.5703 TO 1.2599 - A FACTOR OF**

2.21**, and at $\phi = 3\pi/8$ the peak at 172 is not there at all and the series starts at 380. ****PART 2 WHAT IS ROBUST IS THE SPACING, AND IT IS ROBUSTLY WRONG:**** the mean peak spacing holds at 0.734-0.818 of ℓ_A across the whole scan – a spread of 1.11 against the position's 2.21 – and is ****never 1.0****. *So the disagreement that does not depend on a choice is a ~21% SPACING deficit, not a 22% POSITION one.* ****PART 3 AND 'L-147's F4 FIRES UNDER EVERY PHASE, WHICH IS WHY c54.186 SURVIVES THIS:**** on identical bins the CR arm runs χ^2/dof 202-449 against a control at 0.75, so ****even its BEST phase costs 269x the control****. ****PART 4 THIS REPRODUCES c54.164 ON THE CURRENT INSTRUMENT, WHICH HAD NOT BEEN DONE**** – that revision found ℓ_1 in {150, 165, 315} on the OLD 'ROBUST_p1p2_scan' code, and the finding was never carried across to 'ACOUSTIC_two_arm', which everything since is built on.). *Computes:* ****CRPHI = 0** asserted to reproduce the coded 0.5703, without which the knob is not a control on itself and every comparison is against the wrong baseline**; the position's spread asserted above 1.5x and its maximum pinned at 1.2599, which is the withdrawal; the spacing's spread asserted under 1.35x AND its maximum under 0.95 of ℓ_A , ****since if the spacing ever moved this file would have no positive content****; and the CR arm's BEST phase asserted above 50x the control and above χ^2/dof 150 – ****if that ever falls, c54.186 must be RESTATED rather than defended**** *Bound:* ****scope:** this WEAKENS a number the paper carries and STRENGTHENS a verdict the previous revision landed, and neither is a framework judgement. P15's '21.9% deficit, not an artefact of how last scattering was modelled' is TRUE and is not the sentence a reader takes from it: the deficit is not last scattering's, it IS a function of the seam phase, which is upstream of last scattering and is a choice inside the corpus's own stated datum. 'F5' is not softened – removing a confound makes a measurement cleaner, not a conclusion; 'PO-7' protected; the conversion runs by 'F5's stated procedure. AND WHAT IS STILL OPEN, narrower than before: the phase is ONE degree of freedom in the datum; Theta-hat's flatness in k is another and is NOT scanned here. A scan that moved the SPACING would be the one that mattered, and this file does not run it.** *Run:* python3 receipts/P15_CR_cosmology/P15.the.first.peak.is.the.seam.datum.and.the.spacing.is.not.py

B3.a.dichotomy.stated.is.not.a.dichotomy.proved [OK]

sec:refit-bound (B1 stands but L-235s disposition did not follow: ruling out one horn establishes the other only if the dichotomy is exhaustive, and the truth was a third thing). *Computes:* reads the forks nine phase spectra and asserts the 2.21x position spread against the 1.11x spacing spread (12 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects this lines own r2462 disposition *Run:* python3 receipts/L147_two_arm/B3.a.dichotomy.stated.is.not.a.dichotomy.proved.py

P15.both.datum.freedom.opened.and.the.spacing.still.holds [OK]

sec:coherence (****THE SECOND AND LAST FREEDOM IN THE SEAM DATUM IS OPENED, AND THIS FILE WIDENS ITS OWN PREDECESSOR RATHER THAN CONFIRMING IT.**** Front #2, lead 'L-502'. c54.187 scanned the common PHASE and named the other freedom: what "flat in k" is flat AT. The datum's amplitude is the corpus's C4 shape $T(x) = 3(\sin x - x \cos x)/x^3$; ****the instrument evaluates it at a SINGLE argument $x_e = 1/\sqrt{3}$, which is that shape read at each mode's OWN horizon crossing**** ($x = k \cdot c \cdot s \cdot \eta$ with $k \cdot \eta = 1$ gives exactly $1/\sqrt{3}$) — a self-similar reading. *The alternative evaluates the SAME shape at each mode's own phase AT THE SEAM, $x = k \cdot c \cdot s \cdot \eta_S$. * Both are readings of "one datum per mode"; neither is a new assumption. ****PART 1 THE POSITION IS NO MORE STABLE:**** across eighteen readings ℓ_1/ℓ_A spans ****0.5570–1.2599, 2.26x**** against the phase scan's 2.21x. ****PART 2 THE SPACING IS MEASURED ONLY WHERE THERE IS A SERIES TO MEASURE, WHICH DISQUALIFIES EIGHT OF THE EIGHTEEN:**** those return a spectrum whose higher peaks have COLLAPSED — three features, the last at zero — and *a spacing read off that is not a spacing*. Requirement stated before the numbers: four peaks, the fourth at least 5% of the first. ****PART 3 OVER THE TEN THAT QUALIFY THE SPACING SPANS 0.6764–0.8179 OF ℓ_A — 1.21x, mean 0.772, never above 0.82.**** ***c54.187 reported 1.11x from the phase alone; opening the second freedom widens it to 1.21x, so this file weakens its own predecessor and the ~23% spacing deficit stands at LOWER strength.*** ****PART 4 AND F4 FIRES AT EVERY ONE OF THE EIGHTEEN:**** χ^2/dof runs 157–460 against a control at 0.75 on identical bins, so ****the arm's best reading costs 208x the control****. *And the best-fitting reading is one of the eight DISQUALIFIED ones — it fits better by having less structure to disagree with. Reported because it is the best number in the table and a reader would find it; it is not a direction.*). *Computes:* the position's spread asserted above 1.5x with its maximum pinned at 1.2599, the standing withdrawal from c54.187;

****the spacing's spread asserted under 1.6x AND its maximum under 0.90 of ell_A — the point past which c54.187's conclusion would have to be WITHDRAWN rather than widened****; the usable-series count pinned at 10, since PART 3's range is computed only over those and a silent change in how many qualify would move it without moving the code; and ****the arm's BEST reading asserted above 50x the control and above χ^2/dof 120 — if that falls, c54.186 is restated**** *Bound*: ****scope**: both freedoms the corpus's own phrase leaves open are now scanned and the datum is closed. What has NOT been varied is the background: 'Z.START' is solved so that $\pi \cdot D_M / r_s = 301.6$ exactly, which is a TARGET and not an output — a separate item from the datum, registered rather than resolved. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's. **** Run: python3 receipts/P15_CR.cosmology/P15.both_datum_freedoms_opened_and_the_spacing_still_holds.py**

P15.the.floor.is.a.distance.between.models.not.a.number.from.the.data [OK]
sec:refit-bound (****THE CONTROL'S REMAINING 17% IS NAMED, AND NAMING IT REQUIRED CHANGING THE MEASUREMENT.**** Front #2, lead 'L-503'. c54.186 left it unnamed; ****'F2'** as 'L-147' defines it is a difference of two numbers each computed AGAINST THE SKY, so it mixes how far the two models are apart with where each sits relative to one noise realisation. ****PART 1**** amplitude-matched, the instrument tracks a reference LambdaCDM to ****better than 1.6% everywhere****, the departure running -1.1% at $\text{ell} \sim 200$ through zero near 600 to $+0.7\%$ at 1400 — **a slow undulation, not a mis-phased oscillation**. ****PART 2 F2 IS NOT A CLEAN MEASURE, SHOWN BY MOVING THE REFERENCE:**** three defensible LambdaCDMs give ****+31.2, +21.5, +27.2**** — **a spread comparable to the value**. ****PART 3 SO THE FLOOR IS MEASURED WITHOUT THE DATA**** — $\chi^2_{\text{sep}} = (A_a m_a - A_c m_c)^T F (A_a m_a - A_c m_c)$, a distance between two MODELS: ****21.3 over 185 bins, 0.11 χ^2 per bin, about a third of a standard deviation per bin****. ****PART 4 AND THE ORDERING REVERSES, WHICH IS THE TRAP:**** the massless-neutrino reference gives the ****smallest F2 (21.5) and the largest separation (62.7)**** — **a reference chosen to flatter the instrument through the data is the one it is furthest from as a model**. ****AND c54.178's LIST IS NOW COMPLETE:**** reionisation struck c54.181, lensing built c54.183, the wavenumber range closed c54.186, and ****the neutrino mass — the one item never addressed — is worth about 10 of the 31****, the reference carrying one massive species at 0.06 eV where this instrument carries none.). *Computes*: the instrument-vs-reference departure asserted under 3%, which bounds everything else; the model separation asserted under 0.5 χ^2 per bin, the number that replaces the unnamed 17%; ****THE ORDERING REVERSAL asserted directly — if F2 and χ^2_{sep} ever stop disagreeing, PART 4 is not a trap and the paragraph is unearned****; and ****F4's margin asserted above 100x at the WORST reference, since a verdict that depends on the choice is not one**** *Bound*: ****scope**: 0.11 χ^2 per bin is NOT zero and this file does not say the instrument is exact. It says the residual is at the level where the REFERENCE's own parameter choices matter as much as the instrument's error, and that below that level this statistic cannot arbitrate between two LambdaCDM implementations at all. F3 = 51848 against a floor of 21–63 across every defensible reference, so 'F4's margin is three orders of magnitude whichever is taken — which makes 'L-147's reading SAFER and converts nothing. 'F5' unsoftened, 'PO-7' protected. **** Run: python3 receipts/P15_CR.cosmology/P15.the.floor.is.a.distance.between.models.not.a.number.from.the.data.py**

J1.the.disagreement.no.longer.has.two.sides [OK]
sec:flatlcdm (L-171s internal disagreement no longer has two sides: across 18 datum readings the perturbation route gives no definite first peak, and its 150 is c54.164s number from a retired instrument). *Computes*: reads every c54.187-188 spectrum, asserts 150 and 220 both absent and the 2.26x range, and ties it to the class c54.187 routed as item 35 (12 checks) *Bound*: NOT-A-PAPER-CLAIM — supersedes a register rows content; its next step is untouched *Run*: python3 receipts/L171_two_routes/J1.the.disagreement.no.longer.has.two.sides.py

P15.the.spacing.deficit.survives.the.one.fitted.parameter [OK]
sec:tensions (****THE LAST FREE THING UPSTREAM IS THE CORPUS'S OWN ONE FITTED NUMBER, AND MOVING IT BY 31% MOVES THE SPACING DEFICIT BY 7%**** Front #2, lead 'L-504'. c54.188 closed both seam-datum freedoms and named this: ****'Z.START'** is solved so that $l_A = \pi \cdot D_M / r_s$ hits a TARGET rather than coming out as an output. ****That is declared, not hidden — P15 sec:tensions says "the measured acoustic scale is reproduced at the directly measured H₀ by a single z_{onset}" and calls it "the one fitted number"**. What had not happened is that the number was a literal buried in a 'brentq' call, so nothing downstream could vary it and nothing had.* ****PART 1 MOVING THE PIN MOVES EVERYTHING****

IT SHOULD:** pinning L_A at 260/280/301.6/320/340 drives z_{onset} from 11009 to 5066 and r_s from 157.1 to 120.2 Mpc**. **PART 2 AND THE SPACING DEFICIT DOES NOT FOLLOW IT:** the mean peak spacing stays at $0.7647\text{--}0.8205$ of L_A — a spread of 1.07 against the pin's 1.31, mean 0.798**. *So the $\sim 20\%$ spacing deficit is a property of the construction and not of the value its one fitted parameter was fitted to.* **PART 3** L_1/L_A runs $0.5176\text{--}0.6308$ (1.22x) and **never approaches the sky's 0.7312 ** — and its drift is almost entirely L_A 's, the peak itself moving only 164 $\rightarrow$ 176 while L_A moves 260 $\rightarrow$ 340. **PART 4 THE STRUCTURAL STATEMENT UNDER IT:** in ordinary acoustics L_1 goes as D_M/r_s , so a 24% fall in r_s should raise L_1 by 31%; **here it raises it by 7%** $\Rightarrow$ ***This construction's peaks track its own sound horizon at 24% of the rate acoustics requires, and that — rather than any one ratio — is what the comparison with the sky is registering.*** **CORRECTED c54.190 ('L-505'): the spacing figure in this row is the mean of the FIRST THREE GAPS, which is all four peaks at $L_{\text{MAXL}} = 1000$ can give. At production depth the asymptotic spacing is 0.975 of L_A — 2.5% short, not 21% — and what is wrong is the ACOUSTIC PHASE, by 0.62 pi. The robustness this row measures stands; the name it gives the quantity does not.***). *Computes:* **the spacing's spread under the pin asserted below 1.20x — the point past which the deficit would BE the fitted parameter's and every conclusion resting on it withdrawn**; its maximum asserted below 0.90 of L_A , since a pin taking it near 1.0 would remove the deficit itself; the first-peak position asserted never to reach the sky's ratio at any pin, which is PART 3; **the peaks' sensitivity to r_s asserted in (0, 0.6) of the acoustic rate — if it ever reached that rate the disagreement would be about a NUMBER and not about a mechanism**; and all five pins asserted to give a four-peak series *Bound:* **scope: the acoustic question on this front is now closed as far as the construction's FREEDOMS go — c54.187 scanned the seam phase, c54.188 the amplitude reading, this the fitted parameter, and across all of them the SPACING sits near 0.78–0.80 of L_A and never reaches 0.9 while the POSITION spans 2.26x and states nothing. What is left is not a freedom that can be scanned: whether the peaks SHOULD be tied to the sound horizon is a question about the physics. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's. AND ONE INCONSISTENCY RECORDED RATHER THAN FIXED: the instrument pins $L_A = 301.6$ where 'P15_zonset_determinations' pins the MEASURED 301.76 — 0.05%, and it is why this instrument returns $z_{\text{onset}} = 6761$ where that receipt returns 6797. Named so the two are not read as a disagreement.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_spacing_deficit_survives_the_one_fitted_parameter.py

P15.the.spacing.is.right.and.the.acoustic.phase.is.wrong

[OK]

sec:refit-bound (**THE " $\sim 21\%$ SPACING DEFICIT" OF c54.187-189 IS THE FIRST THREE GAPS AND NOT THE SPACING. AT PRODUCTION DEPTH THE ASYMPTOTIC SPACING IS RIGHT TO 2.4% AND WHAT IS WRONG IS THE ACOUSTIC PHASE.** Front #2, lead 'L-505'. ***THIS FILE CORRECTS THREE OF ITS OWN LINE'S REVISIONS.*** *Those scans ran at $L_{\text{MAXL}} = 1000$ so that eighteen readings and five pins were affordable; at that depth the CR arm has four peaks, so "the mean peak spacing" was a mean of THREE GAPS — and the first three gaps are the only ones where the arms disagree.* **PART 1/2 AT PRODUCTION DEPTH BOTH ARMS CARRY EIGHT PEAKS:** ΛCDM 's last four gaps average 1.002 of L_A and the CR arm's 0.975 — 2.5% short, not 21%**; gap by gap the CR arm runs $0.725, 0.853, 0.875$ of the control's and is $\sim$ level by the fourth (0.973)**. **PART 3 SO THE TWO SERIES ARE PARALLEL LINES WITH DIFFERENT INTERCEPTS, AND THAT IS A PHASE:** fitted on peaks 4–8, slopes 1.003 and 0.976 of L_A (agreeing to 2.6%) and intercepts -0.263 and -0.878 of L_A . *A driven series peaks at $k \cdot r_s = n \cdot \pi - \phi$, so the intercept IS $-(\phi/\pi) \cdot L_A$: $\Rightarrow$ ***the two differ by 0.62 pi in the ACOUSTIC PHASE SHIFT, at a spacing they agree on.*** **PART 4 AND A LOW-ell TRANSIENT THE CONTROL DOES NOT HAVE:** measured each against its OWN asymptotic line, the control's first three peaks sit within 16 of it and the CR arm's sit $+142, +80, +18$, converging — *and that transient is what compressed the early gaps*. ***PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at $\phi = 0$. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the CR asymptotic spacing asserted above 0.90 of L_A and pinned at 0.975 — if it were NOT near L_A there would be nothing to correct and c54.187-189 would stand as written**; the first-three-gap and asymptotic values asserted to STRADDLE 0.85, which IS the diagnosis (one number reported for the other); **the slopes asserted parallel to 6%, without which the offset is not a pure phase and PART 3 is wrong**; the phase offset pinned at 0.615 ± 0.08 , which is

what replaces the retracted number; the transient asserted at more than 3x the control's, since PART 4 is otherwise empty; and the weakest peak's contrast asserted above 1.05, since a series read off non-features is noise *Bound:* ****scope:** WHAT THIS RETRACTS is this line's own — c54.187's "˜21%", c54.188's "˜23%", c54.189's "˜20%" and the paper's "acoustic series some 23% too close" are all the first-three-gap value. WHAT IT DOES NOT RETRACT is the robustness: the first-three-gap spacing IS stable at 0.77–0.82 across eighteen datum readings and five pins and F4 fires at every one; what changes is that the stable quantity was the wrong one to call the disagreement. AND THE CORRECTION MAKES c54.187 CENTRAL rather than a caveat — the disagreement is a PHASE, and the phase is exactly what the seam datum assigns and what "one datum per mode and a COMMON phase" does not fix. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure — and this makes the discrepancy SHARPER, not smaller: a spacing that is right and a phase that is wrong names where to look.** *Run:* python3 receipts/P15_CR.cosmology/P15.the_spacing_is_right_and_the_acoustic_phase_is_wrong.py

B4.the_intercept_is_a_phase_and_the_control_proves_it [OK]

sec:coherence (the intercept IS the acoustic phase by change of variables, and the control validates the reading by reproducing the textbook 0.25-0.27; the 21% spacing figure was a depth artefact). *Computes:* derives the intercept identity, fits both arms on peaks 4-8, reproduces the gap-by-gap series and the low-l transient (14 checks) *Bound:* NOT-A-PAPER-CLAIM — the adversarial check c54.190 requested *Run:* python3 receipts/L147_two_arm/B4.the_intercept_is_a_phase_and_the_control_proves_it

B5.the_scan_cannot_test_the_asymptotic_phase [OK]

sec:coherence (the corrected claims load-bearing quantity cannot be tested by the datum scan: all 23 scan spectra stop at three or four peaks and zero reach the peaks-4-to-8 fit). *Computes:* asserts the depth and peak count of every scan spectrum, that the production pair alone supports the fit, and where the transient sits (11 checks) *Bound:* NOT-A-PAPER-CLAIM — names a small affordable experiment *Run:* python3 receipts/L147_two_arm/B5.the_scan_cannot_test_the_asymptotic_phase.py

P15.the_acoustics_work_and_the_phase_is_the_whole_disagreement [OK]

sec:refit-bound (**THE ASYMPTOTIC SPACING TRACKS THE SOUND HORIZON AT 98% OF THE ACOUSTIC RATE, SO THIS CONSTRUCTION'S ACOUSTICS WORK — AND THE WHOLE DISAGREEMENT IS THE ACOUSTIC PHASE.** Front #2, lead 'L-506'. c54.190 separated spacing from phase and owed both a production-depth test; **the scans that varied them were all at the depth c54.190 showed to be insufficient**. **PART 1 THE SPACING TRACKS r_s AT THE FULL RATE:** moving the pin from l_A = 301.6 to 340 drops r_s by 11%, ordinary acoustics requires the spacing to rise 13%, and **it rises 13% — 98%**. *So the peaks in this construction ARE set by its own sound horizon.* **PART 2 WHICH RETRACTS c54.189's HEADLINE:** that revision reported **"the peaks track their own sound horizon at 24% of the rate acoustics requires"* — **measured on the FIRST peak, which c54.190 showed sits 142 multipoles off the asymptotic line, inside a transient**. *Over this file's pin range the first peak moves at 37% against the series' 98%; the figure is not reproduced and is not meant to be — what is reproduced is the FINDING.* **PART 3 THE PHASE IS ROBUST TO THE FITTED PARAMETER:** the intercept moves $-0.878 \rightarrow -0.899$ of l_A across the same 31% swing — **3% of a discrepancy of 0.62**. **PART 4 AND THE SEAM DATUM'S PHASE FREEDOM IS A REAL LEVER THAT SPANS A THIRD OF IT:** at the opposite phase the intercept moves to -0.671 , closing the gap from 0.615 to 0.408 — **34%, and no further**. $\Rightarrow$ ***Which is the confirmation c54.190's diagnosis wanted: the datum's phase controls the acoustic phase, as a phase diagnosis predicts, and cannot close the gap.*** **PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving CONTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the asymptotic sensitivity to r_s asserted in (0.85, 1.15) — if it were NOT near 1 the acoustics would not work and the phase reading would not be the whole story**; the FIRST peak's sensitivity asserted below 0.6, since PART 2 retracts a number and must show the quantity really is less sensitive; the pin's effect on the phase asserted below 15% of the discrepancy, without which the phase is the fitted parameter's; **the seam phase's closure asserted in (0.15, 0.70) — if it ever closed the WHOLE gap the discrepancy would be the datum's alone and this front's verdict would have to be restated**; the phase discrepancy pinned at 0.615 ± 0.05 ; and at least seven peaks per arm, since every number here is an asymptotic fit *Bound:* ****scope:** the front's acoustic statement in final form — this construction reproduces the

acoustic SPACING (the quantity its one fitted parameter is fitted to, and which it then tracks at 98% of the acoustic rate when that parameter is moved) and disagrees with the sky in the acoustic PHASE by 0.62 pi, robust to the fitted parameter and only a third reachable by the one freedom the seam datum leaves open. That is ONE number and ONE mechanism where the front began with four items and a ratio. ‘F5’ unsoftened, ‘PO-7’ protected, conversion Daryl’s — and the discrepancy is now specific enough to be attacked: an acoustic phase shift is a computable consequence of the driving, so “why 0.62 pi” has an address. This file does not have it.** *Run:* python3 receipts/P15_CR_cosmology/P15_the_acoustics_work_and_the_phase_is_the_whole_disagreement.py

B6_the_acoustics_reproduce_and_my_dichotomy_was_false [OK]

sec:coherence (the acoustics reproduce at 98.2% of the required rate, and r2485s stated dichotomy was not exhaustive: the phase moves and stops at 34%). *Computes:* recomputes the r_s response and the four-row phase table from the forks spectra, and asserts both horns false (13 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.191 and corrects this lines own r2485 *Run:* python3 receipts/L147_two_arm/B6_the_acoustics_reproduce_and_my_dichotomy_was_false.py

B7_the_seam_phase_moves_the_asymptotic_intercept [OK]

sec:coherence (item 38 answered at production depth: the seam phase MOVES the asymptotic intercept by ~0.31 lA (half the 0.615 disagreement) while the spacing holds, so the 0.62pi is datum-contingent not structural). *Computes:* validates the phi=0 control against c54.186, fits three production seam phases on matched acoustic orders 4-7, asserts the intercept span and the stable spacing (5 checks) *Bound:* NOT-A-PAPER-CLAIM — a measurement; informs vein L-202, answers lead L-171; conversion to a PO-7 verdict is unseated *Run:* python3 receipts/L147_two_arm/B7_the_seam_phase_moves_the_asymptotic_intercept.py

K1_a_capability_assumed_from_one_failure [OK]

sec:coherence (the whole acoustic disagreement is the driving: undriven the arms agree to 0.013 in phi/pi and the CR slope is exactly 1.0000 lA, and both lines accepted a capability claim for eight revisions). *Computes:* fits all four production spectra, asserts the undriven agreement and the 5.72x driving ratio, and counts this lines own tarball absorptions (12 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.193-194 *Run:* python3 receipts/L239_capability_assumed/K1_a_capability_assumed_from_one_failure.py

B8_the_zero_velocity_band_is_a_third_of_the_gap [OK]

sec:coherence (the seam datums band across the two ZERO-VELOCITY phases is 0.207 against a 0.615 disagreement — a third, not most — and over that pair the control is NOT inside). *Computes:* solves sin(phi)=0 on [0,2pi), reads both admissible phases off four production spectra from two nodes, and withdraws this lines own L-202 link (13 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows c54.195s withdrawal *Run:* python3 receipts/L147_two_arm/B8_the_zero_velocity_band_is_a_third_of_the_gap.py

P15_the_admissible_band_is_a_third_and_the_two_entries_differ_against_the_potential [OK]

sec:coherence, sec:refit-bound (**ITEM 43 (r2509) ACCEPTED: OVER THE PHASES THE CORPUS ADMITS, THE ACOUSTIC PHASE MOVES 0.207 — A THIRD OF THE 0.615 GAP — AND THE CONTROL’S 0.263 IS OUTSIDE IT. THE DISAGREEMENT IS REAL AND BOUNDED.** Front #2, lead ‘L-513’; VEIN ‘L-202’. **PART 1** the pair is **forced**, not chosen: ‘sec:what-crosses’ says what crosses is **frozen**, a frozen mode has $\dot{\delta}\gamma=0$, continuity gives $\theta\gamma=\frac{34}{Dkcs}\sin\phi$, so $\sin\phi=0$ — the paper’s own transmission argument read at the seam, in print before the scan existed. **PART 2 AND THE REASON c54.199 GAVE 56 IS WRONG:** ‘y0[:,6] = Ph0’ is fixed at \$-1\$ independent of ϕ while ‘dg0’ carries $\cos\phi$, so the two entries are **not** a sign flip of the state — **what reverses is the relative sign of the density against the potential**, a compression against a rarefaction in the same well. **PART 3** 54’s and cc54’s production spectra agree to four decimals at both admissible phases. **PART 4** band 0.2069 vs gap 0.6152 — $2.97\times$, control **outside** [0.6711, 0.8780]. **PART 5** χ^2/dof 281 and 379** against 3.71 — $76\times$ and $102\times$ — so c54.195’s PART 5 conclusion is stronger on the pair than on the span it was illustrated in. **PART 6** and P15 ‘sec:refit-bound’ already carried “spans a third of it” and the 0.408, **two paragraphs above the paragraph that withdrew it**.). *Computes:* the four-step admissibility chain asserted in the instrument’s AND the paper’s source — **if ‘sec:what-crosses’ did not say modes freeze, the pair would be a choice and r2509 weaker than this file claims**; ‘dg0’ asserted to flip while ‘Ph0’ does not, which is what makes item 17’s reason wrong; **54 and cc54 asserted to agree to four decimals — a band measured on one node’s spectra is a band measured once**; the ratio ~3 and, separately, the control asserted OUTSIDE the band, without which c54.195 stands as written; both admissible readings asserted

above $50\times$ the control, without which the withdrawal takes the conclusion with it; and the paper's own earlier 0.408 asserted to reproduce *Bound*: ***scope*: this corrects this line TWICE and in opposite directions — it makes the disagreement firmer than c54.195 left it, and it retracts the explanation c54.199 routed to 56. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure. No new run: five banked production spectra, two nodes.** *Run*: python3 receipts/P15_CR_cosmology/P15_the_admissible_band_is_a_third_and_the_two_entries_differ_against_the

B9_the_admissible_pair_is_forced_not_merely_distinguished [OK]
sec:what-crosses (the admissible pair ϕ in $\{0, \pi\}$ is FORCED by P15s own transmission argument — every mode freezes before the crossing, and a frozen mode has zero velocity). *Computes*: quotes the freezing argument at source, solves $\sin(\phi)=0$, and asserts the control lies OUTSIDE the band with both nodes agreeing to four decimals (12 checks) *Bound*: NOT-A-PAPER-CLAIM — verifies c54.200 and upgrades r2509 *Run*: python3 receipts/L147_two_arm/B9_the_admissible_pair_is_forced_not_merely_distinguished

B10_the_thermal_mechanism_needs_an_exponential_approach [OK]
sec:transmission (L-519 answered from the MECHANISM side: thermality needs an exponential near-horizon approach, the degenerate members is a power law, and P15 already derived that split for its transmission dichotomy). *Computes*: integrates the tortoise coordinate for a simple and a double root, re-verifies the Nariai double root, and asserts P15s exponential/power-law split at source (11 checks) *Bound*: NOT-A-PAPER-CLAIM — a third footing for L-519; no temperature value claimed *Run*: python3 receipts/L147_two_arm/B10_the_thermal_mechanism_needs_an_exponential_approach

P0-7 [OK]
sec:coherence (kill receipt for PO-7 under route (2): object named, four checks answered — and the CHAIN check does not clear, because the disagreement rests on four reproduced things and one ARGUED one). *Computes*: a kill receipt, not a script — the four checks written out with the inversion routes named *Bound*: NOT-A-PAPER-CLAIM — narrows PO-7; no authorisation requested *Run*: python3 receipts/kills/P0-7.md

S1_every_mode_of_interest_freezes_and_the_inversion_is_closed [OK]
sec:what-crosses (PO-7 chain-(4) link (d) reproduced on a second path: every mode of interest ell 28-2475 freezes before the crossing ($r=c_s k/|aH| \rightarrow 0$), so $\sin(\phi)=0$ forces ϕ in $\{0, \pi\}$, the band stays 0.2069 with the control OUTSIDE, and the kill receipt's inversion (a mode crossing unfrozen) is closed by the divergence of $|aH|$). *Computes*: reproduces the progenitor interior independently (matching P16's freeze-out to the digit, $x_{\text{freeze}}=0.002015$ at ell=2475), computes the velocity ratio $\rightarrow 0$ at the crossing for every mode, and anchors *sec:what-crosses* + the kill receipt's chain check (10 checks) *Bound*: NOT-A-PAPER-CLAIM — reproduces PO-7's one non-reproduced link; informs L-171/L-202; F5 unsoftened, no conversion claimed *Run*: python3 receipts/L805_freezing_reproduced/S1_every_mode_of_interest_freezes_and_the_inversion_is_closed.py

S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_branch_point [OK]
sec:what-crosses (A2 (PO-seam / PO-7 inversion 3): a massive mode freezes too, for ANY mass — the branch point $a \rightarrow 0$ annihilates the mass term $m^2 a^2$ while $|aH|$ diverges, so $\omega/|aH| \rightarrow c_s k x \rightarrow 0$ independent of m ; a massive trajectory carries no phase from non-freezing, closing PO-7 route 3 from the massive side). *Computes*: derives the mass term $+m^2 a^2$ from the covariant action (sympy), shows $\omega/|aH|$ at the crossing is m -independent to the digit up to $m=1e5$ across ell 28-2475, freeze-out epoch unmoved (5 checks) *Bound*: NOT-A-PAPER-CLAIM — A2; informs L-202/L-171; CRPHI stays assigned; F5 unsoftened *Run*: python3 receipts/L806_massive_freezing/S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_branch_point

S1_the_peak_estimator_recovers_a_set_phase_independent_of_the_envelope [OK]
sec:coherence (A10 (PO-7 inversion 2): the peak-4-8 estimator recovers a SET asymptotic phase to ± 0.005 in ϕ/π and is envelope-independent (spread ± 0.006) across six damping/width/tilt variations spanning the CR-vs-control transfer differences — so the 0.408 is not an estimator artefact; inversion route 2 closes and the disagreement hardens). *Computes*: injects synthetic acoustic combs with known ϕ in under six envelopes, runs the exact B7/B8 estimator, checks faithfulness and envelope-invariance (4 checks) *Bound*: NOT-A-PAPER-CLAIM — A10; informs L-171; F5 unsoftened, no conversion claimed *Run*: python3 receipts/L807_driven_estimator/S1_the_peak_estimator_recovers_a_set_phase_independent_of_the_envelope

S1_the_seam_phase_is_derived_compression_correlated_on_the_observable_leg [OK]
sec:what-crosses (PO-7/PO-seam: the seam phase CRPHI is DERIVED, not free = CRPHI=0 (a compression correlated with the well), on the observable leg — the pre-onset is pressureless (r2701)

so no oscillation swings the inherited adiabatic overdensity's sign; bracket-checked against the LCDM control's own adiabatic IC; the $\{0,\pi\}$ band collapses to 0.878, disagreement pinned at 0.615, CRPHI= π (0.408, a rarefaction) anti-adiabatic and inadmissible). *Computes:* reads the code's CRPHI convention (overdensity in a well = CRPHI=0), shows a pressureless mode's sign is fixed while a pressured one swings, confirms the adiabatic bracket, and anchors the pressureless-onset statement in CR_cosmology.tex (4 checks) *Bound:* NOT-A-PAPER-CLAIM — closes PO-7 route (3)'s residue; informs L-171/L-202; F5 unsoftened, one assumption (Phi-continuity) flagged *Run:* python3 receipts/L815_seam_phase_derived/S1_the_seam_phase_is_derived_compression_correlated_on_th

S1_the_banked_spectra_are_nk600_not_nk260_and_pass_the_guard [OK]
sec:where-the-likelihood-sits (corrects C52's sampling premise: the banked two-arm spectra the likelihood scores are NOT under-sampled — they are NK=600 (the README command), which PASSES the guard at 5.7 pts/Bessel-period. cc54 (15GB) reproduced c54.178.lcdm BIT-FOR-BIT (max |dDl|=8e-16). C52 read the .npz's 238-point shape as NK=260, but 238=len(arange(100,2000,8)) is set by LSTEP=8/LMAXL=2000, not NK; 56's 3.7GB container OOM'd at NK=600 so tested the guard-failing NK=260. The 7.14/280.09 stand; C51's dropped-bin point is separate and stands). *Computes:* confirms the 238-point shape is arange(100,2000,8) independent of NK, computes guard pts/period= $2\pi(3NK-1)/(LMAXL-12)$ giving 2.46@NK=260 (fails) and 5.69@NK=600 (passes), and asserts the banked npz is bit-identical to cc54's NK=600 reproduction (4 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects C52's provenance premise; the guard itself is right (NK=260 does under-sample); does not touch C51's real dropped-bin blocker *Run:* python3 receipts/L820_banked_spectra_sampling/S1_the_banked_spectra_are_nk600_not_nk260_and_pass_the_guard.

S2_the_lmax2512_extension_recovers_c51s_dropped_bins_and_CR_stays_disfavoured [OK]
sec:where-the-likelihood-sits (discharges C51 (the damping-tail half of 56's resampling route): the LMAXL=2512 extension (302 multipoles to ell=2508) recovers the 30 dropped bins (ell 1759-2508), and including them does NOT rescue CR. Two effects isolated: the wider k-range lifts the control 7.14- $\dot{\iota}$ 3.81/dof on the SAME 185 bins (c54.186's truncation effect), added tail - $\dot{\iota}$ 3.68; the CR arm does not gain (280.1- $\dot{\iota}$ 281.1 overlap) and stays 260.1/dof. F3 widens 50497- $\dot{\iota}$ 51547 — the dropped region makes CR MORE disfavoured, not less). *Computes:* scores the LMAXL=2512 lcdm/cr extensions on the overlap (ell=1996) and full (201) bin sets vs the banked pair, isolating the k-range effect from the damping-tail addition, and computes F3 on both bin sets (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the C51 half of L-820; CR verdict unchanged (both F3 $\sim$ 2400x threshold); control's 3.68/dof still large, truncation-vs-physics not further split *Run:* python3 receipts/L820_banked_spectra_sampling/S2_the_lmax2512_extension_recovers_c51s_dropped_bins_and_CR.

S1_the_cr_residual_is_acoustic_structure_not_truncation_and_that_is_why_it_is_unresponsive_to_lmax [OK]
sec:where-the-likelihood-sits (PO-10 OWED #496 (the row's real remainder): DIAGNOSES the CR arm's 280/dof that r2781/C53 found "unresponsive to L_max and diagnosed by nothing". A per-bin χ^2 decomposition (full plik_lite covariance) localises the residuals: the control's is 75% in the high-ell tail (ell $\dot{\iota}$ 1200, the truncation), the CR arm's is 57% in the acoustic-peak bands (ell $\dot{\iota}$ 850) and 18% tail — so CR is acoustic-structure, not truncation, which is WHY it is unresponsive to L_max. CR comb compressed (172/404/636 vs sky 220/540/810, 11/1A 0.57 vs 0.73 = PO-7's deficit); rescaling doesn't collapse χ^2 (best $\sim$ 200) so position+heights). *Computes:* computes the per-bin χ^2 contribution $r_i(\text{Fr})_i$ with the full covariance and groups by ell-band for both arms, reads the CR peak positions, and scores under an ell-axis rescaling (4 checks) *Bound:* NOT-A-PAPER-CLAIM — OWED #496 CR half; F5 reserves the verdict (PO-7 protected); control third convergence point (L3200) measured separately *Run:* python3 receipts/L822_cr_residual_diagnosed/S1_the_cr_residual_is_acoustic_structure_not_truncation_and_th

S1_the_stored_cr_rs_is_the_from_onset_acoustic_horizon_and_matches_the_arms_own_integral [OK]
sec:acoustic (PO-10 (the code read 56's r2790 routed): the stored CR r_s =135.46 is NOT a book-keeping error. The line is $R.S=rs_from(Z_START)$ with Z_START the FITTED onset; rs_from uses the SAME sound speed/grid/onset the acoustic oscillation uses, so 135.46 IS the arm's acoustic horizon and the peaks asymptote to $\Lambda A=301.6$. r2790's "peaks demand 158" is the low-ell transient (first-4-gap mean - $\dot{\iota}$ r_s =158, high-n gaps - $\dot{\iota}$ $\sim$ 139 bracketing the stored); its "direction requires 245" is the from-a $\sim$ 0 horizon, NOT the acoustic one (pre-onset pressureless, L-815, no oscillations before onset). Smaller than LCDM because CR starts oscillating later — the framework's fitted-onset design). *Computes:* reproduces $R.S=rs_from(Z_START)$ via the brentq fit,

computes r_s from $a \sim 0$ vs the fitted onset for both arms, and reads the CR peak gaps by ell-range (first-4 vs high- n) (4 checks) *Bound*: NOT-A-PAPER-CLAIM — the r2790 code read; not a bug, the fitted-onset design's output; whether that design is forced is F5/PO-10's *Run*: python3 receipts/L823_cr_rs_is_the_onset_horizon/S1_the_stored_cr_rs_is_the_from_onset_acoustic_horizon_and

S1_the_third_point_shows_the_control_converges_by_L3000_and_the_unlensed_floor_is_the_lensing [OK]

sec:acoustic (PO-10 (OWED #496, control half — the THIRD convergence point r2781 asked for): the control CONVERGES by L3000, it does NOT slide down an $L^{-3.4}$ law to "1.1 by L6000". On the same 185 bins the unlensed control goes $7.18 \rightarrow 3.83 \rightarrow 3.73 \rightarrow 3.73$ across LMAXL 2000/2512/3000/3200; the $L^{-3.4}$ law fit to the first two points predicts 2.23 at L3200, the measurement is 3.73 — the two-point rate was a one-time truncation recovery, not a convergence rate. The ~ 3.73 unlensed floor IS the lensing: c54.183's lensed/unlensed ratio takes the SAME converged spectrum $3.73 \rightarrow 1.18$, and the lensed control converges $5.38 \rightarrow 1.33 \rightarrow 1.19 \rightarrow 1.18$ to 1.18 by L3000 vs CAMB lensed's 1.014 (a $\sim 0.16/\text{dof}$ instrument floor). Truncation (LMAXL) and lensing are ORTHOGONAL axes; F3 cancels the shared 1.18 floor). *Computes*: scores the four banked control spectra (c54.178 L2000, L820 L2512, c54.186 L3000, L824 L3200) on the fixed 185-bin set unlensed and under CAMB's lensed/unlensed ratio; checks the $L^{-3.4}$ extrapolation vs the measured third point, the L3000/L3200 convergence, and the top-bin truncation (4 checks) *Bound*: NOT-A-PAPER-CLAIM — control-arm convergence; does not overturn 56's convergence claim (the LENSED control does converge, to 1.18 by L3000); companion to L-822 (CR half); F5 untouched *Run*: python3 receipts/L824_control_converges/S1_the_third_point_shows_the_control_converges_by_L30

S1_the_converged_controls_remainder_is_a_broadband_instrument_floor_not_a_localized_defect [OK]

sec:acoustic (PO-10 (OWED #454, the remainder localized): after L-824's convergence the CONTROL arm's $1.18/\text{dof}$ vs CAMB lensed's 1.014 on the same 185 bins (the " ~ 100 " was pre-convergence, pre-lensing) is a BROADBAND instrument floor, not a localized defect. A per-bin $r_i(\text{Fr})_i$ decomposition shows every ell-band 100-2000 contributes a positive excess ($+4.5/+6.0/+7.4/+1.6/+5.7/+4$ total ~ 30 in χ^2), none dominating, slightly leading in the first two acoustic peaks (300-850 carries 13.4 of 30). So the ~ 1 target is instrument-accuracy-limited (a uniformly better transfer, not a patchable band), and $F3 = \chi^2(\text{CR}) - \chi^2(\text{LCDM})$ cancels this shared floor). *Computes*: per-bin decomposition of the converged lensed control (L824 arm) and CAMB lensed LCDM through plik_lite TT's full covariance on the fixed 185 bins, by ell-band (3 checks) *Bound*: NOT-A-PAPER-CLAIM — control-arm diagnostic; #454's target instrument-limited, still open; F5 untouched; continues L-824 *Run*: python3 receipts/L826_control_floor_broadband/S1_the_converged_controls_remainder_is_a_br

T1_the_transfer_is_half_built [OK]

sec:envelope (PO-12 is HALF BUILT: the debt is two steps and step (1) — the specification of how fluctuations gravitate on the geometrically fixed background — is computed in *sec:envelope*, two sections before the debt is named). *Computes*: asserts the two-step sentence, the envelope-is-derived sentence, the structural-argument clause and the tilt-irreducible-residual clause verbatim (9 checks) *Bound*: NOT-A-PAPER-CLAIM — narrows a register row; step (2) is genuinely not run *Run*: python3 receipts/L202_the_seam/T1_the_transfer_is_half_built.py

C12_the_odd_even_half_has_its_parameter [OK]

sec:envelope (PO-10s odd/even half already has its mechanism and parameter: C5b_baryon_term puts the asymmetry on the expansion leg at $R_b = 0.60$, which is the rows own description of the item). *Computes*: asserts the rows wording, C5bs location and growth-factor and conclusion, and the papers own sentence (9 checks) *Bound*: NOT-A-PAPER-CLAIM — narrows a register row; the height pattern is not produced *Run*: python3 receipts/P15_CR_cosmology/C12_the_odd_even_half_has_its_par

C14_po12_gates_po10 [OK]

sec:envelope (PO-12 GATES both of PO-10s runs: the textbook odd/even ratios at $R_b = 0.60$ are LCDM peak-height formulae that assume a transfer, and both of PO-10s halves are statements about C $_{\ell}$). *Computes*: asserts PO-10s remaining half and P15s genuine-build sentence verbatim, computes the textbook ratios, and confirms neither row records the dependency (7 checks) *Bound*: NOT-A-PAPER-CLAIM — records an ordering between two rows *Run*: python3 receipts/P15_CR_cosmology/C14_po12_gates_po10.py

C15_the_cancellation_does_not_reach_the_ratio [OK]

sec:envelope-consequence (the corpus cancellation does not reach the odd/even ratio: it holds

between two RATES (same quantity, microphysics identical) while odd/even compares two MULTIPOLES on one rate, where the envelope leaves 13-36%). *Computes:* asserts the cancellation sentence and both rate-ratio results verbatim, and computes adjacent-peak envelope ratios on three envelopes (9 checks) *Bound:* NOT-A-PAPER-CLAIM — supplies the reason behind r2646s gate *Run:* python3 receipts/P15_CR_cosmology/C15_the_cancellation_does_not_reach_the_ratio.py

C16.the_transfer_is_six_eighths_built [OK]
sec:envelope (PO-12s step 2 is six-eighths built: the driving verified in closed form, source removal exact, projection calibrated at $L^*=302.2$ vs 301, and a C11 RATIO with no free parameter already in the paper (0.823 recomputed vs 0.82)). *Computes:* verifies the driving satisfies its ODE, is even, and normalises; asserts the genuine-build, projection and no-free-parameter clauses verbatim; recomputes the high-ell ratio (8 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a debt to the absolute spectrum *Run:* python3 receipts/P15_CR_cosmology/C16_the_transfer_is_six_eighths_built.py

C17.the_instrument_already_carries_both [OK]
sec:instrument (CORRECTS r2658: both pieces it marked NOT LOCATED are built and validated — the Peebles recombination history IS the visibility function and the line-of-sight source with its j_l kernel IS the projection). *Computes:* asserts sec:instruments description, its validation figures and its flat-projection wording verbatim, against the debt's geometric-background clause (9 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a prior lines audit; the debt is now what the instrument RUNS ON *Run:* python3 receipts/P15_CR_cosmology/C17_the_instrument_already_carries_both

C18.bespoke_means_two_legs [OK]
sec:envelope (what BESPOKE means for PO-12: the transfer must carry TWO backgrounds joined at the branch point — radiation-dominated collapse leg, geometry-set expansion leg — and the paper forbids mixing the rates). *Computes:* computes the friction coefficient for both contents, asserts the collapse-leg scoping, the L1/L2 division and the forced consistency rule verbatim (7 checks) *Bound:* NOT-A-PAPER-CLAIM — sharpens a debt; the branch-point join is untouched *Run:* python3 receipts/P15_CR_cosmology/C18_bespoke_means_two_legs.py

C19.the_branch_point_join [OK]
sec:transmission (THE BRANCH-POINT JOIN COMPUTED: transmission is conservation of R, so the join is a change of variable worth exactly 9/10 — applied ONCE where LCDM spreads it over equality, which is why CR has no early ISW). *Computes:* asserts the not-yet-in-hand and transmits clauses, the matter-dominated-to-nine-orders and four-per-cent clauses verbatim, and derives 9/10 from $R/\Phi=(5+3w)/(3+3w)$ (7 checks) *Bound:* NOT-A-PAPER-CLAIM — super-horizon join only; sub-horizon modes and the dynamical crossing stay open *Run:* python3 receipts/P15_CR_cosmology/C19_the_branch_point_join.py

C20.the_caveat_named_an_empty_set [OK]
sec:what-crosses (r2661s super-horizon caveat named an EMPTY SET: at the branch point $(rH)^2$ diverges, the comoving horizon goes to zero, and every mode is outside — so the 9/10 join is unrestricted). *Computes:* asserts the every-mode-outside and seam clauses verbatim, computes the divergence of $(rH)^2$ and the vanishing comoving horizon, and confirms the radiation term dominates (7 checks) *Bound:* NOT-A-PAPER-CLAIM — removes a caveat from a prior receipt; the crossing dynamics stay deferred *Run:* python3 receipts/P15_CR_cosmology/C20_the_caveat_named_an_empty_set.py

C21.the_superhorizon_transfer_closes [OK]
sec:envelope (THE SUPER-HORIZON TRANSFER CLOSES: Φ_{-i} Φ_{+i} exactly at the branch point for every k (departure $O(k^2 \eta^2)$), so with r2661s join $\Phi_{\text{exp}}(k) = (9/10) \Phi_{+i}(k)$ scale-invariantly — the branch point does not filter). *Computes:* computes the branch-point limit and the series coefficients, re-derives the join, and asserts the transmits clause verbatim; flags the 1/10 near-miss as different-origin (7 checks) *Bound:* NOT-A-PAPER-CLAIM — scalar super-horizon only; the single end-to-end run stays unrun *Run:* python3 receipts/P15_CR_cosmology/C21_the_superhorizon_transfer_closes.py

C22.the_end_to_end_number [OK]
sec:earlyisw (THE END-TO-END NUMBER: 7.5% low-ell power deficit from the absent early ISW, 18% at 1.D rising to 54% at 2 1.D from damping — and the acoustic peaks BETWEEN them are where the chain stops, correcting r2661s no-further-evolution). *Computes:* asserts the scope qualification, the super-horizon-equation clause, the order-two decay and the not-a-radiation-effect clause verbatim, and computes both ratios (8 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a prior receipts scope; not a prediction against data *Run:* python3 receipts/P15_CR_cosmology/C22_the_end_to_end_number.py

- C23.the_gap_estimated** [OK]
sec:earlyisw (ESTIMATE for the peak-region gap: the decay rate $k^2/(3H)$ is built from the thing that differs, so CRs potential decays $\sim 15\%$ faster and the peak power ratio is ~ 0.813 — BROAD deviation, not tail-confined). *Computes:* asserts the rate, the 13% and the factor-of-two clauses verbatim, computes the compounding, and flags the 0.813/0.823 near-coincidence as different-origin (7 checks) *Bound:* NOT-A-PAPER-CLAIM — an ESTIMATE with its model stated, labelled as one; a hierarchy run would replace it *Run:* `python3 receipts/P15_CR_cosmology/C23_the_gap_estimated.py`
- C24.the_substitution_itemised** [OK]
sec:instrument (PO-12s substitution ITEMISED: conformal time, r_s , the comoving horizon and the diffusion integral carry H while the Thomson rate and x_e do not — and the ratio does NOT cancel because r_D carries a square root that r_s does not). *Computes:* asserts the 13% clause and reconstructs it from $\rho_r/\rho_m=0.3$, asserts the microphysics-outside and θ_* -pinned clauses, and computes both scaling exponents (8 checks) *Bound:* NOT-A-PAPER-CLAIM — sets the bar for the two-leg run; P15s 9.4% is not re-derived *Run:* `python3 receipts/P15_CR_cosmology/C24_the_substitution.py`
- C25.the_gap_closed_by_integration** [OK]
sec:diffusion-scale (the 2.6-point gap closed by INTEGRATION: θ_D/θ_* runs $+7.1\%$ at $z=1200$ to $+14.4\%$ at $z=10000$, with P15s $+9.4\%$ at $z\sim 2500$ — and the dependence is MONOTONIC, which is the papers own two-constraint argument). *Computes:* asserts the branch-point lower-limit clause and the single-datum clause verbatim, integrates the rate difference over five onset redshifts, and checks monotonicity (6 checks) *Bound:* NOT-A-PAPER-CLAIM — reproduces the papers shape on a simplified ρ_r/ρ_m model; the onset is the remaining unknown *Run:* `python3 receipts/P15_CR_cosmology/C25_the_gap_closed_by_integration.py`
- C26.the_onset_is_not_free** [OK]
sec:tensions (PO-12s last free parameter is NOT free: the onset is fitted to the acoustic angle at $z\sim 6797$ and meets the scale at every H_0 — and the residual $+13.1\%$ vs $+9.4\%$ is the WEIGHT $g(R)/x_e$, not the onset). *Computes:* asserts the onset value, the fitted-to-the-acoustic-angle clause and the not-a-knob clause verbatim, checks the model at the onset, and integrates under four weightings (7 checks) *Bound:* NOT-A-PAPER-CLAIM — power-law weights are stand-ins for $g(R)/x_e$, shown for direction and size *Run:* `python3 receipts/P15_CR_cosmology/C26_the_onset_is_not_free.py`
- C27.the_number_bracketed** [OK]
sec:envelope-consequence (P15s $+9.4\%$ BRACKETED from outside its own machinery: Saha $+7.0\%$; Peebles ; unweighted $+13.1\%$, with the ordering FORCED by which history recombines faster — and cancel identically covers the CONSTANTS, not the weight). *Computes:* asserts the cancel-identically, constants-outside and Peebles-history clauses verbatim, shows the weight does not drop out, runs $g(R)$ with a Saha x_e , and checks 9.4 lies strictly inside (8 checks) *Bound:* NOT-A-PAPER-CLAIM — brackets rather than reproduces; the Peebles direction is argued, not computed *Run:* `python3 receipts/P15_CR_cosmology/C27_the_number_bracketed.py`
- C28.length_against_angle** [OK]
sec:envelope-consequence (THERE WAS NEVER A DISCREPANCY: 9.4% is an ANGLE ratio (C10s $r=1.0926$, already netting the 0.7% sound-horizon shift) and 10.8% a LENGTH ratio (C8) — $r_{2686}-r_{2689}$ compared one to the other). *Computes:* asserts both statements and the 0.7% clause verbatim, reads C10s own r , and shows the two do not compose as this line assumed (7 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects four of this lines revisions; C8 and C10 are run, not re-derived *Run:* `python3 receipts/P15_CR_cosmology/C28_length_against_angle.py`
- C29.nothing_to_calculate_across_the_join** [OK]
sec:what-crosses (PO-12s remaining half has NO CONTENT: the pre-onset stretch is pressureless and at $w=0$ the potential equation loses k entirely, so there is no scale-dependent evolution for a single calculation across the join to compute). *Computes:* asserts the no- k , pressureless, boundary-mode and residue clauses verbatim, and verifies symbolically that k is present in the general equation and absent at $w=0$ (7 checks) *Bound:* NOT-A-PAPER-CLAIM — F5 reserves the strike; the residue routes to PO-7 *Run:* `python3 receipts/P15_CR_cosmology/C29_nothing_to_calculate_across_the_join.py`
- A7.the_frontier_has_two_kinds** [OK]
sec:envelope-consequence (THE FRONTIER SORTS INTO TWO KINDS on P15s own $H(a)$ rule: FIVE rows ask whether the theory is DEFINED (PO-2/4/5/6/11) and TWO ask what it PREDICTS (PO-7/10) — and they fail differently, so size does not order them). *Computes:* asserts

the H(a) clause, checks the classification covers exactly the open rows, and pins each rows object against its kind (7 checks) *Bound*: NOT-A-PAPER-CLAIM — the classification is derived here; no paper states it *Run*: `python3 receipts/P12_algebroid/A7_the_frontier_has_two_kinds.py`

C32_half_one_is_model_selection [OK]

sec:frontiers (PO-10s half (1) IS an information-criterion comparison: the corpus names "the likelihood-level model selection" while AIC/BIC/Akaike/Occam/Bayes-factor appear ZERO times — and the penalty is computable now, dAIC=10.0 and dBIC=26.9 in CRs favour). *Computes*: asserts the model-selection phrase and the 215-bin count verbatim, word-boundary-checks five model-selection terms at zero across papers and receipts (excluding itself), and computes both penalties (9 checks) *Bound*: NOT-A-PAPER-CLAIM — k=6 is the standard flat- Λ CDM count and must be stated with any result; the χ^2 is unrun *Run*: `python3 receipts/P15_CR_cosmology/C32_half_one_is_model_selection.py`

C33_the_threshold_corrected [OK]

sec:tensions (THE THRESHOLD CORRECTED: r2709s k=1 was wrong — P15 anchors A.s at the first peak, so CR is k=2 — and against Liddle/Jeffreys (dBIC 2 positive, 6 strong) the threshold is dBIC = 21.5, still 3.5x the strong line). *Computes*: asserts the anchored-amplitude and one-parameter-accommodation clauses verbatim, tables all four parameter-count pairs against the Jeffreys scale, and computes the AICc correction (7 checks) *Bound*: NOT-A-PAPER-CLAIM — no χ^2 is scored; the count must be restated with any result *Run*: `python3 receipts/P15_CR_cosmology/C33_the_threshold_corrected.py`

C34_the_deliverable_is_a_pair [OK]

sec:tensions (PO-10s DELIVERABLE IS A PAIR: the acoustic sector is open as PO-7, its phase freedom moves the gap 0.615 -j 0.408 and cannot close it — but the freedom is DIS-CRETE, so k is unchanged and both branches are reported at the same threshold). *Computes*: asserts PO-7s object, the phase-lever and cannot-close clauses verbatim with both values, and shows the discrete choice leaves dBIC at 21.5 rather than 16.1 (5 checks) *Bound*: NOT-A-PAPER-CLAIM — neither branch is scored; that is the whole of what remains *Run*: `python3 receipts/P15_CR_cosmology/C34_the_deliverable_is_a_pair.py`

C35_the_reference_is_F3 [OK]

sec:tensions (THE REFERENCE FORK SETTLES ITSELF AND AGAINST THIS LINES FRAMING: BIC uses only DIFFERENCES, so F3 (same-instrument) is the score and the CAMB-206.4 comparison charges CR an instrument floor CAMB never pays). *Computes*: asserts L-147s floor definition and pipeline-wiring clauses verbatim, shows dBIC contains only a χ^2 difference, and confirms the threshold is invariant under the pair choice (7 checks) *Bound*: NOT-A-PAPER-CLAIM — corrects r2709-r2711s statement of what the threshold scores against; the threshold itself is unchanged *Run*: `python3 receipts/P15_CR_cosmology/C35_the_reference_is_F3.py`

C36_the_negative_is_the_envelope [OK]

sec:envelope-consequence (SUPERSEDED r2725 (the check shared the runs own n.s-held assumption, so agreeing to the same order confirmed only that both held it fixed) — L-814s NEGATIVE IS NOT AN ARTEFACT: P15s own published envelope, scored against cosmic variance on 215 bins with no pipeline or fit, gives $\chi^2 \sim 1.6e5$ against cc54s measured $5.05e4$ — the same order). *Computes*: evaluates the papers own $\exp[-(1/LD)^2(r^2-1)]$ across the band, scores it against cosmic-variance errors, reads cc54s floor finding from their receipt, and pins L-147s scope clause (6 checks) *Bound*: NOT-A-PAPER-CLAIM — no framework verdict; PO-7 protected and F5 reserves it *Run*: `python3 receipts/P15_CR_cosmology/C36_the_negative_is_the_envelope.py`

C37_the_comparison_is_not_matched [OK]

sec:envelope-consequence (L-814s COMPARISON IS NOT MATCHED and its number is WITH-DRAWN as a result: the CR arm was held at k=2 with n.s fixed at Λ CDMs fitted value, and freeing the tilt alone removes 15.6x). *Computes*: computes the envelope χ^2 with the tilt held and free, shows the threshold barely moves for the extra parameter, and reads cc54s own relative statement from their receipt (5 checks) *Bound*: NOT-A-PAPER-CLAIM — the k=2 spec was this lines at r2710; a matched comparison is not claimed to pass *Run*: `python3 receipts/P15_CR_cosmology/C37_the_comparison_is_not_matched.py`

C38_the_comparison_scores_derivations [OK]

sec:tensions (PO-10s NON-FOREIGN COMPARISON IS A SCORE ON DERIVATIONS: a derivation has no k, so scoring θ_* , r, z_eq and the 9/10 transfer against measurement imports

no parameter vector — and this receipt reports NO score, because the draft that did invented every sigma). *Computes:* pins P15s derived-vs-measured pairs and its own naming of Omega_m and A_s as fitted, and asserts its own refusal to score (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no verdict on CR; the row owes published uncertainties and a derived-quantity list *Run:* python3 receipts/P15_CR_cosmology/C38_the_comparison_scores_derivations.py

C39_the_derived_list_read [OK]

sec:tensions (PO-10s DERIVED-QUANTITY LIST READ from P15s own marker language: four pairs each with the papers own accuracy clause (theta_* 302.2/301, r_s 146.4/145.4, 1+z_eq 3399, eta 1.99 to one per cent) and three exclusions in the papers own words). *Computes:* pins all four derived pairs and all three fitted-exclusions verbatim, and the high-l ratios no-free-parameter clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing is scored; the published uncertainties remain owed and are not this lines to invent *Run:* python3 receipts/P15_CR_cosmology/C39_the_derived_list_read.py

C40_the_pair_was_quoted_against_a_rounding [OK]

sec:tensions (PO-10s FIRST PAIR MUST TAKE THE RECEIPTS VALUE, NOT THE PROSE: P15 writes "the measured ~301" (tilde present, honestly approximate) while P15_zonset_determinations carries and asserts L_A = 301.76 — and the gap is 0.44, not 1.20). *Computes:* quotes both the prose and the receipts assertion, computes both gaps, and carries its own first drafts correction (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no paper edit owed, the tilde is there; nothing is scored and the uncertainty is still not this lines *Run:* python3 receipts/P15_CR_cosmology/C40_the_pair_was_quoted_against_a_rounding.py

C41b_a_tilde_on_a_settled_value_is_a_stale_hedge [OK]

sec:tensions, sec:coherence (A TILDE ON A SETTLED VALUE IS A STALE HEDGE: ~301 became 301.76 (asserted by its receipt) and ~8% became 8.2% in NINE places (computed on CAMBs exact ionization history), three of them while calling it "a real, computed effect"). *Computes:* confirms both figures are asserted by receipts, that no tilde survives on either, and that legitimately approximate tildes are untouched (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing scored; not every tilde is stale, only those on figures a receipt asserts *Run:* python3 receipts/P15_CR_cosmology/C41b_a_tilde_on_a_settled_value_is_a_stale_hedge.py

C41_a_tilde_is_stale_when_nothing_competes [OK]

sec:tensions (A TILDE IS STALE WHEN ONE RECEIPT ASSERTS AND NONE COMPETES: ~301 failed that test and P15 now reads 301.76; ~8% passed it — C8 derives +10.83% analytically and NAMES the disagreement with damping_ratio_cleans +8.2% — so nine edits were reverted). *Computes:* confirms the 301.76 edit landed and the nine tildes remain, pins C8s computed +10.83% and its own naming of the conflict (6 checks) *Bound:* PAPER-CLAIM (301.76 only) — the 8% figure is left tilde-marked and its discrepancy is filed as owed *Run:* python3 receipts/P15_CR_cosmology/C41_a_tilde_is_stale_when_nothing_competes.py

C42_the_cancellation_needs_xe_identical [OK]

sec:coherence (THE DAMPING DISCREPANCY DIAGNOSED: C8s cancellation needs x_e identical in both cosmologies, and a different H(a) changes the recombination HISTORY not merely its epoch — Lambda, the upper limit, the radiation density and arithmetic all eliminated). *Computes:* reproduces C8 to six figures, varies Lambda / z_rec / rho_r and shows none accounts for the gap, and pins C8s own PROVIDED clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — neither number is adjudicated; the x_e response is a calculation neither receipt has done *Run:* python3 receipts/P15_CR_cosmology/C42_the_cancellation_needs_xe_identical.py

C43_the_gap_is_numerical_not_xe [OK]

sec:coherence (r2750 WITHDRAWN — THE DAMPING GAP IS NUMERICAL, NOT x_e: a pure H probe moves z_* by only 0.195% (~0.05pp on the ratio) against a 1.86pp gap, while the CAMB receipt prints a 7.1% miss against CAMBs own thetad on its OWN LCDM arm). *Computes:* runs CAMB at five N_eff values as an H-only probe, scales the response by r2750s own measured sensitivity, and pins the receipts self-reported GATE miss (6 checks) *Bound:* NOT-A-PAPER-CLAIM — C8 is not confirmed, and the tilde stays: two receipts still disagree, one now with a KNOWN defect *Run:* python3 receipts/P15_CR_cosmology/C43_the_gap_is_numerical_not_xe.py

C44_the_gap_is_truncation [OK]

sec:coherence (THE DAMPING GAP IS TRUNCATION: the r_D ratio climbs 8.33% -i 9.94% with the upper limit and converges only by z~5e4, while the CAMB receipts grid stops at z=12000

- and a normalisation factor CANCELS in a ratio, which withdraws r2751). *Computes:* computes the ratio at six upper limits, shows convergence, and demonstrates that a shared constant divides out (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the receipts grid is reasonable for its own purpose; what is claimed is that its RATIO inherits a truncation bias *Run:* `python3 receipts/P15_CR_cosmology/C44.the_gap_is_truncation.py`
- C45_xe_does_not_cancel_from_a_ratio_of_integrals [OK]
sec:coherence (THE DAMPING DISCREPANCY RESOLVED: x.e does NOT cancel from a ratio of INTEGRALS even when identical — toggling it on the same range moves the ratio +9.94% to +8.37%, the whole gap — so C8s cancellation step is a real error). *Computes:* toggles x.e on a fixed range and measure with CAMBs own history, shows a shared CONSTANT does cancel exactly while x.e does not, and pins C8s every-constant-outside claim (6 checks) *Bound:* NOT-A-PAPER-CLAIM — which figure is P15s signature is not settled; C8s reduction and baryon correction stand *Run:* `python3 receipts/P15_CR_cosmology/C45_xe_does_not_cancel_from_a_ratio_of_integrals.py`
- C46_the_nine_are_right_and_the_one_is_wrong [OK]
sec:tensions (THE 9.4% TRACED: P15 -i C10 (+9.26%) -i C9, which divides r.D 1.1015 by r.s 1.0067 — the ANGLE ratio, correctly, but from an r.D that omits the x.e weighting. Corrected it lands at +7.6% to +8.2%, so the nine ~8% were RIGHT and the one 9.4% was the error). *Computes:* pins the citation chain P15-iC10-iC9 and reruns C9s own division on three r.D ratios (6 checks) *Bound:* PAPER-CLAIM — ten instances set to 8.2%, and the tilde removed per r2749s test now that nothing competes *Run:* `python3 receipts/P15_CR_cosmology/C46_the_nine_are_right_and_the_one_i`
- C47_the_neutrinos_were_missing [OK]
sec:coherence (THE +5.66% RECONCILED AND IT WAS THIS LINES ERROR: r2752 used photons alone where the receipt uses Ogama + Onu (a 69% underestimate of what sources H) — and the CR arm has no radiation at all, so the omission hit ONE ARM ONLY. With neutrinos in, the ratio is 1.0896 against the receipts 1.0897). *Computes:* runs the full Hu-Sugiyama for both rates under two radiation densities with CAMBs own x.e, and confirms the absolute r.D difference cancels from the ratio (6 checks) *Bound:* NOT-A-PAPER-CLAIM — only the RATIO is claimed to four figures; the absolute r.D still differ by ~3% from grid and quadrature *Run:* `python3 receipts/P15_CR_cosmology/C47_the_neutrinos_were_missing.py`
- C48_the_pairs_are_three [OK]
sec:tensions (PO-10s PAIRS ARE THREE, NOT FOUR: r.s and 1+z.eq hold against asserting receipts (C9 145.38, DESI 3402), but the onset ratio is an IDENTITY ON INHERITED INPUTS — "the quoted value to one per cent, from standard thermodynamics" — same class as N_eff and A_s). *Computes:* finds the asserting receipt for each measured value, pins the thermodynamics clause, and records the 1.9968 near-miss where another receipt asserts a different quantity with the same digits (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the onset ratio is not wrong, only not a TEST; nothing is scored *Run:* `python3 receipts/P15_CR_cosmology/C48_the_pairs_are_three.py`
- C49_the_row_has_a_likelihood_not_a_lookup [OK]
sec:tensions (PO-10 DOES NOT NEED ERROR BARS — IT HAS A LIKELIHOOD: plik_lite over 215 TT bins with a covariance and a banked LCDM fit the pipeline calls wired. What blocks it is recorded in the same file — the CONTROL arm lands at $\chi^2/\text{dof} \sim 100$). *Computes:* pins the instrument, its calibration standard, its self-recorded F4 defect, and F5s protection of the verdict (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the control defect is not diagnosed here, and no verdict on CR is drawn *Run:* `python3 receipts/P15_CR_cosmology/C49_the_row_has_a_likelihood_not_a_lookup.py`
- C50_the_control_is_at_seven_not_a_hundred [OK]
sec:tensions (THE CONTROL ARM IS AT 7.14, NOT ~100: the receipts F4 prose carried the c54.172 floor while its F6 PRINT reports $1320.5/185 = 7.14$, and F6 says the floor fell 103 -i 22.5 -i 28.6 -i 7.1, a factor of 14). *Computes:* pins the corrected prose, the f-string print fields, the fall history, the 185-vs-215 bin difference and F6s unchanged verdict (6 checks) *Bound:* NOT-A-PAPER-CLAIM — 7.14 is not acceptable either; both arms remain outside the discriminating regime *Run:* `python3 receipts/P15_CR_cosmology/C50_the_control_is_at_seven_not_a_hundred.py`
- C51_the_dropped_bins_are_the_damping_tail [OK]
sec:tensions (THE THIRTY DROPPED BINS ARE ell 1759-2508 — THE DAMPING TAIL: the likelihood arms model ends near ell 1760 and nan-drops the rest, discarding exactly the region P15 names as its exposed edge, where the suppression runs 24% to 42%). *Computes:* reads the bin edges

from the likelihoods own arrays, locates the cut at which 185 survive, and profiles the damping suppression across the dropped range (6 checks) *Bound*: NOT-A-PAPER-CLAIM — the ell limit is a cost not a mistake, and which way the omitted bins would land is the point of running them *Run*: `python3 receipts/P15_CR_cosmology/C51_the_dropped_bins_are_the_damping_tail.py`

C52.the_banked_spectra_fail_the_sampling_guard [OK]
sec:tensions (THE BANKED SPECTRA FAIL THE INSTRUMENTS OWN SAMPLING GUARD: LMAXL=2000 NK=260 gives 2.5 points per Bessel period and prints UNDER-SAMPLED — and the guard says the source comb stays correct while the projection aliases, so no comb-level check could catch it). *Computes*: fixes the banked configuration from the npz shape, computes the sampling ratio and projection memory across four NK values, and pins the guards own wording (6 checks) *Bound*: NOT-A-PAPER-CLAIM — whether aliasing moves the χ^2 enough to matter is UNMEASURED, and measuring it needs a run that will not fit in 3.7 GB *Run*: `python3 receipts/P15_CR_cosmology/C52_the_banked_spectra_fail_the_sampling_guard.py`

C53.the_k_range_fixes_one_arm_only [OK]
sec:tensions (THE k-RANGE FIXES ONE ARM ONLY: raising L_max 26% drops the control 7.14 -; 3.81 on the SAME 185 bins while CR moves 0.4% — a response ratio over twenty to one, so r2760s undiagnosed factor of seven is a k-truncation and CRs 281 is not). *Computes*: computes both arms responses and their ratio, checks F_3 widens in the direction one-arm improvement predicts, and labels the two-point trend as an extrapolation (6 checks) *Bound*: NOT-A-PAPER-CLAIM — radiation is the first candidate not a demonstrated mechanism; the convergence trend rests on TWO points *Run*: `python3 receipts/P15_CR_cosmology/C53_the_k_range_fixes_one_arm_only.py`

C54.the_281_is_the_prediction_being_measured [OK]
sec:tensions (CRs 281 IS THE ORDER P15s OWN PREDICTION PRODUCES: scoring the damping residual over 185 bins gives χ^2/dof from 204 ($\sigma=1\%$) to 5099 ($\sigma=0.2\%$), so 281 sits INSIDE that range and BELOW the value at Plancks quoted errors — the arm is scoring the effect P15 predicts). *Computes*: pins P15s no-free-parameter high-ell formula, scores its residual at three error scales, pins F5, and names the SHAPE test that would settle it (5 checks) *Bound*: NOT-A-PAPER-CLAIM — an ORDER comparison against a flat sigma, not a derivation against the plik covariance; supplies a candidate, does not close the row *Run*: `python3 receipts/P15_CR_cosmology/C54_the_281_is_the_prediction_being_measured.py`

C55.the_shape_test_says_peak_misalignment [OK]
sec:tensions (THE SHAPE TEST RUN: the CR arms residual is neither a damping-profile match nor flat — it OSCILLATES, 0.237 to 2.375, correlation -0.296, dominant period 317 against l_A = 301. The 281 is PEAK MISALIGNMENT, and r2786 is withdrawn by its own named test). *Computes*: computes the banked arms ratio, its correlation with P15s profile, its dominant Fourier period against l_A, and the l_A-agrees / r_s-differs contrast (5 checks) *Bound*: NOT-A-PAPER-CLAIM — the damping signature is not shown absent, only non-dominant; physical vs numerical phase slip is the open question *Run*: `python3 receipts/P15_CR_cosmology/C55_the_shape_test_says_peak_misa`

C56.la_is_computed_and_its_residual_is_too_small [OK]
sec:tensions (l_A IS COMPUTED NOT CALIBRATED (pi D_M/r_s reproduces the stored value in both arms) and its survival is a real cancellation P15 claims in advance — but the +0.075% residual explains under 1% of r2787s measured swing, two orders too small). *Computes*: checks pi D_M/r_s against the stored l_A in both arms, sizes the D_M and r_s moves, pins P15s met-from-the-other-end clause, and propagates the residual against the measured swing (5 checks) *Bound*: NOT-A-PAPER-CLAIM — the cancellation is observed not re-derived; amplitude is what remains after two eliminations, a candidate not a finding *Run*: `python3 receipts/P15_CR_cosmology/C56_la_is_computed_and`

C57.the_cr_peaks_are_not_at_its_own_la [OK]
sec:tensions (THE CR ARMS PEAKS ARE NOT WHERE ITS OWN l_A SAYS: mean spacing 258 against a stored l_A of 301.6 (ratio 0.855) while LCDM sits at 0.995 — so the stored quantities are consistent with each other and NOT with the computed C_ell). *Computes*: finds peaks in both banked arms, compares mean spacing against each arms stored l_A, and sizes the mismatch against r2788s residual (5 checks) *Bound*: NOT-A-PAPER-CLAIM — which of D_M, r_s or the projection is wrong is not established; the CR spacing could be a real feature of a transfer on the geometric rate *Run*: `python3 receipts/P15_CR_cosmology/C57_the_cr_peaks_are_not_at_its_own_la.py`

C58.the_stored_rs_moves_the_wrong_way

[OK]

sec:tensions (THE STORED CR r_s MOVES THE WRONG WAY: removing radiation must RAISE the sound horizon (146.5 - 245.2 Mpc integrated), the peak spacing independently demands 158.35, and the stored value is 135.46 — BELOW LCDMs 144.53). *Computes:* solves the spacing for each candidate, integrates r_s with and without radiation, and shows $\pi D_M/r_s$ on both stored values reproduces the stored ℓ_A exactly (5 checks) *Bound:* NOT-A-PAPER-CLAIM — 158.35 and 245.16 are not proposed as the right value; what is established is the DIRECTION and that the stored value has the wrong sign of deviation *Run:* python3 receipts/P15_CR_cosmology/C58.the_stored

R1.the_correction_stopped_at_the_sentence_and_the_number_it_defines_was_left_behind [OK]

sec:envelope-consequence (r2755 CORRECTED THE SENTENCE AND NOT THE NUMBER THE SENTENCE DEFINES: ‘ θ_D/θ_* ’ was corrected from 9.4% to 8.2% and ‘sec:envelope-consequence’ s r , one paragraph later, IS that ratio and still read 1.093 — so the paper stated one ratio twice, differently, and the second is the one the prediction is made of. Corrected from three agreeing routes to $r=1.0816$: the ratio at ℓ_D moves $0.823 \rightarrow 0.844$, at $1.5\ell_D$ $0.645 \rightarrow 0.682$, $\ell_D(\ell_{\text{rm CR}})$ $1281 \rightarrow 1294$. **FIFTEEN receipts carried it — eight SILENTLY (exiting zero, six of them evaluating the envelope with the wrong value) and seven LOUDLY (failing every full run since r2755 — and they are the seven that PRODUCED the correction).** CONTROL: C10’s gate was built so the paper and the receipt “cannot drift apart silently in either direction”, and it worked — they were consistent with each other and both wrong.). *Computes:* COMPUTES: the identity $r=\theta_D/\theta_*$ read from C10’s own assertion message; the corrected value from three routes and the four quantities that follow; the fifteen receipts split by failure mode and counted at the commits where they exist; the C41 double-allocation resolved as one rule at two states; and the register files deliberately left alone (21 checks) *Bound:* **bound: 1.0816 is NOT re-derived here** — it is read from ‘P15_damping_ratio_clean’, ‘C46’ and the corrected sentence, and checked for agreement. **Not claimed that the envelope is right**, only that it now runs on the ratio the paper’s own preceding paragraph derives. **The correction FAVOURS the construction (16% suppression at ℓ_D rather than 18%) and is flagged for it**; it would apply identically the other way. ‘PO-7’ is ‘sec:refit-bound’/‘sec:coherence’ and is untouched. *Run:* python3 receipts/L557.the_correction_stopped_at_the_sentence/R1.the_correction_stopped_at_the_sentence_and

P1.the_pin_test_run_the_spacing_follows_the_pin_so_the_ratio_does_not

[OK]

sec:tensions (THE PIN TEST, RUN — the runs the observer line’s container killed twice. **The peak spacing tracks the one fitted number PROPORTIONALLY** (‘LATARG’ 280/301.6/320 $\rightarrow$ mean spacing 229.3/245.3/261.3), so spacing/ $\ell_A = 0.8190/0.8134/0.8167$ — **0.69% spread across a 14% pin range**, against the **13.4%** a constant-spacing arm would give: a factor of twenty, so the measurement DISCRIMINATES. $\Rightarrow$ **The routed dichotomy is inverted: following the pin is exactly what makes the RATIO — which is what the deficit is — independent of the pin. THE DEFICIT SURVIVES THE FIT.** Also: ‘KCONT=1’ at NK=320 returns the same four peaks, discharging the instrument’s own standing sampling caveat; and ** $\ell_1 = 172$ at all three pins**, so ℓ_1/ℓ_A is pin-dependent while the spacing ratio is not. LCDM control on the same grid: 0.9909, reproducing the sky to 6 in ℓ). *Computes:* COMPUTES: five banked runs parsed from their own printed blocks; the spacing ratio and its spread; the CONSTANT-spacing alternative computed as the control and the discrimination factor between them; the KCONT identity and its sampling-guard margin; ℓ_1 ’s invariance and ℓ_1/ℓ_A ’s spread; and the LCDM arm checked against the sky before being used (21 checks) *Bound:* NOT-A-PAPER-CLAIM — **and the declaration is deliberate: this is a measurement on the instrument’s own fitted parameter, delivered to a routed question, and whether it lands in ‘sec:tensions’ is the observer line’s call under ‘PO-7’.** A node that banked it into the paper on its own initiative would be taking a step toward the verdict the row protects. **bound: this reports a number and does not convert it into a verdict — ‘PO-7’ is protected.** Not that r2789’s 258 was wrong (different grid; the control was re-run here rather than compared across grids). Not that peaks are known better than the ℓ -grid: ‘LSTEP=8’, ± 4 per peak, ± 8 per spacing — the invariance claim is $20\times$ outside that and the individual spacings are not. Not that the deficit is explained, only that the fitted number does not produce it. *Run:* python3 receipts/L562.the_pin_test/P1.the_pin_test_run_the_spacing_follows_the_pin_so_the_ratio_does_not.py

P15.the_driving_is_bought_by_decay_during_oscillation

[OK]

sec:transmission (THE DRIVING SHIFT IS BOUGHT BY POTENTIAL DECAY OCCURRING

WHILE A MODE OSCILLATES, so a mode whose oscillation begins after the potential has settled inherits the UNDRIVEN phase. ** Coupled photon/matter/potential solve on a matter-and-radiation background, with the potential closed by the ENERGY CONSTRAINT rather than evolved and the acoustic turnover taken at the first zero of the photon velocity that BEGINS A SEQUENCE SPACED BY π – the free oscillator’s own spacing, which the undriven gate validates at 1.0000 on both arms. ** SCOPE: the ASYMPTOTIC statement only, $Q \rightarrow 1$ at large k , which is what P15 and P07 assert; NOT the low- k values, and NOT a magnitude (neutrino free-streaming omitted, $\sim 15\%$ low). A sign flip in the initial amplitude is a π phase shift and moves WHICH ZERO IS FIRST by one half-period, so four initial signs agree to 0.012 at $k=160$). *Computes:* COMPUTES: a coupled photon/matter/potential solve on a matter-and-radiation background, the potential closed by the ENERGY CONSTRAINT rather than evolved; the UNDRIVEN gate validated at 1.0000\$ on both arms before use; the driven arm asserted to FALL toward 1\$ from above and to reach the free-oscillator value to 0.02\$ at large \$k\$; the control asserted well below 1\$ and flat in \$k\$ to 0.10\$; the two shapes asserted qualitatively distinct by a factor of two in range; and the robustness leg showing a factor of two in the radiation budget barely moves it. Re-run r3537, ALL PASS *Bound:* SCOPE, and the receipt states it: the ASYMPTOTIC statement only, \$Q\{ to 1\$ at large \$k\$, which is what ‘P15’ and ‘P07’ assert; **NOT the low-\$k\$ values and NOT a magnitude** — neutrino free-streaming is omitted, some 15\{\}%\$ low *Run:* python3 receipts/P15_CR_cosmology/P15_the_driving_is_bought_by_decay_during_oscillation.py

P03_acceleration_is_slice_curvature

[OK]

sec:curvature · rem:twocritical · the bend (on ANY member of the energy family (E cancels: $(dr/d\tau)^2 = E^2 - f$, and E is a constant of the motion) $d^2r/d\tau^2 = -f'/2 = r \cdot K \cdot G$ identically, so THE SIGN OF THE SLICE’S GAUSSIAN CURVATURE IS THE SIGN OF THE COMOVING ACCELERATION – and the flat locus is the deceleration-to-acceleration turnover, which on Narai is the seam and in standard terms is $\rho_m = 2 \rho_\Lambda$ ($\cosh^2 w = 2$ $\Rightarrow \sinh w = 1/\sqrt{2}$, the seam’s phase). A recent cosmological epoch: 7.06 Gyr at $H_0=73$, 7.65 at 67.4, preceding matter-Lambda equality [borrowed from P3 · P7 · P8]). *Computes:* the identity asserted; the $K \cdot G$ zero solved and matched to $(M \alpha^2)^{(1/3)}$; $\cosh^2 w = 2$ $\Rightarrow \sinh w = 1/\sqrt{2}$ proved by algebra after simplify() failed on the closed forms; epochs at both H_0 *Bound:* bound: identities on the Narai $E=1$ worldline; no causal claim between seam and turnover – they are one locus; dating inherits H_0 *Run:* python3 receipts/P03_SdS_slicing/P03_acceleration_is_slice_curvature.py

K1_the_ladder_waiver_is_checked_against_the_continuum

[OK]

‘P15’ *sec:coherence* (**THE ALIASING WAIVER IS CHECKED — the measurement the gate itself names, never run until now.** ‘alias_gate’ demands four points per Bessel period; the CR arm gets 2.3\$ and the gate waives itself on the ladder being physical, adding “it is only not aliasing if the answer does not depend on it. Run KCONT=1 to check.” $\Rightarrow$ **Run**: the discrete ladder and the KCONT=1 continuum give peaks 171.2, 406.5, 634.6, 917.9\$ against 171.2, 406.6, 634.6, 917.8\$ — **the first peak moves by 0.00\$ and no peak by as much as one multipole.** $\Rightarrow$ *So the conditional the gate states is MET. Not that the sampling is adequate — it is not, 2.3 against 4.0 — but that the answer does not depend on it.* The direct form of the bound ‘L-274’ established analytically, and independent of it. [borrowed from PO-13]). *Computes:* two stored spectra compared peak by peak *Run:* python3 receipts/L280_the_ladder_waiver_is_checked/K1_the_ladder_waiver_is_checked

D1_the_diagnosis_is_the_driving_and_the_driving_is_the_rate

[OK]

‘P15’ *sec:driving*, ‘P7’ *frontier:scalar* (**PO-13 ANSWERED: NONE OF THE THREE LAYERS.** By elimination the offset is not the transfer (a 0.16\{\}%\$-validated line-of-sight transfer leaves the arm unchanged to four figures; across four fidelities the control spans 23\{\}times\$, the arm 5\{\}%\$), not the seam’s freedom (forced to $\{0, \pi\}$), a band of 0.207\$ against 0.615\$, the control outside), and not the spacing (slopes 1.003\$ vs 0.976\$). It is the DRIVING at 98\{\}%\$ — undriven the arms agree to 0.013\$, driven they differ by 0.602\$. $\Rightarrow$ And the mechanism is derived: the k^2 in the forcing cancels the k^{-2} in its duration, so Qk is constant, 0.01040\$ predicted against 0.01037\$ measured. ***The standard shift is universal because every mode crosses during radiation domination; a geometrically fixed rate has none — the Hubble resolution and the phase offset are ONE FACT.*** [borrowed from PO-13]). *Computes:* elimination across four layers; the impulse cancellation symbolic *Run:* python3 receipts/L272_the_misplaced_phase_is_the_radiation_free_rate/D1_the_diagnosis_is_the_driving_and_th

BRANCHPT_transmission_character

[OK]

rem:phase-open (BRANCHPT — the transmission character of the BRANCH POINT $r=0$, on its

own terms. WHY (r2154). P15 1.525 says "The cosmogenesis branch point, BEING the degenerate Narai horizon, is a faithful scale-free transmitter", and prop:transmission argues from the tortoise integral at a DOUBLE ROOT ($p=2$, $\kappa=0$). **PART 2, from the receipt-vs-sentence audit: the PEAK-POSITION EXCLUSION, which is what P15 actually cites this file for and which the file did not compute** — the collapse leg's own conformal time gives $\eta_{\text{bp}} \sim 5e3-1e4$ Mpc, so $r_s^{\text{eff}} = r_s + \eta_{\text{bp}}/\sqrt{3} = 3.0-5.9e3$ Mpc, and reproducing $L_A = 301.8$ would demand $D = 2.9-5.7e5$ Mpc against the actual $1.4e4$, a factor of **21-41**. *PART 1 shows the crossing ADDS no divergent phase; PART 2 shows an inherited phase cannot survive. The paper's sentence needs the second.* [borrowed from P15 (+P7)]. *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15_CR_cosmology/BRANCHPT_transmission_character.py

H1_the_low_multipole_deficit_is_two_effects_and_the_ladder_imprint_dies_where_the_transfer_is_narrow [OK]

corpus integrity (**THE HARMONIC-ANALYSIS BAKE — second of the four fields 'L-272's re-survey left outstanding, thrown r3166.** **BOUNCED, and largely:** the field's first question is why a closed- S^3 source is projected through FLAT spherical Bessel functions, and 'P15' answers it in its own text — the distance slicing is flat, so only the source modes carry the closed- S^3 quantization — while its measure $\$w_L=(L+1)/(L(L+2))\$$ is derived twice, as degeneracy over per-mode power and as $\$d\{\ln k_L/dL\$,$ both agreeing to machine precision. **BIT:** the deficit reported as one geometric effect splits exactly into a **FLOOR** (modes below $L=\{=2\}$ removed) and a **LADDER COARSENESS**. At $\ell=5-6$ the second contributes $\$+0.38\$$ against a floor of $\$0.27\$-\$0.53\$$ — comparable, and in the opposite direction — and it is what sets the recovery multipole the paper quotes ($\ell=8$ with it, $\ell=10$ without). Under $\pm 2\%$ in r_0 the floor moves $\$-2.4\%$ to $\$-6.2\%$ and the full result $\$+27.8\%$ to $\$-43.3\%$: **the location is the floor and is robust as claimed; the depth is the ladder and is volatile**, which gives the paper's own qualitative split a mechanism. **AND A SELF-WAIVED GATE BECOMES A BOUND:** 'P15' records that the ladder "samples the projection below the rate the instrument's own aliasing gate demands, and that gate waives itself on the claim that the ladder is physical." The imprint dies at $\ell \leq k_{2D} \cdot c^3 \cdot \sigma$; to reach the first acoustic peak the transfer would need to draw power over a factor of nine in k . **Safe with a wide margin, by measurement.** [borrowed from —]). *Computes:* COMPUTES: the spherical-Bessel completeness identity $\sum_{\ell}(2\ell+1)j_{\ell}(x)^2=1$ to ten figures and the exact plateau $\int j_{\ell}^2 d\ln x=1/(2(\ell(\ell+1)))$ reproduced on the fixed grid to six, both **before anything is used**, the measure against $\$d\{\ln k_L/dL\$ and against degeneracy-over-power at five L ; the paper's own D_C/r_0 to 0 limit; the floor/discreteness decomposition across $\ell=2-12$; each half's response to $\pm 2\%$ in r_0 ; the recovery multipole with and without the discreteness; and the imprint against transfer width at four σ , with the scaling rule checked against all four rather than extrapolated (15 checks). *Bound:* NOT-A-PAPER-CLAIM — corpus integrity; no paper edited. **NOT claimed:** that the deficit is wrong — every number touched is reproduced or confirmed, the paper's own limit included; that these are predictions about the sky — the decomposition is the pure Sachs–Wolfe limit while the paper's depths come from a genuine Boltzmann transfer it says are milder, and the two are not compared; that the real transfer's width is measured — σ is a Gaussian-in- $\ln k$ stand-in and measuring $\Delta_{\ell}(k)$ is the owed next step; that the aliasing gate should stop waiving — only that the waiver now has a margin; that the low- ℓ oscillation is derived rather than measured. *Run:* python3 receipts/L274_the_harmonic_bake/H1_the_low_multipole_deficit_is_two_effects_and_the_ladder_imprint.

Appendix L The Knowledge Ledgers

Each claim marked ^L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

OPTICS_LENSING_LEDGER.md [field bake]
observational optics and lensing against CR — confirmations, and the dimension clause they forced

SPECTRAL_THEORY_LEDGER.md [field bake]
The spectral-theory field bake — what bit, what bounced, and the boundary. Tier B’s largest never-thrown field ($\times 189$ on the reach measure), verified as NOT covered by the harmonic ledger, which mentions spectral, self-adjoint and deficiency zero times

NUMERICAL_ANALYSIS_LEDGER.md [field bake]
The numerical-analysis field-bake ledger — what bit, what bounced, and where the boundary is

INTEGRABLE_SYSTEMS_LEDGER.md [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is

References

- [1] D. Janzen, *A Critical Look at the Standard Cosmological Picture*, FQXi essay contest (2012); arXiv:1301.5980.
- [2] D. Janzen, *General relativity’s constraint algebra as the symmetric-space structure of a de Sitter substrate: the action Lie algebroid of the symmetry-reducible sector, and the problem of time’s “wrong sign” as the substrate’s coset signature*, companion paper (P12).
- [3] D. Janzen, *The event horizon is a metric singularity: a missing definition in general relativity*, companion paper (P1).
- [4] D. Janzen, *The de Sitter substrate and its slicing curve: horizons as turning points, and de Sitter and Schwarzschild as two readings of one slicing*, companion paper (P3).
- [5] D. Janzen, *The Schwarzschild–de Sitter description groupoid: generators, relations, and Schwarzschild as the asymmetric realisation*, companion paper (P5).
- [6] D. Janzen, *Covariance of geometries over the de Sitter substrate: the slicing operator, the vacuum kernel, and matter as the bend of the cut*, companion paper (P8).
- [7] D. Janzen, *The range of the de Sitter slicing operator: surjectivity onto the symmetry-reducible sector of general relativity, the Kerr–NUT–(A)dS vacuum kernel, and the wall of inhomogeneity*, companion paper (P9).
- [8] D. Janzen, *Collapsed matter must become a universe: the necessary and sufficient augmentation of general relativity into a description of structural existence, proven for gravitational collapse of any symmetry*, companion paper (P7).
- [9] D. Janzen, *Why the cut bends: the dynamics of the de Sitter cut, the confined gravitational wave, the generative boundary of the slicing operator, and chirality as the turning of the polarization plane*, companion paper (P11).
- [10] D. Janzen, *The modern parallax: the redshift-isotropy floor, the empirical forcing of cosmic time and uniform expansion, and the measured resolution of the relativity of simultaneity*, companion paper (P4).
- [11] D. Janzen, *The shadow of existence: scientific theory-choice as an empirically grounded discipline, calibrated on the historical record*, companion paper (P6).
- [12] V. Pavlidou and T. N. Tomaras, *Where the world stands still: turnaround as a strong test of Λ CDM cosmology*, JCAP **09** (2014) 020.

- [13] D. Janzen, *On the ontological foundations of Cosmological Relativity*.
- [14] D. Janzen, *A critical look at the standard cosmological picture*, in *Questioning the Foundations of Physics: Which of Our Fundamental Assumptions Are Wrong?*, Springer (2015) 103–130.
- [15] D. Janzen, *A Solution to the Cosmological Problem of Relativity Theory*, PhD thesis, University of Saskatchewan (2012).
- [16] D. Janzen, *Reached through the imaginary, real by construction: the maximally symmetric substrate as the geometric–ontological core*, companion paper (p0).
- [17] D. Janzen, *Cosmogenesis: the Big Bang as a synthesis of Cosmological Relativity’s deductively forced consequences, and the primordial light-element abundances it produces*, companion paper (P16).
- [18] D. Janzen, *The fermion sector of Cosmological Relativity: three chiral generations from the maximally-symmetric slicing structure*, companion paper (P14).
- [19] R. M. Wald, *Asymptotic behavior of homogeneous cosmological models in the presence of a positive cosmological constant*, Phys. Rev. D **28** (1983) 2118.
- [20] S. Alam et al. (BOSS Collaboration), *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, Mon. Not. R. Astron. Soc. **470** (2017) 2617.
- [21] DESI Collaboration, *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, arXiv:2503.14738 (2025).
- [22] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641** (2020) A6.
- [23] A. G. Riess et al., *A comprehensive measurement of the local value of the Hubble constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ uncertainty from the Hubble Space Telescope and the SH0ES team*, Astrophys. J. Lett. **934** (2022) L7.
- [24] M. Moresco et al., *A 6% measurement of the Hubble parameter at $z \sim 0.45$: direct evidence of the epoch of cosmic re-acceleration*, J. Cosmol. Astropart. Phys. **05** (2016) 014.
- [25] J. Silk, *Cosmic black-body radiation and galaxy formation*, Astrophys. J. **151** (1968) 459.
- [26] R. K. Sachs and A. M. Wolfe, *Perturbations of a cosmological model and angular variations of the microwave background*, Astrophys. J. **147** (1967) 73.
- [27] P. Ginsparg and M. J. Perry, *Semiclassical perdurance of de Sitter space*, Nucl. Phys. B **222** (1983) 245.
- [28] G. Lemaître, *Cosmological Application of Relativity*, Rev. Mod. Phys. **21** (1949) 357.
- [29] E. R. Harrison, *Fluctuations at the threshold of classical cosmology*, Phys. Rev. D **1** (1970) 2726.
- [30] Ya. B. Zel’dovich, *A hypothesis, unifying the structure and the entropy of the Universe*, Mon. Not. R. Astron. Soc. **160** (1972) 1P.
- [31] G. W. Gibbons and S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Phys. Rev. D **15** (1977) 2738.
- [32] A. S. Eddington, *The Expanding Universe*, Cambridge University Press (1933).
- [33] D. Janzen, *The canonical problem of time as a category error: an empirically forced cosmic time, deparametrization, the dissolution of frozen dynamics, and a graviton quantization the de Sitter horizon closes without a free parameter—the seam where $\hbar$ enters gravity—in Cosmological Relativity*, companion paper (P10).
- [34] D. Janzen, *The Schwarzschild and de Sitter circle: the intrinsic geometry of one homogeneous ring—the event horizon and the curvature singularity at $r = 0$ its two poles*, companion paper (P2).
- [35] H. Nariai, *On a new cosmological solution of Einstein’s field equations of gravitation*, Sci. Rep. Tôhoku Univ. Ser. I **35** (1951) 62; reprinted in Gen. Rel. Grav. **31** (1999) 963.
- [36] P. J. E. Peebles, *Recombination of the primeval plasma*, Astrophys. J. **153** (1968) 1.